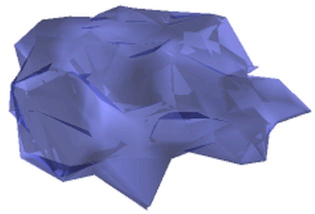
BAYESIAN NONPARAMETRIC GLOTTAL INVERSE FILTERING



by

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Cover image: a three-dimensional latent manifold $\mathcal{D}_Q$ of glottal flow waveforms, learned by q-GPLVM from the EGIFA corpus (Chapter 6). Each point on this surface represents the variational posterior $q(\boldsymbol{x}_n) = \mathrm{Normal}(\boldsymbol{\mu}_n^x, \boldsymbol{S}_n^x)$ of a registered derivative glottal flow waveform in $\mathbb{R}^Q$, $Q = 3$.

Abstract

Speech can be described locally as the convolution of a glottal source wave and a vocal tract filter. We can measure speech directly, but not the source and filter individually. Recovering them from speech alone is a severely underdetermined inverse problem called glottal inverse filtering (GIF), open for over fifty years. Next to its relevance for fundamental speech science, GIF matters for early diagnosis of neurodegenerative diseases such as Parkinson's, because these interfere with speech production long before other motor signs appear.

To handle this underdetermination, we recast GIF as Bayesian inference and call our approach Bayesian nonparametric glottal inverse filtering (BNGIF). We learn a Gaussian process prior over glottal waveforms from a large corpus of physics-based simulations, embed the data in a small latent space, and obtain a compact mixture of surrogate models that serves as the source prior. The vocal tract filter is regularized by a new information-theoretic framework for encoding spectral constraints on the AR coefficients. Both sides are inferred jointly by a fast variational method that runs at $5\times$ realtime on a single GPU.

We validate on two important tasks: glottal waveform recovery and formant estimation. On glottal waveform recovery, BNGIF achieves the strongest speech scores among state of the art and is on par with the best classical method on steady-state vowels. On formant estimation, it reduces overall RMSE by 30% relative to the best classical tracker. Both results come from the same posterior: better source modeling yields better filter estimates, and vice versa.

These scores are encouraging, but the thesis does not so much "solve" GIF as push it to what seems a more natural setting for it. That is, our main contribution is a coherent attempt to cast GIF as Bayesian inference without suffering a practical cost in terms of usage: inference is stable and still fast enough such that BNGIF could be part of daily practice. The Bayesian handle we develop below opens the door to transparent model criticism, data-driven learning, principled regularization, and calibrated uncertainty – properties that the current state of the art, for all its practicality and ingenuity, does not yet offer.

Lay summary

In the past decade, we have made enormous strides in recognizing and synthesizing speech using large “black-box” models trained on massive datasets. Today, these models have learnt to speak and understand major world languages such as English, Chinese, and Spanish with impressive accuracy. Remarkably, they remain largely agnostic to the physical mechanisms underlying speech, relying instead on large-scale pattern recognition and vast quantities of data.

In contrast, “white-box” models grounded in accumulated scientific knowledge do focus explicitly on these physical mechanisms. While they are not remotely competitive with black-box models in speech recognition or synthesis, they do offer the valuable ability to extract biologically meaningful information about the speaker from the signal itself.

This is important. A growing body of evidence shows that quantitative features of voice and speech can act as digital biomarkers of neurological and clinical conditions. In Parkinson’s disease, for example, changes in speech and language have been reported to precede diagnosis by as much as a decade, and acoustic measures such as voice quality, articulation rate, and prosody have shown promise for early detection and monitoring. Similarly, alterations in specific acoustic parameters have been linked to mood disorders, suggesting that speech analysis could complement clinical assessment of depression.

Nevertheless, white-box models often lack the flexibility of black-box models needed to deal with real-world speech. I therefore propose a “grey-box” approach that attempts to combine the best of both worlds. By integrating physical insight with data-driven modeling, I derive a class of grey-box models that allows for more accurate analysis and decomposition of speech into separate contributions from the vocal folds and the vocal tract. These models are also capable of indicating how certain they are of their measurements, while running in real-time speed on modern hardware.

This was achieved by combining new theoretical insights in glottal flow and vocal tract models with customized machine learning models. I hope that this work can contribute to a more reliable and holistic use of speech analysis in clinical practice, forensic work, and the scientific study of vocalization as a whole.

Preface

TODO

About navigating the thesis:

- The thesis is structured into four parts: main chapters, datasets, algorithms and appendices.
- The main chapters contain the narrative and main results; the rest are additional details.
- Each main chapter is structured as an introduction and a summary. The last main chapter contains a list of contributions.
- Most figures are clickable: when clicked, an interactive version will open in the browser.
- Upon hovering over hyperlinks such as equations, chapters, citations, etc., most PDF viewers will show the context of the reference. For example, this reference to equation (174).

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TODO

1 Overview

Every time we make a voiced sound, a stream of air from the lungs is shaped twice: first by the vibrating vocal folds, then by the resonances of the throat and mouth. The vocal folds produce the *glottal flow* $u(t)$; the throat and mouth act as a filter $h(t)$ that sculpts that flow into the speech pressure waveform $s(t)$ we hear. In the source-filter model (Fant, 1960) that still underpins most of acoustic phonetics (Maurer, 2016), this chain is

$$s(t) = u(t) * h(t) * r(t), \tag{1}$$

where $r(t)$ is the radiation characteristic of the lips, assumed known.

Glottal inverse filtering (GIF) is the problem of undoing that chain: given only the speech signal $s(t)$, recover both the source $u(t)$ and the filter $h(t)$. Figure 1 illustrates this schematically. It is a hard inverse problem because many different source-filter pairs can produce the same speech (Johnson, 2016). This thesis develops a new approach called *Bayesian nonparametric glottal inverse filtering* (BNGIF), which treats GIF as a Bayesian grey-box inference problem in which regularization of the GIF solution is provided by data-driven priors.

1.1 Motivation and state of the art

1.1.1 Why GIF matters

The voice carries information about the body that produced it. Sustained use, aging, and neurological disease all leave measurable traces in how the vocal folds vibrate and how the vocal tract shapes sound. A striking example is Parkinson's disease, where speech impairment affects up to 90% of patients and may precede the defining motor signs by a decade (Cao et al., 2025). But the quantities that matter for such diagnoses – fundamental frequency, jitter, shimmer, open quotient, harmonics-to-noise ratio, formant trajectories – are not properties of the acoustic waveform $s(t)$ directly. They are properties of the glottal flow $u(t)$ and the vocal tract filter $h(t)$ hidden beneath it.

Extracting these quantities from a recording is a two-stage process (Alku et al., 2026). The first stage is GIF itself: recover $u(t)$ and $h(t)$ from the speech signal. The second stage is parameterization: reduce the estimated waveforms to the handful of numbers that carry the phonetically or clinically relevant information. Since GIF operates on a standard microphone recording, the entire pipeline is noninvasive, and in principle a phone call suffices (Alku et al., 2026). This makes it attractive not only for laboratory phonetics but also for remote screening, where the alternative is a subjective clinical rating scale (Corcoran et al., 2019).

Three scientific communities make routine use of the measurements this pipeline aims to recover. Acoustic phonetics needs physiologically grounded source and tract measurements (Stevens, 2000). Clinical voice assessment needs GIF-derived biomarkers of pathology and vocal effort – parameters with direct physiological meaning that black-box acoustic features do not provide (Alku et al., 2026). Forensic speaker comparison needs formant-based evidence

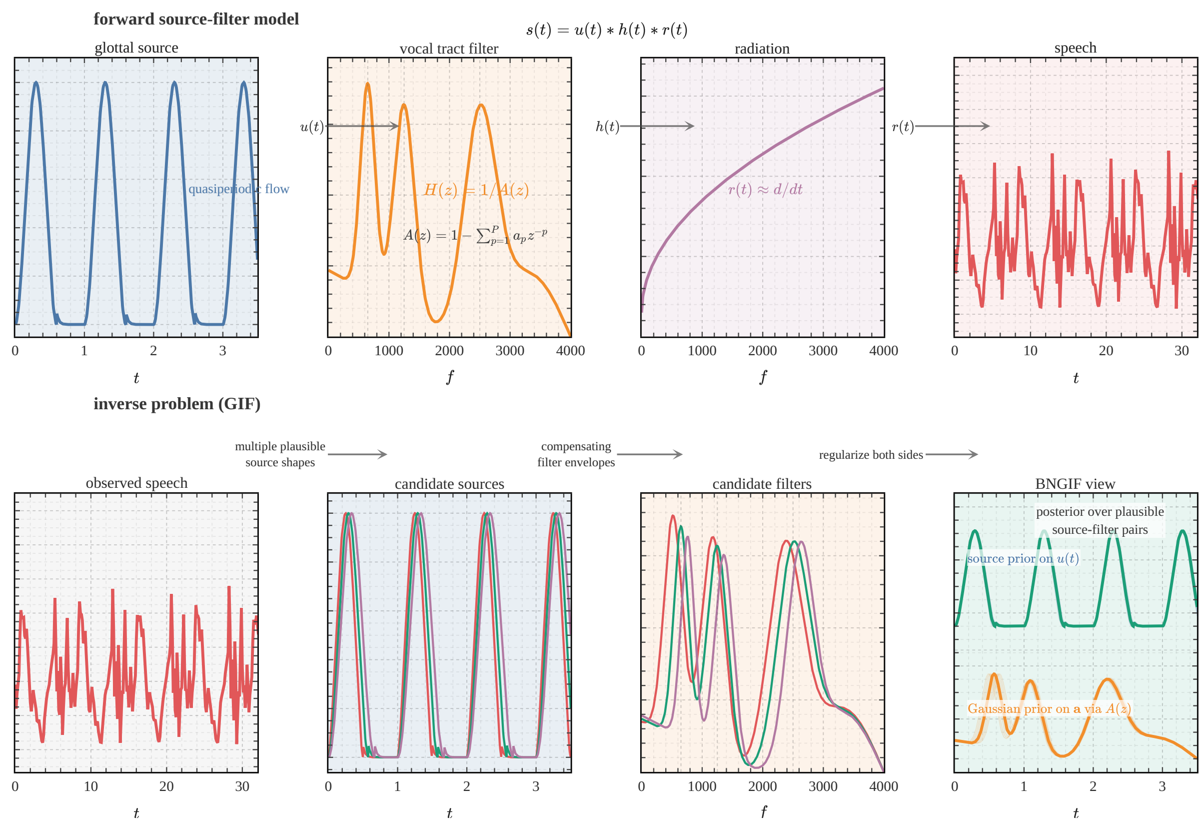

Figure 1 | **Schematic overview of source-filter separation and GIF.** Top row: the forward model $s(t) = u(t) * h(t) * r(t)$ ((1)). A glottal source waveform $u(t)$ passes through a vocal-tract filter $H(z) = 1/A(z)$, then through the radiation characteristic $r(t)$, producing the observed speech waveform $s(t)$. Bottom row: the inverse problem. Given only the speech waveform, multiple source shapes and filter envelopes can explain the same data. BNGIF attempts to handle this by regularizing both sides: a Gaussian process prior on the source waveform and a Gaussian prior on the AR coefficients $\mathbf{a}$.

within the Bayesian likelihood-ratio framework now standard in that field (Becker et al., 2008; Bonastre et al., 2015). In all three cases, the quality of the downstream measurement is limited by the quality of the inverse filter that feeds it.

1.1.2 Two traditions of GIF

The relatively long history of research into GIF naturally produced several lines along which the problem may be approached, but, following two recent reviews (Drugman et al., 2019a; Kadiri et al., 2021), it is possible to separate these approaches loosely into two broad classes: inverse filtering and joint source-filter optimization.[1]

[1]For the sake of the argument we ignore a here smaller third class which uses mixed-phase decomposition methods (Degottex, 2010; Drugman et al., 2019a; Kadiri et al., 2021) such as complex cepstrum decomposition (CCD) (Drugman et al., 2011). We do compare the latter on a GIF task in Chapter 7.

Inverse filtering Miller (1959) founded the basis for the first and larger class of GIF methods, which holds most of the popular methods used today such as quasi-closed phase (QCP) analysis (Airaksinen et al., 2014) and iterative adaptive inverse filtering (IAIF) (Alku, 1992). The inverse filtering approach aims to recover the glottal source $u(t)$ by estimating the vocal tract filter from the speech data $\mathbf{x}$ and then applying its inverse. In practice, the filter is modeled as the all-pole transfer function $H(z) = 1/A(z)$ where $A(z) = 1 - \sum_{p=1}^{P} a_p z^{-p}$ is the AR polynomial whose coefficients $\mathbf{a}$ are estimated by LP ((7)); the methods differ mainly in how that estimation is done (Kadiri et al., 2021). No explicit parametric form of $u(t)$ is assumed: the estimated source is simply the LP residual $\boldsymbol{\varepsilon}$ that falls out once the filter has been removed.

Joint source-filter optimization Milenkovic (1986) proposed a different approach for the second class of GIF methods, a logical consequence of the increased computing power available in the 1980s. With joint source-filter optimization methods the fit of a parametric model to the speech data $\mathbf{x}$ is optimized to estimate both source $u(t)$ and filter $H(z) = 1/A(z)$ simultaneously (as opposed to consecutively as in the first class). These methods involve the nonconvex optimization of an objective function to determine the parameters $\boldsymbol{\theta}$ of some assumed functional form of the DGF $u'(t) = f(t; \boldsymbol{\theta})$, typically an approximation of the LF model ((50)) (Alzamendi & Schlotthauer, 2017; Degottex, 2010; Fu & Murphy, 2006; Schleusing, 2012). This optimization is computationally costly compared to the inverse filtering methods in the first class, and a variety of tricks are used to reduce runtime and improve robustness, such as two-stage optimization (Fu & Murphy, 2006) and regressing $\boldsymbol{\theta}$ to a single parameter (Degottex, 2010).

Although we are not aware of any study that compares the performance of these two approaches systematically, it is possible to make a few general statements. Inverse filtering methods are generally efficient and nonparametric, while joint source-filter optimization methods have the distinct advantage of cleanly separating $u(t)$ and $H(z)$ from the outset. In doing so they avoid having to derive $u(t)$ directly from a filter estimate $\hat{H}(z)$ that inevitably contains pronounced effects from the former (Auvinen et al., 2014), to which Schroeder (1999, p. 94) refers as a "suspiciously circular sounding 'bootstrap' method."

So, where does that leave BNGIF? An interesting property of BNGIF is that it mixes something of both classes: it is a joint source-filter optimization method capable of the flexibility of inverse filtering methods through its nonparametric model for $u(t)$. In addition, the Bayesian framework adopted by BNGIF handles uncertainty in a straightforward way through sampling the posterior for $u(t)$, which can be seen as generalization of the nonconvex optimization that characterizes (and plagues) the class of joint source-filter optimization methods it belongs to. In contrast, inverse filtering methods pay for their efficiency by having no principled way of dealing with uncertainty, other than operationalising it as input variability.

1.1.3 Existing approaches and the current state of the art

The recent literature suggests that classical LP-based signal processing remains the practical reference standard for GIF. An evaluation study by Freixes et al. (2023) explicitly benchmarks IAIF and QCP as "state-of-the-art GIF techniques". They also report that QCP-based techniques consistently outperform IAIF-based ones, something we also see in our own benchmark tests in Chapter 7. From the applied medicine side, Corcoran et al. (2019) write that clinical workflows mainly rely on IAIF and QCP as primary estimation tools.

Comparative analyses continue to focus on IAIF, QCP, and LP-based variants without inclusion of neural alternatives (Palaparthi & Titze, 2020). Despite the success of machine learning in speech processing broadly, there seems to be no neural "takeover" for GIF in clinical settings. No end-to-end neural inverse filtering method has achieved widespread adoption, and no benchmark has demonstrated neural approaches outperforming IAIF or QCP on glottal waveform recovery. Recent work still proposes model-based or probabilistic extensions rather than neural ones (Shi et al., 2017). Sasou & Chen (2023) write that "GIF is hardly realized through supervised deep learning."

Machine learning is highly effective in clinical speech analysis, but its role is downstream of GIF rather than a replacement for it (Moell et al., 2025). It operates on acoustic features or on features derived from GIF estimates, and physically-grounded glottal features can outperform learned speech representations in pathology detection — a structured HMM method built on GIF-derived features outperformed self-supervised learned representations in one such setting (Sasou & Chen, 2023). Machine learning has thus largely bypassed GIF rather than replacing it.

Three structural reasons may contribute to the persistence of classical methods. First, GIF is ill-posed: $u(t)$ must be inferred jointly with $h(t)$ from $s(t)$ alone (1), and without strong inductive bias the problem admits many solutions. Second, interpretability is important in clinical contexts: glottal parameters are physiologically meaningful and can be interpreted by skilled acousticians; black-box models are deemed more unsuitable (Alku et al., 2026). Three, deep learning methods typically need plenty of data, and ground truth $u(t)$ data cannot be measured easily (Sasou & Chen, 2023). One needs to rely in practice on synthetic corpora generated by physics-based synthesizers like VocalTractLab (Dataset 1).

However, there are some early signs of movement. Physics-informed neural networks can simultaneously estimate glottal flow and related physiological variables from speech, beginning to integrate physical constraints with data-driven inference (Yokota et al., 2025). DNN-based GIF shows improved accuracy over IAIF in specific constrained settings such as coded speech (Narendra et al., 2019). But none of these has yet displaced IAIF or QCP as the default in standard benchmarks or clinical practice.

Finally, there have been earlier Bayesian attempts at GIF. Auvinen et al. (2014) were the first to apply Bayesian inversion to the problem, followed by Wang et al. (2016) and Rao & Ghosh (2018). All three produce point estimates – posterior mean, MAP sequence, and MAP coefficients respectively – and none propagate posterior uncertainty to downstream quantities such as formant estimates. More critically, the computational cost is prohibitive: Auvinen et al. (2014) report 2.5 hours per 25 ms frame using MCMC, and Wang et al. (2016) require a 1255-dimensional per-frame sampler in a two-layer particle filter scheme. Practitioners in acoustic phonetics are accustomed to near-instantaneous workflows, so a Bayesian method that takes minutes or hours per short frame is not yet in the running for daily use.

1.2 Proposed approach

The route taken in this thesis is to keep the inference engine fixed and place the modeling burden on the priors. The engine is IKLP, the nonparametric Bayesian extension of LP proposed by Yoshii & Goto (2013). It keeps the familiar $\mathrm{AR}(P)$ model on the vocal-tract side, but replaces white excitation by a bank of source hypotheses that can be learned, refined, and eventually evaluated inside one variational model (Chapter 2, Algorithm 1). That arrangement makes the framework modular. The same posterior gives source and filter estimates together, so improving the source prior improves formant estimation too, while structured filter priors feed back into the source estimate.

The thesis then asks a practical question: what source prior should such an engine carry if it is to be useful for voiced speech? The answer is built in stages (Figure 2). We start from classical glottal models, generalize them into kernels, calibrate those kernels on a synthetic corpus, compress the resulting amplitude posteriors into a compact surrogate, and only then insert that object into the inverse problem. The same pattern holds on the filter side: instead of leaving the AR coefficients under an isotropic default prior, we replace that by Gaussian spectral priors that encode what is already known about vocal tract structure.

This is why the thesis has the shape it does. Each chapter refines one part of the prior structure while keeping the same inverse problem in view. The aim is not to claim that GIF is solved, but to place it in a setting where assumptions are explicit, uncertainty belongs to the model itself, and failure modes can be examined rather than worked around.

On the source side, the classical parametric glottal flow models of Chapter 4 are generalized into a family of nonparametric kernels, calibrated on EGIFA in Chapter 5, and compressed in Chapter 6 into the surrogate object $\mathcal{M}_R$. On the filter side, Chapter 3 replaces the isotropic prior of Yoshii & Goto (2013) by spectral shaping priors that remain Gaussian and therefore fit the same inference engine. Appendix Appendix A gives the only GP fact needed for what follows: every GP in this thesis is either already a finite-rank Bayesian linear model or is used that way in practice. Chapter 7 finally assembles both sides and tests the result on glottal waveform recovery and formant estimation.

1.3 Outlook

From here the route is straightforward. We first take the inference engine seriously and ask what IKLP actually buys over ordinary LP on voiced speech (Chapter 2). We then regularize

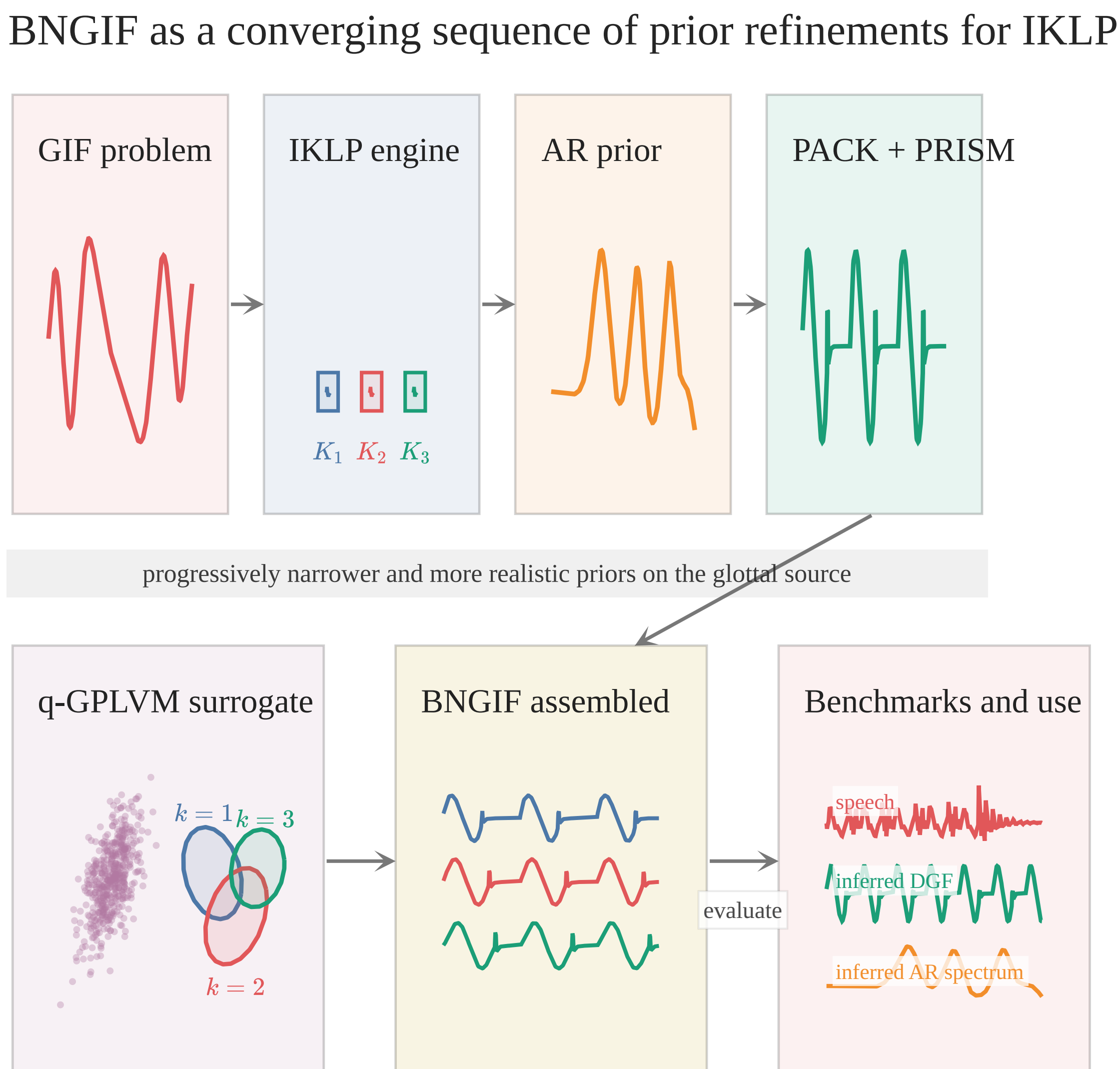

Figure 2 | **The thesis as a sequence of prior refinements.** Each step keeps the same voiced-speech inverse problem in view, but narrows the glottal source prior from a broad kernel bank to a compact surrogate that can finally be conditioned frame by frame in BNGIF.

the filter side with Gaussian spectral priors (Chapter 3) and build the source side in the opposite direction: from classical glottal models (Chapter 4), through aligned synthetic training data (Dataset 1), to kernel calibration (Chapter 5) and surrogate compression (Chapter 6). The APLAWD and VTRFormants dataset chapters define the pitch-track prior and the formant benchmark that this main line depends on.

By the time Chapter 7 is reached, the main ingredients have all been introduced separately. The source prior, the filter prior, the inference engine, and the evaluation contract are then already on the table, so the final chapter can be read mainly as a test of the assembled framework rather than as the introduction of yet another method. Chapter 8 then returns to the desiderata stated here and asks how far that framework actually carries.

2 Infinite kernel linear prediction

The inference engine at the heart of BNGIF is infinite kernel linear prediction (IKLP), the nonparametric Bayesian extension of linear prediction proposed by Yoshii & Goto (2013). This chapter develops the conceptual picture: why IKLP matters for GIF, what it buys over ordinary LP, and why the vanilla model is still not enough. The full CAVI derivation, Woodbury–Cholesky reformulation, and prior reparameterizations are given separately in Algorithm 1.

We begin with source-filter theory, which gives IKLP its physical interpretation. Then the story is simple. LP models the vocal-tract envelope and treats everything else as white excitation. That works well enough in practice, but on voiced speech the estimated envelope tends to chase pitch harmonics instead of formants. IKLP addresses this by replacing white excitation with a bank of kernel hypotheses. What remains, and what drives the rest of the thesis, is the question of what those hypotheses should be.

2.1 Source-filter theory

Source-filter theory[2] models speech production as a linear time-invariant (LTI) system (Antsaklis & Michel, 2006) – a linear approximation to the underlying nonlinear dynamics, roughly valid up to 4–5 kHz (Doval et al., 2006).

The input to the system is the source $u(t)$, with Fourier transform

$$\tilde{u}(f) = \mathcal{F}[u(t)](f) = \int_{-\infty}^{\infty} u(t) \exp\{-i2\pi f t\} \,\mathrm{d}t. \tag{2}$$

Two transfer functions determine how the frequency content of $\tilde{u}(f)$ is modified to produce the output. Source and filter are assumed decoupled: the transfer functions are independent of the input.

The vocal tract transfer function $\tilde{h}(f)$ describes the resonances and antiresonances of the vocal tract, while the radiation characteristic $\tilde{r}(f)$ describes the lip radiation effect. Their time-domain equivalents are the impulse responses

$$\begin{aligned} h(t) = \mathcal{F}^{-1}\left[\tilde{h}(f)\right](t) &= \int_{-\infty}^{\infty} \tilde{h}(f) \exp\{i2\pi f t\} \,\mathrm{d}x \\ r(t) = \mathcal{F}^{-1}[\tilde{r}(f)](t) &= \int_{-\infty}^{\infty} \tilde{r}(f) \exp\{i2\pi f t\} \,\mathrm{d}x. \end{aligned} \tag{3}$$

[2]As the dominant paradigm in acoustic phonetics (Maurer, 2016), the source-filter theory of voice production is commonly attributed to (Fant, 1960) but appears to date back to the early 19th century (Chen, 2016). Its strength lies in its extremely useful and compact description of the acoustics of phonation, summarized by (Fant, 1960, p. 15) as “the speech wave is the response of the vocal tract filter systems to one or more sound sources.” A modern perspective on source-filter theory is given by Svec et al. (2021).

Unlike $u(t)$ and $h(t)$, $r(t)$ is assumed fixed and known: the lip radiation effect is modeled as a 6 dB/octave high-pass filter (Stevens, 2000, p. 128) with transfer function $\tilde{r}(f) = i2\pi f$, which corresponds to differentiation in the time domain: $r(t) = \delta'(t)$.[3]

The output is the speech pressure signal $s(t)$, assumed measured in the free field (Stevens, 2000):

$$\begin{aligned} & s(t) = u(t) * h(t) * r(t) && \text{(time domain)} \\ \implies \quad & \tilde{s}(f) = \tilde{u}(f) \;\; \tilde{h}(f) \;\; \tilde{r}(f) && \text{(frequency domain).} \end{aligned} \tag{4}$$

The meaning is best understood in the frequency domain. The source provides an initial energy distribution $\tilde{u}(f)$. The vocal tract filter $\tilde{h}(f)$ reacts: frequencies near its resonances are emphasized,[4] while those near antiresonances are suppressed. The result is radiated from the lips via $\tilde{r}(f)$, which boosts high frequencies – the lips act as small passive tweeters (Schroeder, 1999). What we hear is the spectral imprint of the speaker's vocal tract on the source energy, and that imprint is what allows us to distinguish speech sounds (Klatt, 1986).

Since $r(t) = \delta'(t)$, we substitute into (4) and differentiate $u(t)$ (by convention (Doval et al., 2006)) to obtain

$$s(t) = u'(t) * h(t) \tag{5}$$

where $u'(t)$ is the *glottal flow derivative* (DGF), now acting as the sole source driving the filter. The original glottal flow is recovered by integration:

$$u(t) = \int_{-\infty}^{t} u'(\tau)\,\mathrm{d}\tau. \tag{6}$$

(5) is the basis for the generative model used throughout: we separate speech $s(t)$ into source $u'(t)$ and filter $h(t)$.

We now look at the source and filter individually. The effective regularization of GIF depends on realistic priors for both, as argued in Chapter 1.

2.1.1 The glottal flow waveform $u(t)$

The glottal flow $u(t)$ is the source: the initial acoustic energy that the vocal tract and lips subsequently filter. In voiced speech, the vocal folds vibrate at the fundamental frequency F_0 while modulating a stream of air from the lungs. The waveform $u(t)$ measures the flow rate of this airstream – a quasiperiodic string of pulses, each a "puff of air" through the glottis (Schroeder, 1999). The quasiperiodicity transfers directly to the speech signal $s(t)$ because the articulators move much more slowly than the vibrating folds.

[3]Convolution with the derivative of the Dirac delta is differentiation: $f(t) * \delta'(t) = f'(t)$, justified by distribution theory (Lighthill, 1958).

[4]The vocal tract filter is passive ($|\tilde{h}(f)| < 1$ for all f) (Flanagan, 1965): resonances do not amplify, they merely dampen less than neighboring bands.

The most widely used parametric model for the DGF $u'(t)$ is the Liljencrants-Fant model of Fant et al. (1985). Chapter 4 takes it as the starting point for a family of parametric and nonparametric glottal flow models, showing that they are unified by the arc cosine kernel of Cho & Saul (2009). Chapter 5 calibrates these kernels to a large corpus of synthetic glottal flow data; Chapter 6 compresses the result into the surrogate mixture $\mathcal{M}_R$ that enters IKLP's kernel bank.

2.1.2 The impulse response of the vocal tract $h(t)$

The resonances and antiresonances of the vocal tract appear as peaks and valleys in the power spectrum $|\tilde{h}(f)|^2$. These are *formants* and *antiformants*: their frequencies and bandwidths give a compact description of how the vocal tract shapes sound. During speech production, we manipulate them on the fly by changing the vocal tract configuration – rounding the lips, moving the tongue, opening the jaw – to produce the desired vowel or consonant.

The standard approach in source-filter theory models $\tilde{h}(f)$ as a rational transfer function whose poles and zeros correspond to formants and antiformants (Stevens, 2000). We adopt this assumption but without a strict one-to-one correspondence: a single formant may be modeled by more than one pole, and some poles serve as spectrum-shaping terms rather than corresponding to a physical resonance. The transfer function is thus a parametric rational expansion of the true underlying vocal tract response, controlled by its order, a gain factor, and a vector of poles and zeros.

In practice, all-pole expansions are generally thought to be sufficiently expressive for the vocal tract (Flanagan, 1965; Fulop, 2011) and have been the computationally preferred choice since the beginning of digital acoustic phonetics (Atal & Hanauer, 1971), mainly because they admit linear prediction (Markel & Gray, 1976). This thesis follows that convention: the vocal tract is modeled throughout as an $\mathrm{AR}(P)$ process, and Chapter 3 develops the informative priors that replace the isotropic default of Yoshii & Goto (2013).

2.2 Linear prediction

The source-filter equation (5) asserts that speech is a convolution of source and filter. Linear prediction is its workhorse discrete approximation.

Let $\mathbf{x} = (x_1, ..., x_M)^\top \in \mathbb{R}^M$ be M consecutive speech samples drawn from a short analysis frame. The autoregressive (AR) model of order P asserts that each sample is a linear combination of the P most recent ones plus an excitation residual:

$$x_m = \sum_{p=1}^{P} a_p x_{m-p} + \varepsilon_m. \tag{7}$$

The vector $\mathbf{a} = (a_1, ..., a_P)^\top \in \mathbb{R}^P$ collects the predictor coefficients and $\boldsymbol{\varepsilon} = (\varepsilon_1, ..., \varepsilon_M)^\top$ is the excitation. In source-filter terms the two play distinct roles: $\mathbf{a}$ encodes the resonance structure of the vocal tract, while $\boldsymbol{\varepsilon}$ carries the source.

Two matrices appear repeatedly. The AR operator $\boldsymbol{\Psi}(\mathbf{a}) \in \mathbb{R}^{M \times M}$ is approximately lower-triangular with unit diagonal and sub-diagonal entries given by $-a_1, ..., -a_P$; it satisfies $\boldsymbol{\varepsilon} = \boldsymbol{\Psi}\mathbf{x}$, or equivalently $\mathbf{x} = \boldsymbol{\Psi}^{-1}\boldsymbol{\varepsilon}$. The lag matrix $\mathbf{X} \in \mathbb{R}^{M \times P}$ has m-th row $(x_{m-1}, ..., x_{m-P})$ and is the design matrix for the normal equations.

The classical probabilistic LP model assumes white Gaussian excitation,

$$\boldsymbol{\varepsilon} \sim \texttt{Normal}(\mathbf{0}, \nu\mathbf{I}), \tag{8}$$

which yields the likelihood

$$\mathbf{x} \sim \texttt{Normal}\left(\mathbf{0}, \nu\boldsymbol{\Psi}^{-1}\boldsymbol{\Psi}^{-\top}\right). \tag{9}$$

Maximum-likelihood estimation under (9) gives the familiar normal equation

$$\mathbf{X}^\top\mathbf{X}\mathbf{a} = \mathbf{X}^\top\mathbf{x}, \tag{10}$$

which is what standard LPC implementations solve.

The model is not useless on voiced speech; in practice it often works surprisingly well. But its failure mode is clear. During voiced phonation the excitation is periodic, not white, and fitting (9) to a pitched signal forces the estimated envelope to track the harmonic partials of the spectrum: the poles of the all-pole filter chase the pitch harmonics rather than the formants. This is the defect that motivates everything that follows.

2.3 Kernel linear prediction

The natural fix is to replace the white-excitation model by a richer one. We model the excitation as a continuous function $\varepsilon(t)$ over time, approximated by a weighted sum of basis functions plus a broadband residual:

$$\varepsilon(t) = \boldsymbol{\varphi}(t)^\top\mathbf{w} + \eta(t), \tag{11}$$

where $\boldsymbol{\varphi}(t) = \left(\varphi_1(t), ..., \varphi_{J(t)}\right)^\top$ and $\eta(t)$ is the residual. Sampling at times $t_1, ..., t_M$ and writing $\boldsymbol{\Phi} \in \mathbb{R}^{M \times J}$ for the design matrix with rows $\boldsymbol{\varphi}(t_m)^\top$ gives $\boldsymbol{\varepsilon} = \boldsymbol{\Phi}\mathbf{w} + \boldsymbol{\eta}$ in vector form.

With independent Gaussian priors

$$\mathbf{w} \sim \texttt{Normal}(\mathbf{0}, \nu_w\mathbf{I}), \quad \boldsymbol{\eta} \sim \texttt{Normal}(\mathbf{0}, \nu_e\mathbf{I}), \tag{12}$$

marginalizing over $\mathbf{w}$ yields

$$\boldsymbol{\varepsilon} \sim \texttt{Normal}(\mathbf{0}, \nu_w\mathbf{K} + \nu_e\mathbf{I}), \quad \mathbf{K} = \boldsymbol{\Phi}\boldsymbol{\Phi}^\top. \tag{13}$$

The matrix $\mathbf{K}$ is a kernel matrix with entries $K_{mm'} = \boldsymbol{\varphi}(t_m)^\top\boldsymbol{\varphi}(t_{m'})$, and the kernel trick lets us specify it directly as $K_{mm'} = k(t_m, t_{m'})$ without ever constructing the basis functions explicitly.

Substituting (13) into $\mathbf{x} = \mathbf{\Psi}^{-1}\boldsymbol{\varepsilon}$ gives the kernelized LP likelihood,

$$\mathbf{x} \sim \texttt{Normal}\big(\mathbf{0}, \mathbf{\Psi}^{-1}(\nu_w \mathbf{K} + \nu_e \mathbf{I})\mathbf{\Psi}^{-\top}\big). \tag{14}$$

Classical LP (9) is the special case $\mathbf{K} = \mathbf{I}$ with $\nu = \nu_w + \nu_e$. A periodic kernel with period T concentrates excitation power at harmonics and thereby decorrelates the envelope estimate from the pitch structure. That is the key move of Yoshii & Goto (2013).

Excitation semantics

It is worth being precise about how the structured and broadband parts of $\boldsymbol{\varepsilon}$ map onto GIF. We write the excitation as $\varepsilon(t) = s(t) + \eta(t)$, where $s(t) = \boldsymbol{\varphi}(t)^\top \mathbf{w}$ is the structured GP component (not to be confused with the speech signal of (4)) and $\eta(t)$ is broadband noise. We distinguish the full excitation $\varepsilon(t)$ from $u'(t)$, the true physical DGF. In this thesis $u'(t)$ is the theoretical DGF in the source-filter picture, and inferred $\varepsilon(t)$ is our practical DGF estimate. During voiced regions these are close but not equal, and we always report $\varepsilon(t)$ rather than $s(t)$ alone. The exact post-CAVI posterior split is given in Algorithm 1: under the current variational family the full residual $\boldsymbol{\varepsilon}$ is fixed at the observed LP residual, while only its decomposition into structured signal and broadband noise carries Gaussian posterior uncertainty.

This is deliberate. Even ordinary LPC — the $s = 0$ limit where all excitation comes from $\eta(t)$ — already recovers recognizable DGF estimates from long enough analysis frames, because $\eta(t)$ is free to inject the sharp broadband transients needed to excite all formant resonances simultaneously. Those transients also model the broadband energy injections associated with glottal closure. IKLP enriches what $s(t)$ can model, but it leaves $\eta(t)$ active and useful.

2.4 Infinite kernel linear prediction

The kernelized model (14) requires choosing one kernel $\mathbf{K}$ to match the excitation structure. For voiced speech the fundamental period T is itself unknown, so this is exactly the wrong place to commit too early. The Bayesian response is multiple kernel learning: prepare a bank of candidate kernels $\mathbf{K}_1, \ldots, \mathbf{K}_I$ and define the excitation kernel as their weighted sum,

$$\mathbf{K} = \sum_{i=1}^{I} \theta_i \mathbf{K}_i, \quad \theta_i \geq 0. \tag{15}$$

Each weight θ_i quantifies the degree to which hypothesis i explains the excitation. In the original formulation of Yoshii & Goto (2013), the bank indexes candidate fundamental periods $T_1, \ldots, T_I$ on a grid, so the weights encode a posterior over F0 candidates.

That is the right starting point, but too narrow as a final interpretation. The index i is really a generic hypothesis axis: it can index candidate periods, candidate glottal-flow morphologies, candidate phonation modalities, or any structured excitation model that can be expressed as a kernel. Later chapters exploit this generality directly. The bank will no longer contain ad hoc periodic kernels but learned GP priors over DGF shape.

The resulting multiple-kernel likelihood is

$$\mathbf{x} \sim \texttt{Normal}\left(\mathbf{0}, \boldsymbol{\Psi}^{-1}\left(\nu_w \sum_{i=1}^{I} \theta_i \mathbf{K}_i + \nu_e \mathbf{I}\right)\boldsymbol{\Psi}^{-\top}\right). \tag{16}$$

MAP estimation of $\boldsymbol{\theta}$ under a regularizing prior does not yield truly sparse weights. Yoshii & Goto (2013) solve this by taking $I \to \infty$ and placing a gamma-process prior on the infinite weight sequence: for any finite truncation level I,

$$\theta_i \sim \texttt{Gamma}(\alpha/I, \alpha), \tag{17}$$

and as $I \to \infty$ the vector $\boldsymbol{\theta}$ converges to a draw from a GaP with concentration α. The effective number of active elements remains almost surely finite.

The full IKLP likelihood is therefore

$$\mathbf{x} \sim \texttt{Normal}\left(\mathbf{0}, \boldsymbol{\Psi}^{-1}\left(\nu_w \sum_{i=1}^{\infty} \theta_i \mathbf{K}_i + \nu_e \mathbf{I}\right)\boldsymbol{\Psi}^{-\top}\right). \tag{18}$$

This single model simultaneously infers the spectral envelope through $\mathbf{a}$, the dominant excitation hypothesis through $\boldsymbol{\theta}$, and voiced/unvoiced status through the ratio $\mathbb{E}[\nu_w]/\mathbb{E}[\nu_w + \nu_e]$.[5]

IKLP turns LP into a hypothesis-testing engine. The kernel bank is the only interface between the inference engine and the prior on the excitation. Once that interface is in place, changing the source model does not require changing the variational architecture. This is why the method remains central throughout the thesis even though its priors change completely.

It also removes a great deal of DSP staging. Pitch, voiced/unvoiced structure, and envelope estimation are no longer separate heuristic steps but consequences of a single probabilistic model. That does not make the problem easy. It does make the assumptions explicit.

Inference is coordinate ascent variational inference (CAVI): each latent variable is updated in closed form while the others are held fixed. The updates are deterministic – no sampling, no stochastic gradients – which makes the algorithm stable and reproducible. The dominant cost per iteration is an $O(M^3)$ solve involving the excitation covariance; the Woodbury-Cholesky reformulation in Algorithm 1 reduces this to $O(L^3)$ where L is the number of active kernel hypotheses, typically $L \ll M$. The implementation uses JAX in float32 on GPU; the full variational machinery – mean-field family, auxiliary lower bound, CAVI updates, and hyperparameter reparameterizations – is derived in Algorithm 1.

[5] Setting $P = 0$ is also valid: the AR operator becomes $\boldsymbol{\Psi} = \mathbf{I}$ and IKLP reduces to a pure multi-kernel Bayesian inference engine over the excitation process with no envelope component.

Two priors drive the rest of the thesis. The *excitation prior* enters through the kernel bank $\mathbf{K}_1, \mathbf{K}_2, \ldots$: each kernel encodes a hypothesis about DGF shape, and IKLP selects among them. Designing this bank is the subject of Chapter 4, Chapter 5 and Chapter 6. The *filter prior* enters through $\mathbf{a}$: IKLP as stated places an isotropic Gaussian on the AR coefficients, which is uninformative about formant structure. Chapter 3 replaces this with a structured prior that encodes known spectral constraints.

Figure 3 is worth keeping in mind when reading the benchmark below. The vanilla periodickernel model already does the right *kind* of Bayesian work: it uses the kernel bank to localize a plausible F_0, transfers that periodicity into the excitation estimate, and keeps the tract and source coupled in one fit. Formally, the plotted green and grey curves are the posterior mean split $\boldsymbol{\mu}_s$ and $\boldsymbol{\mu}_n$ of Algorithm 1. So the problem is not that IKLP has the wrong inference architecture. The problem is that the periodic hypothesis family is still too crude.

2.5 Early benchmark on EGIFA

Table 1 gives an early benchmark preview on EGIFA. The exact evaluation protocol is developed more carefully later in Dataset 1 and Appendix E; for now only the ranking matters. Higher is better.

The message is not subtle. The periodic prior of Yoshii and Goto is a real improvement over white excitation, so the direction is correct. Figure 4 shows exactly that: the periodic-kernel density is shifted to the right of the white-excitation density. But the shift is small. The QCP-variant density sits in a completely different region of score space.

Figure 5 makes the same point in waveform space. The three rows are, respectively, a modal frame at 1414 Pa, a pressed frame at 1414 Pa, and a slightly breathy frame at 2000 Pa. The periodic kernel often cleans up the gross periodic structure and can materially outperform white excitation on individual frames. But it still leaves too much broadband mismatch around the main closures, and that is precisely where the QCP variant remains far stronger. The gap is therefore not a quirk of one summary score: it is visible in the estimated DGFs themselves.

Vanilla IKLP is therefore not yet a competitive GIF method simply by being Bayesian.

This is also why the thesis does not stay in the world of hand-designed periodic kernels. The engine is sound. The priors are not.

2.6 Summary

IKLP turns linear prediction into a modular Bayesian hypothesis engine. The kernel bank is the single interface between inference and the prior on excitation; the variational architecture (Algorithm 1) stays fixed while the priors change. The early EGIFA benchmark confirms that the architecture works – structured excitation beats white noise – but exposes two concrete design targets.

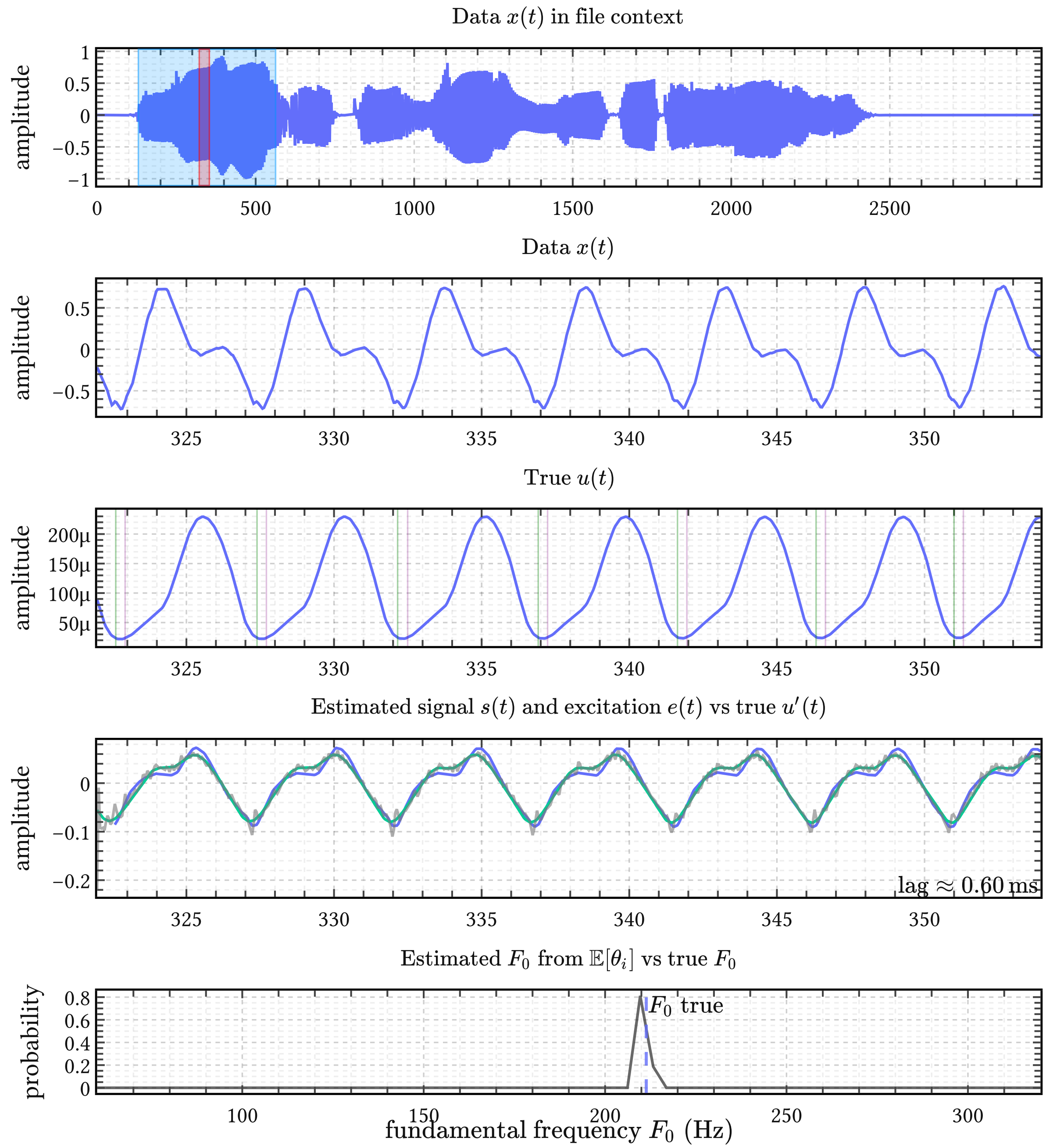

Figure 3 | **One periodickernel IKLP fit in detail.** Top: the analyzed EGIFA speech file and the selected 32 ms frame. Middle: the frame-local speech signal, the aligned true DGF $u'(t)$, and the inferred structured signal $s(t)$ together with the full excitation estimate $\varepsilon(t)$. Bottom: the posterior over candidate F_0 values derived from $\mathbb{E}[\theta_i]$. The intended mechanism is already visible: weight mass concentrates near one fundamental frequency and the excitation estimate is pulled toward that periodic structure while the tract envelope is fitted jointly.

On the excitation side, the periodic kernel is too crude: it can express smooth periodicity, but not the quasiperiodicity, open/closed asymmetry, or learned morphology of real DGF waveforms. Chapter 4 develops the parametric and nonparametric glottal flow models that

Method	Vowel median	Speech median
White excitation	0.325	0.338
Periodic kernel	0.344	0.349
QCP variant	0.957	0.910

Table 1 | **Early EGIFA preview for vanilla IKLP.** The first two rows are IKLP with white-noise and periodic-kernel excitation, respectively. The third row is the AME-weighted QCP variant using the LBGID backend (Algorithm 2). Values are median centered cosine similarity on the aligned excitation waveform.

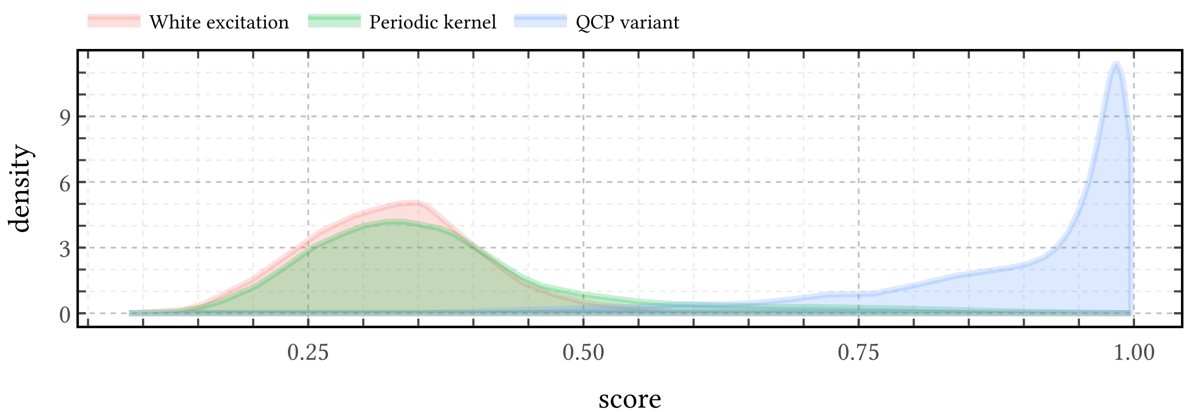

Figure 4 | **Score distributions on the early EGIFA preview.** The periodic kernel shifts the white-excitation distribution slightly to the right, confirming that structured periodic excitation is the correct direction. But the shift is small, while the QCP-variant distribution is concentrated much further to the right.

will eventually populate the kernel bank; Chapter 5 calibrates them; Chapter 6 compresses the result into a compact prior.

On the filter side, the isotropic AR prior is too weak to push the tract estimate toward physically plausible regions when the source-filter tradeoff becomes difficult. Chapter 3 replaces it with a structured Gaussian prior on the AR coefficients that encodes known spectral constraints.

With those two repairs IKLP becomes the BNGIF inference engine of Chapter 7.

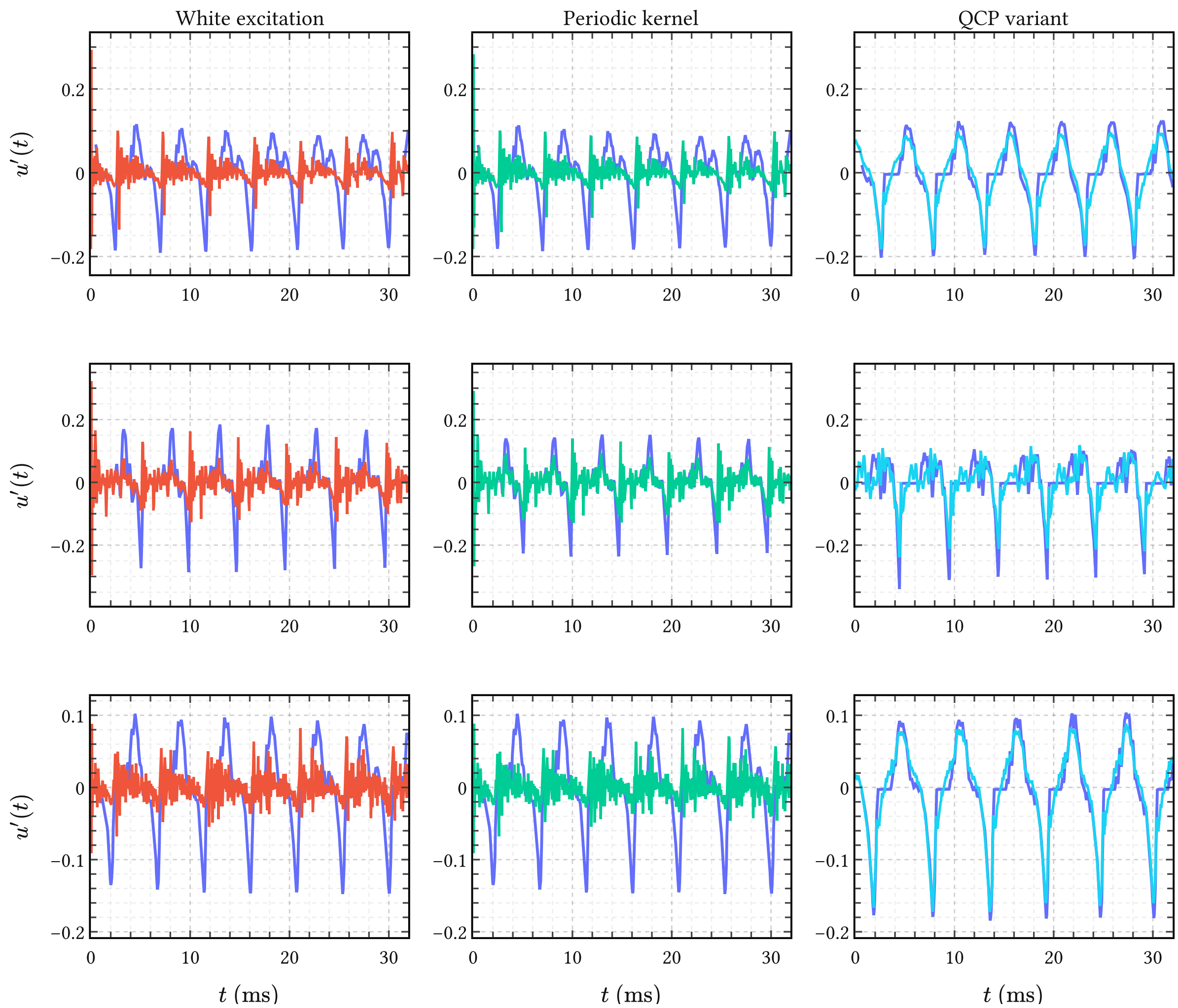

Figure 5 | **Three representative EGIFA speech frames.** In each panel, the blue curve is the aligned true DGF $u'(t)$ and the colored curve is the estimated excitation. Left: white excitation. Center: Yoshii and Goto's periodic kernel. Right: the QCP variant. The first row shows the typical case: periodic structure helps, but only modestly. The second row shows a frame where the periodic kernel already gives a clearly better voiced fit than white excitation. The third row shows the real failure mode: both vanilla IKLP variants still miss badly, while the QCP variant locks onto the closure pattern.

3 Priors for autoregressive filters

The vocal tract filter is modeled throughout this thesis as an $\mathrm{AR}(P)$ process: a rational all-pole transfer function whose P coefficients $\mathbf{a} = (a_1, ..., a_P)^\top$ encode the resonant structure of the vocal tract. Estimating $\mathbf{a}$ from speech data is done traditionally by LPC methods, or in our case, by IKLP. Like any Bayesian model, IKLP requires a prior $p(\mathbf{a})$, and throughout their paper Yoshii & Goto (2013) simply assert

$$p(\mathbf{a}) = \texttt{Normal}(\mathbf{0}, \lambda \mathbf{I}_P), \quad \text{with a fixed } \lambda := 0.1. \tag{19}$$

This choice, which we will refer to as the *isotropic prior*, is clearly something more of a blunt regularizer than an informative prior. It's there to prevent the normal matrix in (215) from becoming singular, not to encode anything known a priori about the vocal tract.

While it gets the job done, two things about (19) are perhaps unsatisfying. First, the isotropic prior with $\lambda = 0.1$ implies a standard deviation of about 3.16 per coefficient. This is a very loose prior under which most induced AR filters will almost surely be unstable. Second, we actually know a great deal about vocal tract filters: their resonances are spaced roughly 500 Hz apart, their spectral tilt is on the order of $-6\ \mathrm{dB/octave}$, and, plain from the physics of speech production, the filter must be at least be passive and causal. Except for causality none of this is encoded in (19).

This chapter then develops two priors that attempt to do better in this regard. The first, the *Monahan prior*, is the closest Gaussian to the uniform distribution over stable AR coefficient vectors; at first sight, this sounds like exactly the thing to do, but we show that it doesn't work. The second, the *soft spectral shaping prior*, is a new construction that encodes a number of spectral features useful in speech – resonances and spectral rolloff – directly into the prior mean and covariance through a closed-form projection. Both priors remain Gaussian and are therefore compatible with IKLP's variational inference without modification, and we test their merit in Chapter 7.

3.1 The isotropic prior

3.1.1 Expected output power

To understand the inductive bias that $\lambda = 0.1$ represents, consider what an isotropic Gaussian prior

$$\texttt{Isotropic}(\lambda) := \texttt{Normal}(\mathbf{0}, \lambda \mathbf{I}_P) \tag{20}$$

does to the expected output power of an $\mathrm{AR}(P)$ process driven by white noise with variance ν as in equations (7) and (8) from the previous chapter:

$$x_m = \sum_{p=1}^{P} a_p x_{m-p} + \varepsilon_m, \quad \text{where} \quad \begin{cases} \mathbf{a} & \sim \texttt{Normal}(\mathbf{0}, \lambda \mathbf{I}_P) \\ \varepsilon_m & \sim \texttt{Normal}(0, \nu) \end{cases} \tag{21}$$

Up to scale, then, the output power is determined by the frequency response

$$H(e^{i\omega}) = \frac{1}{A(e^{i\omega})} \quad \text{where} \quad A(z) = 1 - \sum_{p=1}^{P} a_p z^{-p}. \tag{22}$$

To second order, the Maclaurin expansion of $|H(e^{j\omega})|^2$ in a_p gives

$$|H(e^{j\omega})|^2 \approx 1 + 2\sum_{p=1}^{P} a_p \cos(p\omega) - \sum_{p=1}^{P}\sum_{\ell=1}^{P} a_p a_\ell \cos((p-\ell)\omega) \tag{23}$$

$$+ 4\sum_{p=1}^{P}\sum_{\ell=1}^{P} a_p a_\ell \cos(p\omega)\cos(\ell\omega) + \dots \tag{24}$$

Taking the expectation over the zero-mean isotropic prior (19), the linear terms and the cross-terms $p \neq \ell$ vanish and the expected spectral power becomes

$$\mathbb{E}_{\mathbf{a}}[|H(e^{j\omega})|^2] \approx 1 - P\lambda + 4\lambda \sum_{p=1}^{P} \cos^2(p\omega). \tag{25}$$

Integrating over frequency and applying Parseval's identity gives the expected output variance of (21):

$$\mathbb{E}[\text{var}(x_m)] \approx \nu(1 + P\lambda). \tag{26}$$

With $\nu := 1$, $P = 30$ and $\lambda = 0.1$ as in Yoshii & Goto (2013), this gives an expected standard deviation of $\sqrt{1+3} = 2$, which is indeed $O(1)$. Nevertheless, power scales linearly with P, which contradicts the prevalent idea that increasing P just prepares the model for more fine-structured data, not "louder" data. The lesson from (26) is thus: to keep power approximately controlled under an isotropic prior, one needs $\lambda \propto 1/P$. This rule is a known heuristic in signal processing literature (Rabiner et al., 1993, for example; Sørbye & Rue, 2017), here we have derived it from a probabilistic argument.

3.1.2 Stability

Beyond raw power, there is a more fundamental problem: `Isotropic`(λ) places most of its mass outside the stability region $\mathcal{A}_P \subset \mathbb{R}^P$, which is the set of coefficient vectors for which all roots of $A(z) = 1 - \sum_p a_p z^{-p}$ lie strictly inside the unit disk. Physically, *stability* means the filter's impulse response decays over time: the vocal tract is a passive resonator that dissipates energy, so its transfer function cannot have poles on or outside the unit circle. An unstable AR filter would predict an impulse response that grows without bound, which is physically impossible for a tube of flesh and air (the vocal tract).

For $P = 1$ the stability region is the interval $(-1, 1)$, and a unit Gaussian already places about 32% of its mass outside. For larger P the stability region is a complicated manifold and the fraction of unstable draws tends to grow rapidly – see Figure 6.

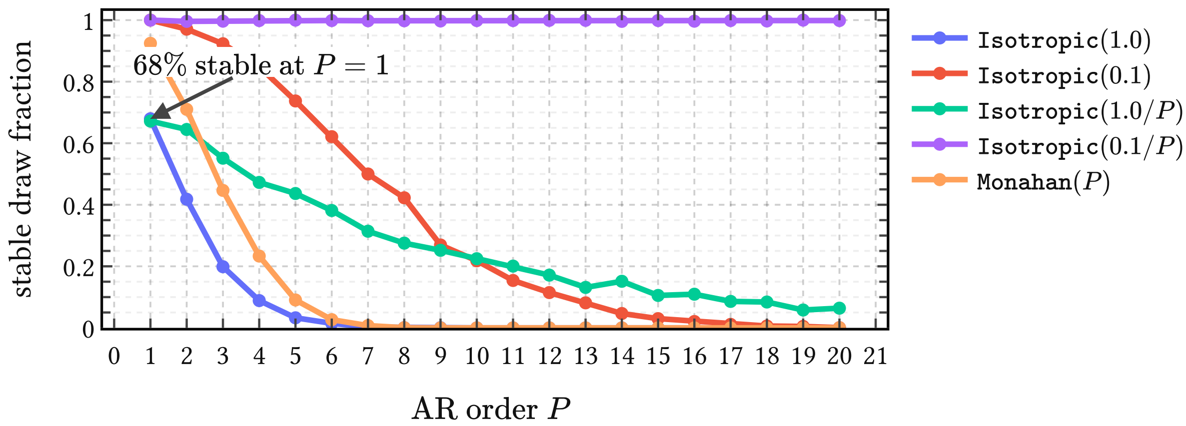

Figure 6 | **Probability that a prior draw lies in the stable AR region $\mathcal{A}_P$** for different values of λ with the isotropic prior (20). Stable prior mass collapses quickly under isotropic Gaussian AR priors, unless the $\lambda \propto 1/P$ rule is used with a $O(0.1)$ proportionality constant; the Monahan prior (30) performs underwhelmingly.

This figure also reveals that the proportional constant in the $\lambda \propto 1/P$ rule does seem to matter, because a Gaussian prior for $\mathbf{a}$ entangles expected power output and stability. At one extreme, $\lambda \to 0$ yields a filter than is trivially stable but will only produce signals with vanishing power. However, turning up expected power output to $O(1)$ by increasing λ then gradually decreases stability. The `Isotropic`$(0.1/P)$ prior is seen to strike a reasonable balance here. But can't we encode our preference for stability directly?

3.2 The Monahan prior

3.2.1 The Monahan map

A classical result from time series analysis (Monahan, 1984) is that the partial autocorrelation function (PACF) parameters $\boldsymbol{\varphi} = (\varphi_1, ..., \varphi_P)^\top$ provide a smooth bijection between the hypercube $(-1,1)^P$ and the stability region $\mathcal{A}_P$:

$$T : (-1,1)^P \to \mathcal{A}_P, \quad \mathbf{a} = T(\boldsymbol{\varphi}). \tag{27}$$

The map T is defined by the so-called Levinson recursion which we illustrate below in Figure 7, but whose details are unimportant for our purpose. Its main value is that drawing $\varphi_p \overset{\text{iid}}{\sim} \texttt{Uniform}(-1,1)$ produces only stable AR coefficient vectors. The push-forward of this uniform through T defines a density

$$g(\mathbf{a}) = 2^{-P} \ |\det \nabla_{\mathbf{a}} T^{-1}(\mathbf{a})|, \quad \mathbf{a} \in \mathcal{A}_P, \tag{28}$$

on the stability region, where 2^{-P} is just the density of the uniform base measure on $(-1,1)^P$. This g is the natural "uniform over stable filters" measure: it assigns equal mass to equal volumes in PACF space, mapped into coefficient space by T.

Since tractable inference in AR models usually requires Gaussian priors, which (28) is not, the best we can do then is to approximate g by a Gaussian $p(\mathbf{a}) = \texttt{Normal}(\boldsymbol{\mu}_P, \mathbf{S}_P)$ that envelops $\mathcal{A}_P$ in some optimal way.

3.2.2 Deriving the prior

We want the Gaussian to encompass $g(\mathbf{a})$, such that it is neither overconfident about any subregion of $\mathcal{A}_P$, nor wasteful in placing mass far outside it. We thus minimize $D_{\text{KL}}\left(g \parallel \texttt{Normal}(\boldsymbol{\mu}_P, \mathbf{S})_P\right)$ over all Gaussians and solve for $\boldsymbol{\mu}_P$ and $\mathbf{S}_P$. This is a standard convex problem whose unique solution is the moment-matched Gaussian:

$$\boldsymbol{\mu}_P = \mathbb{E}_g[\mathbf{a}] = \mathbf{0}, \quad \mathbf{S}_P = \mathbb{E}_g\left[\mathbf{a}\mathbf{a}^\top\right]. \tag{29}$$

Surprisingly, the mean is identically zero by the sign-symmetry of the uniform PACF law and the fact that the Monahan map is odd in each coordinate. The Monahan prior is therefore

$$\texttt{Monahan}(P) := \texttt{Normal}(\mathbf{0}, \mathbf{S}_P), \tag{30}$$

where $\mathbf{S}_P$ captures the full geometry of the stability region in P dimensions within the approximation of a Gaussian family.

$\mathbf{S}_P$ has no closed form but is easy to estimate by Monte Carlo: draw N samples $\boldsymbol{\varphi}^{(n)} \overset{\text{iid}}{\sim} \texttt{Uniform}(-1,1)^P$, compute $\mathbf{a}^{(n)} = T\left(\boldsymbol{\varphi}^{(n)}\right)$, and average the outer products,

$$\hat{\mathbf{S}}_P = \frac{1}{N}\sum_{n=1}^{N} \mathbf{a}^{(n)}\mathbf{a}^{(n)\top} \to \mathbf{S}_P. \tag{31}$$

This converges at standard $\mathcal{O}\left(N^{-1/2}\right)$ rate and can be precomputed for any desired P. At first sight this all appears promising: the Monahan map guarantees stability by construction, and (30) provides a convenient, tractable prior for IKLP that one might hope captures the shape of $\mathcal{A}_P$ sufficiently well.

3.2.3 Why it does not work

Sampling from $\texttt{Monahan}(P)$, however, leads to a rapidly vanishing fraction of stable filters as P increases, as shown by the last curve in Figure 6. The issue is geometric: g is supported entirely on $\mathcal{A}_P$, which is a highly non-elliptical subset of $\mathbb{R}^P$, while any nondegenerate Gaussian has full support on $\mathbb{R}^P$. Figure 7 illustrates this in one and two dimensions.

What if we tuck $\texttt{Monahan}(P)$ inside the support of $g(\mathbf{a})$ and make sure every draw leads to a stable filter? This would be done by minimizing the reverse projection $D_{\text{KL}}(\texttt{Normal}(\boldsymbol{\mu}_P, \mathbf{S}_P) \parallel g)$. Unfortunately, it is unbounded for any nondegenerate Gaussian since g assigns zero density outside $\mathcal{A}_P$. There is thus no Gaussian approximation that is simultaneously tractable and faithful to the stability constraint: the stable set is simply too irregular to be well-approximated by an ellipsoid.

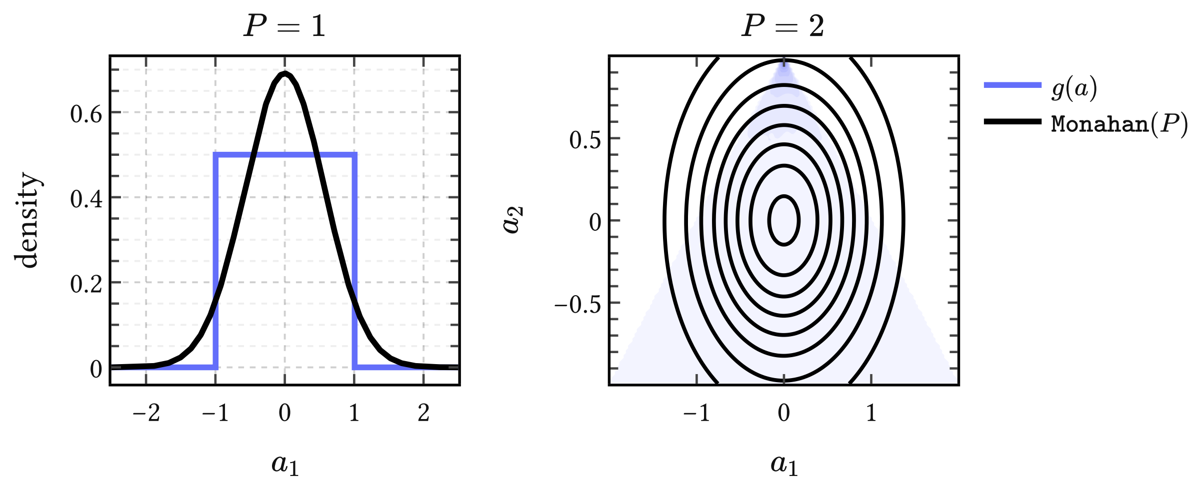

Figure 7 | **Geometry mismatch.** For $P = 1$ the Monahan map is just $a_1 = \varphi_1$ and $\mathcal{A}_1 = (-1, 1)$. For $P = 2$ it gives $(a_1, a_2) = (\varphi_1(1 - \varphi_2), \varphi_2)$, so uniform PACF draws induce a density $g(\mathbf{a})$ supported on the triangular stable region. The Monahan prior $\texttt{Monahan}(P)$ matches the first two moments of g, but it cannot reproduce the bounded, non-elliptical shape of that support faithfully, and mass leaks out of the region of stable filters.

3.3 Spectral shaping priors

Setting worries about power and stability aside for a moment, suppose we expect that the AR filter has a resonance near 500 Hz, or that its spectrum rolls off at a particular rate, or that it has a formant structure typical of a particular vowel. All of these are relevant for acoustic phonetics. How can we "fold" such beliefs into a the first and second moments of a simple Gaussian prior?

The spectral shaping priors that we introduce in this section tackle this problem in some generality. We present a recipe for encoding "spectral beliefs" into the prior mean of $p(\mathbf{a})$ through a single closed-form projection (44). This works by shifting the burden on the user to encode their desired *spectral features* as *roots of* $A(z)$, which are then imposed on average.

3.3.1 Spectral features as root constraints

Features of an AR filter's spectral envelope $|H(e^{i\omega})|^2 = 1/\ |A(e^{i\omega})|^2$ can be traced to the pole structure of $H(z)$, or, equivalently, roots of $A(z)$. That is, controlling where $A(z)$ has its roots is an operational handle on the spectral shape of the filter.

For example, a conjugate pair of roots close to the unit circle at angle ω_0 produces a resonance peak at that frequency; the closer the roots sit to the unit circle, the sharper the peak. A real root near $z = 1$ boosts low frequencies and therefore induces negative spectral tilt, which is exactly the speech-like case of interest. These and other examples of spectral features are collected in Table 2.

One interesting way to encode such spectral features is through *polynomial divisibility*. Saying "we want $A(z)$ to have a root at z_0" is the same as saying "the factor $(1 - z_0^{-1} z)$ divides

Spectral feature	$Q(z)$	deg
Resonance at (ρ, ω_0)	$1 - 2\rho\cos(\omega_0)z + \rho^2 z^2$	2
Real pole at ρ	$1 - \rho z$	1
Negative spectral tilt $-6k$ dB/oct	$(1 - \rho z)^k$	k
Seasonal pole of period s	$1 - z^s$	s
Repeated resonance at (ρ, ω_0)	$(1 - 2\rho\cos(\omega_0)z + \rho^2 z^2)^k$	$2k$

Table 2 | **Spectral features and their polynomial encodings.** Each feature corresponds to a polynomial $Q(z)$ whose roots impose the desired pole structure on $|H(z)|^2 = 1/\ |A(z)|^2$.

$A(z)$." A collection of desired roots defines a polynomial $Q(z)$, and the spectral constraint becomes

$$Q(z) \mid A(z), \tag{32}$$

which is read as "Q divides A". For instance, a resonance near frequency ω_0 with damping $1 - \rho$ corresponds to the conjugate pair $z = \rho e^{\pm i\omega_0}$, giving the quadratic

$$Q(z) = 1 - 2\rho\cos(\omega_0)z + \rho^2 z^2. \tag{33}$$

Generalizing, assume $K \geq 1$ features are desired simultaneously. The individual $\{Q_k(z)\}_{k=1}^K$ are combined into a single *constraint polynomial*

$$M(z) = \texttt{lcm}\{Q_1(z), ..., Q_K(z)\}. \tag{34}$$

Over $\mathbb{C}$ this amounts to collecting the union of desired roots with their maximum multiplicities, and the effective constraint degree is $L_{\text{eff}} = \deg M$. For example, a prior that expects both (1) a resonance and (2) a 18 dB/octave negative rolloff uses $M(z) = Q_1(z) \cdot Q_2(z)$ with $L_{\text{eff}} = 2 + 3 = 5$, where the degrees can be read off the last column of Table 2.

3.3.2 Divisibility is linear

The observation that makes (32) a tractable ansatz for encoding inductive bias towards given spectral features is that polynomial divisibility is a *linear* condition on the coefficients of A. $M(z) \mid A(z)$ if and only if the remainder of the polynomial division $A \bmod M$ vanishes, and this remainder depends linearly on $\mathbf{a}$.

More precisely, for a constraint polynomial $M(z)$ of degree L_{eff}, there exists a matrix $\mathbf{F} \in \mathbb{R}^{L_{\text{eff}} \times P}$ and a vector $\mathbf{c} \in \mathbb{R}^{L_{\text{eff}}}$ such that the remainder

$$\mathbf{f}(\mathbf{a}; M) \coloneqq \mathbf{F}\mathbf{a} + \mathbf{c} = \mathbf{0} \quad \Leftrightarrow \quad M \mid A. \tag{35}$$

The matrix $\mathbf{F}$ and offset $\mathbf{c}$ are determined by the coefficients of $M(z)$ and the monic leading term $a_0 = 1$ of $A(z)$. The set of AR(P) coefficient vectors compatible with the constraint is therefore an affine subspace of $\mathbb{R}^P$ of dimension $P - L_{\text{eff}}$.

Worked example To see this concretely, consider an AR(4) polynomial

$$A(z) = 1 - a_1 z - a_2 z^2 - a_3 z^3 - a_4 z^4 \tag{36}$$

and suppose we want to encode a resonance at (ρ, ω_0) via the quadratic $Q(z) = 1 - cz + dz^2$ where $c = 2\rho\cos(\omega_0)$ and $d = \rho^2$. Divisibility means there exists a quotient $R(z) = 1 - r_1 z - r_2 z^2$ such that $A(z) = Q(z)R(z)$. Expanding and matching the coefficients of z, z^2, z^3, z^4 gives

$$a_1 = c + r_1, \qquad a_2 = r_2 - c r_1 - d, \tag{37}$$

$$a_3 = d r_1 - c r_2, \quad a_4 = d r_2. \tag{38}$$

The last two equations pin down the quotient: $r_2 = \frac{a_4}{d}$ and $r_1 = \frac{a_3 + c a_4 / d}{d}$. Substituting back into the first two gives two linear equations in $(a_1, ..., a_4)$ alone,

$$\underbrace{\begin{pmatrix} 1 & 0 & -\frac{1}{d} & -\frac{c}{d^2} \\ 0 & 1 & \frac{c}{d} & \frac{c^2}{d^2} - \frac{1}{d} \end{pmatrix}}_{\mathbf{F}} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix} + \underbrace{\begin{pmatrix} -c \\ d \end{pmatrix}}_{\mathbf{c}} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \tag{39}$$

from which the matrix $\mathbf{F}$ and vector $\mathbf{c}$ in (35) are read off. The resonance constraint thus carves out a $(P - L_{\text{eff}}) = 2$-dimensional affine subspace of $\mathbb{R}^4$.

3.3.3 The soft spectral shaping prior

We can now state derive the soft spectral shaping prior for $\mathbf{a}$. Let $p_0(\mathbf{a}) = \texttt{Normal}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$ represent our initial state of knowledge of the AR coefficients and let $M(z)$ be the constraint polynomial (34) encoding the desired spectral features. For example, the assignments

$$p_0(\mathbf{a}) \coloneqq \texttt{Isotropic}(0.1/P), \quad M(z) \coloneqq Q(z), \tag{40}$$

where $Q(z)$ is the resonance defined in (33), express a state of knowledge where we expect the AR filter to have unit power output, be stable, and have a single resonance.

We now update $p_0(\mathbf{a})$ to $p^*(\mathbf{a})$ by imposing the new piece of information that the statistic $\mathbf{f}(\mathbf{a}; M)$ has zero expectation under the new state of knowledge expressed by p^* – that is,

$$\mathbb{E}_{p^*}[\mathbf{f}(\mathbf{a}; M)] = \mathbf{0}. \tag{41}$$

Knuth & Skilling (2012) have shown that the unique variational potential for updating information is the Kullback-Leibler divergence, so our task is to

$$\text{find} \quad p^* = \operatorname*{argmin}_p D_{\text{KL}}(p \parallel p_0) \quad \text{given} \quad \mathbb{E}_p[\mathbf{f}(\mathbf{a}; M)] = \mathbf{0}. \tag{42}$$

Since $p_0(\mathbf{a})$ is Gaussian and $\mathbf{f}(\mathbf{a}; M) = \mathbf{F}\mathbf{a} + \mathbf{c}$ is linear in $\mathbf{a}$, we can solve this in closed form:

$$p^*(\mathbf{a}) = \texttt{Normal}(\boldsymbol{\mu}_M, \boldsymbol{\Sigma}_M), \tag{43}$$

which is the unique normalized density closest to p_0 under the constraint (41). The solution keeps $\mathbf{\Sigma}_0$ unchanged, so $\mathbf{\Sigma}_M = \mathbf{\Sigma}_0$, but it does shift the mean to

$$\boldsymbol{\mu}_M = \boldsymbol{\mu}_0 - \mathbf{\Sigma}_0 \mathbf{F}^\top \left(\mathbf{F}\mathbf{\Sigma}_0\mathbf{F}^\top\right)^{-1}(\mathbf{F}\boldsymbol{\mu}_0 + \mathbf{c}). \tag{44}$$

Wrapping up, we write the resulting prior formally as

$$\texttt{SoftSpectral}(\boldsymbol{\mu}_0, \mathbf{\Sigma}_0; M) := \texttt{Normal}(\boldsymbol{\mu}_M, \mathbf{\Sigma}_0) \equiv p^*(\mathbf{a}), \tag{45}$$

and refer to it as the *soft spectral shaping prior.*

An example is shown in the left and middle panels of Figure 8, where we encode three resonances that have center frequency and bandwidth typical for the first three vowel formants. The prior spectrum induced by the base prior (left panel) has no discernable resonance structure. The middle panel shows the prior spectrum obtained after updating our state of information per (42). We can see the three resonances clearly imprinted, but what is more interesting is that the prior is noncommital with respect to their actual locations: the zoom on the F_1 region shows clearly that the actual peak locations “happen around” the mean: this is the maximum entropy principle at work. There is also some interesting tail structure which indicates that the prior is anticipating extra resonances at higher frequencies.

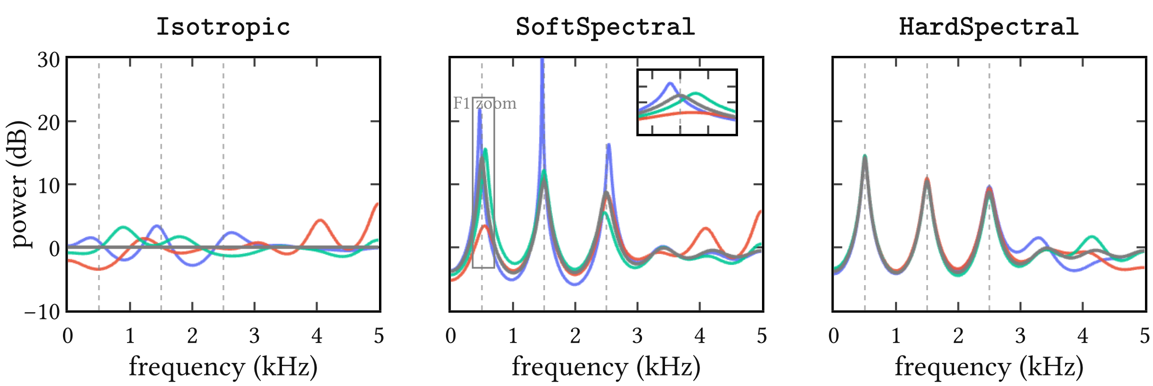

Figure 8 | **Example of imposing spectral features** on an isotropic base prior leading to the soft and hard spectral shaping priors from (45) and (48). We used a constraint polynomial $M(z) = Q_1(z) \cdot Q_2(z) \cdot Q_3(z)$ to encode the spectral features for this example. These are three resonances at $(F_1, F_2, F_3) = (500, 1500, 2500)$ Hz as indicated by the dashed vertical lines. Each panel shows the spectrum mean (grey) together with three samples (colors) of the associated prior. From left to right, these are (1) the base prior $p_0(\mathbf{a}) := \texttt{Isotropic}(0.1/P)$ with $P = 12$; (2) the soft spectral shaping prior $\texttt{SoftSpectral}(\mathbf{0}, (0.1/P)\mathbf{I}_P; M)$ with an inset that zooms in on the F_1 region; and (3) the hard spectral shaping prior $\texttt{HardSpectral}(\mathbf{0}, (0.1/P)\mathbf{I}_P; M)$.

3.3.4 The hard spectral shaping prior

If one prefers to enforce divisibility $M \mid A$ exactly rather than in expectation – for instance when a particular formant location is known with certainty – this can derived to by solving a related D_{KL} problem. However, since we already know $p^*(\mathbf{a})$ to be Gaussian, we can just condition it on the observation

$$\mathbf{f}(\mathbf{a}; M) = \mathbf{0}. \tag{46}$$

Then the standard conditioning formula gives the hard version of (45): the same mean shift, but the covariance is additionally deflated along the constrained directions,

$$\boldsymbol{\Sigma}_M = \boldsymbol{\Sigma}_0 - \boldsymbol{\Sigma}_0 \mathbf{F}^\top \left(\mathbf{F} \boldsymbol{\Sigma}_0 \mathbf{F}^\top\right)^{-1} \mathbf{F} \boldsymbol{\Sigma}_0, \tag{47}$$

and we write the *hard spectral shaping prior* as

$$\texttt{HardSpectral}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0; M) := \texttt{Normal}(\boldsymbol{\mu}_M, \boldsymbol{\Sigma}_M). \tag{48}$$

Since we are rarely certain of spectral features in advance, this prior might be overconfident so less useful for applications.

An example is shown in the last panel of Figure 8. Compared to the soft spectral shaping prior spectra (middle panel), it is seen that the hard spectral shaping prior collapses the uncertainty around the first three resonances, which is allowed to increase again towards the tail end of the spectrum.

3.4 Modeling or expanding?

Jaynes (1991) recounts in a memorable passage the confusion of the renowned experimental physicist R. W. Wood, who in 1911 proposed optical experiments to determine whether the sidebands $\omega \pm \nu$ in the amplitude-modulated wave

$$(1 + m \cos \nu t) \cos \omega t = \cos \omega t + \frac{m}{2} \cos(\omega + \nu)t + \frac{m}{2} \cos(\omega - \nu)t \tag{49}$$

are "physically real frequencies actually present," or whether "there is in reality only one frequency ω, but with varying amplitude." Today it is clear that the above is a mathematical identity; the two descriptions are two ways of saying the same thing, and "no experiment could have any bearing on it," as Jaynes wrote.

$\mathrm{AR}(P)$ filters raise similar questions. The polynomial $A(z)$ has P roots, and each conjugate pair near the unit circle produces a resonance peak in the spectral envelope $|H(e^{i\omega})|^2$. Now, the vocal tract genuinely has resonances – the formants $F_1, F_2, F_3, \ldots$ with their bandwidths $B_1, B_2, B_3, \ldots$ – and these are as physically real as anything in acoustics: they correspond to standing wave patterns in the cavities of the vocal tract and show up as broad peaks in the spectrum.

When P is chosen small, say $P \approx 2N_F$ where N_F is the number of formants in the analysis bandwidth, each conjugate pole pair tends to line up with one formant, and the correspondence is fairly direct. In this regime, $A(z)$ is acting as a *physical model* of the vocal tract, and it makes sense to constrain it: stability is a physical requirement (the vocal tract is a passive resonator), spectral tilt reflects the glottal source coupling, and the formant locations themselves are knowable a priori within a vowel class (Peterson & Barney, 1952).

But this is not the only way to read $A(z)$. In signal processing practice, P is routinely set much larger than $2N_F$ – the well-known rule of thumb $P \approx f_s/1000 + 2$ for a sampling rate f_s gives $P = 2$ at 10 kHz, which is generous enough that many poles will have no individual formant counterpart. These "extra" poles shape spectral tilt, model subglottal coupling, smooth over source harmonics, or simply approximate fine spectral structure that no small set of resonances could capture on its own.[6] In this regime, $A(z)$ is no longer a model but just a *rational expansion* of whatever the true transfer function happens to be – a Padé-type approximant whose individual poles are artifacts of the expansion order, not of the physics (Press et al., 1992).

This distinction underlies frequent confusions between poles and formants in acoustic phonetics (Titze et al., 2015). It also matters from the Bayesian point of view. If we are *modeling*, then encoding formant structure into the prior is encoding real physics and we should expect it to help inference. If we are *expanding*, then the same prior could easily turn into a straitjacket on what is essentially a flexible basis, and we should expect it to hurt inference. These two situations map cleanly on whether we assume informative or noninformative priors.

The same tension applies to the concept of stability itself. In the modeling regime, stability is a genuine physical constraint and enforcing it through the prior is well motivated. But in the expansion regime, a root landing slightly outside the unit circle need not signal physical instability – it may simply mean that the rational approximation of order P is locally imperfect, and penalizing it amounts to penalizing the approximation for being an approximation.

3.4.1 Modeling *and* expanding?

Where does the soft spectral shaping prior (45) stand on all this? In theory at least, rather well, because it operates at what one might call the *suggestion* level.

The constraint polynomial $M(z)$ encodes beliefs about the few poles that *are* likely to have physical counterparts – the formants F_1, F_2, F_3 with their bandwidths, perhaps a spectral tilt factor – and (44) shifts the prior mean to a coefficient vector $\boldsymbol{\mu}_M$ that is compatible with these features in expectation. But note that the covariance $\boldsymbol{\Sigma}_0$ is left untouched. Indeed, the remaining $(P - L_{\text{eff}})$ degrees of freedom, corresponding to the "expansion" poles that have

[6] A single broad formant, for instance, can easily be shaped by two or three poles acting together rather than a single conjugate pair (Van Soom & de Boer, 2021). Conversely, a pole may sit far from the unit circle and contribute little more than a nearly flat spectral factor. The correspondence between poles and formants is thus many-to-many, not one-to-one, except approximately in the parsimonious regime.

no individual physical meaning, are entirely free to go wherever the data require. So we are simultaneously modeling *and* expanding.

Nevertheless, even this soft, suggestive encoding of spectral structure may not always help. The $\lambda \propto 1/P$ scaling rule derived in (26) keeps expected power controlled under an isotropic prior, and does so without imposing any spectral shape at all. Whether the soft spectral shaping prior (45) actually improves inference over this simple baseline – or whether the mean shift inadvertently constrains the expansion poles in ways that hurt the fit – is ultimately an empirical question that depends on the data, the phonation type, and the model order P.

3.5 Summary

The IKLP machinery from the previous chapter requires us to choose a prior $p(\mathbf{a})$ for the AR coefficients. In this chapter we discuss three families of such priors, each attempting to encode progressively more structure about the vocal tract filter.

We began with the *isotropic prior* and showed that it entangles two things one would rather control independently: expected output power, which requires $\lambda \propto 1/P$ to stay bounded, and stability, which tends to collapse with P. In practice, the choice $\lambda := 0.1/P$ appears to be a good comprise, however.

The *Monahan prior* attempts to address the stability issue in a more principled way by approximating the uniform distribution over stable AR coefficient vectors, but the geometry of the stability region is too irregular to be faithfully captured by any Gaussian, and the attempt does not survive increasing P.

The *soft* and *hard spectral shaping priors* take a different approach entirely: rather than constraining where the coefficients may live, they encode what the resulting spectrum should roughly look like. These priors answer the question of how to anticipate complex speech-like spectra with something as simple as a Gaussian prior.

The proposed construction rests on the idea that an operational handle on "$A(z)$ should have certain roots" is given by the concept of polynomial divisibility. That is, we can encode the desired spectral features as roots of $A(z)$, and ask that they divide the latter "on average." This turns out to be a linear constraint on the coefficients $\mathbf{a}$, and we can solve the associated D_{KL} update in closed form. We present soft and hard flavors of the principle, where the former merely suggests spectral features and the latter insists on them.

This is, as far as we know, a new possibility to have it both ways in general AR modeling: *model* what you know, *expand* where you don't. The soft and hard spectral shaping priors have broad applicability beyond speech processing and can be used for modeling any time series, whether finance or EEG signals. Because they are derived from a well-posed maximum entropy problem, they form the bridge between work in the literature on *model-based* (informative) priors on the one hand, and *expansion-based* (uninformative) priors on the other (Cuevas et al., 2021; Gibson, 2018; Sørbye & Rue, 2017).

4 From parametric to nonparametric glottal flow models

Glottal flow models (GFMs) describe the source signal $u(t)$ as it drives the vocal tract during voiced speech. These models operate in the time domain because the delicate phase characteristics of the *glottal cycle* are an integral part of vocal communication.[7]

Decades of empirical work have over time produced a "model zoo" of GFMs: handcrafted waveforms of $u(t)$ with handfuls of carefully chosen parameters. Because these are *parametric* models, they all share the same basic trait: interpretable, but inevitably inflexible (Schleusing, 2012). Their parametric nature limits the range of time domain information they can encode. This motivates the construction of a family of *nonparametric* models, which are more expressive as they are able to grow "parametric capacity" depending on the data at hand.

With the many glottal flow models to follow, it is all too easy to lose connection to the actual physical phenomenon being studied. After a short look at the glottal cycle, we begin with a review of the classic Liljencrants-Fant model. Then we revisit the parametric piecewise polynomial models and generalize them to the nonparametric regime, where they are identified with the *arc cosine kernel* of Cho & Saul (2009). The latter is argued to combine the advantages of both worlds: the interpretability of parametric models with the flexibility of nonparametric ones. This then sets the stage for Chapter 5, which derives the *(quasi)periodic arc cosine kernel* as a learnable surrogate model of synthetic glottal flow data.

The glottal cycle is the periodic opening and closing of the vocal folds, depicted in Figure 9. During one *pitch period*, a single cycle is completed, in this case spanning $T = 6$ ms. GFMs

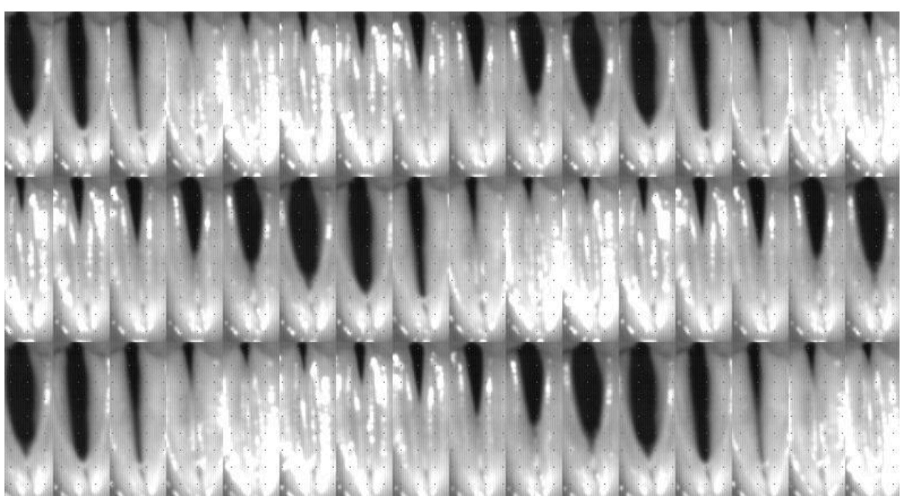

Figure 9 | **A look at the glottal cycle** via laryngeal stroboscopy of the vocal folds. Each frame here is a picture taken at intervals spanning different cycles to give the illusion of steady-state behavior. The glottis is the opening between the folds. After Chen (2016); Jopling (2009).

[7] For example, plosives: glottal stops are micro-events at the millisecond level that underlie semantic information two orders of magnitude higher up the scale. The timing of the vocal fold movement is so precisely controlled by our brain that the slightest deviations are studied as biomarkers for neurodegenerative diseases like Parkinson's (Ma et al., 2020).

describe the rate of airflow $u(t)$ [typically expressed in cm³/s] through the opening between the vocal folds called the *glottis*, visible as a black slit in Figure 9. Importantly, it is not at maximum glottis aperture that the acoustic output is greatest; it is at the sudden *glottal closure instant* (GCI), as first shown by Miller (1959) in modern literature. At this point, the moving air column in the vocal tract is abruptly interrupted, and kinetic energy is converted efficiently into acoustic energy which then excites the vocal tract much like plucking a harp or clapping in a resonant room (Chen, 2016, Section 4.6). The sharpness of this transition governs much of the clarity and perceived strength of voiced speech (Fant, 1979). Thus, efficiency demands that the glottal cycle be strongly asymmetric: a slow buildup of flow during the open phase, followed by a rapid, almost impulsive closure to the closed phase.

4.1 The Liljencrants-Fant model

A model that has been very useful in describing how $u(t)$ varies during the glottal cycle is the *Liljencrants-Fant (LF) model* proposed by Fant et al. (1985). It has been the GFM of choice for many joint-inverse filtering approaches[8] but also been studied in its own right.[9]

The model states that $u'(t)$ in a single pitch period of length T consists of three parts, or phases. The *open phase* (O) is modeled as a rising and falling exponential sinusoid part, and transitions to the *closed phase* (C) via an exponential return during the *return phase* (R). The latter is thus part of (C). Mathematically, this is defined piecewise as

$$u'_{\mathrm{LF}}(t) = \begin{cases} -E_e e^{a(t-t_e)} \frac{\sin(\pi t/t_e)}{\sin(\pi t_e/t_p)} & 0 \leq\ t \leq t_e \qquad \text{(O)} \\ -\frac{E_e}{\varepsilon T_a}\left(e^{-\varepsilon(t-t_e)} - e^{-\varepsilon(t_c-t_e)}\right) & t_e < t \leq t_c \qquad \text{(C, R)} \\ 0 & t_c \leq t \leq T \qquad \text{(C)} \end{cases} \tag{50}$$

where $\boldsymbol{\theta}_{\mathrm{LF}} = \{E_e, t_p, t_e, t_c, T_a, T\}$ are the six model parameters explained below in (53) and $\{a, \varepsilon\}$ are determined implicitly from the *closure constraint*

$$\int_0^T u'(t)\,\mathrm{d}t = 0 \quad \text{such that} \quad u(0) = u(T), \tag{51}$$

which holds for any GFM, and the continuity constraint

$$\lim_{t \underset{<}{\to} t_e} u'(t) = \lim_{t \underset{>}{\to} t_e} u'(t) \tag{52}$$

which is peculiar to GFMs with smooth return phases rather than hard, discontinuous GCI events (Veldhuis, 1998). Note that like most GFMs, the LF model is defined in $u'(t)$ not $u(t)$ due to the radiation effect (5), though $u(t)$ can be recovered easily via integration (6). Figure 10 shows the waveforms for $u'_{\mathrm{LF}}(t)$ and $u_{\mathrm{LF}}(t)$ for a typical setting of $\boldsymbol{\theta}_{\mathrm{LF}}$. When varying $\boldsymbol{\theta}_{\mathrm{LF}}$

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[8] See **Joint source-filter optimization** (page 3).

[9] See Degottex (2010, Section 2.4) for a discussion. Research into the LF model falls mainly into study of its spectral characteristics (Doval et al., 2006) and (re)parametrization, notably Fant (1995); Perrotin et al. (2021).

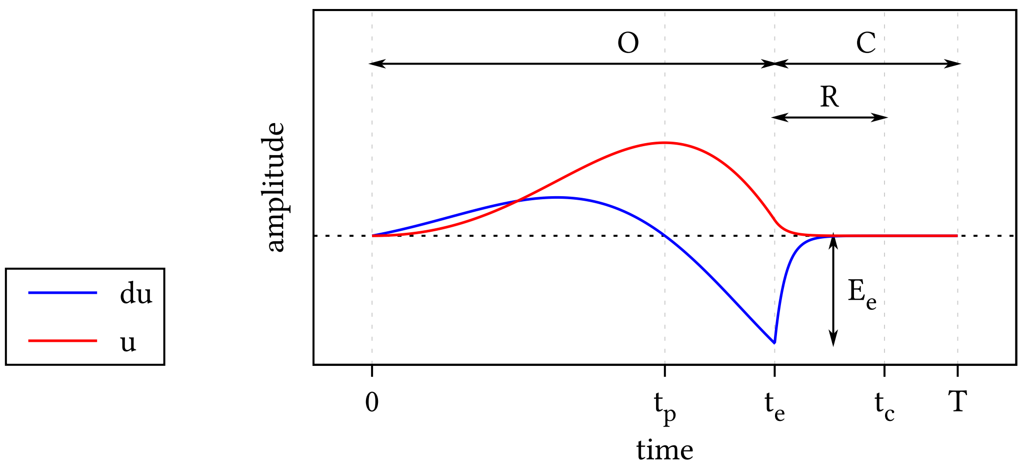


Figure 10 | **The Liljencrants-Fant model** for —— $u'_{\mathrm{LF}}(t)$ and —— $u_{\mathrm{LF}}(t)$ during a single period of length T. The open phase (O), closed phase (C) and return phase (R) are marked at the top and are determined by the changepoints $\{t_p, t_e, t_c, T\}$. The amplitude at peak excitation t_e is E_e. Not shown here is time constant of the return phase T_a, which during (R) determines the steepness of exponential decay towards zero in $u'(t)$.

the LF waveforms trace a family of broad to narrow asymmetric pulses with smooth to abrupt closure encoding positive ($u(t) \geq 0$) glottal flow.

Parameters of the LF model The LF model achieved broad adoption because it strikes a balance between simplicity and empirical accuracy. After several iterations, Fant et al. (1985) arrived at this parametrization:

$$\boldsymbol{\theta}_{\mathrm{LF}} = \begin{cases} E_e & = \min_t u'(t) = u'(t_e) & \text{peak excitation amplitude} \\ t_p & = \operatorname{argmax}_t u(t) & \text{instant of peak glottal flow} \\ t_e & = \operatorname{argmin}_t u'(t) & \text{instant of peak excitation} \\ t_c & & \text{instant of glottal closure (GCI)} \\ T_a & & \text{time constant of the return phase} \\ T & & \text{fundamental period} \end{cases} \tag{53}$$

which turned out to be effective in capturing the glottal cycle in both modal and more extreme phonations.[10] Note that, apart from E_e, which sets the vertical scale, these are all time domain parameters which determine *changepoints* in the glottal cycle in Figure 9. These tend to covary significantly and can in fact be regressed into a single convenient parameter R_d (Fant, 1995).

[10]Actually, Fant et al. present LF as a "four-parameter model". They ignore E_e and set $T := t_c$ for convenience, thus arriving at a count of four not six as in (53). In their unifying theory of common GFMs, Doval et al. (2006) also collapse T to t_c. It is common too in applications (for example, O Cinneide, 2012). Of course, nothing is really lost because a closed phase of any length can be appended to any GFM by piecewise concatenation. We choose to keep t_c distinct from T explicitly because this becomes significant for short return phases and is experimentally observed to be important, especially in clinical scenarios (Kreiman et al., 2007; Rosenberg, 1971).

From the modeling perspective, the most important changepoints are t_e and t_c: when these lie close together and return is swift (T_a small and vanishing return phase), the glottal cycle is at peak efficiency and $\tilde{u}(t)$ has broadband spectral energy. This results in a clear voice and sharp features in the speech waveform. Conversely, a smooth return (T_a medium to large) results in a longer return phase with t_c tending to T and the GCI becoming ill-defined. The result here is a more mellow voice and more smooth features in the speech waveform. Both modes are common in voiced speech (Maurer, 2016) and GIF algorithms must be able to handle both cases.

Implementation In practice, the LF model is unwieldy to compute (Gobl, 2017; Veldhuis, 1998). The closure and continuity constraints in (51) and (52) lack an analytical solution, meaning that for each evaluation of $u'(t)$ one must numerically solve for the auxiliary constants $\{a, \varepsilon\}$ by a bisection or root-finding routine. This seems like a high price to pay for an analytical model, and worse, this optimization is brittle. As noted by Perrotin et al. (2021), simplified LF models exist that are perceptually equivalent to LF and easier to work with.

Nevertheless, it remains the de facto standard in parametric source-filter modeling and we therefore use it here. Our `JAX` implementation of the LF equations (Bradbury et al., 2020), with compiled bisection routines and end-to-end differentiability, can be found **on this web page**. The code also implements four LF parametrizations, including the previously mentioned R_d formulation of Fant (1995).

4.1.1 One of many: the glottal flow "model zoo"

Over the years, a large number of glottal flow models have been proposed in acoustic phonetics. A sample of these is shown in Figure 11, already an impressive lineup by 1986. Despite their visual differences, these models are, essentially, variations on a single theme: they all describe a glottal pulse that tracks the periodic opening and closing of the glottis.

Doval et al. (2006) demonstrated that the most common GFMs[11] share a common mathematical structure and can be unified under a single framework; the $\boldsymbol{\theta}_{\text{LF}}$ parameters in (53) can in fact express all of them. These GFMs all model a glottal flow $u(t)$ that is positive or null, continuous, and typically differentiable except at the glottal closure instant (GCI). Within

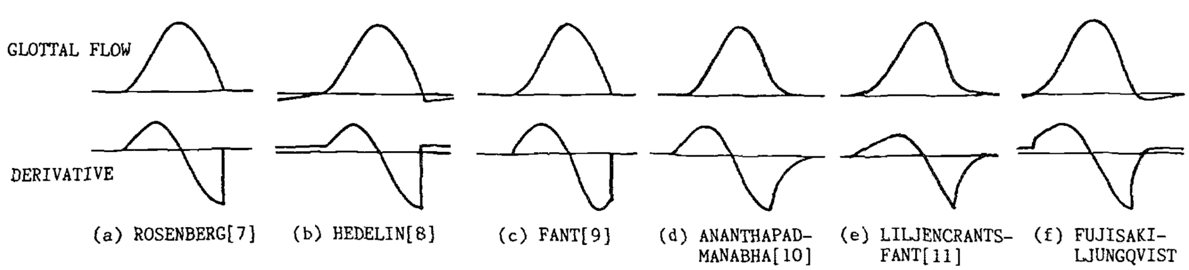


Figure 11 | **A lineup of GF models** back from 1986 (handdrawn?), together with their derivatives (DGFs). These models are visually very similar. From Fujisaki & Ljungqvist.

[11] These are LF, KLGLOTT88 (Klatt & Klatt, 1990), Rosenberg C (Rosenberg, 1971), R++ (Veldhuis, 1998).

each period, $u(t)$ rises during the opening phase, falls during the closing phase, and returns to zero during the closed phase, possibly through a short return phase that smooths the closure. The derivative $u'(t)$ alternates positive, zero, and negative in the expected order and integrates to zero over one period, satisfying the closure constraint (51). It is continuous in case of a nonempty return phase and discontinuous otherwise (meaning a jump at t_c as in Figure 12), known as hard closure.

Allow partial closure? The strict closure constraint in (51) is of course just a convenient modeling choice and must be expected to be violated to varying degrees in reality, as real folds often leak, especially in breathy or pathological voices. On the one hand, it clearly is a defining feature[12] of the periodic nature of the glottal cycle. On the other, empirical data show the limits of this hard constraint. For example, Kreiman et al. (2007, p. 601) observe that

> in many cases in our data, returning flow derivatives to 0 at the end of the cycle conflicted with the need to match the experimental data and conflicted with the requirement for equal areas under positive and negative curves in the flow derivative. [...] These conflicts between model features, constraints, and the empirical data were handled by abandoning the equal area constraint.[13]

The approach we take in Chapter 5 is to learn a softened version of the closure constraint (51) from data rather than set it a priori. This means we learn both whether the flow typically returns to zero and, if not, to what extent this constraint is plausibly violated.

Allow negative flow? Most GFMs enforce a positive glottal flow and assert $u(t) \geq 0$. But there are exceptions such as Fujisaki & Ljungqvist (1986). They allow for transient negative flow which could represent a lowering of the vocal folds after GCI (presumably when air is sucked back into the lungs). As with the closure constraint, we will learn this effect from data in Chapter 5 and do not forbid $u(t) < 0$ a priori.

4.2 Classic piecewise polynomial models

Polynomial GFMs occupy an interesting corner of the "model zoo." They fit into the same piecewise framework we have been using for the LF model, but with each part modeled as a polynomial of degree d. We list several GFMs proposed in the literature in Table 3.

This class of GFMs mainly appeal to simplicity and spectral strength. Piecewise polynomial models admit an analytical solution to the closure constraint while remaining computationally cheap. More importantly, the sharp transitions generated by low-order polynomials have

[12]The pioneering study on GIF by Miller (1959) had to deconvolve taped speech waveforms with analogue resistor networks and manual optimization. A review article on GIF methods notes the part played by the defining quality of the closure constraint in this search for $u(t)$: "the fine-tuning of the resistors of the inverse network was conducted by searching for such values that yielded zero flow after the instant of glottal closure" (Alku, 2011, p. 626).

[13]That "equal area constraint" is just another name for the closure constraint (51).

Degree d	Name	Reference
0	Triangular pulse model	Alku et al. (2002)
1	—	Titze (2000); Verdolini & Titze (1995)
2	KLGLOTT88	Klatt & Klatt (1990) Chang-Sheng Yang & Kasuya (1998)
3	R++	Veldhuis (1998)
3	FL model	Fujisaki & Ljungqvist (1986)
6	—	Childers & Hu (1994)

Table 3 | **Classic piecewise polynomial models** from the literature. Of these, KLGLOTT88 and R++ remain popular today (Doval et al., 2006).

bright spectra with slow decay, which is exactly what it takes to model hard GCIs (Childers & Hu, 1994) and score well on perceptual tests of synthetic speech (Rosenberg, 1971).

To get a feel for these models, we now turn to the simplest case: $d = 0$.

4.2.1 The triangular pulse model

The simplest polynomial model is the *triangular pulse model* proposed by Alku et al. (2002). It takes $u(t)$ piecewise linear and $u'(t)$ piecewise constant, as shown in Figure 12. Mathematically, the triangular pulse model is given by:

$$u'_\Delta(t) = \begin{cases} +E_e\left(\frac{t_c}{t_p} - 1\right) & 0 < t \leq t_p \quad \text{(O)} \\ -E_e & t_p < t \leq t_c \quad \text{(O)} \\ 0 & t_c < t \leq T \quad \text{(C)} \end{cases} \tag{54}$$

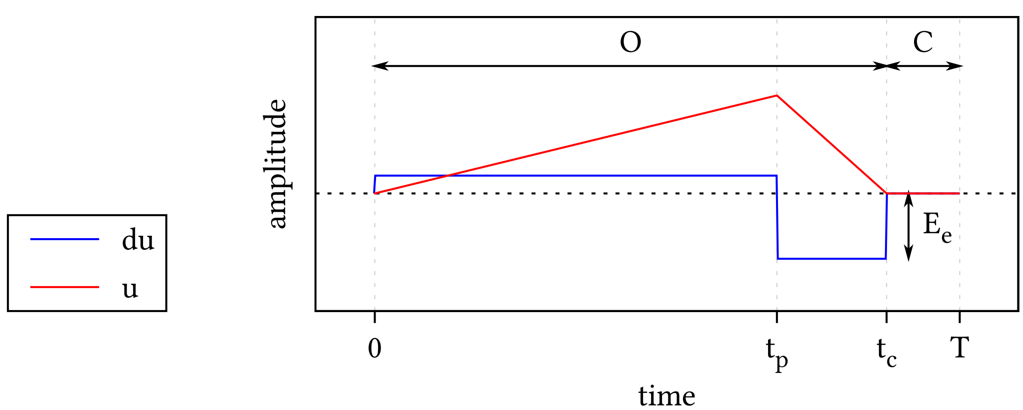


Figure 12 | **The triangular pulse model** for $u'_\Delta(t)$ and $u_\Delta(t)$ during a single period of length T. The parameters used in this model are identical to Figure 10, except for the lack of a return phase and that the 'instant' of maximum excitation t_e now spans the interval $[t_p, t_c]$.

The model has no return phase and only four parameters: $\boldsymbol{\theta}_\Delta = \{E_e, t_p, t_c, T\}$. Note that the amplitude of the first case in (54) is fixed because the closure constraint (51) removes one degree of freedom. Alku et al. (2002) point out that this enables a difficult-to-measure time domain quantity to be expressed as the ratio of two easy-to-measure quantities in the amplitude domain and exploit this fact to measure the open quotient more robustly.

4.3 Parametric piecewise polynomial models

The triangular pulse model (54) contains two jumps in the derivative $u'_\Delta(t)$, so we can write the latter more compactly as a linear combination of two Heaviside functions

$$u'_\Delta(t) = \begin{cases} a_1(t-b_1)_+^0 + a_2(t-b_2)_+^0 & 0 < t \leq t_c \quad \text{(O)} \\ 0 & t_c < t \leq T \quad \text{(C)} \end{cases} \tag{55}$$

where $(t-b)_+^0 = \texttt{Heaviside}(t-b)$ and

$$\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = E_e \begin{pmatrix} \frac{t_c}{t_p} - 1 \\ -\frac{t_c}{t_p} \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} 0 \\ t_p \end{pmatrix}. \tag{56}$$

During the open phase, we can interpret the rewritten triangular pulse model as a *parametric piecewise polynomial model* of *degree* $d = 0$ and *order* $H = 2$. It has parameters $\boldsymbol{\theta}_2 = \{\mathbf{a}, \mathbf{b}\}$ which describe $H = 2$ jumps with amplitudes $\mathbf{a}$ at locations $\mathbf{b}$. Note that in this formulation the amplitudes $\mathbf{a}$ differ from the piecewise amplitudes in (54) but are still linearly dependent given the changepoints $\mathbf{b}$ and t_c.

Continuing, nothing stops us from generalizing both H and d. The open phase part of (55) becomes:

$$u'_H(t) = \sum_{h=1}^{H} a_h (t-b_h)_+^d \qquad 0 < t \leq t_c \qquad \text{(O)} \tag{57}$$

This is the general parametric piecewise polynomial model of degree d and order H. Again, its parameters $\boldsymbol{\theta}_H = \{\mathbf{a}, \mathbf{b}\}$ are the amplitudes $\mathbf{a} = (a_1, ..., a_H)^\top \in \mathbb{R}^H$ and the changepoints $\mathbf{b} = (b_1, ..., b_H)^\top \in \mathbb{R}^H$. These scale and shift H piecewise monomials of degree d:

$$(t-b)_+^d = (t-b)^d \, \texttt{Heaviside}(t-b) = \begin{cases} 0 & t < b \\ (t-b)^d & t \geq b \end{cases} \tag{58}$$

We recognize (58) as the family of activation functions called *rectified power units* (RePUs). These include Heaviside for $d = 0$ and the influential ReLU function for $d = 1$, both shown among higher degree variants in Figure 13. It may sound like a stretch to relate polynomial GFMs to neural nets, but this view of things will pay off greatly in Section 4.4 when we take the limit $H \to \infty$.

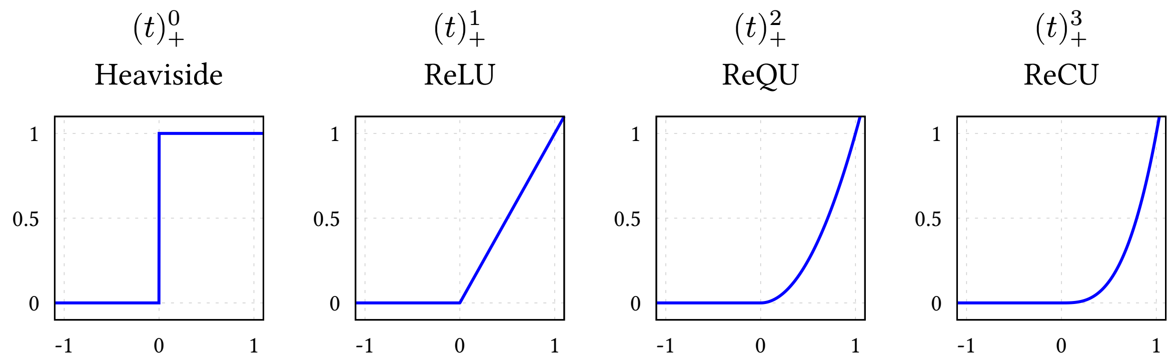


Figure 13 | **The RePU activation functions** —— $(t)_+^d$ for $d \in \{0, 1, 2, 3\}$.

4.3.1 Regression view

GFMs ultimately play the role of regression models in the GIF problem. From this point of view, the general parametric model in (57) is just a standard *linear model*, and we will now develop it into a Bayesian regression model following MacKay (1998).

Suppose we observe N time samples $\boldsymbol{t} = \{t_n\}_{n=1}^N$ during one glottal cycle, and fix H basis functions of the form

$$\phi_h(t) = (t - b_h)_+^d, \tag{59}$$

where the changepoints $\mathbf{b}$ and t_c are treated as given. Collecting these into the design matrix $\boldsymbol{\Phi} \in \mathbb{R}^{N \times H}$ with

$$[\boldsymbol{\Phi}]_{nh} = \phi_h(t_n), \tag{60}$$

the general parametric model during the open phase becomes simply

$$\mathbf{u}' = \boldsymbol{\Phi}\mathbf{a}. \tag{61}$$

This is linear in the amplitudes $\mathbf{a}$ given $\boldsymbol{\Phi}$, so the model is rather constrained at this point. Indeed, $\mathbf{a}$ is currently the only source of variability and power in this model because we assume all other parameters fixed. Which prior then for $\mathbf{a}$ should we choose? If we want to maximize model support while adhering to the known additivity and power constraints, the optimal choice is the canonical Gaussian prior over amplitudes (Bretthorst, 1988):

$$p(\mathbf{a} \mid \mathbf{b}) = \texttt{Normal}(\mathbf{a} \mid \mathbf{0}, \sigma_a^2 \mathbf{I}). \tag{62}$$

Moving on to data space, the induced prior probability of $\mathbf{u}'$ is also Gaussian,

$$p(\mathbf{u}' \mid \mathbf{b}) = \int \delta(\mathbf{u}' - \boldsymbol{\Phi}\mathbf{a})\, p(\mathbf{a} \mid \mathbf{b})\, \mathrm{d}\mathbf{a} = \texttt{Normal}(\mathbf{u}' \mid \mathbf{0}, \sigma_a^2 \boldsymbol{\Phi}\boldsymbol{\Phi}^\top), \tag{63}$$

due to closure of the Gaussian under linear combinations, and

$$\mathbb{E}_{\mathbf{a}|\mathbf{b}}[\mathbf{u}'] = \mathbf{\Phi}\,\mathbb{E}_{\mathbf{a}|\mathbf{b}}[\mathbf{a}] = 0, \tag{64}$$

$$\mathbb{E}_{\mathbf{a}|\mathbf{b}}\left[\mathbf{u}'\mathbf{u}'^{\top}\right] = \mathbf{\Phi}\,\mathbb{E}_{\mathbf{a}|\mathbf{b}}[\mathbf{a}\mathbf{a}^{\top}]\,\mathbf{\Phi}^{\top} = \sigma_a^2\mathbf{\Phi}\mathbf{\Phi}^{\top}. \tag{65}$$

Because $\operatorname{rank}(\mathbf{\Phi}\mathbf{\Phi}^{\top}) \leq H$, prior samples of $\mathbf{u}'$ are intrinsically confined to an H-dimensional subspace of $\mathbb{R}^N$. Assuming $H < N$, we thus see that linear models like (61) are inherently low-rank. Their expressivity grows with H and saturates when H reaches N.

Closure constraint The closure constraint (51) becomes

$$\int_0^T u'_H(t)\,\mathrm{d}t = \sum_{h=1}^{H} a_h \int_0^{t_c} \phi_h(t)\,\mathrm{d}t = \sum_{h=1}^{H} a_h r_h = 0, \tag{66}$$

with $r_h = (t_c - b_h)^{d+1}/(d+1)$. Observe that the prior (62) already satisfies the closure constraint in expectation:

$$\mathbb{E}_{\mathbf{a}|\mathbf{b}}\left[\sum_{h=1}^{H} a_h r_h\right] = \sum_{h=1}^{H} \mathbb{E}_{\mathbf{a}|\mathbf{b}}[a_h]\,r_h = 0. \tag{67}$$

This motivates setting the prior mean of the $\mathbf{a}$ to zero, otherwise than symmetry of ignorance around $\mathbf{a} = \mathbf{0}$. We can also enforce it:[14]

$$\sum_{h=1}^{H} a_h r_h = 0 \Longrightarrow p(\mathbf{a} \mid \texttt{closure}, \mathbf{b}) = \texttt{Normal}\left(\mathbf{a} \mid \mathbf{0}, \sigma_a^2(\mathbf{I} - \mathbf{q}\mathbf{q}^{\top})\right) \tag{68}$$

where $\mathbf{r} = (r_1, ..., r_H)^{\top}$ and $\mathbf{q} = \frac{\mathbf{r}}{\|\mathbf{r}\|}$. A convenient way to a sample from this updated prior makes use of the fact that this is a rank-one projection:

$$\mathbf{a} = (\mathbf{q}\mathbf{q}^{\top})\mathbf{a} + (\mathbf{I} - \mathbf{q}\mathbf{q}^{\top})\mathbf{a} \quad \sim \quad p(\mathbf{a} \mid \mathbf{b}) \tag{69}$$

$$\Longrightarrow \mathbf{a}_{\perp} = \qquad (\mathbf{I} - \mathbf{q}\mathbf{q}^{\top})\mathbf{a} \quad \sim \quad p(\mathbf{a} \mid \texttt{closure}, \mathbf{b}). \tag{70}$$

How does this look in data space? By the same calculation (64), marginalizing over $\mathbf{a}$ yields

$$p(\mathbf{u}' \mid \texttt{closure}, \mathbf{b}) = \texttt{Normal}\left(\mathbf{u}' \mid \mathbf{0}, \sigma_a^2\mathbf{\Phi}(\mathbf{I} - \mathbf{q}\mathbf{q}^{\top})\mathbf{\Phi}^{\top}\right). \tag{71}$$

We can then obtain $\mathbf{u}$ from $\mathbf{u}'$ via integration using (6). Compared to samples from the isotropic prior (63), the anisotropic prior (71) induces glottal flows $u(t)$ that tend to look more pulse-like, as shown in Table 4. Indeed, moving to the bottom right corner, it is seen that increasing H and d tends to produce LF-like pulses out of the box!

Regression with (71) ensures that posterior mass vanishes at solutions $\mathbf{u}$ that violate the closure constraint. This illustrates how linear constraints on the amplitudes $\mathbf{a}$ in the linear model (61) can encode GFM properties at the cost of just a single rank-one downdate per constraint.

[14]The general solution is given by $\operatorname{argmin}_f D_{\mathrm{KL}}[f(\mathbf{a}) \parallel p(\mathbf{a} \mid \mathbf{b})]$ under the constraint (66), but the closure under linear conditioning of Gaussian distributions allow us to take a shortcut here.

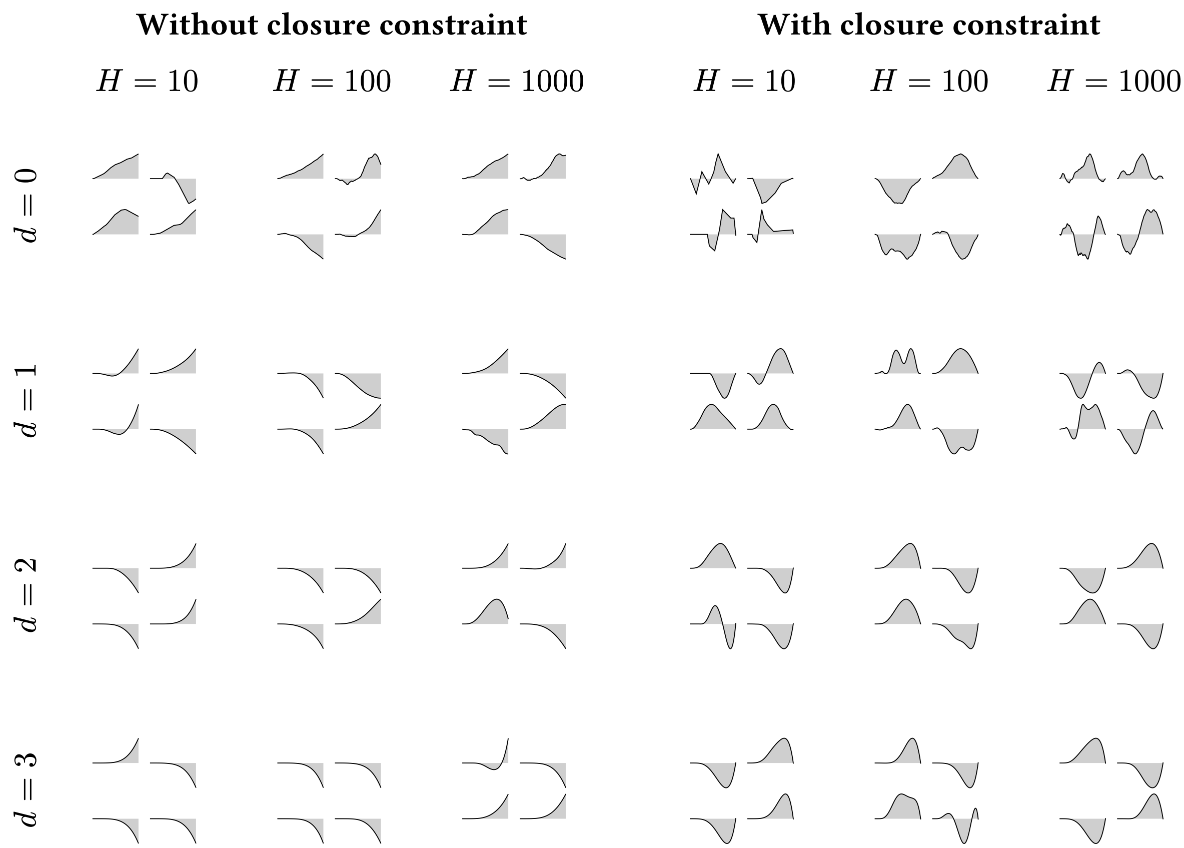


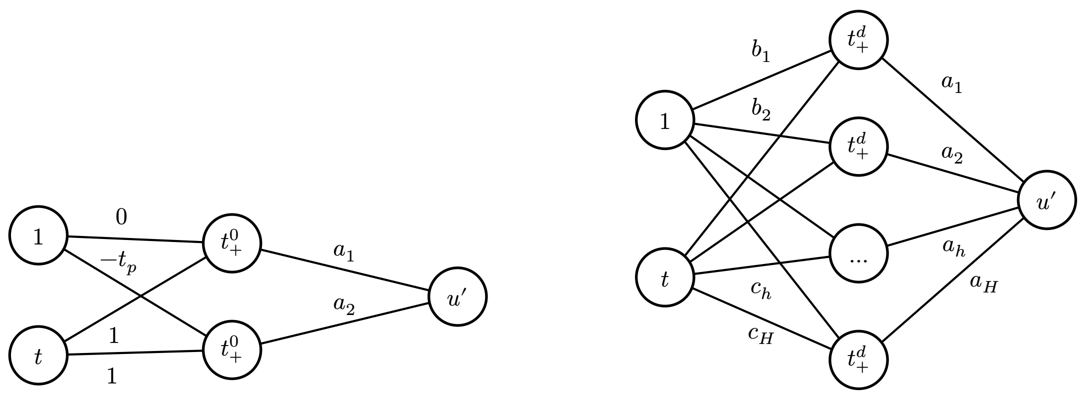
Table 4 | **Samples of $u(t)$** from the general parametric piecewise polynomial model (57) with or without the closure constraint (66). For each combination of degree d and order H, a total of H changepoints $b_h \sim \texttt{Uniform}(0, t_c)$ were drawn randomly, and four instances of $\mathbf{u}'$ conditioned on these were sampled either from (63) [**Without closure constraint**] or from (71) [**With closure constraint**]. Then $u(t) = \int_0^t u'(\tau)\,\mathrm{d}\tau$ was obtained by quadrature of $\mathbf{u}'$.

Moving on from linear models to Gaussian processes, these linear constraints become linear functionals known as *interdomain features*, which may be used to impose structure directly in function space, without touching rank, but more challenging mathematically.

Connection to classic polynomial models Observe that the analytical solution (68) to the closure constraint is valid for any degree $d \geq 0$ and order $H \geq 1$, so each classic polynomial model listed in Table 3 can be expressed in our linear model framework given suitable choices of $\mathbf{b}$ and t_c, without specialized derivations.

4.3.2 Neural network equivalence

Figure 14 illustrates that the triangular pulse model of Section 4.2.1 can be seen as a tiny neural network. Likewise, the general parametric piecewise polynomial model (57) can be recast as a single hidden layer of width H with a RePU activation and a linear readout:



(a) (b)

Figure 14 | **Neural network view.** (a) The triangular pulse model as a tiny neural network with a single hidden layer, linear readout and t_+^0 activation. (b) The general parametric polynomial model of degree d with order H with hidden weights $\{\mathbf{b}, \mathbf{c}\} \in \mathbb{R}^{2H}$, output weights $\mathbf{a} \in \mathbb{R}^H$ and RePU activation function t_+^d.

$$u'_{\text{NN}}(t) = \sum_{h=1}^{H} a_h (c_h t - b_h)_+^d. \tag{72}$$

Each unit computes $\phi_h(t) = \left(\mathbf{w}_h^\top \mathbf{x}_t\right)_+^d$ with hidden weights $\mathbf{w}_h = (c_h, -b_h)^\top$ and $\mathbf{x}_t = (1, t)^\top$, exactly like a neuron with a bias input. The $3H$ parameters of the neural net are thus $\mathbf{a}$ and $\mathbf{b}$ (as before) and the newly acquired degrees of freedom $\mathbf{c} = (c_1, ..., c_H)^\top$.

Seen from this point of view, a natural symmetry appears. We argued in Section 4.3.1 that in linear regression context the output weights $\mathbf{a}$ are best assigned an uninformative Gaussian prior, so

$$p(\mathbf{a}) = \texttt{Normal}(\mathbf{a} \mid \mathbf{0}, \sigma_a^2 \mathbf{I}). \tag{73}$$

But the neural network equivalence makes very clear that the hidden weights $\{\mathbf{b}, \mathbf{c}\}$ actually play similar roles: they enter linearly inside the activation, so the same argument applies, and we are encouraged to assign Gaussians again:

$$p(\mathbf{b}) = \texttt{Normal}(\mathbf{b} \mid \mathbf{0}, \sigma_b^2 \mathbf{I}), \quad p(\mathbf{c}) = \texttt{Normal}(\mathbf{c} \mid \mathbf{0}, \sigma_c^2 \mathbf{I}). \tag{74}$$

This step is not obvious from the viewpoint of any of the GFMs we've discussed before, where changepoints $\mathbf{b}$ were nonlinear hyperparameters fixed a priori; here they become "active" and "linear" degrees of freedom. Assuming t_c is known, the remaining hyperparameters to describe the open phase are thus the three scales $\boldsymbol{\theta}_{\text{NN}} = \{\sigma_a, \sigma_b, \sigma_c\}$ controlling amplitude, typical changepoint location, and horizontal stretch, respectively.

Turning to data space again and having assigned priors to all parameters, we can now do a full marginalization unlike the conditional in (63). Updating the design matrix $\boldsymbol{\Phi}$ to

$$[\mathbf{\Phi}]_{nh} = \phi_h(t_n) = (c_h t_n - b_h)_+^d, \tag{75}$$

and assuming the isotropic priors (73) and (74) we have

$$p(\mathbf{u}') = \int \delta(\mathbf{u}' - \mathbf{\Phi}\mathbf{a})\, p(\mathbf{a})\, p(\mathbf{b})\, p(\mathbf{c})\, \mathrm{d}\mathbf{a}\, \mathrm{d}\mathbf{b}\, \mathrm{d}\mathbf{c} \tag{76}$$

$$= \int \mathtt{Normal}\big(\mathbf{u}' \mid \mathbf{0}, \sigma_a^2 \mathbf{\Phi}\mathbf{\Phi}^\top\big)\, p(\mathbf{b})\, p(\mathbf{c})\, \mathrm{d}\mathbf{b}\, \mathrm{d}\mathbf{c}. \tag{77}$$

This is a Gaussian mixture with varying covariances (not means) as $\mathbf{\Phi}$ depends on the hidden weights $\{\mathbf{b}, \mathbf{c}\}$. A closed form for this integral exists only for $H \to \infty$ as $p(\mathbf{u}')$ converges to a Gaussian process, as we will see in Section 4.4.

Closure constraint The closure constraint (51) given $\{\mathbf{b}, \mathbf{c}\}$ remains a linear condition on $\mathbf{a}$, so we can define

$$p(\mathbf{a}, \mathbf{b}, \mathbf{c} \mid \mathtt{closure}) = p(\mathbf{a} \mid \mathtt{closure}, \mathbf{b}, \mathbf{c})\, p(\mathbf{b})\, p(\mathbf{c}), \tag{78}$$

where $p(\mathbf{a} \mid \mathtt{closure}, \mathbf{b}, \mathbf{c})$ is identical to (68) but with $r_h = (c_h t_c - b_h)_+^{d+1} / c_h(d+1)$. This prior yields a mixture in data space that is similar to (76):

$$p(\mathbf{u}' \mid \mathtt{closure}) = \int \mathtt{Normal}\big(\mathbf{u}' \mid \mathbf{0}, \sigma_a^2 \mathbf{\Phi}(\mathbf{I} - \mathbf{q}\mathbf{q}^\top)\mathbf{\Phi}^\top\big)\, p(\mathbf{b})\, p(\mathbf{c})\, \mathrm{d}\mathbf{b}\, \mathrm{d}\mathbf{c}. \tag{79}$$

Here both $\mathbf{q} = \frac{\mathbf{r}}{\|\mathbf{r}\|}$ and $\mathbf{\Phi}$ depend on $\{\mathbf{b}, \mathbf{c}\}$. This integral has no closed form for $H \to \infty$, but we can still impose the closure constraint on (76) indirectly in the spectral domain via interdomain features (Chapter 5).

Connection to LF and other GFMs The neural network view of the parametric polynomial model also clarifies expressivity. It is well known that a single RePU layer of sufficient width can approximate any continuous waveform on a bounded interval (Hornik, 1993). Therefore, with its analytical closure constraint (78) and the ability to represent sharp changepoint behavior, the parametric piecewise polynomial model in the form (72) effectively unifies the entire family of GFMs encountered in the literature, from triangular pulses to LF-type exponentials and sinusoids, given sufficiently large H.

4.4 Nonparametric piecewise polynomial models

The correspondence of piecewise polynomials GFMs with a single hidden layer shows that their expressivity depends mostly on the order H, as model order is equivalent to layer width in the neural network picture. Indeed, *parametric* linear models have a finite representation capacity determined primarily by rank because the modeled function can only explore the subspace spanned by the basisfunctions.[15] *Nonparametric* models grow capacity as the

[15]More precisely: a higher rank naturally enlarges the subspace of attainable functions, but the prior over amplitudes still constrains where probability mass is placed within that subspace. Geometrically, the model

number of datapoints increases; in the case of Gaussian processes, posterior inference given N datapoints is equivalent to Bayesian linear regression with N linearly independent basis-functions, hence full-rank (Rasmussen & Williams, 2006).

It thus makes sense to consider the limit $H \to \infty$ in our search for more expressive GFMs. Continuing where we left, we already saw in (76) that $p(\mathbf{u}')$ is a nontrivial Gaussian covariance mixture weighed by the isotropic priors defined in (74). This can be made explicit:

$$p(\mathbf{u}') = \int \texttt{Normal}(\mathbf{u}' \mid \mathbf{0}, \sigma_a^2 \mathbf{\Phi}\mathbf{\Phi}^\top)\, p(\mathbf{b})\, p(\mathbf{c})\, \mathrm{d}\mathbf{b}\, \mathrm{d}\mathbf{c} \tag{80}$$

$$= \int \texttt{Normal}(\mathbf{u}' \mid \mathbf{0}, \mathbf{Q})\, p(\mathbf{Q})\, \mathrm{d}\mathbf{Q}, \tag{81}$$

where we defined the push forward

$$p(\mathbf{Q}) = \int \delta(\mathbf{Q} - \sigma_a^2 \mathbf{\Phi}\mathbf{\Phi}^\top)\, p(\mathbf{b})\, p(\mathbf{c})\, \mathrm{d}\mathbf{b}\, \mathrm{d}\mathbf{c} \tag{82}$$

which essentially counts the number of $\{\mathbf{b}, \mathbf{c}\}$ configurations that give rise to a given $\mathbf{Q}$. Thus each random sample of $\{\mathbf{b}, \mathbf{c}\} \in \mathbb{R}^{2H}$ drawn from (74) induces in data space a random covariance matrix $\mathbf{Q} \in \mathbb{R}^{N \times N}$,

$$\{\mathbf{b}, \mathbf{c}\} \mapsto \mathbf{Q} = \sigma_a^2 \mathbf{\Phi}\mathbf{\Phi}, \qquad \operatorname{rank}(\mathbf{Q}) \leq H \tag{83}$$

and hence a Gaussian component with weight $p(\mathbf{Q})$ in the mixture (80). Below we will show that by choosing $\sigma_a \propto 1/\sqrt{H}$ and letting $H \to \infty$, the density of states (82) degenerates to

$$p(\mathbf{Q}) \longrightarrow \delta(\mathbf{Q} - \mathbf{K}), \qquad \operatorname{rank}(\mathbf{K}) = N \text{ almost surely}, \tag{84}$$

for some $\mathbf{K} \in \mathbb{R}^{N \times N}$ derived below in (120), causing the mixture to collapse to a single Gaussian and thus completing the transition to an indexed Gaussian process with covariance matrix $\mathbf{K}$.

4.4.1 Mean and covariance description

While the mixture (80) has no closed form for finite H, progress can be made by computing its mean and covariance as before in (64):

$$\mathbb{E}[\mathbf{u}'] = \mathbb{E}_{\mathbf{b},\mathbf{c}}\left[\mathbb{E}_{\mathbf{a}|\mathbf{b},\mathbf{c}}[\mathbf{u}']\right] = \mathbf{0}, \tag{85}$$

$$\mathbb{E}\left[\mathbf{u}'\mathbf{u}'^\top\right] = \mathbb{E}_{\mathbf{b},\mathbf{c}}\left[\mathbb{E}_{\mathbf{a}|\mathbf{b},\mathbf{c}}\left[\mathbf{u}'\mathbf{u}'^\top\right]\right] = \sigma_a^2\, \mathbb{E}_{\mathbf{b},\mathbf{c}}[\mathbf{\Phi}\mathbf{\Phi}^\top], \tag{86}$$

explores an H-dimensional ellipsoid like (63) or (71) rather than the entire hyperplane. When data accumulate, the prior constraints are eventually overwhelmed and the model can move freely on that plane. Random kitchen sinks (Rahimi & Recht, 2008) take these two ideas to the extreme: forgo specialized modeling; just use H *random* basis functions with uninformative priors and rely on the fact that an entire H-dimensional hyperplane is reachable with high probability.

The mean is zero everywhere due to our zero-mean prior $p(\mathbf{a})$. The covariance matrix, however, is nontrivial. To investigate, define

$$\mathbf{C} := \mathbb{E}\Big[\mathbf{u}'\mathbf{u}'^{\top}\Big] \in \mathbb{R}^{N\times N} \tag{87}$$

with elements

$$[\mathbf{C}]_{nm} = \sigma_a^2\,\mathbb{E}_{\mathbf{b},\mathbf{c}}\left[\sum_{h=1}^{H}\phi_h(t_n)\phi_h(t_m)\right], \tag{88}$$

where $\phi_h(t) = (c_h t - b_h)_+^d$ for the neural network model $u'_{\mathrm{NN}}(t)$ was defined in (75). Since the $\phi_h(t)$ are statistically equivalent under i.i.d. priors like the ones we adopted in (74), define a single "mother wavelet" *without index h* as

$$\phi(t; b, c) = (ct - b)_+^d, \quad b \sim \texttt{Normal}(0, \sigma_b^2), \quad c \sim \texttt{Normal}(0, \sigma_c^2). \tag{89}$$

Then the elements of $\mathbf{\Phi}\mathbf{\Phi}^{\top}$ become sums of H i.i.d. random variables, and their expectation factorizes across h:

$$\begin{aligned}
[\mathbf{C}]_{nm} &= \sigma_a^2\sum_{h=1}^{H}\mathbb{E}_{\mathbf{b},\mathbf{c}}[\phi_h(t_n)\phi_h(t_m)] && (90)\\
&= \sigma_a^2\sum_{h=1}^{H}\mathbb{E}_{b,c}[\phi(t_n; b, c)\phi(t_m; b, c)] && (91)\\
&= \sigma_a^2\,H\,\mathbb{E}_{b,c}[\phi(t_n; b, c)\phi(t_m; b, c)] && (92)\\
&= \sigma_a^2\,H\,\texttt{TACK}(t_n, t_m). && (93)
\end{aligned}$$

Here we anticipate the *temporal arc cosine kernel* of degree d, defined below in (108):

$$\begin{aligned}
\texttt{TACK}(t, t') &:= \mathbb{E}_{b,c}[\phi(t; b, c)\phi(t'; b, c)] && (94)\\
&\equiv \int \phi(t; \mathbf{w})\phi(t'; \mathbf{w})\ \texttt{Normal}(\mathbf{w} \mid \mathbf{0}, \mathbf{\Sigma})\,\mathrm{d}\mathbf{w}, && (95)
\end{aligned}$$

where $\mathbf{\Sigma} := \begin{pmatrix}\sigma_b^2 & 0\\ 0 & \sigma_c^2\end{pmatrix}$ describes the covariance of the hidden weights $\mathbf{w} = (-b, c)^{\top}$. For legibility we suppress the dependence of TACK, written $\texttt{TACK}(t, t')$, on both the degree d and the covariance matrix $\mathbf{\Sigma}$.

To recapitulate, the first and second central moments of the mixture (80) are known for any H:

$$\begin{aligned}
\mathbb{E}[\mathbf{u}'] &= \mathbf{0}, && (96)\\
\mathbb{E}\Big[\mathbf{u}'\mathbf{u}'^{\top}\Big] &\equiv \mathbf{C} = \sigma_a^2\,H\,\texttt{TACK}(t, t'). && (97)
\end{aligned}$$

The third, fourth, ... central moments are in general nonzero and difficult to compute. They will vanish, however, when $H \to \infty$ and we choose $\sigma_a \propto 1/\sqrt{H}$. Nevertheless, even before taking any limit, the first two moments of the prior already mirror something kernel-like. Note that this result hinges critically on the independence assumption for $\{b_h, c_h\}$ in (89). The closure-constrained prior of (79) would couple these parameters nonlinearly, destroying that independence and making the derivation intractable, which is why we temporarily set it aside.

4.4.2 The temporal arc cosine kernel

Of course, I wouldn't have had the courage to attempt this whole calculation had I not already known that a very similar limit has been evaluated succesfully in the neural-network-as-a-Gaussian-process literature. The marginalization we performed above is, in fact, a one-dimensional variant of the classic infinite-width limit of feedforward networks studied by Neal (1996); Williams (1998) and later generalized by Cho & Saul (2009).

In these works, the expectation over random weights $\mathbf{w}$ with $\texttt{Normal}(\mathbf{w} \mid \mathbf{0}, \mathbf{\Sigma})$ priors gives rise to a family of kernels which describe an infinitely wide Bayesian network, and which differ only with respect to the activation function used and the imposed a priori covariance $\mathbf{\Sigma}$. This same reasoning will now allow us to identify our expected covariance (94) with a time-domain specialization of the *arc cosine kernel family* of degree d.

Table 5 organizes the two kernels we will introduce shortly. From here on we refer to these kernels in prose simply as ACK and TACK, and suppress their dependence on the listed parameters for legibility.

The arc cosine kernel (ACK) of degree d proposed by Cho & Saul (2009) is defined as the expected inner product between infinite-dimensional random RePU features of D-dimensional inputs $\mathbf{x}, \mathbf{x}' \in \mathbb{R}^D$:

$$\texttt{ACK}(\mathbf{x}, \mathbf{x}') = 2 \int (\mathbf{w}^\top \mathbf{x})_+^d (\mathbf{w}^\top \mathbf{x}')_+^d \; \texttt{Normal}(\mathbf{w} \mid \mathbf{0}, \mathbf{I}_D) \, \mathrm{d}\mathbf{w}, \tag{98}$$

where the hidden weights $\mathbf{w} \in R^D$, and $\mathbf{I}_D$ is the $D \times D$ identity matrix. This integral representation makes it clear that this is indeed a valid positive definite kernel in any input dimension $D \geq 1$. It admits a beautiful closed form:

$$\texttt{ACK}(\mathbf{x}, \mathbf{x}') = \frac{1}{\pi} \, \|\mathbf{x}\|^d \, \|\mathbf{x}'\|^d \; J_d(\theta), \tag{99}$$

Name	**Form**	**Parameters**	**Eq.**
arc cosine kernel	$\texttt{ACK}(\mathbf{x}, \mathbf{x}') = \frac{1}{\pi} \, \|\mathbf{x}\|^d \, \|\mathbf{x}'\|^d \; J_d(\theta)$	d	(99)
temporal arc cosine kernel	$\texttt{TACK}(t, t') = \frac{1}{2\pi} \, \|\mathbf{\Sigma}^{\frac{1}{2}} \mathbf{x}_t\|^d \, \|\mathbf{\Sigma}^{\frac{1}{2}} \mathbf{x}_{t'}\|^d \; J_d(\theta)$	$d, \mathbf{\Sigma}$	(108)

Table 5 | **Arc cosine kernels** introduced in this section. The ACK defined on arbitrary input dimension $D \geq 1$ was originally proposed by Cho & Saul (2009). The TACK is a simple application of the ACK for 1D time series.

where

$$\theta = \arccos\left(\frac{\mathbf{x}^\top \mathbf{x}'}{\|\mathbf{x}\| \ \|\mathbf{x}'\|}\right) \in [0, \pi] \tag{100}$$

is the positive angle between $\mathbf{x}$ and $\mathbf{x}'$, and $J_d(\theta)$ is given by Cho & Saul (2009, Eq. 4) as the generator expression

$$J_d(\theta) = (-1)^d (\sin\theta)^{2d+1} \left(\frac{1}{\sin\theta} \frac{\mathrm{d}}{\mathrm{d}\theta}\right)^d \left(\frac{\pi - \theta}{\sin\theta}\right). \tag{101}$$

The first few instances of which are

$$J_0(\theta) = \pi - \theta, \tag{102}$$
$$J_1(\theta) = \sin\theta + (\pi - \theta)\cos\theta, \tag{103}$$
$$J_2(\theta) = 3\sin\theta\cos\theta + (\pi - \theta)\left(1 + 2\cos^2\theta\right), \tag{104}$$
$$J_3(\theta) = 15\sin\theta - 11\sin^3\theta + (\pi - \theta)\left(9\cos\theta + 6\cos^3\theta\right). \tag{105}$$

Compared to the familiar Matérn kernels, the arc-cosine kernel is a rather atypical kernel, and probably only sparingly used in machine learning applications. It is, for one, blatantly nonstationary, and unlike stationary kernels such as the RBF, it can represent asymptotic behavior: its posterior mean need not revert to the prior mean outside the data regime, much like a neural network's response. The closed form (99) shows this structure explicitly: the integral representation factors into a polynomial term $\|x\|^d \ \|x'\|^d$, which controls the overall dynamic range and absorbs asymptotic variance, and an angular term $J_d(\theta)$, which captures the nonlinear thresholding of the RePU activation and thus allows for modeling the kind of changepoint structure we know is essential for GFMs.

The temporal arc cosine kernel (TACK) is the kernel which describes $\mathbf{C} \equiv \mathbb{E}\left[\mathbf{u}'\mathbf{u}'^\top\right]$ in (85). We can cast it as an affine shift of the bias augmented ACK on the input dimension $D = 2$. Define the auxiliary vectors

$$\mathbf{x}_t = \begin{pmatrix} 1 \\ t \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} -b \\ c \end{pmatrix}. \tag{106}$$

and the covariance matrix for the hidden weights $\{b, c\}$

$$\mathbf{\Sigma} = \begin{pmatrix} \sigma_b^2 & 0 \\ 0 & \sigma_c^2 \end{pmatrix}. \tag{107}$$

Then, by substituting these into (99) and integrating with respect to the Gaussian weight prior $\texttt{Normal}(\mathbf{w} \mid \mathbf{0}, \mathbf{\Sigma})$, we obtain the TACK:

$$\begin{aligned}
\texttt{TACK}(t, t') &= \mathbb{E}_{b,c}[\phi(t; b, c)\phi(t'; b, c)] && (108)\\
&= \int (\mathbf{w}^\top \mathbf{x}_t)_+^d (\mathbf{w}^\top \mathbf{x}_{t'})_+^d \ \texttt{Normal}(\mathbf{w} \mid \mathbf{0}, \mathbf{\Sigma})\, \mathrm{d}\mathbf{w} && (109)\\
&= \frac{1}{2} \texttt{ACK}\left(\mathbf{\Sigma}^{\frac{1}{2}}\mathbf{x}_t, \mathbf{\Sigma}^{\frac{1}{2}}\mathbf{x}_{t'}\right) && (110)\\
&= \frac{1}{2\pi} \ \|\mathbf{\Sigma}^{\frac{1}{2}}\mathbf{x}_t\|^d \ \|\mathbf{\Sigma}^{\frac{1}{2}}\mathbf{x}_{t'}\|^d \ J_d(\theta), && (111)
\end{aligned}$$

where the factor $1/2$ compensates for the convention used in Cho & Saul (2009). Here,

$$\theta = \arccos\left(\frac{\mathbf{x}_t^\top \mathbf{\Sigma} \mathbf{x}_{t'}}{\sqrt{\mathbf{x}_t^\top \mathbf{\Sigma} \mathbf{x}_t}\sqrt{\mathbf{x}_{t'}^\top \mathbf{\Sigma} \mathbf{x}_{t'}}}\right), \tag{112}$$

is the angle between $\mathbf{x}_t$ and $\mathbf{x}_{t'}$ in the geometry induced by $\mathbf{\Sigma}$. This anisotropic variant was also used in the context of D-dimensional feature extraction by Pandey & Dukkipati (2014).

Figure 15 makes the role of the degree parameter visible before we move on to the infinite-width limit. Across all three covariance settings, low degrees produce rougher draws with sharper local bends, whereas higher degrees regularize these bends into smoother open-phase trajectories. The covariance parameters then control where this structure is concentrated: smaller σ_b imply a more localized changepoint location and lead to sharp changepoint behavior. Smaller values of σ_c just smear out time, which have the effect of longer timescales. Note that $d = 2$ and $d = 3$ are practically indistinguishable.

4.4.3 Taking the Gaussian-process limit

We turn again to the mixture (80). Recall from (83) that each random draw $\{\mathbf{b}, \mathbf{c}\} \in \mathbb{R}^{2H}$ from the prior (74) produces a random covariance matrix $\mathbf{Q} \in \mathbb{R}^{N \times N}$, which elementwise decomposes as a sum of H i.i.d. terms:

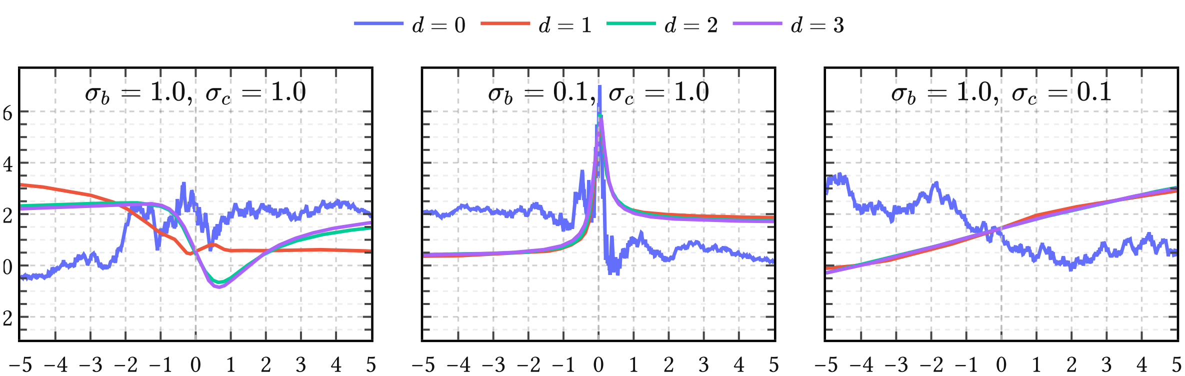


Figure 15 | **Prior samples from TACK for different degrees d.** Each column fixes a different covariance setting (σ_b, σ_c). Within each column the four curves correspond to different values of d. Increasing d progressively softens the hingepoint-like behavior, while decreasing σ_b relative to σ_c makes the prior more localized around the center.

$$[\mathbf{Q}]_{nm} = \sigma_a^2 \sum_{h=1}^{H} \phi_h(t_n)\phi_h(t_m). \tag{113}$$

This sum tends to grow as $\mathcal{O}(H)$, since from (90)

$$\mathbb{E}_{\mathbf{b},\mathbf{c}}[\mathbf{Q}] = \mathbf{C} \propto H. \tag{114}$$

Such growth of the marginal variance of $p(\mathbf{u}')$ makes the $u'_{\text{NN}}(t)$ model (72) useless as a prior when we want strong inductive bias even for large H, which we do. Therefore, we couple σ_a and H by choosing $\sigma_a \propto 1/\sqrt{H}$, in accordance with the variance-preserving initialization principle (Glorot & Bengio, 2010). Thus, substituting $\sigma_a \to \sigma_a/\sqrt{H}$ in (113), we get

$$[\mathbf{Q}]_{nm} = \sigma_a^2 \times \left[\frac{1}{H}\sum_{h=1}^{H} \phi_h(t_n)\phi_h(t_m)\right]. \tag{115}$$

Observe that the quantity between brackets is a Monte Carlo estimate of the TACK (108). Therefore, by the strong law of large numbers we conclude that

$$\left[\frac{1}{H}\sum_{h=1}^{H} \phi_h(t_n)\phi_h(t_m)\right] \longrightarrow \mathbb{E}_{b,c}[\phi(t;b,c)\phi(t';b,c)] \qquad \text{as } H \to \infty, \tag{116}$$

with deviations vanishing as $\mathcal{O}\left(1/\sqrt{H}\right)$. Therefore from (94)

$$[\mathbf{Q}]_{nm} \longrightarrow \sigma_a^2\, \texttt{TACK}(t_n, t_m) \qquad \text{as } H \to \infty. \tag{117}$$

Equivalently, the induced density $p(\mathbf{Q})$ of covariance matrices collapses to

$$p(\mathbf{Q}) \longrightarrow \delta(\mathbf{Q} - \mathbf{K}) \qquad \text{as } H \to \infty, \tag{118}$$

which in turn collapses the mixture (80) to

$$p(\mathbf{u}') \longrightarrow \texttt{Normal}(\mathbf{0}, \mathbf{K}) \qquad \text{as } H \to \infty. \tag{119}$$

Here the *kernel matrix* is the symmetric PSD matrix $\mathbf{K} \in \mathbb{R}^{N\times N} \propto \mathbf{C}$ given by

$$[\mathbf{K}]_{nm} = \sigma_a^2\, \texttt{TACK}(t_n, t_m). \tag{120}$$

In other words, the marginal $p(\mathbf{u}')$ converges entirely to a Gaussian and the first and second moments in (96) become *sufficient statistics* (Jaynes, 2003). All higher central moments vanish as $\mathcal{O}\left(1/\sqrt{H}\right)$ because the covariance fluctuations in (116) themselves decay at that rate. Since this argument holds for any finite collection of times $\boldsymbol{t} = \{t_n\}_{n=1}^{N}$, the limiting process is indeed officially a Gaussian process:

$$u'_{\text{NN}}(t) \sim \texttt{GaussianProcess}\left(0, \sigma_a^2\, \texttt{TACK}(t, t')\right). \tag{121}$$

Except for H, it has the same hyperparameters as the neural network model of Section 4.3.2: the scales $\boldsymbol{\theta}_{\text{NN}} = \{\sigma_a, \sigma_b, \sigma_c\}$ describing overall power, typical changepoint location and horizontal stretch, respectively; and the two higher-level hyperparameters $\{d, t_c\}$, which determine the polynomial degree and the instant of glottal closure, respectively.

Integrating out changepoints Note that t_c is the only changepoint parameter that survived the Gaussian-process limit. All other changepoints [each changepoint being defined by a pair of amplitude and location parameters $\{a_h, b_h\}$] have been analytically integrated out. That act of marginalization removed the parametric bottleneck carried by finite rank models like (50), (54), (57), (72) completely. This may come as a surprise, because we pointed out before that changepoint parameters as in (53) constitute the main "control points" of any GFM expressed in the time domain: what previously required a discrete arrangement of control points is now averaged into a single continuous covariance function capable of learning the "timing relationships [which] are very important for modelling the glottal flow signal" (Doval et al., 2006, p. 1).

More aspects of (121) are discussed in greater detail in Appendix C.

4.5 Summary

Great care was taken to argue that any *parametric* glottal flow model of the open phase of the glottal cycle can be expressed as a RePU network with a single hidden layer of width H, given H large enough.

Then we showed that in a Bayesian regression context with noninformative priors, the limit $H \to \infty$ corresponds to a *nonparametric* glottal flow model: a zero-mean GP with the temporal arc cosine kernel. This well-known limit, borrowed from classical GP literature, is applied here to unify the two dominant approaches to GIF. These are [parametric $\Leftrightarrow$ joint source-filter estimation] and [nonparametric $\Leftrightarrow$ inverse filtering], and we attempt here to combine the best of both worlds: parametric interpretability and nonparametric expressivity, respectively.

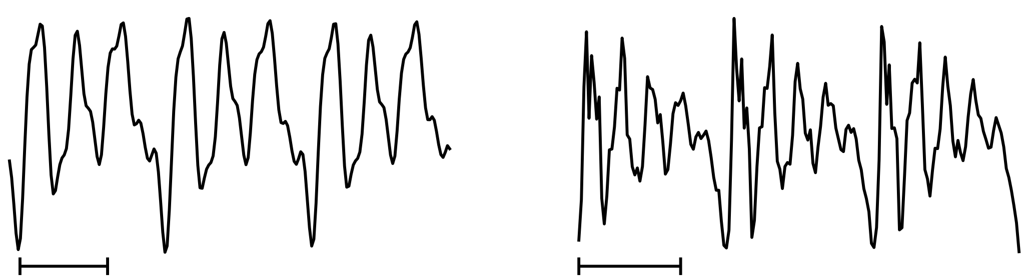
That is, the proposed probabilistic model has support for any parametric glottal flow model while retaining capacity to "let the data speak for itself" if need be. In the next chapters we will refine this initially tiny support gradually into a strong inductive bias by learning features from synthetic glottal flow simulations.

5 The periodic and quasiperiodic arc cosine kernel

Chapter 4 established parametric and nonparametric glottal flow models for a *single period* of the glottal cycle. In this chapter, our goal is to extend this basic building block to *multiple periods.* Recordings of voiced speech often visually exhibit self-similarity on timescales of $O(10 \text{ ms})$ because in normal speech the glottal cycle often reaches steady-state, and a clear pitch is perceived as a consequence.[16] An illustrative example for the case of steady-state vowels is given in Figure 16.

To model this steady-state source behavior we proceed in three steps. We begin with an idealized periodic kernel, the *periodic arc cosine kernel* (PACK), which is stationary and perfectly periodic. We then relax this to the *quasiperiodic arc cosine kernel* (QPACK), which retains the same within-cycle carrier while allowing slow amplitude variation and residual baseline structure across cycles. Finally we introduce *gated variants* (g-PACK and g-QPACK) in which the closed phase is incorporated explicitly. This last step breaks stationarity, but it turns out to be a useful one in practice.

The second half of the chapter turns from construction to learning. It is well known from functional data analysis that learning a single shared basis for a large corpus of training data requires aligning the data points, the process of which is called *registration.* (Ramsay & Silverman, 1997). We show how to do this accurately for the EGIFA data by aligning the



(a) (b)

Figure 16 | **Examples of steady-state vowels** during three pitch periods. Here (a) is perfectly periodic because it was synthesized with (50); and (b) is quasiperiodic because it is taken from a real recording.[17] The timescale is indicated by the horizontal bars ⊢⊣ below the waveforms, each bar having a length of 5 msec.

[16]Results from clinical trials (Little et al., 2007) indicate that normal (non-pathological) voices are usually something called Type I (Titze, 1995); that is, phonation results in nearly periodic sounds, which supports the notion that uttered vowels in normal speech typically reach a steady-state before "moving on". Thus when a vowel is perceived with a clear and constant pitch, it is reasonable to assume that the vowel attained steady-state at some point (though perceptual effects forbid a one-to-one correspondence). In practice, steady-state vowels are identified simply by looking for such quasiperiodic strings, which typically consist of about 3 to 5 pitch periods in everyday speech (Rabiner & Schafer, 2007).

[17]Taken from a recorded steady-state instance of the vowel /æ/ at a fundamental frequency of 138 Hz. Source: `bdl/arctic\_a0017.wav:0.51-0.54sec` (Kominek & Black, 2004).

examples to GCI (and optional GOI) events. This also allows us to extract voiced groups from the data, which then serve as high-quality training data to *calibrate* the global kernel hyperparameters. These hyperparameters determine the shared basis seen by the whole corpus, and we learn and evaluate them directly from the registered data.

Three algorithms do the practical work of the chapter. LBGID (Algorithm 2) detects the GCI and GOI events needed for registration; PRISM (Algorithm 3) calibrates a single global basis on the registered corpus; and t-PRISM (Algorithm 4) robustly filters raw period tracks into clean voiced groups before calibration. For the sake of brevity their full details and development have been deferred to the algorithm chapters; here we just accept them as given tools and only motivate their usage.

5.1 The periodic arc cosine kernel

The TACK or temporal arc cosine kernel derived in (108) provides a nonstationary covariance structure for the open phase of a *single* glottal cycle. That covariance reflects the change-point geometry inherited from piecewise polynomial GFMs, and it does so in a principled, nonparametric manner. But a single pitch period cannot exist in isolation: voiced speech is characterized by the imperfect repetition of this basic unit, and so we must now try to extend TACK to periodic signals.

Arguably the most natural route to periodicity is to replace the linear time coordinate in the bias-augmented input of TACK, namely $\mathbf{x}_t = (1, t)^\top$, by a set of *harmonic features* that are periodic simply by construction. This is the classical idea of embedding time on a circle (MacKay, 1998): functions of the embedded coordinates are automatically periodic in the original time variable. The construction proceeds in three steps: define the embedding, apply the arc cosine kernel, and normalize.

5.1.1 Harmonic embedding

Let $\omega_0 = 2\pi/T$ denote the fundamental angular frequency for a period $T > 0$, and let $J \geq 1$ be the *harmonic order*, which controls the richness of the embedding. Define the harmonic feature map

$$\boldsymbol{\varphi}(t) = \begin{pmatrix} \sigma_b \\ \sigma_{c,1} \cos(\omega_0 t) \\ \sigma_{c,1} \sin(\omega_0 t) \\ \vdots \\ \sigma_{c,J} \cos(J\omega_0 t) \\ \sigma_{c,J} \sin(J\omega_0 t) \end{pmatrix} \in \mathbb{R}^{2J+1}, \tag{122}$$

where $\sigma_b > 0$ and $\sigma_{c,j} > 0$ for $j = 1, \ldots, J$ are scale parameters playing analogous roles to σ_b and σ_c in Section 4.3.2, but act on the embedded space rather than linear time. These scale parameters modulate the harmonic basis (122) and thereby shape the angular similarity seen by the kernel. The constant coordinate σ_b is not a DC component of the waveform itself;

after normalization it becomes the zeroth harmonic weight and shifts the carrier toward or away from perfect self-alignment. The remaining scales control how strongly each harmonic pair participates in that comparison. In any case, where TACK lived in $\mathbb{R}^2$, we now work in $\mathbb{R}^{2J+1}$ with $(J+1)$ independent harmonic scales, which gives the kernel much greater spectral flexibility, while still inheriting (some of) the changepoint behavior from its ancestor.

An interesting property of this particular embedding is that its squared norm is constant in t:

$$\|\varphi(t)\|^2 = \sigma_b^2 + \sum_{j=1}^{J} \sigma_{c,j}^2 (\cos^2(j\omega_0 t) + \sin^2(j\omega_0 t)) = \sigma_b^2 + \sum_{j=1}^{J} \sigma_{c,j}^2 := R^2. \tag{123}$$

Every point $\varphi(t)$ therefore lies on a sphere of radius R in $\mathbb{R}^{2J+1}$. For the inner product between two embedded points, the cosine addition formula $\cos\alpha\cos\beta + \sin\alpha\sin\beta = \cos(\alpha - \beta)$ collapses the harmonic pairs:

$$\varphi(t)^\top \varphi(t') = \sigma_b^2 + \sum_{j=1}^{J} \sigma_{c,j}^2 \cos(j\omega_0(t-t')). \tag{124}$$

This depends on t and t' only through their difference, and is manifestly periodic with period T. Define θ as the angle between the two embedded points,

$$\theta(t-t') := \arccos\left(\frac{\varphi(t)^\top \varphi(t')}{R^2}\right), \tag{125}$$

which takes values in $[0,\pi]$ and satisfies $\theta(0) = 0$.

5.1.2 Definition

Applying the ACK (99) to $\varphi(t)$ and $\varphi(t')$ gives

$$\mathtt{ACK}(\varphi(t), \varphi(t')) = \frac{1}{\pi} R^{2d} J_d(\theta(t-t')), \tag{126}$$

which depends on t and t' only through their difference since R is constant. The prefactor R^{2d}/π couples the overall differentiability of sample paths (controlled by d) to the output scale. To circumvent this, we apply standard normalization and define the *periodic arc cosine kernel of degree d* as

$$\mathtt{PACK}(t,t') := \sigma_a^2 \frac{J_d(\theta(t-t'))}{J_d(0)}, \tag{127}$$

where $J_d(\theta)$ was defined in (101) and $J_d(0) = \pi(2d-1)!!$ is the value of the angular kernel at zero angle. Positive definiteness is preserved because composing a positive definite kernel with a deterministic feature map yields another positive definite kernel. By construction, the variance $\mathtt{PACK}(t,t) = \sigma_a^2$, so σ_a is the only parameter that controls the overall amplitude

Symbol	Name	Role
σ_a	amplitude	marginal standard deviation
σ_b	bias amplitude	DC component of the embedding
$\sigma_{c,1}, \ldots, \sigma_{c,J}$	harmonic scales	modulate each harmonic pair
T	period	fundamental period of the kernel
d	degree	smoothness class (RePU order)
J	harmonic order	number of harmonics in embedding

Table 6 | **Hyperparameters of PACK.** Of these, σ_a, σ_b and T are positive scalar, the harmonic scales $\sigma_{c,j}$ are J-dimensional, and d and J are discrete.

scale. Note that for legibility we generally suppress the dependence of $\texttt{PACK}(t, t')$ on its hyperparameters; we list them in Table 6 for an oversight.

How come PACK is stationary even though ACK and TACK were not? This is simply because we assigned equal weight to each cosine-sine pair in (124). It is a practical choice to ease training the kernel, and does not impair learning nonstationary phenomena. The thresholding behavior of the RePU activations is still present, but it is now replicated periodically.[18] This means that PACK can still express very sharp within-cycle structure even though it is now stationary and periodic.

5.1.3 Hyperparameters and the weight simplex

The definition (127) involves $J + 1$ scale parameters (σ_b and $\sigma_{c,1}, \ldots, \sigma_{c,J}$) that only enter through the angle θ in (125). Since θ depends on the normalized inner product $\varphi(t)^\top \varphi(t')/R^2$, only the *ratios* of these scales matter. This motivates introducing the *harmonic weights*

$$w_0 = \frac{\sigma_b^2}{R^2}, \quad w_j = \frac{\sigma_{c,j}^2}{R^2} \quad (j = 1, \ldots, J), \tag{128}$$

which satisfy the simplex constraint $w_0 + \sum_{j=1}^J w_j = 1$ with $w_j \geq 0$. In terms of these weights, the cosine of the angle (125) becomes

$$\cos\theta(t - t') = w_0 + \sum_{j=1}^{J} w_j \cos(j\omega_0(t - t')), \tag{129}$$

and the kernel (127) can be evaluated by computing $\theta(t - t')$ from (129) and substituting into J_d. We learn the harmonic weights directly in the simplex space to avoid degeneracies.

[18]The neural network interpretation of Section 4.3.2 carries over directly. PACK is the $H \to \infty$ limit of a single hidden layer with RePU activations applied to the $(2J + 1)$-dimensional harmonic input (122) rather than the 2-dimensional bias-augmented input.

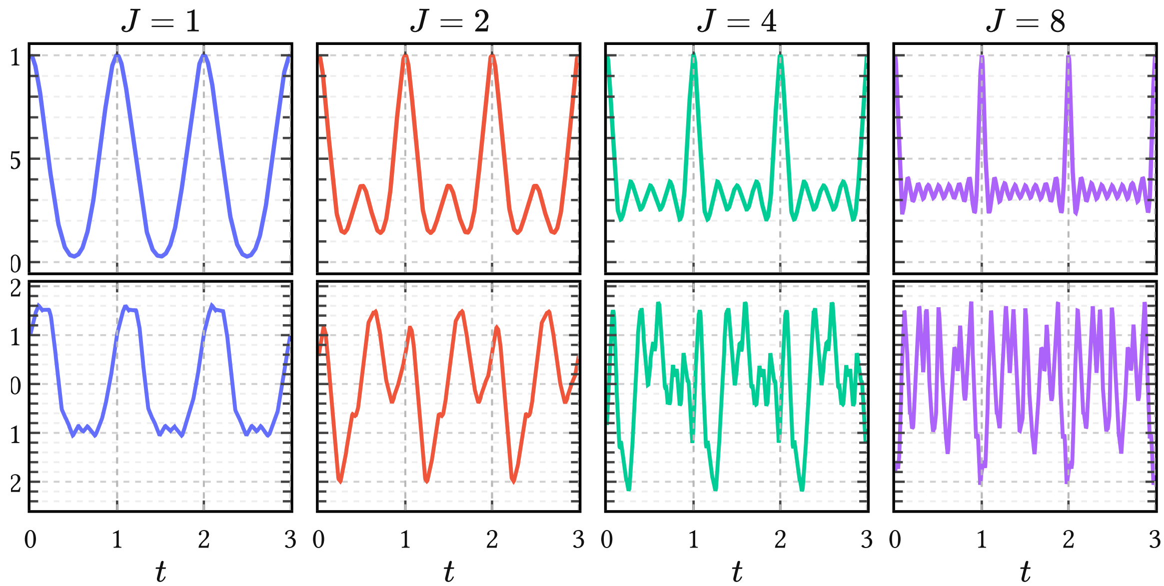


Figure 17 | **Effect of harmonic order J on PACK.** The top row shows the kernel section $\texttt{PACK}(t, 0)$ and the bottom row a prior sample. At fixed degree $d = 1$, increasing J decreases the effective within-cycle lengthscale and yields progressively wigglier covariance structure and sample paths. The remaining hyperparameters are fixed at $T = 1$, $\sigma_a^2 = 1$, $w_0 = 0.2$, and uniform harmonic weights $w_1, \ldots, w_J$ over the remaining mass.

The zeroth order harmonic weight w_0 plays an interesting role. It descends directly from the bias amplitude σ_b in the TACK construction, but its behavior is altered nontrivially by the harmonic embedding. When $w_0 = 1$ (all harmonic weights zero), the angle θ is identically zero and the kernel degenerates to a constant σ_a^2, describing a DC signal. Intermediate values of w_0 let the cosines oscillate about zero in (124) so that θ can reach $\pi/2$ and beyond, where $J_d(\theta)$ is able to produce steep, changepoint-like transitions of the type we need to model sharp glottal closures. If $w_0 = 0$, the autocovariance $\texttt{PACK}(t, 0)$ is allowed to reach zero periodically, which for higher d effectively creates kernels with compact support, although it isn't immediately clear what this implies for modeling speech signals.

At least J has a clear geometric effect. Increasing J inserts higher harmonics directly into the embedding and therefore decreases the effective within-cycle lengthscale of the process. In prior draws this appears as progressively finer oscillatory detail, as seen in Figure 17. Practically, increasing J while keeping all $w_j \sim O(1/J)$ gives rise to smaller effective lengthscales.

5.1.4 Spectral decay and smoothness

Because PACK is both stationary and periodic with period T, its covariance is an even T-periodic function that admits a Fourier cosine expansion

$$\texttt{PACK}(t, t') = a_0 + \sum_{m=1}^{\infty} a_m \cos(m\omega_0(t - t')), \tag{130}$$

with coefficients

$$a_0 = \frac{1}{T}\int_0^T \texttt{PACK}(t,0)\,\mathrm{d}t, \quad a_m = \frac{2}{T}\int_0^T \texttt{PACK}(t,0)\cos(m\omega_0 t)\,\mathrm{d}t \quad (m \geq 1). \quad (131)$$

By Bochner's theorem, positive definiteness of PACK is equivalent to $a_m \geq 0$ for all $m \geq 0$, so we can interpret these coefficients as encoding power. The Fourier expansion (130) has a direct interpretation as a *Bochner line spectrum* with spectral density given by

$$\tilde{k}(\xi) = 2\pi a_0\,\delta(\xi) + \pi\sum_{m=1}^{\infty} a_m[\delta(\xi - m\omega_0) + \delta(\xi + m\omega_0)]. \quad (132)$$

The decay of the harmonic weights a_m determines both the spectral character of the prior and the mean square differentiability of GP sample paths. For stationary kernels, this is governed by the behavior of the covariance near the origin: the more derivatives $k(t) = k(t,0)$ has at $t = 0$, the more mean square derivatives the process admits. A familiar example is the Matérn family, whose sample paths are r times mean square differentiable for every integer $r < \nu$. Now, it can be shown that the sample paths of a degree-d PACK kernel are exactly d times mean square differentiable, independent of J. This implies that the harmonic weights decay in power as

$$a_m = \mathcal{O}\big(m^{-(2d+2)}\big), \quad (133)$$

corresponding to an approximate rolloff in magnitude of $-6(d+1)$ dB per octave, as illustrated for the $d = 1$ case in Figure 19.

This behavior is also visible in Figure 18: increasing d removes the cusp at the origin and produces progressively smoother kernels with faster spectral decay. Low degrees ($d = 0$ or $d = 1$) therefore yield broad spectra with slowly decaying harmonics, appropriate for modeling sharp glottal closure events.

Classical source models report a spectral tilt of roughly -12 to -18 dB per octave for the glottal flow $u(t)$ itself (Doval et al., 2006; Rosenberg, 1971). Differentiation increases spectral slope by $+6$ dB per octave, so the corresponding glottal flow derivative $u'(t)$ exhibits a tilt of approximately -6 to -12 dB per octave, which corresponds to PACK kernels of degree $d = 0$ and $d = 1$.

5.1.5 Comparison to the standard periodic kernel

The standard periodic kernel in GP literature is the benign periodic squared exponential, which MacKay (1998) obtained by composing the RBF kernel with a circular embedding:

$$k_{\text{per}}(t,t') = \sigma^2 \exp\left(-2\sin^2\left(\pi\frac{t-t'}{T}\right)/\ell^2\right). \quad (134)$$

Its exact cosine expansion is (Riutort-Mayol et al., 2020, Eq. B.2):

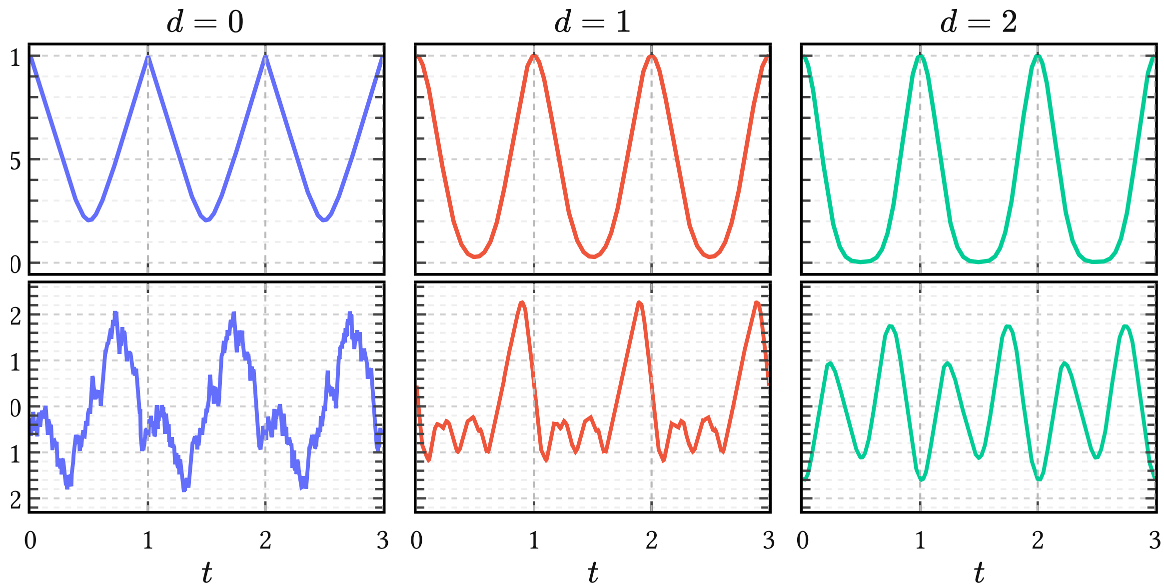


Figure 18 | **Effect of degree d on within-cycle smoothness.** The top row shows the kernel section $\texttt{PACK}(t, 0)$ and the bottom row a single prior sample. Holding J and the harmonic weights fixed, increasing d softens the changepoint-like geometry and produces smoother periodized waveforms. The remaining hyperparameters are fixed at $J = 1$, $T = 1$, $\sigma_a^2 = 1$, $w_0 = 0.2$, and $w_1 = 0.8$.

$$k_{\text{per}}(t, t') = \sigma^2 e^{-\frac{1}{\ell^2}} \left[I_0(\ell^{-2}) + 2 \sum_{m=1}^{\infty} I_{m(\ell^{-2})} \cos(m\omega_0(t - t')) \right]. \tag{135}$$

Thus its entire harmonic envelope is controlled by a single lengthscale ℓ. Changing ℓ only broadens or narrows a smooth low-pass spectrum.

Figure 19 illustrates how PACKs are different in two ways. First, the harmonic weights w_j can redistribute mass freely across harmonics rather than merely broadening a fixed envelope. Second, the coupling between degree d and the asymptotic decay allows the model to learn more realistic, non-exponential spectral rolloff.

5.2 The quasiperiodic arc cosine kernel

Of course, perfect periodicity is too rigid for actual glottal flow; even synthesized glottal flow contains all sorts of drifts and interesting imperfections, which add organic warmth to the synthesized voice. Therefore the next step is to relax PACK into a quasiperiodic variant.

We thus define the *quasiperiodic arc cosine kernel* by

$$\texttt{QPACK}(t, t') := \texttt{env}(t, t') \cdot \texttt{PACK}(t, t') + \texttt{base}(t, t'). \tag{136}$$

This is a composite kernel modeling amplitude modulation of a periodic carrier, plus a second term that adds a small low pass correction. The full parameterization of (136) is listed below

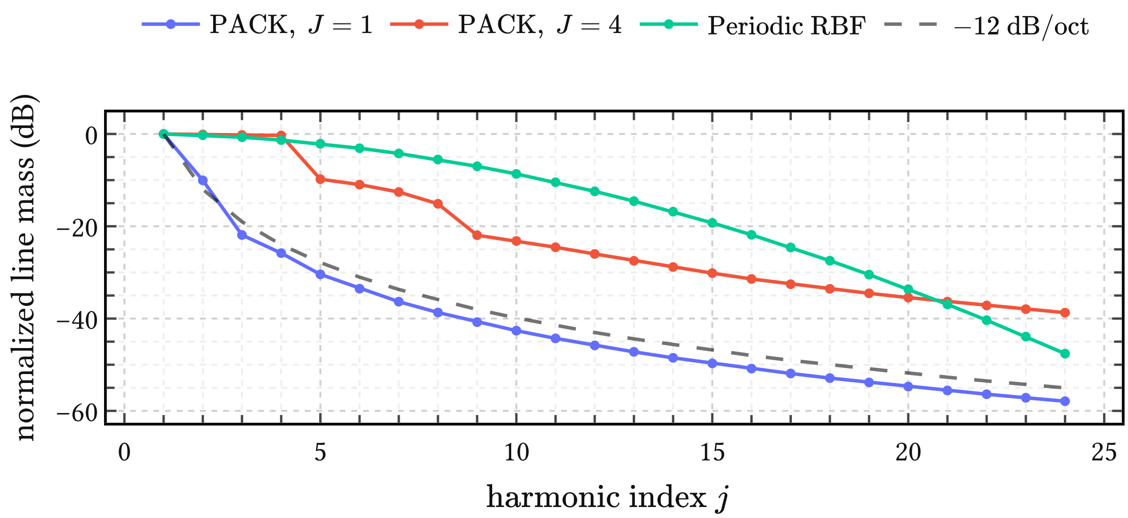


Figure 19 | **PACK versus the periodic squared exponential in the line spectrum.** The curves show the harmonic weights a_m from (132) in dB. PACK with $d = 1$ exhibits the expected -12 dB per octave tail, while the periodic squared exponential traces out a one-parameter low-pass envelope. The PACK curves use $T = 1$, $\sigma_a^2 = 1$, and $w_0 = 0.2$, with uniform weights over the remaining harmonic mass; the periodic squared exponential uses lengthscale $\ell = 0.20$. The dashed reference line indicates -12 dB per octave.

in Table 7. In what follows, we choose the envelope $\texttt{env}(t, t')$ and baseline kernel $\texttt{base}(t, t')$ to be stationary kernels. This makes sure that QPACK is still stationary such that it has a well defined Bochner spectrum. We give an example of the latter in Figure 20 and now turn to each of the three components that make up QPACK.

Carrier kernel The carrier is a PACK of degree d from the previous section. The amplitude modulation introduced by the envelope does not materially alter the high-frequency rolloff, so the degree d still governs closure sharpness and spectral slope, while the weights and harmonic order J determine the detailed within-cycle geometry.

Envelope kernel The envelope kernel, written $\texttt{env}(t, t')$, modulates this carrier on a much longer timescale ℓ_{env} than the fundamental period T – typically on the order of a dozen periods. We use a unit-variance RBF,

$$\texttt{env}(t, t') := \exp\left(-\frac{(t - t')^2}{2\ell_{\text{env}}^2}\right), \tag{137}$$

for the envelope because this does not interfere with the differentiability class set by d. In the frequency domain, the product of a slowly varying RBF envelope with the PACK carrier broadens each of the harmonic lines into narrow bands like the ones on display in Figure 20.

Baseline kernel The baseline kernel, written $\texttt{base}(t, t')$, is a small additive correction. We choose the rational quadratic (RQ) kernel

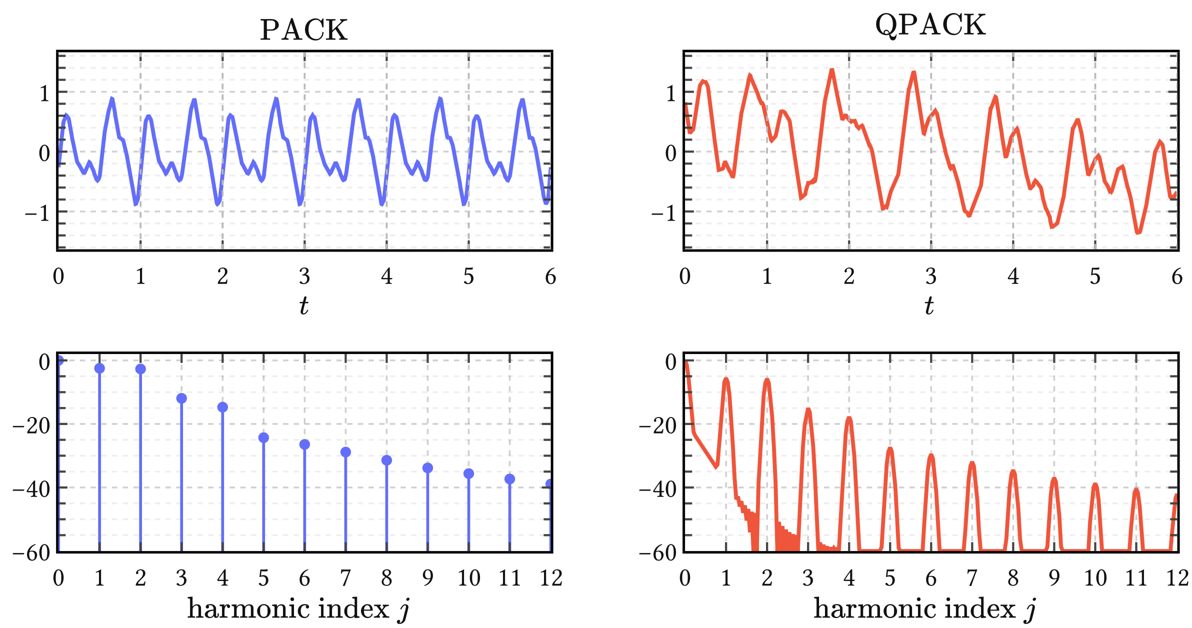


Figure 20 | **From PACK to QPACK in time and frequency.** The top row shows prior samples over six periods; the bottom row shows the corresponding normalized spectra in dB against harmonic index j. The shared carrier hyperparameters are $d = 1, J = 2, T = 1, \sigma_a^2 = 1$, and $w = (0.2, 0.4, 0.4)$. For QPACK the envelope is a unit-variance RBF with lengthscale $\ell_{\text{env}} = 3$, and the baseline is a rational quadratic kernel with $\sigma_{\text{base}}^2 = 0.02$, $\ell_{\text{base}} = 0.5$, and $\alpha_b = 1$.

$$\texttt{base}(t, t') := \sigma_{\text{base}}^2 \left(1 + \frac{(t - t')^2}{2\alpha_b \ell_{\text{base}}^2}\right)^{-\alpha_b}, \tag{138}$$

with scale $\sigma_{\text{base}} \ll \sigma_a$, lengthscale $\ell_{\text{base}} \ll \ell_{\text{env}}$, and a general shape parameter $\alpha_b > 0$. Like the RBF, the RQ kernel is infinitely differentiable and will not influence the differentiability of the composite kernel. Its role is to capture weak residual structure which is not phase-locked enough to belong in the carrier but also not unstructured enough to be dismissed as white noise. This includes open-phase ripple caused by glottal-vocal-tract interaction (Ananthapadmanabha & Fant, 1982), as well as violations of the closure condition that occur frequently in more irregular or pathological voices (Kreiman et al., 2007). In spectral terms the baseline adds a broad low-frequency "pedestal" underneath the broadened carrier bands.

5.3 Gated variants

PACK and QPACK are manifestly stationary kernels. After conditioning on data they can produce phase-dependent (nonstationary) posterior means, but *a priori* they assign the same variance to every phase of the cycle, whereas we know that the excitation energy depends strongly on the phase of the glottal cycle.

Symbol	Component	Role
σ_a	carrier	PACK amplitude scale; fixed to 1 in the whitened calibration experiments
$w_0, \ldots, w_J$	carrier	harmonic weight simplex inherited from PACK
d	carrier	within-cycle smoothness and spectral slope
J	carrier	harmonic order; effective within-cycle lengthscale control
T	carrier	fundamental period
ℓ_{env}	envelope	timescale of cycle-to-cycle amplitude modulation
σ_{base}	baseline	amplitude of the additive correction
ℓ_{base}	baseline	correlation lengthscale of the additive correction
α_b	baseline	rational quadratic shape parameter

Table 7 | **Hyperparameters of QPACK.** QPACK inherits the full PACK carrier and adds a long-scale envelope and a weak baseline. In the EGIFA calibration experiments the envelope is taken to have unit variance, so only its lengthscale is learned.

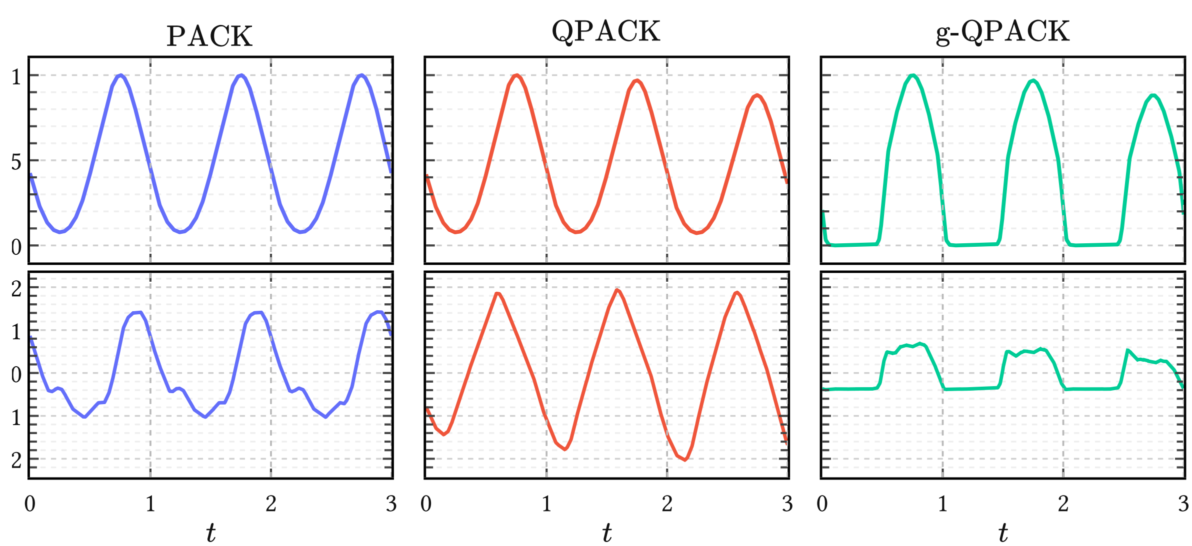


Figure 21 | **From PACK to QPACK to g-QPACK.** The top row shows kernel sections $k(t, t')$ at anchor time $t' = 0.75$, which is halfway the first open phase. The bottom row shows prior samples, all with matched carrier hyperparameters. QPACK relaxes strict periodicity by adding a slow envelope; g-QPACK then suppresses covariance in the closed phase of the cycle. The shared carrier settings are $d = 1$, $J = 1$, $T = 1$, and $w = (0.2, 0.8)$. To emphasize the closed phase the baseline correction was suppressed by setting $\sigma_{\text{base}} = 0$. The envelope is a unit-variance RBF with $\ell_{\text{env}} = 4$, and the gate sharpness of g-QPACK is $\beta = 12$.

The next step in taking advantage of prior information is therefore to encode the open/closed asymmetry directly in the kernel itself. This can be done easily by gating the kernel covariance (Alvarado & Stowell, 2016). Define therefore the two *gated variants*

$$\texttt{g-PACK}(t, t') \coloneqq g_\beta(t)\, g_\beta(t')\, \texttt{PACK}(t, t'), \tag{139}$$

$$\texttt{g-QPACK}(t, t') \coloneqq g_\beta(t)\, g_\beta(t')\, \texttt{QPACK}(t, t'). \tag{140}$$

These inherit all properties and hyperparameters of their respective base kernel except the property of stationarity.[19] The periodic gate is defined as

$$g_\beta(t) = \sigma(-\beta \sin(2\pi t/T)), \tag{141}$$

where $\beta > 0$ is a new hyperparameter that controls the sharpness and $\sigma(z) = (1 + e^{-z})^{-1}$ is the standard logistic sigmoid. This is a smooth approximation to a square wave: over one half-cycle $g_\beta(t) \approx 0$, over the other $g_\beta(t) \approx 1$, and the transitions are smooth.

Figure 21 illustrates the behavior of g-QPACK and compares it to its ancestors. As the gate $g_\beta(t)$ is itself periodic, g-PACK remains a proper extension of PACK (and incidentally a rare example of a periodic-but-nonstationary kernel). In practice, however, there is little reason to give up on the extra flexibility of QPACK, so the real model of interest is g-QPACK, not g-PACK, and we will focus on the former from here on.

Note that (141) implies that the first half of the period is closed phase and the second half open phase, with the two subintervals equally long. This means that g-QPACK always generates GF waveforms with fixed open quotient $\text{OQ} = 0.5$ (unless $\beta \to 0$). We relax this in the next section by warping physical time t into normalized time τ which can stretch and compress the phases and achieve any value of interest for OQ.

5.4 Registration with EGIFA

In the next section, we intend to learn the hyperparameters of the kernels introduced so far from the EGIFA dataset (Dataset 1). EGIFA provides a large number of high-quality glottal flow waveforms $\{u(t)\}$ simulated from a physics-based articulatory speech production model called VocalTractLab (Birkholz, 2013), which is the current open-source state of the art (Gao et al., 2024). We can use the generated $\{u(t)\}$ waveforms as examplars to calibrate the PACK family on, but this requires aligning them beforehand – a procedure known as *registration* in functional data analysis, hence the title of this section.

In order to maximize learning signal, we therefore

1. extract only clean voiced groups from the raw $\{u(t)\}$ waveforms;
2. calculate their derivatives $\{u'(t)\}$ in a physiologically grounded way;[20]
3. normalize power across the DGF exemplars;

[19] This also means that the gated variants lack well-defined Bochner spectra. They are still in the family of harmonizable kernels, however (Altamirano & Tobar, 2022).

[20] See Appendix D for details.

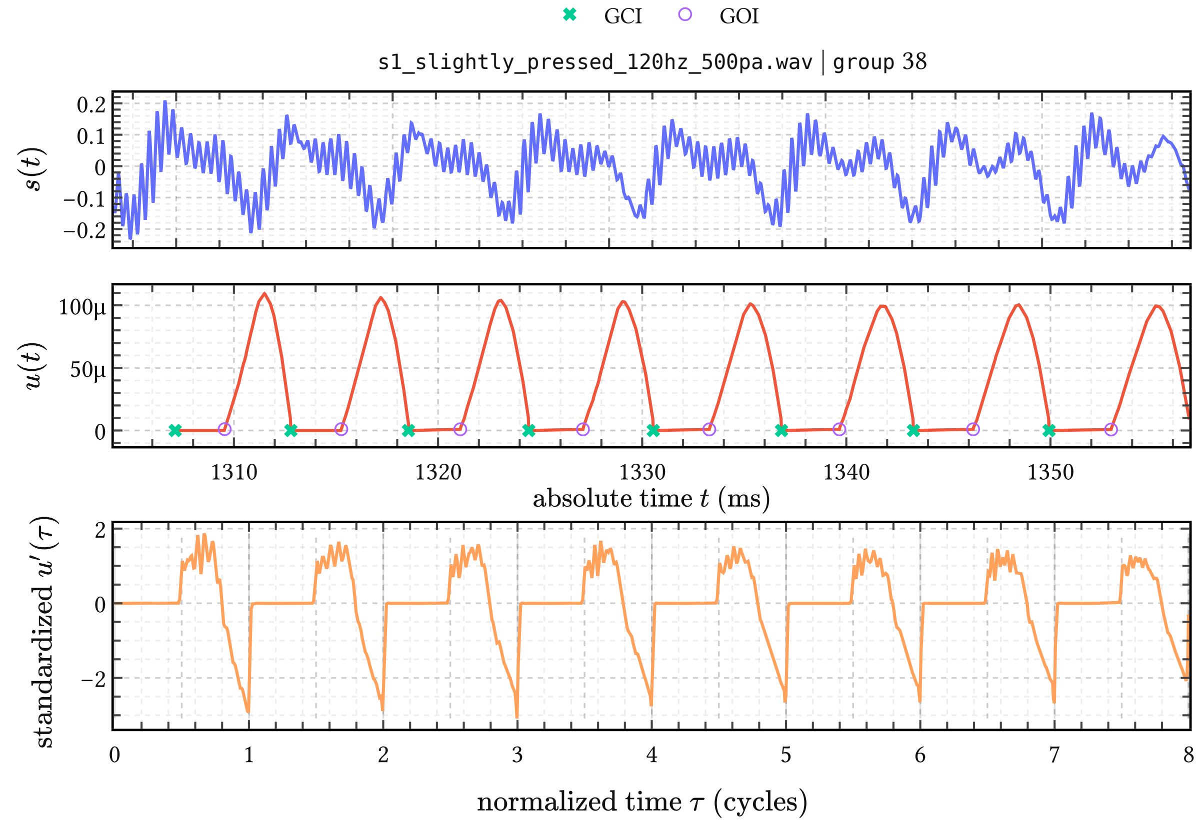


Figure 22 | **Registration of one retained EGIFA voiced group.** Top: the raw speech waveform $s(t)$ in physical time, extracted from the indicated `.wav` file and voiced group. Middle: the corresponding raw glottal flow $u(t)$, again in physical time, with LBGID-detected GCI and GOI landmarks. Bottom: the same group after differentiation, amplitude normalization and registration to normalized time, now shown as $u'(\tau)$. In this representation the GCIs land at integers $\tau = 0, 1, 2, \ldots$ and the GOIs at half-integers $\tau = 0.5, 1.5, 2.5, \ldots$.

4. and finally align them in time such that the GCI and GOI events across different $u'(t)$ waveforms align in *normalized time* τ.

The procedure is illustrated in Figure 22, which follows a single EGIFA group through the registration process. With regard to step 4, time alignment, it is seen that the upper two rows live in physical time t, where period length and OQ vary continually and the GCIs and GOIs appear as irregularly spaced events. The bottom row lives in "subdued" normalized time τ, where the glottal closing and opening instants have been mapped to integer and half integer values of τ, respectively, and $\mathrm{OQ} = 0.5$ everywhere.

Rather unfortunately, steps 1 and 4 above are nontrivial data preprocessing steps, and difficult to get right consistently, even for 1D time series. Because data quality is crucial, and with the prospect of reusing these transformations for future large scale benchmarks, we have had to resort to custom algorithms in order to guarantee meaningful alignment. LBGID (Algorithm 2) is a simple heuristic to reliably extract GCIs and GOIs from ground truth glottal flow $u(t)$.

$\{(\mathbf{t}_n, \mathbf{u}_n)\}_{n=1}^{N}$

raw voiced groups in physical time

$\xrightarrow{\text{registration}}$

$\mathcal{D} = \{(\boldsymbol{\tau}_n, \mathbf{u}'_n)\}_{n=1}^{N}$

registered DGF corpus in shared cycle time

$\xrightarrow{\text{calibration}}$

$\mathcal{D}_R = \{(\mathbf{m}_n, \mathbf{S}_n)\}_{n=1}^{N}$

PRISM posteriors in amplitude space

Figure 23 | **From EGIFA waveforms to the calibrated PRISM object.** Left: retained voiced groups in physical time before differentiation, normalization, and registration. Center: the registered DGF corpus $\mathcal{D}$ used for PACK calibration in shared cycle time. Right: the calibrated PRISM output $\mathcal{D}_R$, that is, one Gaussian amplitude posterior per registered waveform. The compression from $\mathcal{D}_R$ to $\mathcal{D}_Q$, $\mathcal{M}_Q$, and finally $\mathcal{M}_R$ is continued in Figure 28.

With these in hand, t-PRISM (Algorithm 4) then clusters consistent progressions of such GCIs into voiced groups; that is, it segments the raw $\{u(t)\}$ waveforms into regions where the glottal cycle has reached something resembling steady state.

In this manner, most of the amplitude and phase variations in the raw data $\{u(t)\}$ are normalized to an orderly collection of waveforms with reliable signal to learn from.

We leave a detailed discussion of the registration procedure to Dataset 1 and now proceed to the task of *calibration*; that is, learning the hyperparameters of the PACK kernels from the aligned $\{u'(\tau)\}$ exemplars.

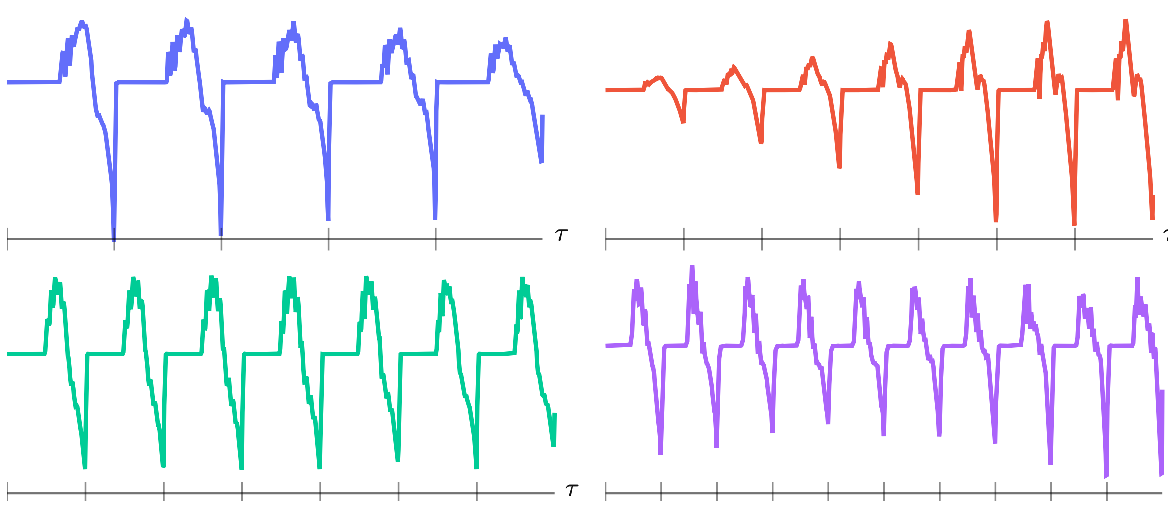


Figure 24 | **Four samples from the registered EGIFA corpus.** From left to right and top to bottom these are waveforms with index $n = 19, 601, 1452, 5892$ out of a total $N = 7697$, with $\left(M_n, \tau_{nM_n}\right) = (875, 5), (884, 7), (860, 7), (759, 10)$.

5.5 Calibration on EGIFA

In ordinary GP regression we have one data set, learn a distribution over latent functions parametrized by the hyperparameters, and predict in the context of that same data set. Our setting here is quite different: we are given a large collection of N aligned DGF waveforms

$$\mathcal{D} = \{(\boldsymbol{\tau}_n, \mathbf{u}'_n)\}_{n=1}^{N}, \tag{142}$$

where waveform n has its own registered grid

$$\boldsymbol{\tau}_n = \left(\tau_{n1}, ..., \tau_{nM_n}\right) \tag{143}$$

and standardized DGF samples

$$\mathbf{u}'_n = \left(u'_{n1}, ..., u'_{nM_n}\right). \tag{144}$$

As Figure 24 illustrates, each waveform (144) is a data set in its own right, with its own length M_n, its own irregularly sampled grid (143) and its own number of completed cycles τ_{nM_n}.

Given one of the above kernels $k_{\boldsymbol{\theta}}(\tau, \tau')$, the task is then to learn a single set of kernel hyperparameters $\boldsymbol{\theta}$ that maximizes the support for $\mathcal{D}$. To distinguish from ordinary GP inference, we call this process *calibration*: learning a common basis from the corpus without yet "finetuning" on individual data points.

This is similar to basis learning in functional data analysis, but here we incorporate GP features like kernel regularization; minibatching; principled handling of irregularly sampled data. Algorithm 3 gives the derivation and further details. The basic idea is to extend the well-known sparse variational inference machinery of Titsias (2009) from a single function to many independent waveforms. We refer to this extension as PRISM, for *process-induced surrogate modeling*, and use it to accomplish the calibration task. The output is a learned shared feature map $\boldsymbol{\varphi}(\tau)$ together with the amplitude-space collection

$$\mathcal{D}_R = \{(\mathbf{m}_n, \mathbf{S}_n)\}_{n=1}^{N}, \tag{145}$$

which is the exact object handed on to refinement in Chapter 6.

Quantile	0%	25%	50%	75%	95%
Cycles covered	5	5	6	8	9

Table 8 | **Quantiles of covered pitch cycles at width $W = 2048$** across the registered EGIFA corpus. At the 95th percentile, the padded width accommodates about 9 cycles per frame. At the standard IKLP frame length of 32 ms and a 99th-percentile F_0 of roughly 250 Hz, the maximum expected number of pitch periods per frame is about $32\ \text{ms} \times 0.25\ \text{kHz} = 8$; this table confirms that $W = 2048$ covers 95% of observed cycle counts comfortably.

5.5.1 Experimental setup

All calibration experiments use the registered EGIFA corpus from Dataset 1. The raw objects there are registered voiced groups; the calibration unit here is a *frame*. Each voiced group is first resampled to 20 kHz, smoothed, differentiated, and standardized within group, and is then cut without overlap into frames of width $W = 2048$. Only incomplete terminal frames are padded, purely for GPU batching, and padded positions are masked so they do not contribute to the PRISM objective. At this width, Table 8 shows that 95% of retained frames cover at most about nine glottal cycles $\tau_{\max} \sim 9$.

The framing step produces $N = 7697$ registered frames in total, from which we retain 6500 for training and a disjoint held-out set of 1000 for testing. For ungated models only GCIs are aligned to integers; for gated models GOIs are additionally aligned to half-integers, so that the gate transition in (141) remains phase-consistent across the corpus.

Each run maximizes the minibatch PRISM objective (270), that is, a stochastic estimate of the collapsed corpus objective (260). We optimize with Adam (Kingma & Ba, 2015) at learning rate 10^{-3} for 2500 iterations with a batch size of $B = 128$ complete frames. Learning is stable as judged from ELBO curves, and we optimize on the GPU in float32 with Cholesky jitter 10^{-5} throughout.

All runs use the same RBF envelope (137) and rational quadratic baseline (138) inside QPACK (136) or g-QPACK (139). What varies is the carrier, whether that carrier is gated, the harmonic order J, and the inducing rank R. Here `pack:0` and `pack:1` mean that the carrier is the $d = 0$ or $d = 1$ PACK kernel (127), while `periodic` replaces that carrier by the standard periodic kernel (134), which acts as a baseline. The values of J and R investigated here were chosen from early exploratory runs, and the present grid is meant to decide which prior structure is worth carrying forward given limited compute.

Concretely, we compare carrier names `pack:0`, `pack:1`, and `periodic`, harmonic orders $J \in \{16, 32\}$ and inducing basis sizes $R \in \{128, 192\}$. Each configuration is run from 32 random initializations. For evaluating model performance we use the held-out reduced-rank log evidence per effective sample

$$\mathcal{S}_{\text{test}}^{c(k)} := \frac{1}{|\mathcal{D}_{\text{test}}|} \sum_{n \in \mathcal{D}_{\text{test}}} \frac{1}{M_n^{\text{eff}}} \log p_{\text{RR}}(\mathbf{u}_n' \mid \boldsymbol{\tau}_n, k), \tag{146}$$

where k denotes the kernel being tested, M_n^{eff} denotes the number of non-padded samples in held-out frame n, and

$$p_{\text{RR}}(\mathbf{u}_n' \mid \boldsymbol{\tau}_n, k) = \mathtt{Normal}\Big(\mathbf{u}_n' \mid \mathbf{0}, \mathbf{Q}_{nn} + \sigma^2 \mathbf{I}_{M_n^{\text{eff}}}\Big) \tag{147}$$

is exactly the reduced-rank Gaussian term from (259) in Algorithm 3, evaluated on the held-out grid $\boldsymbol{\tau}_n$ at the fitted global parameters of kernel k. It should be noted that good

carrier	R	J	QPACK	g-QPACK
`pack:1`	192	16	0.451 ± 0.109	0.560 ± 0.071
`pack:1`	192	32	0.298 ± 0.136	0.538 ± 0.083
`pack:1`	128	16	0.353 ± 0.121	0.480 ± 0.091
`pack:1`	128	32	0.221 ± 0.145	0.412 ± 0.169
`pack:0`	192	16	0.161 ± 0.081	0.349 ± 0.086
`pack:0`	192	32	0.040 ± 0.055	0.255 ± 0.064
`pack:0`	128	16	0.169 ± 0.059	0.305 ± 0.096
`pack:0`	128	32	0.011 ± 0.104	0.161 ± 0.094
`periodic`	192	1	−0.107 ± 0.284	0.138 ± 0.240
`periodic`	128	1	0.020 ± 0.245	0.239 ± 0.251

Table 9 | **Held-out calibration scores on EGIFA.** Each entry reports mean ± sd of (146) across 32 independent random restarts. Indicated are the QPACK's carrier, rank R and the harmonic order J, which is 1 for the standard periodic kernel.

performance on this test is not necessarily indicative of good performance in the GIF task, though we expect these to correlate.

As a sanity check, every trained model is also compared to a white noise model with kernel $k_{\text{null}}(\tau, \tau') = \sigma^2 \delta_{\tau\tau'}$ with matched variance σ^2 and observation noise scale. On the scale of (146), the best null models achieve scores $\mathcal{S}_{\text{test}}^{c(\text{null})} \approx -6$, which indicates that the calibration fits pick up significant structure from the data.

5.5.2 Results

The results are summarized in Table 9, which reports (146) across random restarts.

The strongest mean configuration is `g-QPACK`, `pack:1`, `R=192`, `J=16`, with a held-out score of 0.560 ± 0.071. Its ungated counterpart, `QPACK`, `pack:1`, `R=192`, `J=16`, reaches 0.451 ± 0.109. Across the matched grid, g-QPACK scores higher mean scores than QPACK for every carrier and basis size reported here, and also has a smaller deviation around that mean. Therefore the gated variant of QPACK seems to be worthwile, and Chapter 7 will confirm this for GIF.

The effects of basis size R and harmonic order J are as follows: a higher R increases performance for the g-QPACK kernels, but not always for the QPACK kernels, and never for the periodic carrier. A similar tempering effect is seen for J: at least on this test, $J = 32$ seems to be overfitting. This is comparable to kernels with too short lengthscales overfitting data and generalizing badly. It thus appears that $J = 16$ is a better choice here.

The held-out fits in Figure 25 show that the strongest mean configuration has learnt bases that are able to extrapolate beyond the observed data (left column) and model very sharp GCI spikes (right column). It is also evident in the top left pane that QPACK loses memory of the

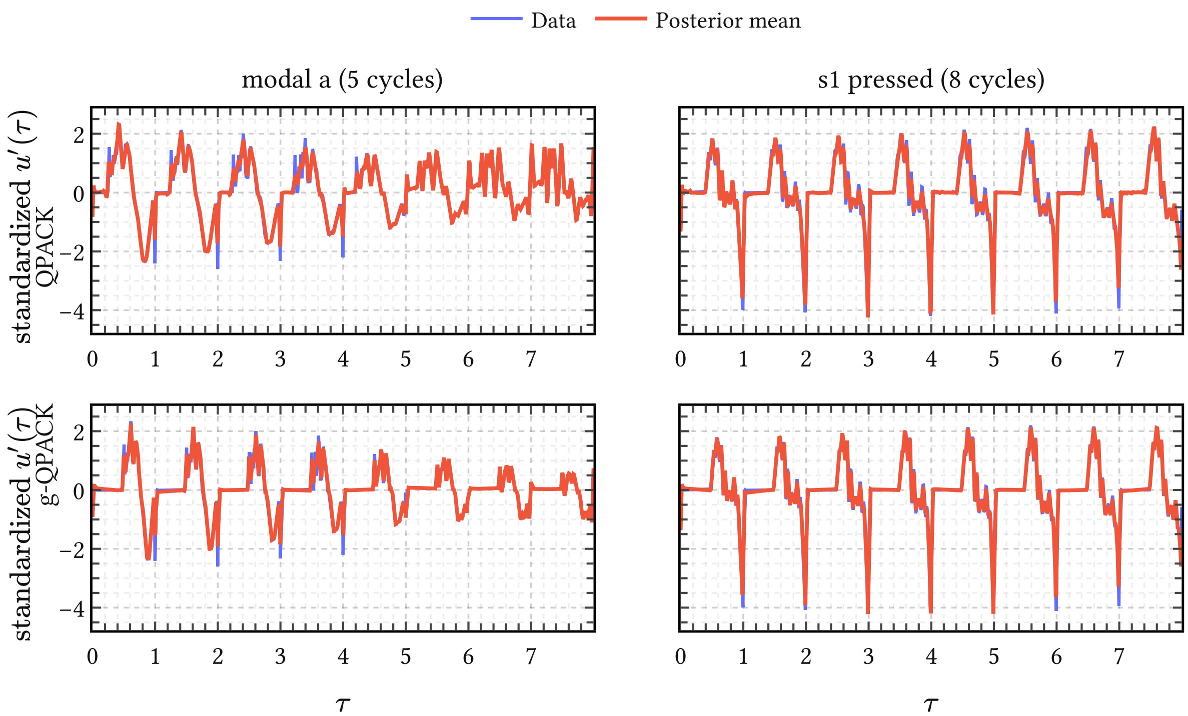


Figure 25 | **Held-out fits from the strongest matched QPACK and g-QPACK configurations.** The best QPACK and g-QPACK restarts from the `pack:1`, `R=192`, `J=16` family fit held-out data and extrapolate from $\tau = 4$ up until $\tau_{\text{max}} = 8$ in the left column.

closed phase quickly, while g-QPACK is able to retain it. Note also the amplitude modulation by the envelope in the bottom left pane modeling a decrease in volume.

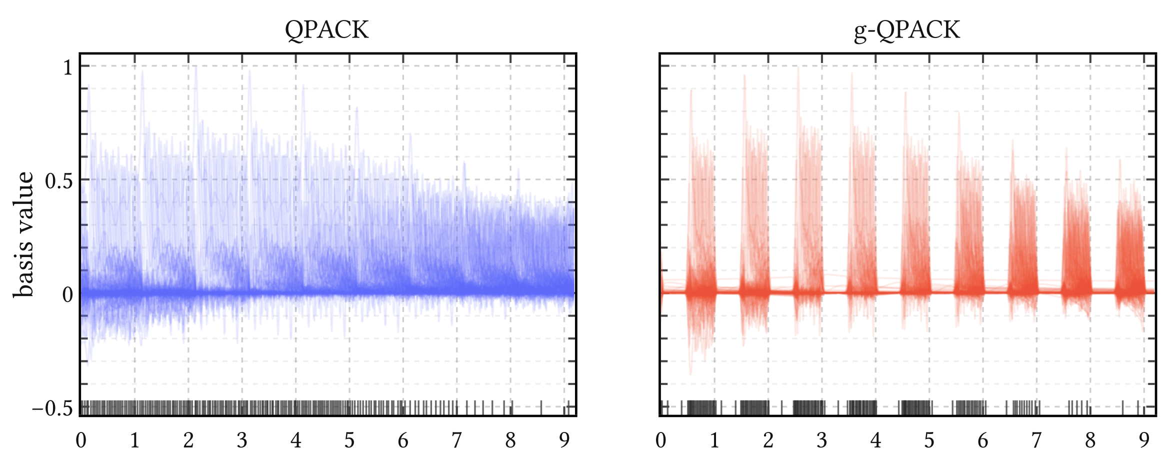


Figure 26 | **Learned bases for the strongest matched QPACK and g-QPACK configurations.** Left: the best QPACK restart within the `pack:1`, `R=192`, `J=16` family. Right: the best g-QPACK restart within the same family. Each panel shows the learned basis functions $\varphi(\tau) \in \mathbb{R}^R$ on a τ-grid. The locations of the inducing points are also shown as a carpet plot.

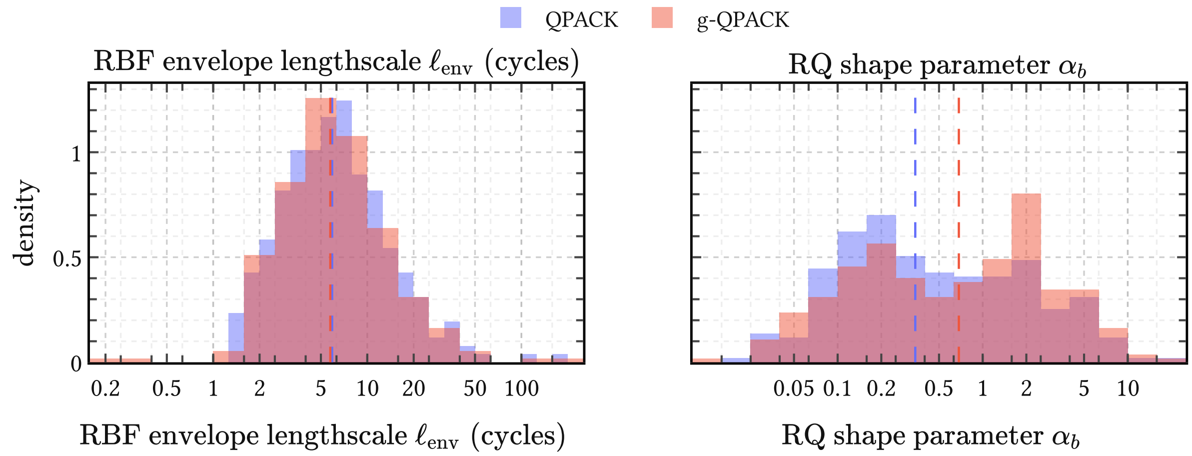


Figure 27 | **Distributions of learned long-scale hyperparameters across restarts.** Left: the RBF envelope lengthscale ℓ_{env}, which measures cycle-to-cycle coupling in units of glottal cycles. Right: the rational quadratic shape parameter α_b of the additive baseline.

An impression of the learned bases in Figure 26 make the effect of gating more tangible. In QPACK, both the basis functions and the inducing locations are spread across the full cycle. In g-QPACK, covariance is suppressed in the closed phase, and the inducing locations are correspondingly pushed out of it and packed more densely into the open phase. So gating not only improves the aggregate score; it redistributes basis resolution towards the more dynamic phase of the glottal cycle.

Some future work One obvious extension would be to relax the envelope itself. The histograms in Figure 27 show a broad distribution of learned envelope lengthscales across random restarts, which suggests that the simple RBF envelope may still be too rigid and that an RQ envelope may be worth testing. The current grid is also incomplete: `pack:2` is still missing, though early pilot runs and the known tail decay of the DGF suggest it is unlikely to beat `pack:0` or `pack:1`; and the ranges $J = 1, 2, 4, 8$ and $R = 16, 32, 64$ remain to be checked in order to quantify where the present gains first appear. We did not push these directions further here simply because of limited compute.

5.6 Summary

We began with PACK, the periodic extension of TACK obtained by replacing the linear time coordinate with a harmonic embedding on the circle. Its stationarity and exact line spectrum made it a convenient starting point, and its harmonic weights provide spectral flexibility that the standard periodic squared exponential cannot match: where the latter traces a one-parameter low-pass envelope, PACK can redistribute mass freely across harmonics while the degree d independently sets the asymptotic rolloff.

QPACK then relaxed strict periodicity to more realistic quasiperiodic behavior by modulating the carrier with a slow RBF envelope and adding a weak rational quadratic baseline. The decisive step, however, was gating: g-QPACK suppresses covariance in the closed phase directly through a periodic sigmoid gate rather than relying on posterior inference alone. This breaks stationarity, but the calibration experiments show it is worth the trade: g-QPACK consistently outperforms its ungated counterpart across all carriers and basis sizes, and redistributes inducing resolution from the closed phase into the more informative open phase. This comes with the downside of requiring GOI information next to GCI information.

To make calibration possible at all, the raw EGIFA waveforms first had to be registered. LBGID and t-PRISM turned a large and heterogeneous collection of simulated glottal flows into aligned voiced groups in shared cycle time, compressing the corpus into a form amenable to learning. On that aligned corpus, PRISM then calibrated a single low-rank surrogate basis shared across thousands of waveforms, with stable optimization throughout.

The resulting basis extrapolates cleanly beyond observed data and reproduces sharp GCI spikes, confirming that the registration and calibration pipeline does extract meaningful structure from the corpus. This calibrated basis is the endpoint of the present chapter; the next chapter takes it as input and refines it into richer latent models of the DGF.

6 Surrogate glottal flow models

In the last two chapters we have been working towards building a suitable DGF prior $p(\mathbf{u}' \mid \boldsymbol{\tau}, \mathcal{M}_R)$. Chapter 4 derived ACK and TACK from changepoints in classical glottal flow models, in order to move from the parametric to the nonparametric regime. Chapter 5 specialized the TACK to PACK, QPACK and g-QPACK and calibrated the resulting kernels on registered EGIFA waveforms. That calibration step gave us a useful global basis, able to extrapolate succesfully, but not yet a compact generative model of the DGF.

More precisely, after learning, PRISM yields a global basis of rank R that is efficient in representing a large corpus of DGF examplars. Each examplar can then be projected on that basis via (273) to give the amplitude-space collection

$$\mathcal{D}_R = \{(\mathbf{m}_n, \mathbf{S}_n)\}_{n=1}^{N}, \tag{148}$$

whose means $\{\mathbf{m}_n\}$ are coordinates in *amplitude space*. PRISM thus maps the registered corpus

$$\mathcal{D} = \{(\boldsymbol{\tau}_n, \mathbf{u}'_n)\}_{n=1}^{N} \tag{149}$$

from variable-size data space to one shared fixed-dimensional amplitude space in $\mathbb{R}^R$.

This further manipulation is where the current chapter comes in. q-GPLVM compresses $\mathcal{D}_R$ into $K \ll N$ compact components in a nonlinear *manifold space* of dimension $Q \ll R$. Each k-th component thus captures a discovered DGF "mode" or archetype. We shall see below that these modes mainly capture smooth variation in the open quotient (OQ) for the QPACK kernels, and additional fine structure for the g-QPACK kernels. We then push each k-th component from manifold space via amplitude space to data space directly, and thus end up with a mixture of K sparse GP models that define the generative prior $p(\mathbf{u}' \mid \boldsymbol{\tau}, \mathcal{M}_R)$ for DGF waveforms. We call these sparse GP models *surrogate glottal flow models* because they cheaply emulate a corpus of expensive-to-simulate glottal flow examples.

We explore $p(\mathbf{u}' \mid \boldsymbol{\tau}, \mathcal{M}_R)$ in this chapter via generative samples, combining it with *time warping* and the $\mathrm{AR}(P)$ prior from Chapter 3 for generating short voiced speech fragments. In the next chapter, we use it for GIF by plugging the mixture directly into IKLP, which then knows about what kind of DGFs to expect in real-world speech, and how to regularize accordingly.

6.1 Refinement

Refinement on EGIFA (Dataset 1) is the last step after registration (Section 5.4) and calibration (Section 5.5) on that same corpus.

Figure 28 shows that the full pipeline has an hourglass, VAE-like structure:

- PRISM compresses variable-length waveforms to a fixed amplitude space in $\mathbb{R}^R$;

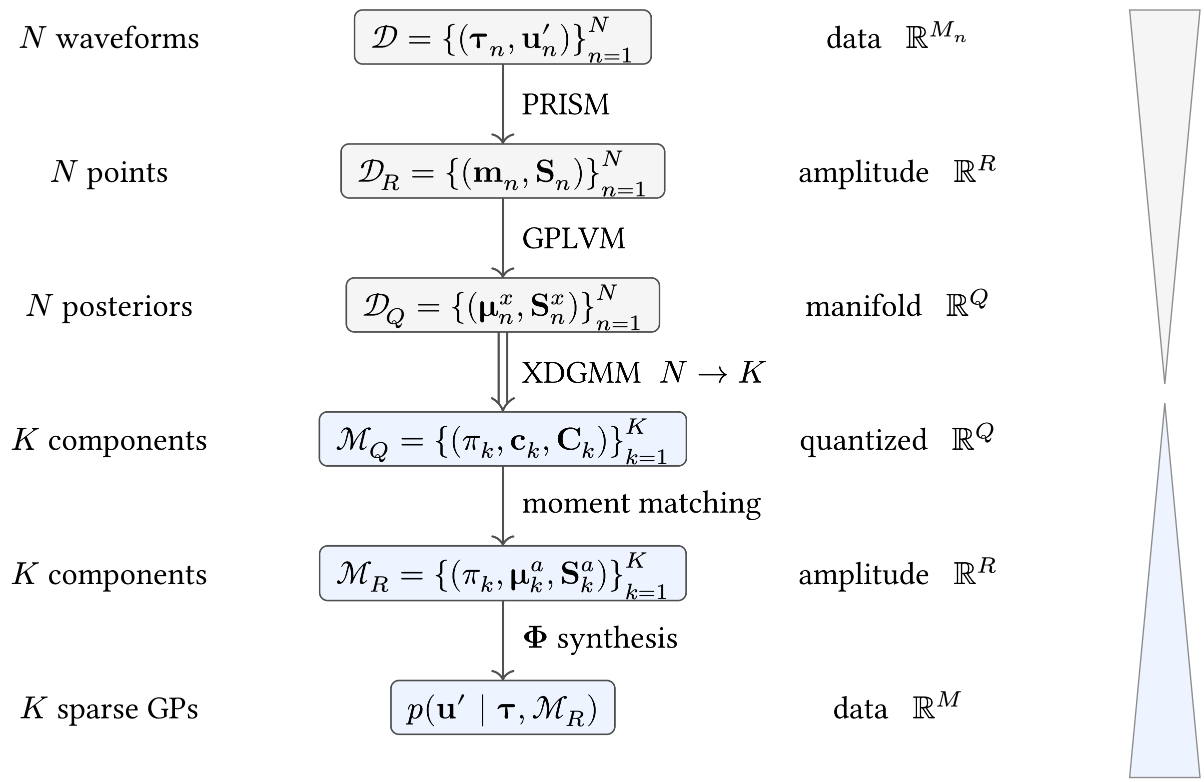


Figure 28 | **From examplars to surrogate.** The hourglass on the right tracks the imploding and expanding dimensionalities. The orders of magnitude involved are

$$M_n \sim M \sim 10^3 > R \sim 10^2 > K \sim 10^1 > Q \sim 10^0. \tag{150}$$

Top half: compression of N variable-length DGF waveforms to N posteriors in a small Q-dimensional manifold. The double arrow marks the $N \to K$ quantization step. Bottom half: synthesis of K sparse GP surrogates in data space from the quantized manifold components.

- q-GPLVM discovers a low-dimensional manifold in $\mathbb{R}^Q$ at the bottleneck;
- XDGMM quantizes that manifold into K Gaussian mixture components;
- and those components are pushed back through amplitude space to data space as sparse GPs with again variable dimension M.

We now proceed to define each object below. Refer to Algorithm 3 and Algorithm 5 for detailed derivations and more details on PRISM and q-GPLVM, respectively.

6.1.1 The three spaces

Data space Each registered DGF waveform $\mathbf{u}'_n \in \mathbb{R}^{M_n}$ lives in a potentially different-dimensional space because cycle counts and sampling grids vary across utterances. PRISM (Algorithm 3) absorbs this variability by learning a shared feature map $\boldsymbol{\varphi}(t)$ of rank R, under which any waveform is synthesized as

$$u'(t) \approx \boldsymbol{\varphi}(t)^\top \mathbf{a}, \quad \mathbf{a} \in \mathbb{R}^R. \tag{151}$$

Amplitude space Projecting each waveform onto this basis yields the amplitude-space object

$$\mathcal{D}_R = \{(\mathbf{m}_n, \mathbf{S}_n)\}_{n=1}^N. \tag{152}$$

By interpreting the projection (151) as inference, this induces the empirical density

$$p(\mathbf{a} \mid \mathcal{D}_R) = \frac{1}{N} \sum_{n=1}^{N} \mathtt{Normal}(\mathbf{a} \mid \mathbf{m}_n, \mathbf{S}_n). \tag{153}$$

This is not yet a compact model: it has N components and remains tied to the training set. Note that, unlike similar adaptor approaches in functional data analysis, PRISM actually infers a latent *density* in the amplitude space, being the extension of a well-posed Bayesian inference problem (Titsias, 2009). We talk about point clouds $\{\mathbf{m}_n\}$ here and discard the uncertainty in the $\{\mathbf{S}_n\}$ because q-GPLVM demands point estimates.

Manifold space q-GPLVM (Algorithm 5) discovers a Q-dimensional nonlinear manifold within the amplitude cloud. Each training waveform n receives a variational posterior, collected as

$$\mathcal{D}_Q = \{(\boldsymbol{\mu}_n^x, \mathbf{S}_n^x)\}_{n=1}^N, \tag{154}$$

equivalently

$$q\big(\mathbf{X} \mid \mathcal{D}_Q\big) = \prod_{n=1}^{N} \mathtt{Normal}(\mathbf{x}_n \mid \boldsymbol{\mu}_n^x, \mathbf{S}_n^x). \tag{155}$$

This is the same collection viewed in a shared manifold coordinate system. Note that the S_n^x covariance matrices are diagonal. We set $Q := 3$, motivated by the six parameters of the LF model (53) with amplitude, open quotient and period already normalized at registration time, leaving three residual degrees of freedom. This also keeps the manifold directly visualizable.

6.1.2 Quantization and pushforward

The manifold posteriors (154) are quantized into $K \ll N$ components by XDGMM (Algorithm 5.4.):

$$\mathcal{M}_Q = \{(\pi_k, \mathbf{c}_k, \mathbf{C}_k)\}_{k=1}^K, \tag{156}$$

equivalently

$$q\big(\mathbf{x} \mid \mathcal{M}_Q\big) = \sum_{k=1}^{K} \pi_k \, \mathtt{Normal}(\mathbf{x} \mid \mathbf{c}_k, \mathbf{C}_k), \quad \mathbf{c}_k \in \mathbb{R}^Q, \quad \mathbf{C}_k \in \mathbb{R}^{Q \times Q}. \tag{157}$$

Because q-GPLVM returns Gaussian posteriors rather than point estimates, XDGMM accounts for the latent uncertainties $\mathbf{S}_n^x$ directly: this time we do not have to discard it.

Each quantized component is then pushed forward through the learned GPLVM map to amplitude space via moment matching ((323), (328)), giving the compact object

$$\mathcal{M}_R = \{(\pi_k, \boldsymbol{\mu}_k^a, \mathbf{S}_k^a)\}_{k=1}^K, \quad \boldsymbol{\mu}_k^a \in \mathbb{R}^R, \quad \mathbf{S}_k^a \in \mathbb{R}^{R\times R}. \tag{158}$$

and hence the amplitude-space density

$$p(\mathbf{a} \mid \mathcal{M}_R) = \sum_{k=1}^K \pi_k \, \texttt{Normal}(\mathbf{a} \mid \boldsymbol{\mu}_k^a, \mathbf{S}_k^a). \tag{159}$$

This is where *uncertainty calibration* enters. q-GPLVM returns uncertain manifold posteriors $\mathcal{D}_Q$, XDGMM fits $\mathcal{M}_Q$ against both their means and covariances, and moment matching then pushes that spread forward into $\mathcal{M}_R$. The resulting surrogate is therefore intentionally less rigid than a pure point-estimate fit, which matters both for generation here and for avoiding overconfident priors later in Chapter 7.

6.1.3 The DGF prior

Before refinement, the same synthesis map already turns the empirical PRISM density $p(\mathbf{a} \mid \mathcal{D}_R)$ into the unrefined data-space prior

$$p(\mathbf{u}' \mid \boldsymbol{\tau}, \mathcal{D}_R) = \frac{1}{N} \sum_{n=1}^N \texttt{Normal}(\mathbf{u}' \mid \boldsymbol{\Phi}(\boldsymbol{\tau})\mathbf{m}_n, \boldsymbol{\Phi}(\boldsymbol{\tau})\mathbf{S}_n\boldsymbol{\Phi}(\boldsymbol{\tau})^\top + \sigma^2\mathbf{I}). \tag{160}$$

Refinement replaces this N-component empirical prior by the compact K-component surrogate below.

Composing $\mathcal{M}_R$ with the PRISM feature map $\boldsymbol{\Phi}(\boldsymbol{\tau})$ yields the sought object: a mixture of K sparse Gaussian process priors over DGF waveforms,

$$p(\mathbf{u}' \mid \boldsymbol{\tau}, \mathcal{M}_R) = \sum_{k=1}^K \pi_k \, \texttt{Normal}(\mathbf{u}' \mid \boldsymbol{\mu}_k^u, \boldsymbol{\Sigma}_k^u) \tag{161}$$

$$\text{where} \qquad \boldsymbol{\mu}_k^u = \boldsymbol{\Phi}(\boldsymbol{\tau})\boldsymbol{\mu}_k^a, \tag{162}$$

$$\boldsymbol{\Sigma}_k^u = \boldsymbol{\Phi}(\boldsymbol{\tau})\mathbf{S}_k^a\boldsymbol{\Phi}(\boldsymbol{\tau})^\top + \sigma^2\mathbf{I}. \tag{163}$$

Each component is a sparse (degenerate) GP: its covariance $\boldsymbol{\Phi}\mathbf{S}_k^a\boldsymbol{\Phi}^\top$ has rank at most R, inherited from the PRISM basis, plus isotropic noise $\sigma^2\mathbf{I}$. The k-th component's mean function $\boldsymbol{\Phi}\boldsymbol{\mu}_k^a$ is a learned DGF archetype; its covariance encodes the plausible variation around that archetype.

6.1.4 Learning

Refinement starts from a retained calibrated PRISM source on EGIFA. That source fixes the learned feature map $\boldsymbol{\varphi}(\tau)$ and the amplitude-space object $\mathcal{D}_R$.

We then fit a Bayesian GPLVM at fixed $Q = 3$, obtaining $\mathcal{D}_Q$, quantize its latent posteriors by XDGMM at $K \in \{1, 3, 5, 7\}$ to obtain $\mathcal{M}_Q$, and push those components back to amplitude space as $\mathcal{M}_R$ and then to data space as the surrogate prior (161). The two optimizations are separate:

- The GPLVM stage optimizes the variational bound (313) by full-batch Adam on all 6500 training waveforms in $\mathcal{D}_R$ at once, with learning rate $5 \cdot 10^{-2}$ for 4000 iterations and otherwise standard settings.
- The XDGMM stage then fits $\mathcal{M}_Q$ by a separate EM optimization of the standard extreme-deconvolution likelihood (Algorithm 5.4.), using the implementation of (Holoien et al., 2017) for 100 EM iterations.

As in the calibration experiments, all runs were carried out on the GPU in float32 and both stages proved uneventful and predictable across the full sweep.

For one held-out EGIFA waveform $(\boldsymbol{\tau}_n, \mathbf{u}'_n)$, the refinement score is the q-GPLVM log evidence per effective sample,

$$\mathcal{S}_{\text{test}}^{q(\mathcal{M}_R)} = \frac{1}{N_{\text{test}}} \sum_{n=1}^{N_{\text{test}}} \frac{1}{M_n} \log p(\mathbf{u}'_n \mid \boldsymbol{\tau}_n, \mathcal{M}_R), \tag{164}$$

where M_n is the number of observed samples in waveform n. Here $N_{\text{test}} = 1000$. This score gauges how realistic the refined surrogate prior is, but it does not guarantee strong performance when the same surrogate is used for IKLP inference in Chapter 7.

6.2 Experiments

We refine the top-5 best-scoring PRISM calibration fits on EGIFA for each family (QPACK, g-QPACK) and EGIFA collection (vowel, speech). After deduplication this gives 17 calibrated

Family	Source runs	q-GPLVM fits	Improved	Median gain	Best q-GPLVM score
`QPACK`	8	32	32/32	+0.057	0.579
`g-QPACK`	9	36	36/36	+0.086	0.588

Table 10 | **Refinement sweep on EGIFA. Source runs** counts distinct calibrated PRISM sources; **q-GPLVM fits** counts all refinements over $K \in \{1, 3, 5, 7\}$. **Median gain** is the median increase in held-out score (164) over the corresponding PRISM score, and **Best q-GPLVM score** is the best absolute value attained within the family.

Family	Carrier	R	J	Iteration	K
`QPACK`	`pack:0`	128	16	15	5
`g-QPACK`	`pack:1`	128	32	31	3

Table 11 | **Configurations of representative refinements.** Both use $Q = 3$.

sources in total, 8 from the QPACK family and 9 from the g-QPACK family. Each is compressed at $Q = 3$ and $K \in \{1, 3, 5, 7\}$, for 68 q-GPLVM fits overall. Table 10 shows the result. Every refinement improves the held-out q-GPLVM score (164) over the score attained by its PRISM source. In other words, refinement helps in modeling more realistic DGF waveforms.

Since most refinement runs behave very similarly we choose two representative runs for illustration below: one QPACK source shown at $K = 5$, and one g-QPACK source at $K = 3$. Table 11 lists the precise configurations used.

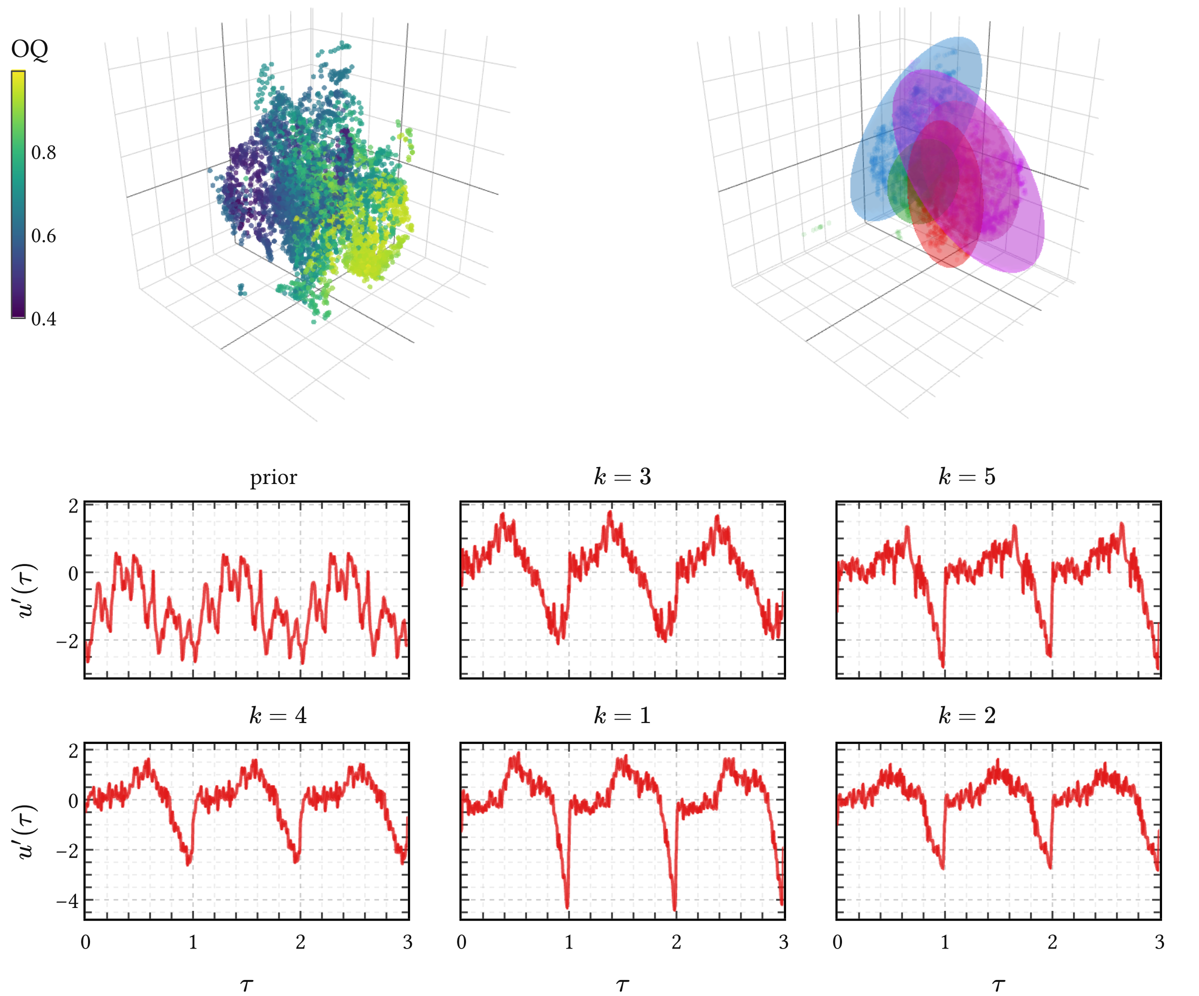


Figure 29 | **QPACK surrogate: manifold space, quantization, and data-space samples.** Top left: posterior means $\{\boldsymbol{\mu}_n^x\}_{n=1}^N$ from $\mathcal{D}_Q$, shown as a manifold point cloud and colored by ground-truth OQ. Top right: the $K = 5$ XDGMM ellipsoids of $\mathcal{M}_Q$ fit to the data in the top left panel. Bottom: the left panel samples the unrefined PRISM prior $p(\mathbf{u}' \mid \boldsymbol{\tau}, \mathcal{D}_R)$ from (160); the numbered panels sample the k-th component conditionals of $p(\mathbf{u}' \mid \boldsymbol{\tau}, \mathcal{M}_R)$ induced by (161).

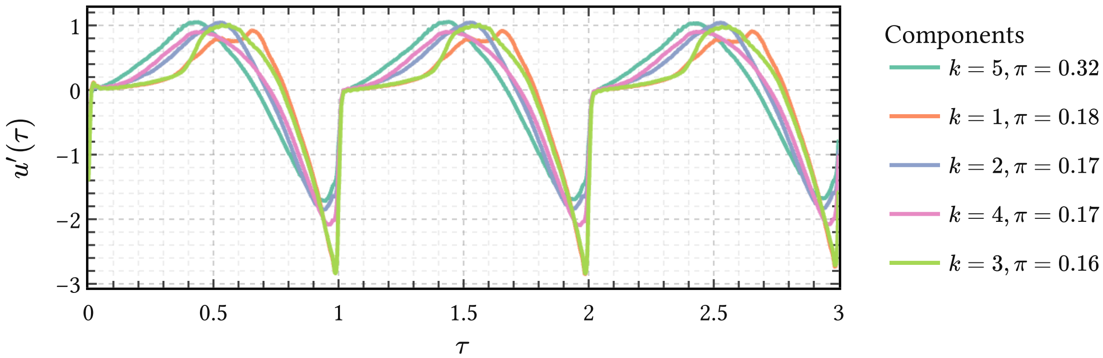


Figure 30 | **Learned QPACK component means in data space.** Shown are the data-space means $\boldsymbol{\mu}_k^u = \boldsymbol{\Phi}(\boldsymbol{\tau})\boldsymbol{\mu}_k^a$ implied by $\mathcal{M}_R$ in (161). Their smooth progression in OQ corresponds to a strongly nonlinear trajectory in amplitude space and sharpens closure from rounder to peakier cycles.

6.2.1 QPACK

The performance and nature of the surrogate prior $p(\mathbf{u}' \mid \boldsymbol{\tau}, \mathcal{M}_R)$ depend on which kernel family PRISM used to produce the amplitude-space object $\mathcal{D}_R$. We begin with the representative QPACK refinement at $K = 5$ in Figure 29.

The top-left panel shows the posterior means $\{\boldsymbol{\mu}_n^x\}_{n=1}^N$ from $\mathcal{D}_Q$ as the means of diffuse Gaussian-blob manifold, colored by ground-truth OQ. Note that the manifold is organized mainly by OQ, with only a small number of diffuse outliers around its edge. The manifold also appears connected, because OQ variation is naturally smooth in the glottal cycle. Therefore, the XDGMM fit in the top-right panel is not so much clustering as expanding the latent density in manifold space.[21]

Nevertheless, the components $\mathcal{M}_Q$ discovered by XDGMM do show modal structure. Figure 30 shows that the learned means $\boldsymbol{\mu}_k^u = \boldsymbol{\Phi}(\boldsymbol{\tau})\,\boldsymbol{\mu}_k^a$ associated with each component k mainly track OQ, together with intensity of the GCI event. This is remarkable because the OQ degree and GCI event intensity are two highly nonlinear phenomena in amplitude space; nevertheless, q-GPLVM learned them and managed to carry them across 3 spaces "in one piece."

Of course, these are just the means, only the first moment of the density of the learned surrogate mixture $\mathcal{M}_M$. To visualize the latter properly, we should also sample from it. The bottom row in Figure 29 shows a single sample for each component k Each numbered panel k drew

$$\mathbf{a} \sim \texttt{Normal}(\boldsymbol{\mu}_k^a, \mathbf{S}_k^a) \tag{165}$$

[21]This is akin to the "modeling or expanding?" discussion in Section 3.4.

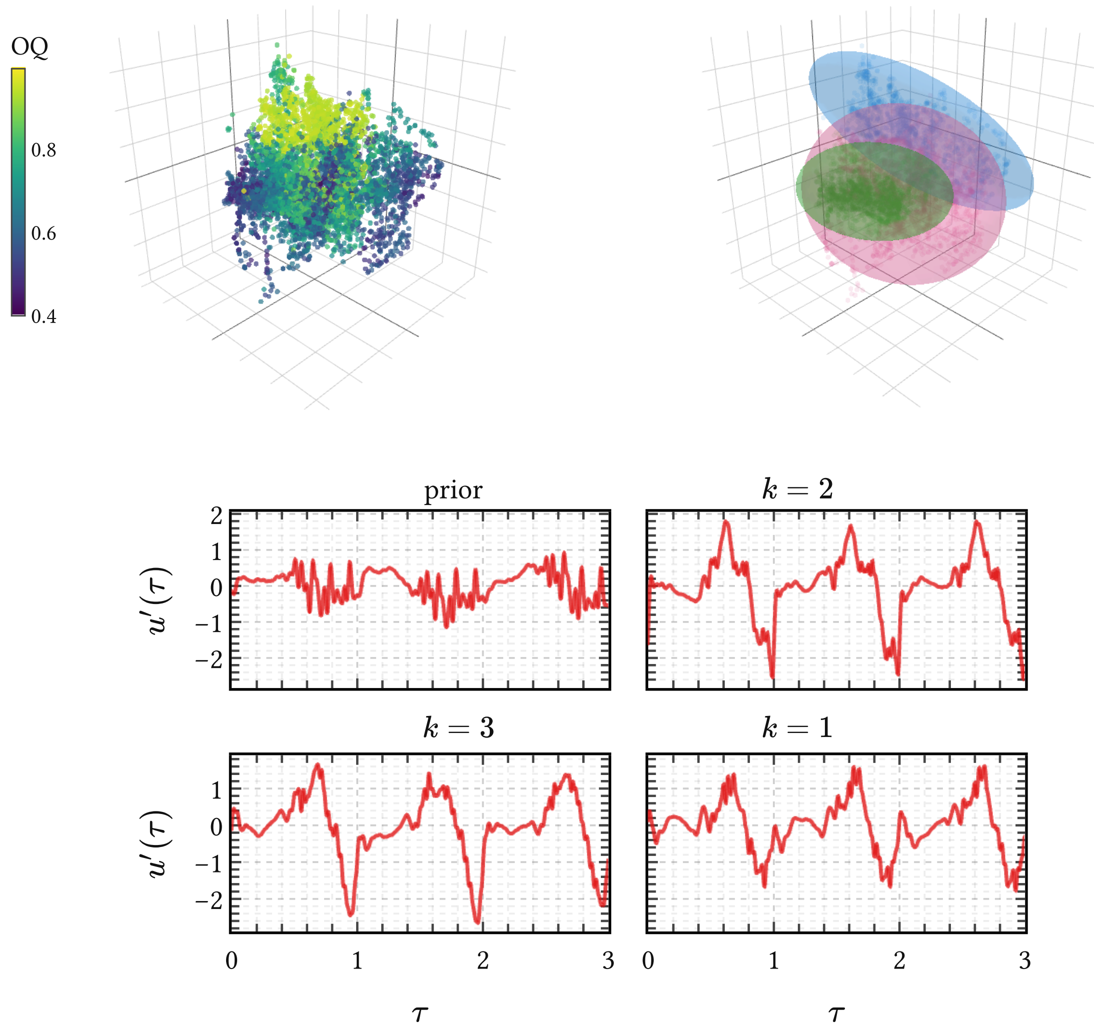


Figure 31 | **g-QPACK surrogate: manifold space, quantization, and data-space samples.** Same three-step layout as Figure 29, but here the fitted quantization uses only $K = 3$ components. After uncertainty calibration the closed phase is no longer perfectly flat, but its variance is still substantially suppressed by the gate.

from component k of $p(\mathbf{a} \mid \mathcal{M}_R)$ in (159) and maps it to data space through (161). The leftmost **prior** panel shows draws from the unrefined PRISM-induced prior $p(\mathbf{u}' \mid \boldsymbol{\tau}, \mathcal{D}_R)$ in (160). It is evident that the prior panel contains much less structure than the component panels: this is the difference between calibration and refinement.

6.2.2 g-QPACK

Compared to QPACK, the g-QPACK manifold in Figure 31 is less organized by OQ, but still partly so. Since OQ is effectively normalized away during registration, this ordering implies that there must be strong correlations between OQ and other features, incidentally a feature of the LF model (50), which GPLVM picked up on. The samples in the bottom row still show a notion of the closed phase, but with more distortion because the entire chain $\mathcal{D}_R \to \mathcal{D}_Q \to \mathcal{M}_Q \to \mathcal{M}_R$ retains uncertainty; see the uncertain manifold posteriors in (317), the XDGMM fit in (318), and the moment-matched pushforward in Algorithm 5.5.2.. Comparing the samples from the mixture components to the PRISM prior, it is again clear that refinement picks up plenty of extra structure from the data.

6.3 Generating voiced speech

The object $\mathcal{M}_R$ constructed above has two distinct uses.

First, it defines a tractable prior over DGF waveforms,

$$p(\mathbf{u}' \mid \boldsymbol{\tau}, \mathcal{M}_R), \tag{166}$$

which is a mixture of sparse Gaussian processes and can therefore be used directly inside IKLP without further approximation. This is the primary motivation for the pipeline: every step from $\mathcal{M}_Q \to \mathcal{M}_R$ was chosen to preserve that Gaussian structure.

Second, the same object is generative. It defines a conditional distribution over functions given $\boldsymbol{\tau}$, from which DGF cycles can be sampled cheaply. Classical surrogate models typically emphasize fast forward evaluation or sampling, while Bayesian formulations introduce a separate prior. Here both coincide: the surrogate *is already* the prior.

We now use the refined source model as a genuine generator. The construction has three layers. The QPACK mixture $\mathcal{M}_R$ supplies normalized DGF shape in cycle time τ, the pitch-track prior $\mathcal{P}$ from Dataset 2 turns cycle time into physical time t, and the $\mathrm{AR}(P)$ prior from Chapter 3 shapes the resulting source into speech. The figures below follow that assembly in order. For this final step we continue with QPACK: the warp supplies GCIs through the period trajectory, but not GOIs, so QPACK interfaces cleanly whereas g-QPACK would require an additional GOI model.

6.3.1 Time warping

Registration earlier solved the map from physical time to cycle time. Generation needs the inverse one. Once a source has been sampled on a fixed grid $\boldsymbol{\tau}$, the remaining question is how quickly that grid is traversed in physical time.

That timing is supplied by the APLAWD-trained pitch-track prior $\mathcal{P}$, defined in Dataset 2 through (187) and (188). Working on the latent log-period function $g(\tau)$ keeps periods positive after exponentiation and turns multiplicative pitch variation into additive GP structure. A draw $g(.) \mid \mathcal{P}$ therefore yields a smooth period trajectory $T(\tau) = \exp(g(\tau))$, and hence a smooth monotone warp into physical time. Because $\mathcal{P}$ was learned from hand-corrected GCI events, it already carries realistic jitter-scale variability on the timescale of speech.

Instantaneous phase and pitch Much of the nonstationary variation of voiced speech can be captured by a single monotone phase function $\tau(t)$ satisfying

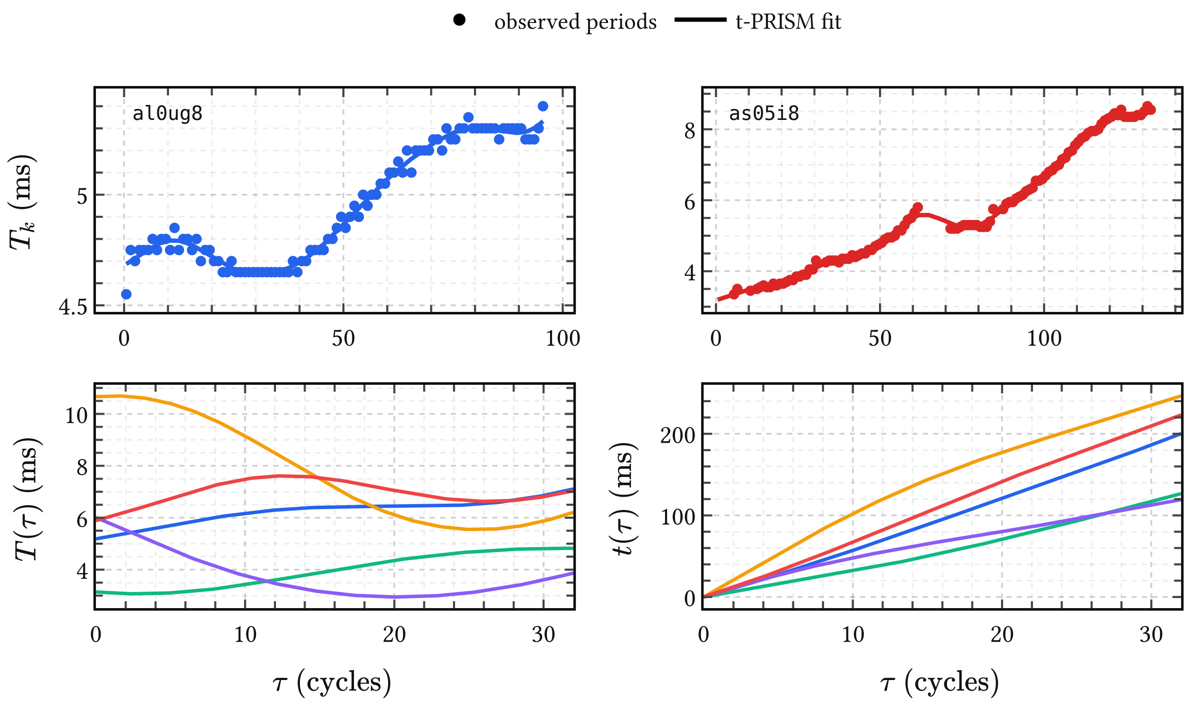


Figure 32 | **From the learned pitch-track prior $\mathcal{P}$ to time warp.** Top: two retained APLAWD period tracks together with their t-PRISM posterior means. Bottom left: prior samples of the instantaneous period function $T(\tau)$ induced by $\mathcal{P}$. Bottom right: the corresponding induced warps $t(\tau) = t_0 + \int_0^\tau T(u)\,\mathrm{d}u$. This is the object needed later to turn refined DGF surrogates in cycle time into voiced signals in physical time.

$$\frac{\mathrm{d}\tau}{\mathrm{d}t} = f_0(t), \tag{167}$$

where $f_0(t)$ is instantaneous pitch (Hz) and τ counts cycles (has units of cycles).

During registration, $\tau(t)$ was inferred from observed GCIs. For generation the useful view is the inverse one: once τ has been fixed on a regular grid, (167) tells us how that grid should be stretched into physical time. For constant pitch the result is just $t(\tau) = t_0 + T\tau$: advance by T for every unit interval covered by τ. In general, defining the instantaneous period $T(\tau) := 1/f_0(\tau)$ and integrating (167) gives

$$t(\tau) = t_0 + \int_0^\tau T(u)\,\mathrm{d}u. \tag{168}$$

A draw from $\mathcal{P}$ therefore induces a monotone warp directly. Once $T(\tau)$ has been sampled, (168) converts a regular grid in cycle time into an irregular grid in physical time. Pitch variation enters through that warp, while within-cycle OQ and amplitude variation remain with the surrogate $\mathcal{M}_R$.

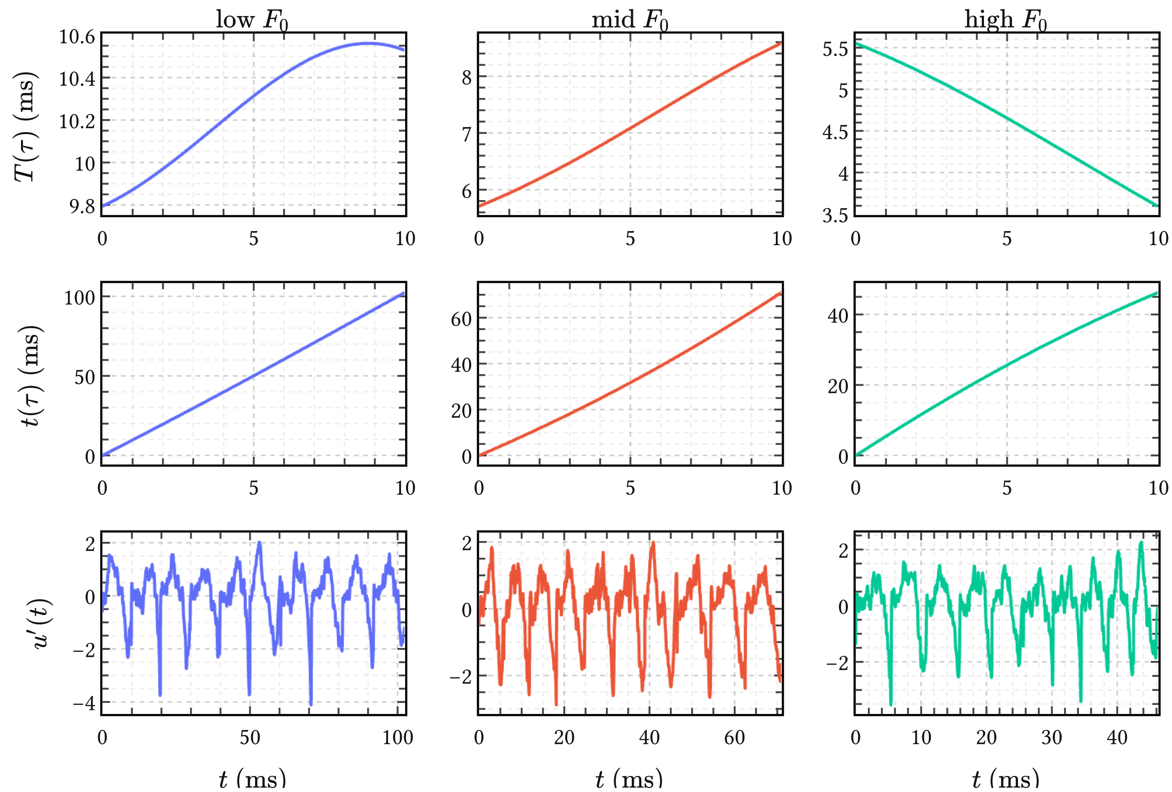


Figure 33 | **From data-space surrogate to voiced source.** Top: three sampled period trajectories $T(\tau) = \exp(g(\tau))$ obtained from draws $g(.) \mid \mathcal{P}$, representing low, mid, and high F_0 regimes. Center: the induced monotone warps $t(\tau) = t_0 + \int_0^\tau T(u)\,\mathrm{d}u$ from (168). Bottom: QPACK source draws $\mathbf{u}' \mid \boldsymbol{\tau}, \mathcal{M}_R$, transported through those warps and resampled on regular physical-time grids; each spans ten pitch periods.

6.3.2 Generating DGF waveforms

For the chapter figures we work on a finite interval of about ten cycles and choose a dense regular grid $\boldsymbol{\tau}$ on that interval. First sample the timing object

$$g(.) \mid \mathcal{P}, \quad T(\tau) = \exp(g(\tau)), \quad \mathbf{t} = t(\boldsymbol{\tau}), \tag{169}$$

with $t(\tau)$ from (168). Then sample a normalized source on the same cycle-time grid,

$$\mathbf{u}' \mid \boldsymbol{\tau}, \mathcal{M}_R. \tag{170}$$

Operationally this means drawing amplitudes from the Gaussian mixture $\mathcal{M}_R$ in amplitude space and evaluating the shared PRISM synthesis map (301) on a dense within-cycle grid. The pair $(\boldsymbol{\tau}, \mathbf{u}')$ is then relabeled as $(\mathbf{t}, \mathbf{u}')$; interpolating that irregularly timed curve back to a regular t grid yields the physical-time DGF waveform $u'(t)$.

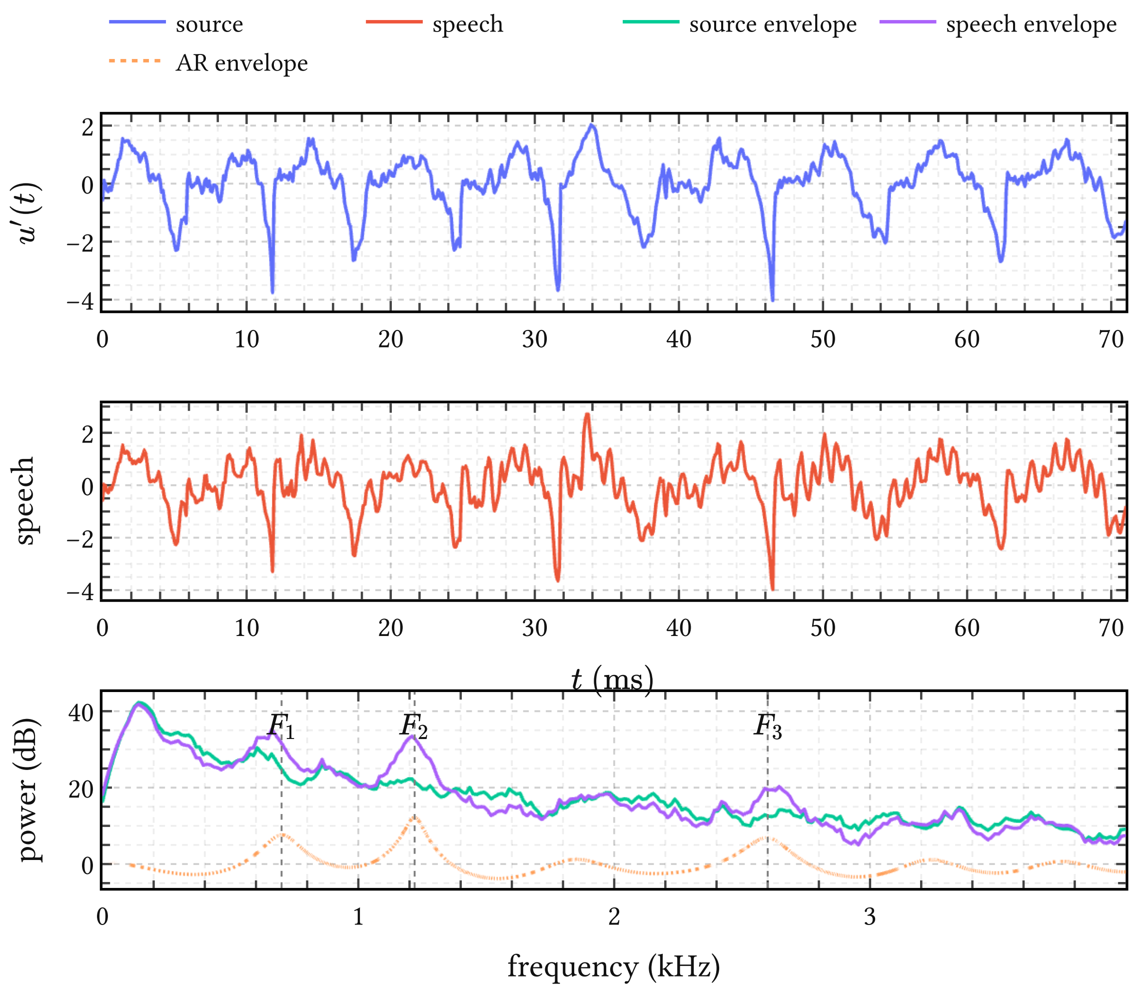


Figure 34 | **Generated voiced speech fragment.** Top: source waveform $u'(t)$ obtained by drawing $\mathbf{u}' \mid \boldsymbol{\tau}, \mathcal{M}_R$ and transporting it through the warp induced by $\mathcal{P}$. Middle: speech waveform after applying the AR filter with coefficients from (171), conditioned on an open-vowel formant pattern. Bottom: spectral envelopes of source and speech together with the AR envelope; F_1, F_2, F_3 targets are marked.

Figure 33 illustrates the procedure: sampled $T(\tau)$, induced $t(\tau)$, and the resulting source fragments. Pitch modulation comes from $\mathcal{P}$, while OQ, closure sharpness, amplitude variation, and quasi-periodic within-cycle shape come "holistically" from $\mathcal{M}_R$.

6.3.3 Generating voiced speech

A voiced speech fragment requires one more draw, now on the filter side. The AR coefficients are drawn from the SoftSpectral prior (45) conditioned on a target formant structure:

$$\mathbf{a}_{\mathrm{AR}} \sim \texttt{SoftSpectral}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0; M), \tag{171}$$

where the constraint polynomial $M(z)$ (44) encodes resonances at the desired (F_1, F_2, F_3) frequencies and bandwidths. Applying the AR operator $\mathbf{\Psi}(\mathbf{a}_{\mathrm{AR}})$ to the generated source $u'(t)$ produces the speech waveform. Figure 34 should be read top to bottom. The first row shows the sampled source, the second row the resulting voiced speech fragment, and the third row the corresponding spectral envelopes. The division of labour is visible there: $\mathcal{M}_R$ supplies source harmonics and closure structure, while the AR prior redistributes that energy into formant structure.

Longer generated sources and vowel-filtered fragments can be heard **on this web page**. The amount of variation of the vowel structure allowed by the `SoftSpectral` prior is interesting here, and sounds plausible. Nevertheless, the samples have a distinct lack of organic warmth and their timbre lands somewhere in the middle between LF-based synthetic speech (Alku et al., 2019) and physics-based speech simulation (Krug et al., 2021).

The examples on that page extend the same local generator by overlap-conditioned continuation, because the surrogate is learned on a finite interval in normalized cycle time. Note that our approach goes beyond codebook-style source models such as Thomas et al. (2009): on its local support in τ, the surrogate is process-valued and can therefore be sampled at arbitrary temporal resolution before warping. Hence the name of PRISM: *process-induced* surrogate modeling.

6.4 Summary

The chapter compresses thousands of calibrated DGF waveforms into a mixture of K sparse GP components through an information bottleneck in $\mathbb{R}^Q$. Each step of the hourglass – PRISM projection, GPLVM embedding, XDGMM quantization, moment-matched pushforward – was carefully chosen to preserve Gaussian structure, so the resulting object $\mathcal{M}_R$ can enter IKLP without further approximation. In other words, the refinement procedure was not about dimensionality reduction for its own sake, but about a specific dimensionality reduction method that stays compatible with the IKLP inference engine.

We showed that the manifold discovered by q-GPLVM is not an arbitrary latent space. Its dominant axis tracks the open quotient, the single most important scalar summary of glottal flow shape. The surrogate thus inherits physically meaningful phonation variation from the data rather than imposing it through a parametric model; refinement pushed data-driven features inside a *nonparametric* model.

A classical surrogate optimizes for fast forward evaluation; a Bayesian prior encodes inductive bias for inference. Here both coincide: $\mathcal{M}_R$ is the surrogate *and* the prior. Combined with the pitch-track prior $\mathcal{P}$ from Dataset 2 and the AR prior `SoftSpectral` from Chapter 3, we found it to be a useful generative model of voiced speech, which can be used for synthetic data generation at scale. We now at last turn to Chapter 7 to explore the learned surrogate mixture as a prior for GIF.

7 Bayesian nonparametric glottal inverse filtering

Chapter 2 set up the variational inference engine, Chapter 3 developed informative priors for the AR filter, Chapter 4 and Chapter 5 derived custom kernels for the glottal flow derivative, and Chapter 6 compressed them into a compact learned surrogate mixture $\mathcal{M}_R$. This chapter assembles everything into BNGIF and evaluates the result on two tasks: *glottal inverse filtering* on EGIFA and *formant estimation* on VTRFormants.

7.1 Assembly

The refined surrogate mixture from Chapter 6 is

$$\mathcal{M}_R = \{(\pi_i, \boldsymbol{\mu}_i^a, \mathbf{S}_i^a)\}_{i=1}^{I}, \tag{172}$$

a set of I weighted Gaussian components in amplitude space, each with nonzero mean $\boldsymbol{\mu}_i^a$ and covariance $\mathbf{S}_i^a$. These encode distinct DGF archetypes discovered from the EGIFA corpus. We write I for the number of kernel components rather than K to avoid collision with the kernel matrices $\mathbf{K}_i$; in the q-GPLVM setting of Chapter 6, I coincides with the number of mixture components, whereas in the original formulation of Yoshii & Goto (2013) it indexed the frequency hypotheses in the kernel bank.

IKLP, however, consumes zero-mean low-rank kernels of the form $\mathbf{K}_i = \boldsymbol{\Phi}_i \boldsymbol{\Phi}_i^\top$ (225). A zero-mean requirement may seem like a loss, but it is not: the mean can be folded into the second moment. For each component, we compute

$$\mathbf{L}_i \mathbf{L}_i^\top = \mathbf{S}_i^a + \boldsymbol{\mu}_i^a (\boldsymbol{\mu}_i^a)^\top, \quad \boldsymbol{\Phi}_i = \boldsymbol{\Phi}(\boldsymbol{\tau}) \mathbf{L}_i, \tag{173}$$

so that $\mathbf{K}_i = \boldsymbol{\Phi}_i \boldsymbol{\Phi}_i^\top$ preserves the full second-moment structure of the original surrogate component under a zero-mean law. The isotropic nugget of the surrogate is not carried over separately; it is already represented by the broadband excitation term $\nu_e \mathbf{I}$ in (174) below. The mixture weights π_i provide the natural initialization for the IKLP kernel weights θ_i. One could in principle extend IKLP to accommodate nonzero-mean kernels, but the zero-mean formulation is preferable: any nonzero mean would presuppose a known polarity of the glottal flow, and in real speech the polarity is unknown (Drugman et al., 2019b, Sec. 3.2).

IKLP (Algorithm 1) then assembles the kernel bank into the full BNGIF likelihood

$$\mathbf{x} \sim \texttt{Normal}\left(\mathbf{0}, \boldsymbol{\Psi}^{-1}\left(\nu_w \sum_{i=1}^{I} \theta_i \mathbf{K}_i + \nu_e \mathbf{I}\right) \boldsymbol{\Psi}^{-\top}\right), \tag{174}$$

which is (18) with the concrete kernel bank and AR prior substituted in. This is worth pausing on, because that equation (174) is the entire model, and reduces to several classical GIF approaches as special cases. The speech frame $\mathbf{x}$ is explained as an AR-filtered mixture of structured source components $\mathbf{K}_i$ plus broadband noise $\nu_e \mathbf{I}$, with the filter encoded by the AR operator $\boldsymbol{\Psi}(\mathbf{a})$ and the source structure encoded by the kernel bank. Classical LP-based methods solve a version of this problem too, but regularize through the likelihood: they choose a weight function w_m in (181) that downweights samples near the glottal closure. In contrast, BNGIF regularizes through the prior: the learned kernel mixture $\sum_i \theta_i \mathbf{K}_i$ in (174) and the Gaussian prior on $\mathbf{a}$ (201) encode what is known about source and filter before any data is seen. The comparison is therefore between two ways of bringing prior knowledge to bear on the same underdetermined inverse problem.

Inference on (174) proceeds by coordinate-ascent variational inference (CAVI; Algorithm 1.1.), updating the AR coefficients, the kernel weights, and the noise precisions in closed form. The Woodbury-Cholesky reformulation keeps the cost $\mathcal{O}(M)$: linear in frame length for fixed inducing rank.

7.1.1 Variational posterior

After CAVI converges (Algorithm 1.1.4.), the posterior has two qualitatively different parts.

The filter is a point estimate. The AR coefficients converge to the MAP value $\mathbf{a}^*$ (202); there is no posterior uncertainty on $\mathbf{a}^*$. The vocal tract spectrum is the all-pole envelope $|H(z)|^2 = \frac{1}{|}A(z)|^2$ where $A(z) = 1 - \sum_{p=1}^{P} a_p^* z^{-p}$ (7), and the formant frequencies and bandwidths follow from its poles.

The excitation carries posterior uncertainty. The kernel weights θ_i and noise precisions ν_w, ν_e each carry a GIG posterior (210), and together they determine how the LP residual decomposes into structured signal and broadband noise. Fixing $\mathbf{a}^*$ gives the observed residual $\boldsymbol{\varepsilon}^* := \boldsymbol{\Psi}(\mathbf{a}^*)\mathbf{x}$ (216), which splits as

$$\boldsymbol{\varepsilon} = \mathbf{s} + \mathbf{n}, \tag{175}$$

where $\mathbf{s}$ is the structured signal and $\mathbf{n}$ is broadband noise (217). Their posterior means are (223)

$$\boldsymbol{\mu}_s = \boldsymbol{\Sigma}_s \boldsymbol{\Omega}^{-1} \boldsymbol{\varepsilon}^*, \quad \boldsymbol{\mu}_n = \boldsymbol{\varepsilon}^* - \boldsymbol{\mu}_s, \tag{176}$$

where $\boldsymbol{\Sigma}_s := \mathbb{E}[\nu_w] \sum_i \mathbb{E}[\theta_i] \mathbf{K}_i$ is the structured prior covariance and $\boldsymbol{\Omega} := \boldsymbol{\Sigma}_s + \mathbb{E}[\nu_e]\mathbf{I}$ (206). The posterior covariance $\boldsymbol{\Sigma}_{s|\varepsilon} = \boldsymbol{\Sigma}_s - \boldsymbol{\Sigma}_s \boldsymbol{\Omega}^{-1} \boldsymbol{\Sigma}_s$ (221) quantifies how uncertain the decomposition is at each sample.

7.2 Evaluation on EGIFA

7.2.1 Experimental setup

All tested GIF methods operate on 32 ms analysis frames, except CCD, which operates at the cycle level (roughly 7 ms on average). CAVI iterations continue until the improvement on the ELBO bound (203) falls below 10^{-4}; convergence is typically reached in about 20 iterations, with outliers up to 50 for less easily resolved speech frames.

Strategy	Update	Targets	F_i (Hz)	B_i (Hz)
`none`	isotropic	—	—	—
`F1`	soft	F_1	500	80
`all`	soft	F_1–F_4	500, 1500, 2500, 3500	80, 90, 110, 180
`hard-F1`	hard	F_1	500	80
`hard-all`	hard	F_1–F_4	500, 1500, 2500, 3500	80, 90, 110, 180

Table 12 | **AR-prior strategies used in the BNGIF experiments.** The soft strategies apply `SoftSpectral` (45), which shifts the prior mean toward the target formants F_i with bandwidths B_i while leaving the covariance unchanged. The hard strategies apply `HardSpectral` (48), which additionally contracts the covariance along the constrained spectral directions.

The dataset and registration are described in Dataset 1. EGIFA has two collections, EGIFA/vowel and EGIFA/speech, corresponding to the vowel and speech subsets in Table 19. The classical baselines are CPCA, SLP, QCP, IAIF, and CCD, exactly as defined there; among these, QCP is the strongest classical reference point (see Section 1.1.3). All results use LBGID-estimated GCIs.

Scoring Because different GIF algorithms resolve the source-filter scale and delay ambiguity differently (Appendix E), raw waveform comparison is misleading. We therefore align each estimated frame to the ground truth by jointly fitting an affine map and a nonnegative integer lag via (345), which absorbs scale, polarity, mean offset, and unknown propagation delay in one step. The score is then the centered cosine similarity (347) between the aligned estimate and the reference, a shape measure bounded in $[-1, 1]$ that is monotonically related to NRMSE after alignment. All reported scores are computed on the structured posterior mean $\boldsymbol{\mu}_s$ (176), which is the Bayes-optimal DGF estimate under squared error.

Configurations

BNGIF has two independent design dimensions.

The first is the **source**: which kernel family populates the bank $\{\mathbf{K}_i\}$ in (174). The configurations tested here use either QPACK or its gated variant g-QPACK (Chapter 5), compressed into a small mixture by the q-GPLVM of Chapter 6. Table 13 shows what the source side alone buys before any AR prior enters the picture: gating helps, and q-GPLVM compression helps again. The final system continues with the strongest g-QPACK lineage, enlarged to $I = 7$ components at inference time.

The second is the **filter**: the Gaussian AR prior $p(\mathbf{a}) = \texttt{Normal}(\boldsymbol{\mu}_a, \boldsymbol{\Sigma}_a)$ (201). Every strategy starts from the isotropic base prior $\texttt{Isotropic}(\lambda) = \texttt{Normal}(\mathbf{0}, \lambda \mathbf{I}_P)$ (20) and optionally reshapes it via the spectral shaping priors of Chapter 3. Table 12 lists the five strategies tested: `SoftSpectral` (45) shifts the prior mean toward the target formants, `HardSpectral` (48) additionally contracts the covariance.

Kernel	Carrier	Algorithm	Best config	Vowel	Speech
QPACK	`pack:0`	PRISM	$R = 128$, $J = 16$	0.745	0.803
g-QPACK	`pack:0`	PRISM	$R = 128$, $J = 16$	0.888	0.849
QPACK	`pack:1`	PRISM	$R = 128$, $J = 16$	0.697	0.757
g-QPACK	`pack:1`	PRISM	$R = 128$, $J = 32$	0.888	0.867
QPACK	`pack:0`	q-GPLVM	$K = 7$, $Q = 3$	0.805	0.839
g-QPACK	`pack:1`	q-GPLVM	$K = 5$, $Q = 3$	**0.928**	**0.915**

Table 13 | **Source-prior ladder on EGIFA before AR assembly.** Each row reports the best available gauge-invariant median score for that source lineage in downstream IKLP runs on EGIFA. The final BNGIF source bank follows the strongest q-GPLVM-refined g-QPACK lineage.

7.2.2 Results

All results below use the best-performing configuration: g-QPACK source kernels with $R = 192$ and $J = 16$, combined with the `hard-F1` AR prior at $\lambda = 1$.

Figure 35 tells the story. On EGIFA/speech, BNGIF achieves the strongest scores among all tested methods. On EGIFA/vowel, it matches QCP statistically while slightly exceeding SLP. The asymmetry across collections is the most informative signal in the plot. On sustained vowels the glottal cycle repeats almost identically, and classical methods designed to exploit that quasi-periodicity – QCP's AME weighting in particular – leave little room to improve. In continuous speech the excitation changes shape within each frame as the speaker moves between phonemes, and that is where a kernel bank covering varied DGF archetypes pays off against a periodic assumption.

Table 14 and Table 15 make the ranking precise with paired Wilcoxon tests.[22] On speech, BNGIF is decisively ahead of every classical baseline ($p < 0.001$). On vowels, QCP has the highest median but the difference from BNGIF is not significant ($p = 0.635$); BNGIF in turn edges out SLP ($p = 0.026$). QCP's vowel advantage is consistent with its design: the AME weighting in (181) was built for exactly this quasi-stationary regime (Airaksinen et al., 2014).

7.2.3 Stratification by F_0

Figure 36 breaks the scores down by nominal fundamental frequency. The speech advantage is broad rather than concentrated in one lucky region of the corpus. The remaining losses, however, are not random: they concentrate in the high-F_0 bins, especially for vowels, which is precisely where the first formant frequency F_1 approaches F_0 and the source-filter separation becomes hardest. This $F_1 \approx F_0$ regime is a long-standing difficulty in GIF (Chien et al., 2017) that affects all methods; Figure 38 examines it directly.

[22] The one-sided p-value is the probability of seeing the observed score difference under the null that both methods perform equally; small p means the advantage is unlikely to be due to sampling alone.

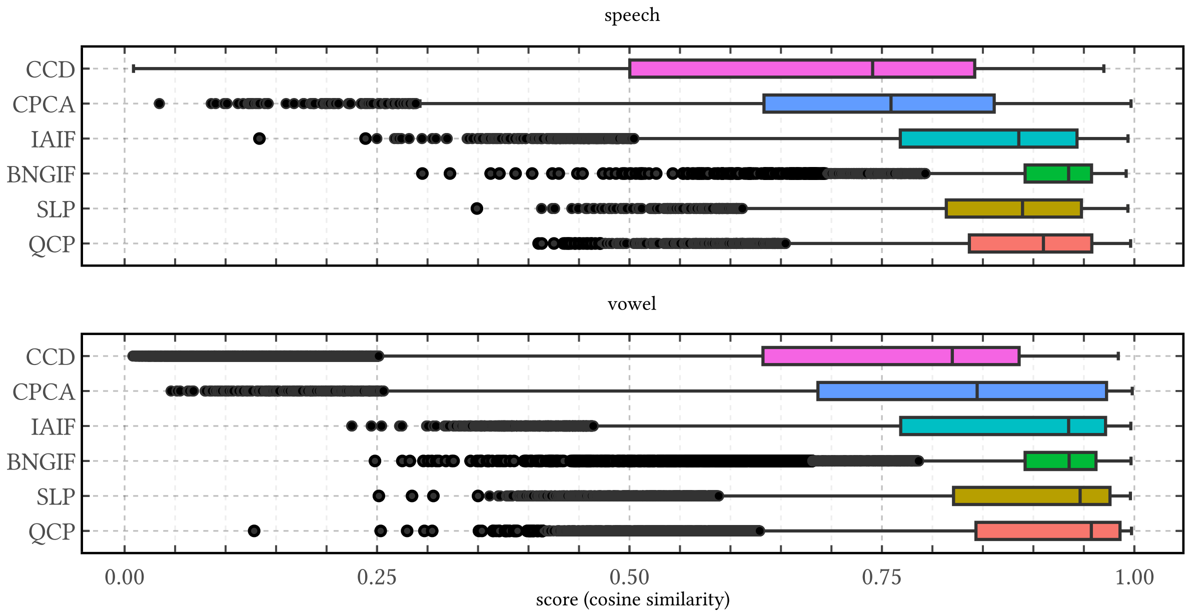


Figure 35 | **EGIFA benchmark scores by method and collection.** Scores are centered cosine similarities after affine+lag alignment (Appendix E). The classical baselines are CPCA, CCD, IAIF, SLP, and QCP; the Bayesian result is BNGIF. On speech, BNGIF leads; on vowels, QCP is numerically strongest but not significantly ahead of BNGIF. Note that BNGIF has smaller spread than other methods and a more diffuse tail.

7.2.4 What the estimates look like

Figure 37 walks through one speech frame from the same benchmark run. The structured posterior mean $\boldsymbol{\mu}_s$ tracks the aligned ground truth $u'(t)$ through several glottal cycles, including the sharp closure transients that carry most of the diagnostic information. The AR envelope below recovers clear formant peaks, and the kernel weights $\mathbb{E}[\theta_i]$ concentrate on a single surrogate component – the model needed only one archetype to explain this frame. When the decomposition works, source waveform, spectral envelope, and kernel selection all tell a coherent story.

Speech method	Median score	One-sided p
BNGIF	**0.935**	–
QCP	0.910	< 0.001
SLP	0.889	< 0.001
IAIF	0.886	< 0.001
CPCA	0.759	< 0.001
CCD	0.741	< 0.001

Table 14 | **Speech ranking and paired Wilcoxon tests against BNGIF.** P-values are one-sided paired Wilcoxon tests. On EGIFA/speech, BNGIF is decisively ahead of all classical baselines.

Vowel method	Median score	One-sided p
QCP	**0.957**	0.635
SLP	0.946	0.026
BNGIF	0.935	–
IAIF	0.935	< 0.001
CPCA	0.844	< 0.001
CCD	0.820	< 0.001

Table 15 | **Vowel ranking and paired Wilcoxon tests against BNGIF.** QCP is numerically strongest but not significantly better than BNGIF; BNGIF slightly exceeds SLP and remains well ahead of IAIF, CPCA, and CCD.

7.2.5 Failure modes

Two failure modes appear at the low-score tail, both visible in Figure 38.

The first is $\boldsymbol{F_1 \approx F_0}$ **congruence** (left panel). When the first formant frequency is close to F_0, the AR filter begins sculpting the broadband noise component of the excitation into a quasi-harmonic structure instead of modeling the true DGF. The bottom-left spectrum shows what happens: the estimated AR envelope peaks right at F_0, absorbing the fundamental into the filter and leaving the excitation estimate nearly flat. The DGF estimate shrinks below the

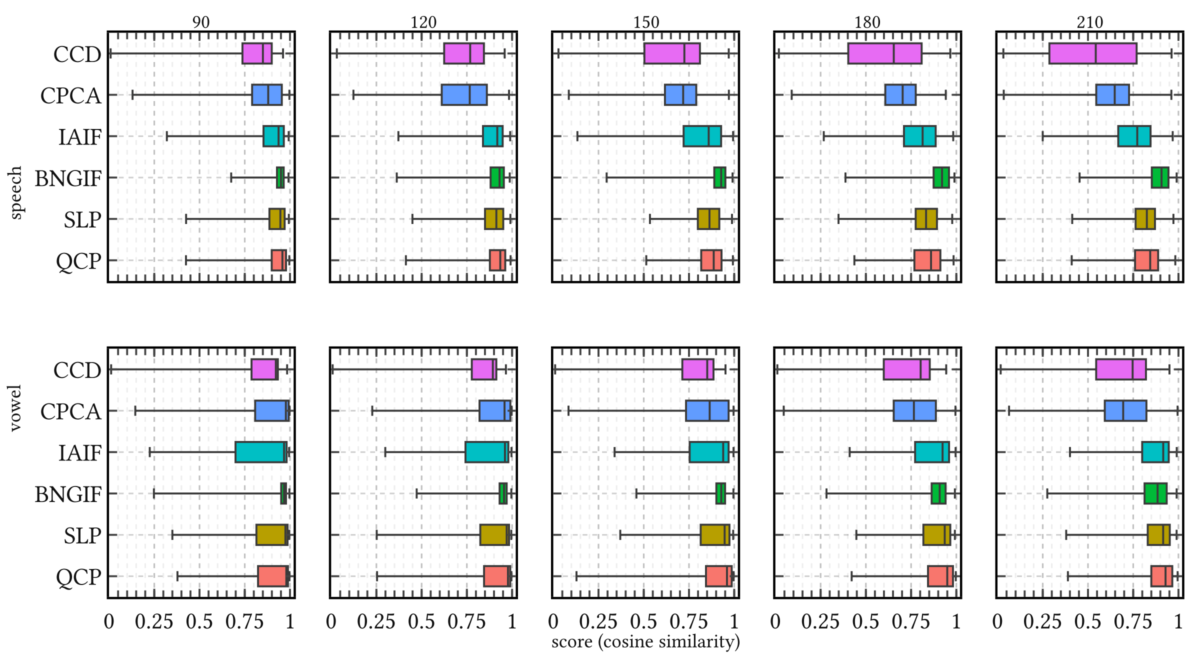


Figure 36 | **EGIFA scores stratified by nominal $\boldsymbol{F_0}$ (columns: 90, 120, 150, 180, 210 Hz) and collection (rows: speech, vowel).** BNGIF holds the strongest speech scores across all bins. At high F_0 (180, 210 Hz) all methods spread out, especially on vowels, where $F_1 \approx F_0$ interference is most acute.

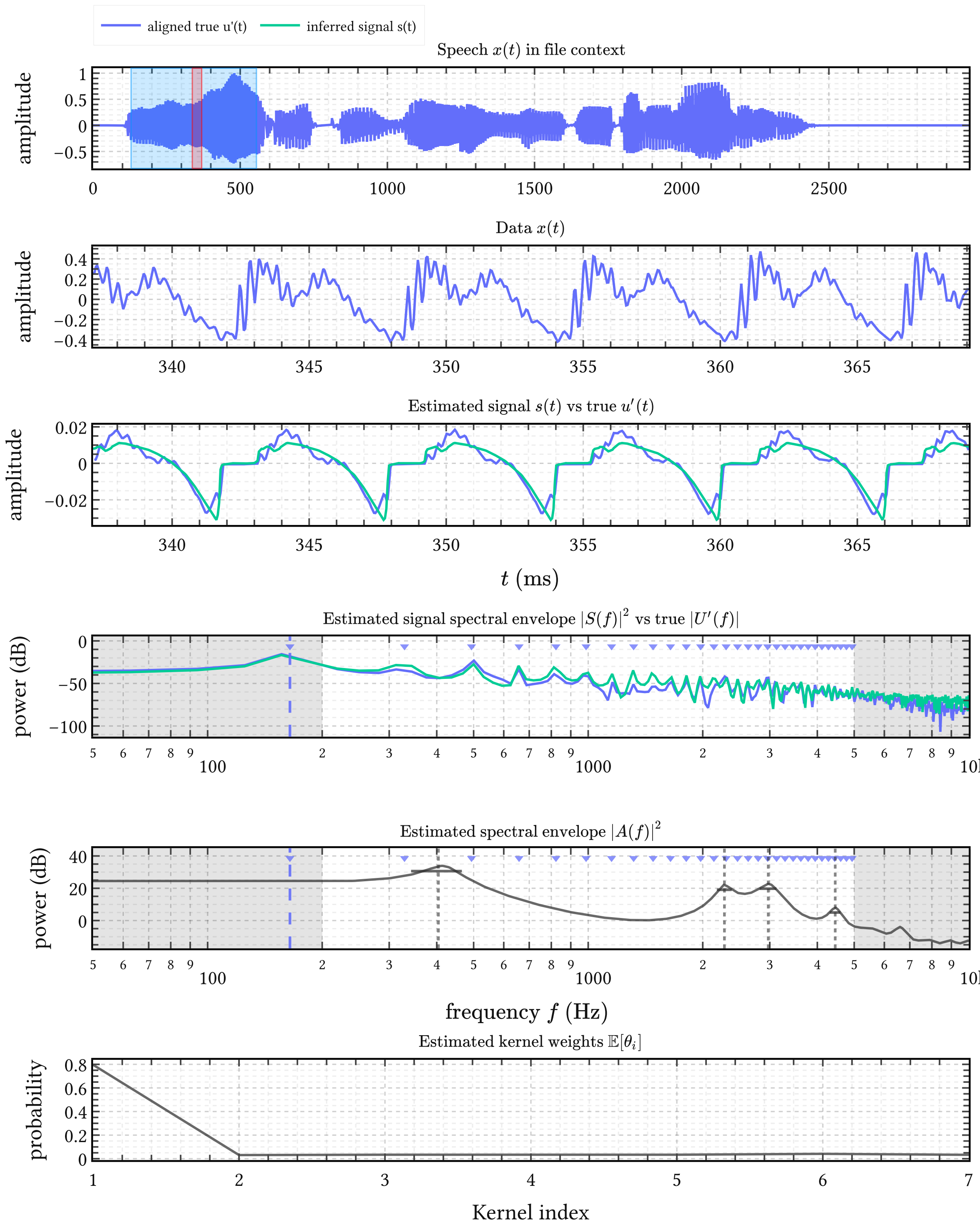


Figure 37 | **Representative BNGIF output on one speech frame.** Top: full-file speech waveform $x(t)$ with the analysis window highlighted. Second panel: the data $\mathbf{x}$ at analysis resolution. Third panel: the structured posterior mean $\boldsymbol{\mu}_s$ (green) overlaid on the aligned ground truth $u'(t)$ (blue). Fourth and fifth panels: estimated source spectrum $|S(f)|^2$ and AR envelope $|H(f)|^2$ with detected formant peaks. Bottom: posterior kernel weights $\mathbb{E}[\theta_i]$ – component $i = 1$ dominates.

noise floor, though it often still traces the gross pulse shape. This is not a failure of inference but a fundamental ambiguity: if $F_1 \approx F_0$, the filter can "explain" the source harmonic, and the data alone cannot distinguish the two decompositions.

The second is a **genuine alternative solution** (right panel). At very low F_1, BNGIF occasionally finds a structurally different posterior mode: a plausible low-noise explanation that is internally coherent but does not match the ground truth. This is a consequence of the inherent nonidentifiability of GIF (Appendix E) rather than an inference failure: the model has found a different valid decomposition of the same signal.

Both failure modes concentrate in the same region of the corpus: high F_0, low F_1. That is also where the AR prior begins to matter. Figure 39 shows the effect of softly biasing the AR prior toward expected formant locations via the spectral shaping priors from Chapter 3. The nudge moves scores in the right direction, but the effect stays small: median improvements on the order of 10^{-3}. The mechanism is promising; the current effect size is not yet decisive.

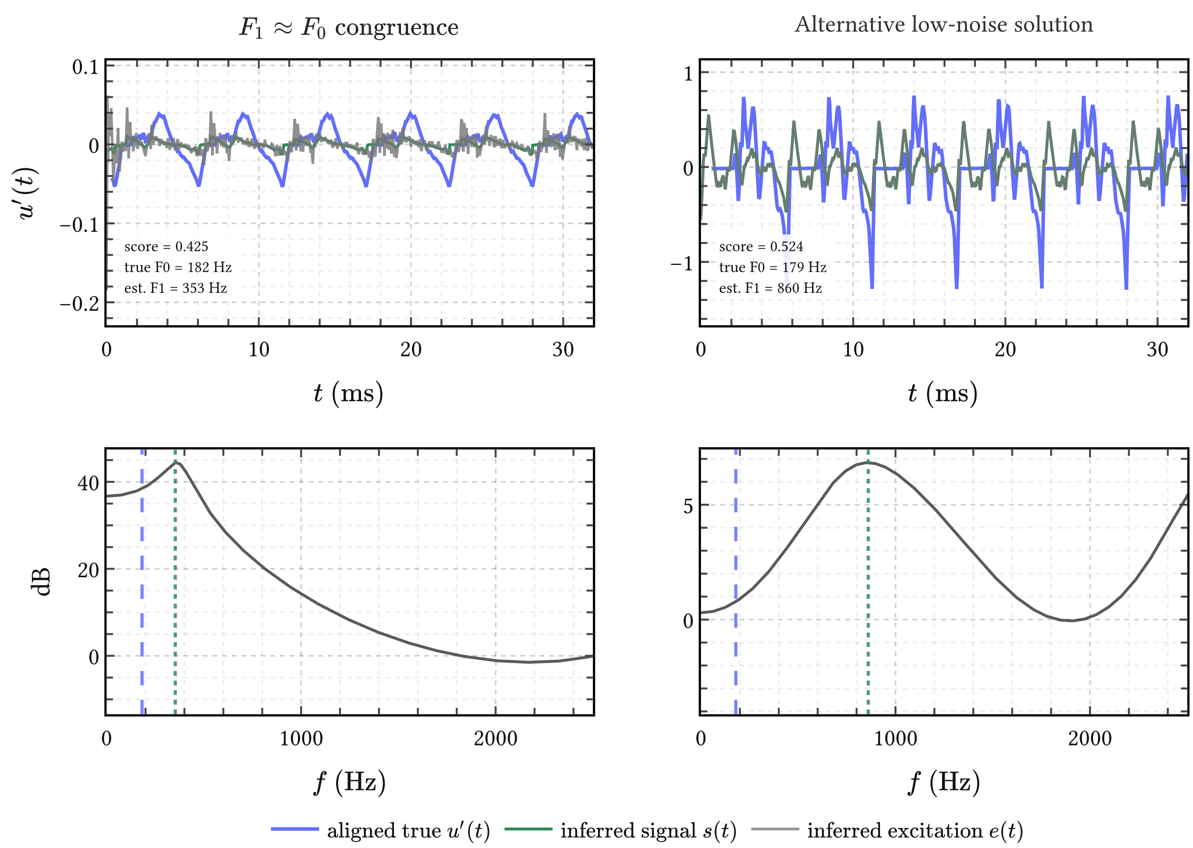


Figure 38 | **Two failure modes from the low-score tail.** Top: aligned ground truth $u'(t)$ (blue), structured posterior mean $\boldsymbol{\mu}_s$ (green), and full LP residual ε^* (grey). Bottom: estimated AR envelope $|\hat{H}(f)|^2$ with true F_0 (dashed blue) and estimated $\hat{F}_1$ (dotted green). Left: $F_1 \approx F_0$ congruence – the filter absorbs the fundamental, leaving $\boldsymbol{\mu}_s$ nearly flat. Right: the model finds an internally coherent alternative decomposition that does not match the ground truth.

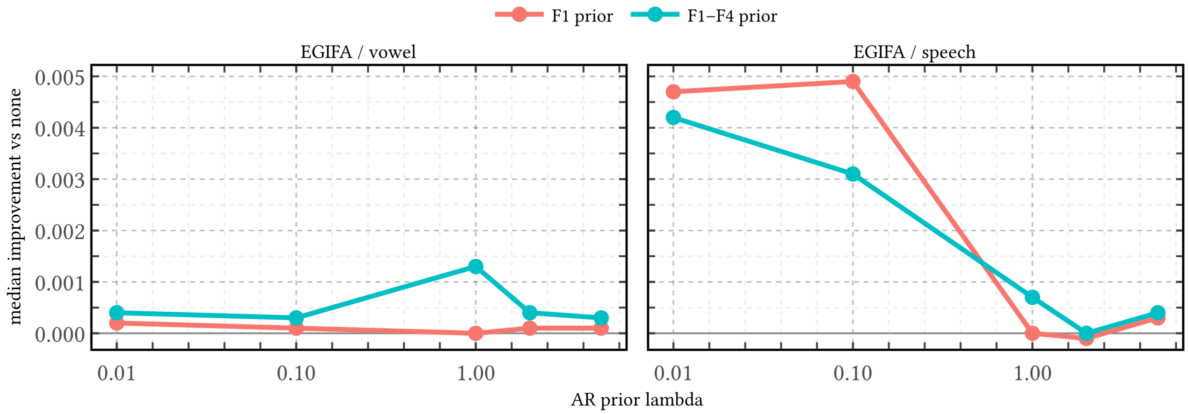


Figure 39 | **Effect of soft AR priors on EGIFA.** Median score improvement over the isotropic baseline for the F_1 and F_1–F_4 strategies at varying λ. On speech (right), both strategies peak near $\lambda \approx 0.1$ and fall back at stronger regularization. On vowels (left), neither moves the score meaningfully.

The next step for the AR prior should probably not be a stronger global spectral template but a *local* prior conditioned on the current frame, for instance an estimate of F_1 from a fast pre-pass that steers the prior away from F_0. The present experiments point to the mechanism; what is still missing is enough local adaptivity to resolve the $F_1 \approx F_0$ cases without replacing one failure mode by another.

7.2.6 Runtime and numerical stability

Memory scales as $\mathcal{O}(BMR + IR^2)$ and compute as $\mathcal{O}(MR^2 + IR^2)$, where R is the inducing rank and B the batch size; both are linear in frame length M for fixed R. At 20 kHz (native EGIFA sampling rate) and float32 precision, BNGIF runs at $5\times$ realtime on a single RTX 2080. The full EGIFA corpus processes in approximately 3 minutes, excluding JAX tracing and I/O. Float64 is roughly an order of magnitude slower and is therefore mainly useful as a diagnostic reference rather than the operative benchmark setting.

7.3 Evaluation on VTRFormants

7.3.1 Experimental setup

A grey-box model of speech should yield good vocal tract estimates as a byproduct of source estimation, not only the other way around. In BNGIF, the posterior over the AR coefficients $\mathbf{a}$ directly induces posterior formant estimates, so formant tracking falls out of GIF inference

Precision	Throughput	Full EGIFA pass
x32	$5\times$ realtime	3 min
x64	$0.5\times$ realtime	30 min

Table 16 | **Runtime on EGIFA.** Float32 is the operative precision for the benchmark; float64 is retained only as a slower diagnostic reference.

System	P	λ	AR strategy	F1	F2	F3	Overall
KARMA	—	—	—	114 Hz	226 Hz	320 Hz	220 Hz
Praat	—	—	—	185 Hz	254 Hz	303 Hz	247 Hz
WaveSurfer	—	—	—	170 Hz	276 Hz	383 Hz	276 Hz
BNGIF (best)	16	1	F1	74 Hz	137 Hz	286 Hz	180 Hz
BNGIF (worst)	16	0.1	hard-all	101 Hz	294 Hz	329 Hz	261 Hz
BNGIF (best)	**20**	**1**	**hard-F1**	**82 Hz**	**129 Hz**	**223 Hz**	**153 Hz**
BNGIF (worst)	20	0.1	hard-all	94 Hz	265 Hz	309 Hz	241 Hz

Table 17 | **Formant RMSE on VTRFormants (Hz).** Classical baselines from (Mehta et al., 2012, Table IV). BNGIF rows show the best and worst configurations at each AR order P. Better source modeling lowers formant RMSE substantially, but overly aggressive global AR constraints (`hard-all` at low λ) can hurt.

with no dedicated post-processing. If the excitation is modeled better, one should expect the tract estimate to improve too.

The VTRFormants protocol used here is defined in Dataset 3: the standard TIMIT-derived benchmark, restricted in this thesis to vowel events, evaluated on the native 10 ms reference grid, and interpreted mainly comparatively because visible labeling errors remain. Here we simply reuse that contract with QuickGCI for event detection and $P \in \{16, 20\}$ LP poles. The classical baselines are KARMA, WaveSurfer, and Praat as reported in (Mehta et al., 2012, Table IV). The BNGIF rows all use the same g-QPACK source lineage as on EGIFA and differ only in P, AR strategy, and λ; the strategy names and formant targets are the ones listed in Table 12.

7.3.2 Results

At $P = 20$, the best BNGIF configuration reaches 82/129/223/153 Hz on $F_1/F_2/F_3$/overall RMSE, against KARMA's 114/226/320/220 Hz – an overall reduction of about 30%. The largest gain is on F_3, which is the formant most easily contaminated by unmodeled excitation structure in classical LP: when the source prior absorbs the harmonic energy that LP would otherwise need poles for, those poles become available to track the actual resonances.

The table also contains a quieter lesson. Once $P = 20$ is in place, the AR prior changes aggregate RMSE only mildly in the good configurations, but aggressive global constraints (`hard-all` at low λ) degrade performance sharply. The source side does most of the work; the AR prior helps as a stabilizer, not a driver.

7.4 Summary

BNGIF assembles the thesis: learned source priors, informative AR priors, and fast variational inference in one model. On EGIFA/speech it achieves the strongest scores among all tested methods; on vowels it is competitive with QCP but does not beat it. On VTRFormants it reduces overall formant RMSE from 220 Hz (KARMA) to 153 Hz, with the largest gains on F_3 – the formant most contaminated by unmodeled excitation structure in classical LP. Both results come from the same posterior: a better source model frees AR poles to track the actual resonances.

The limitations are equally clear. The $F_1 \approx F_0$ regime remains a genuine ambiguity that no prior on either side alone can resolve; the AR prior experiments show the right mechanism but the effect size is small, and a local, frame-adaptive version is needed. On sustained vowels the kernel bank does not outperform the targeted closed-phase heuristics that QCP was designed around, which suggests that the source prior still lacks the steady-state specificity those methods exploit. And the mean-field factorization in CAVI means the filter $\mathbf{a}^*$ carries no posterior uncertainty at all, which limits the model's ability to express doubt about the decomposition in the hard cases where doubt would be most informative.

What BNGIF does provide is modularity. The inference engine is fixed; the source bank and AR prior can be replaced independently. The failure modes are now visible and concentrated rather than diffuse, which is the precondition for the next round of work.

8 Conclusion

The overview presented this thesis as an attempt to place GIF in a different scientific posture. We reformulated GIF as a Bayesian grey-box inference problem in which priors, uncertainty, and model criticism are explicit; while source and filter models are data-driven, but at the same time physically interpretable. It appears then that a proper conclusion should assess how far that program was actually carried through, and we will undertake that here. We end with future work and a final reflection.

8.1 Summary of contributions

Each chapter tried to model an aspect of speech production a little more closely, developing general-purpose tools along the way and solving practical problems as they came up. Table 18 gives the map, and we now list the main contributions.

IKLP The groundbreaking work of Yoshii & Goto (2013) replaced the white-noise LP excitation prior with a mixture of GP kernels within a tractable variational inference scheme. Our contribution is a modern Woodbury–Cholesky reformulation of their algorithm in JAX which enables 5× realtime throughput at float32 on commodity GPU hardware. Earlier Bayesian approaches to GIF (Auvinen et al., 2014; Rao & Ghosh, 2018; Wang et al., 2016) relied on MCMC, produced point estimates only, and none propagated posterior uncertainty to downstream quantities. The computational cost was the central obstacle: Auvinen et al. (2014) report 2.5 hours per 25 ms frame. BNGIF replaces MCMC with CAVI, achieving a speedup of roughly 10^6, and is as far as we know the only Bayesian GIF method with a data-driven GP prior on the source waveform. This matters because practitioners in acoustic phonetics are accustomed to near-instantaneous FFT-based tools; if a Bayesian alternative is to be taken seriously, it has to meet them on that ground, and we believe this objective is now met.

AR priors replaced the isotropic default on the LP coefficients with a more informative Gaussian prior. The soft and hard spectral shaping priors are, as far as we know, a novel information-based framework which can encode rich prior information about the spectral properties of the AR process as soft or hard constraints, respectively. These priors have broad applicability beyond speech alone: any application of an AR process could benefit from them.

Glottal flow kernels Starting from the classical parametric glottal flow models, we showed that the Bayesian nonparametric limit leads to the arc cosine kernel of Cho & Saul (2009), unifying the parametric and nonparametric traditions in GIF within a single GP family. We then developed a host of more detailed kernels modeling periodicity; quasiperiodicity; and the closed phase of the glottal cycle. Our contribution here was to show that custom-tailored kernels can really pay off; and how their hyperparameters can be calibrated to a large corpus of example instances.

Surrogate models Gaussian processes are popular as surrogate models in engineering, but typically for a single dataset at a time. Here we had thousands of variable-length DGF waveforms and needed to learn a shared basis from all of them at once. PRISM does this: it learns a kernel-regularized basis from arbitrarily many irregularly sampled exemplar functions, and projects each to a Gaussian embedding in a fixed-dimensional space $\mathcal{D}_R$. We then used q-GPLVM to compress the collection into a compact sparse GP mixture $\mathcal{M}_R$. Because these are just sparse GPs, they serve simultaneously as generative surrogates *and* as priors in IKLP. Both methods are completely general and work unchanged for any input dimension. PRISM scales as $\mathcal{O}(BR^2 + R^3)$, where B is minibatch size and R is chosen by the user to accomodate their available compute. Anyone who has a large dataset of expensive examplars lying around might find it useful.

Pitch-track prior A single GP prior $\mathcal{P}$ over pitch trajectories, learned from APLAWD's hand-corrected GCI events, serves three downstream roles at once: voiced-group segmentation, time warping for generation, and analytical jitter prediction. We contribute the tool that made this possible: t-PRISM, a robust variant of PRISM that replaces the Gaussian likelihood with a Student-t, allowing it to learn from data contaminated by outliers; in our case, GCI annotations.

BNGIF is our proposed grey-box Bayesian model of voiced speech, assembling all of the above into a single framework for both GIF and formant estimation. It achieved the strongest speech scores on EGIFA among the state of the art ($p < 0.001$ against QCP), remained competitive on vowels, and reduced formant RMSE on VTRFormants by 30% over KARMA — with the largest gains on F_3, the formant most affected by excitation artifacts in classical LP.

8.2 What worked

Variational inference is a feasible approach to GIF and can deliver Bayesian regularization at scale and with a low-enough cost for daily use. It also appears the VI approximation itself is benign for GIF and formant estimation, and MCMC is not needed.

Learned GPs can serve as both surrogates and priors It is possible to learn not just the hyperparameters of a GP, but to refine the entire kernel structure from large exemplar datasets and then use those same sparse GPs directly as priors in IKLP. That this works – that the learned surrogate is informative enough to improve both GIF and formant estimation – was not at all obvious at the outset.

Contribution	Chapters	Algorithms
IKLP	Chapter 2	Algorithm 1
AR priors	Chapter 3	
Glottal flow kernels	Chapter 4, Chapter 5	
Surrogate models	Chapter 6	Algorithm 3, Algorithm 5
Pitch-track prior	Dataset 2	Algorithm 4
BNGIF	Chapter 7	

Table 18 | **Overview of contributions** and their location in the thesis.

The kernel bank as hypothesis bank Speech is highly modal: the glottal cycle can be voiced or unvoiced, pressed, breathy, pathological, etc. Inference over such signals needs to make either/or decisions in the space of hypotheses, and IKLP's kernel mixture can really provide that mechanism via its kernel bank. For example, BNGIF can handle formant estimation in the case of unvoiced speech because it will select the "unvoiced" kernel $\nu_e I$. The mechanism is also open-ended: one can extend the kernel bank without touching the inference engine, so that any failure modes can be addressed simply by learning new surrogate models on failure cases and adding them to the kernel bank.

Source priors matter more than one might expect Perhaps the biggest lesson of the thesis is yet another example of the fact that: *priors matter*. Replacing white or weakly structured excitation by learned surrogate models is what most clearly moved the model into the right operating regime. The AR prior does help, visibly in the difficult $F_1 \approx F_0$ cases, but it did not matter as much as was expected.

Inference in float32 GP libraries typically assume float64 arithmetic due to "brittle" Cholesky decompositions. In our experiments we found that, with some care, float32 works equally well and yields a huge speedup on GPU. In float32 there are also many advantages to SVD over Cholesky decompositions, and future work should switch to these.

8.3 Limitations

LTI within frame Like essentially all GIF methods, BNGIF assumes a time-invariant vocal tract filter within each analysis frame. That approximation is imperfect during the open phase, where source-filter coupling slightly shifts formant locations and bandwidths (Walker & Murphy, 2005). The gated source priors partly mitigate the resulting ripple but do not remove the approximation.

$F_1 \approx F_0$ **congruence** This remains the dominant failure mode. When the first formant approaches the pitch frequency, the AR filter begins to sculpt the excitation itself and the separation becomes genuinely ambiguous. This is a classical difficulty of LP-based GIF, and we have not resolved it. However, we might have the keys in our hand with the AR prior: we can use it to bias the filter spectrum away from F_0. It currently seems this is nontrivial to do, as our obvious attempts failed. This needs more looking into.

Synthetic and event-annotated supervision The learned source priors are calibrated on synthesis corpora; the pitch-track prior depends on corpora with accurate GCI annotations. That gives a clean experimental setting, but it limits how strongly one can claim independence from synthetic ground truth.

Approximate inference BNGIF is Bayesian but not exact. All runtime gains come with the usual price of variational approximation. While it seems we pay a small price, it remains worth asking which remaining failures come from the model and which come from the approximation. This can be done with more expensive MCMC-based inference such as nested sampling (Skilling, 2004).

Evaluation scope The current evaluation covers waveform and formant benchmarks. Derived phonatory measures such as OQ, H1-H2, jitter, shimmer (Drugman et al., 2019a) and real clinical tasks remain largely outside the present validation. For those nonlinear quantities, it would be interesting to see what posterior calibration brings to table, especially in medical and forensic applications.

8.4 Scientific and clinical significance

A full probabilistic model of speech matters for three communities at once: fundamental research in acoustic phonetics, clinical voice assessment, and forensic speaker comparison (Section 1.1). The features all three care about – F_0, jitter, shimmer, OQ, HNR, formant trajectories – are exactly what GIF recovers from a single short recording, and formant estimation alone is the daily bread and butter of thousands of phoneticians worldwide. Yet the standard tools remain decades-old LP implementations that return point estimates with no uncertainty and no explicit model of the glottal source. Their regularization is implicit in weighting curves, window choices, and pipeline conventions, which makes it difficult to analyze or improve in any principled way (Jaynes, 2003; MacKay, 2005).

BNGIF offers something different, at least in principle. Its output is a posterior over source and filter jointly, from which derived measurements inherit uncertainty automatically. That enables errors-in-variables analyses, longitudinal tracking with calibrated confidence, and cleaner separation of measurement noise from genuine vocal change. But importantly, it is competitive with the strongest classical baselines on both waveform and formant benchmarks, and does so with a grey-box model whose parameters remain physically interpretable. The combination of speed, accuracy, uncertainty, and interpretability in a single framework is, as far as we know, new.

We have not validated BNGIF as a clinical biomarker system, and we are careful not to overstate what the current results imply. But there are hopeful signs, and the pieces are there: a fast, interpretable model that produces calibrated posteriors and scores well against the state of the art. Whether that translates to meaningful advances in clinical, forensic, and fundamental practice is an empirical question that now, at least, seems worth asking.

8.5 Future work

Local AR priors The AR-prior experiments in Chapter 7 suggest that the right direction is not a stronger global spectral template but a local one: priors conditioned on fast preliminary formant estimates may prove to mitigate $F_0 \approx F_1$ congruence and improve performance on sustained synthetic vowels compared to QCP.

Reducing event dependence BNGIF currently requires GCI and GOI annotations; IAIF and CCD share that dependence, while SLP and QCP do not. In spite of that, the latter pairs still proceeds to score formidably on EGIFA. Can BNGIF do the same? The hypothesis-grid-née-kernel-bank mechanism of IKLP for inferring an unknown pitch extends naturally to uncertain OQ: a discrete set of plausible values can be evaluated in parallel at $\mathcal{O}(I)$ cost via shared-feature kernel mixtures, and the same applies to uncertain GCIs. A more radical step would be to skip the careful registration of exemplar DGF waveforms altogether and let alignment be learned jointly (Kazlauskaite et al., 2018), relying on IKLP's sparse gamma prior to pick out the right hypothesis from a larger kernel bank. This increases runtime by a factor I but would be a natural generalization of IKLP. It would also automatically *transfer uncertainty in GCI and GOI directly to the output DGF*, a most desireable effect.

End-to-end optimization The present pipeline is staged: PRISM and q-GPLVM learning and IKLP inference are done in separate steps. The optimizations involved with each stage were stable and benign, but the overall procedure required manual orchestration. Since our IKLP extension is written in JAX, differentiation through the entire pipeline is in principle possible – one could optimize EGIFA's NRMSE loss directly rather than through proxy losses. Joint or partially joint training, online learning, and frame-to-frame propagation of posterior information are all natural extensions; but these will require more compute than we had handy.

Richer model classes Assuming the inference engine stays fixed, failure modes can be addressed systematically by refining the kernel bank. On the source side, different physics-based simulators such as those of Zhang et al. (2020) and Schoder et al. (2024) can serve as alternative training corpora. On the filter side, ARMA or pole-zero variants, mixture AR priors, and the coupled AR-GP model of Yoshii & Goto (2012) are natural directions.

Clinical and forensic validation BNGIF scored well on EGIFA and VTRFormants, and it runs fast enough for daily use. The natural next question is whether that improved accuracy and the calibrated uncertainty it carries can actually translate to meaningful gains in downstream clinical tasks – early detection of neurodegenerative diseases or heart failure, longitudinal monitoring of voice quality, forensic speaker comparison. Taking the model to real clinical and forensic data is the obvious next step.

8.6 Final reflection

Jaynes (1999, p. 6) wrote:

> But every Bayesian problem is open-ended; no matter how much analysis you have completed, this only suggests still other kinds of prior information that you might have had, and therefore still more interesting calculations that need to be done, to get still deeper insight into the problem. A person who tries to present a Bayesian solution, being obliged to produce a finite sized manuscript in a finite time, must forego mentioning many other interesting things about the problem that he became aware of while writing it. […] In effect, anyone writing about a Bayesian solution must draw a kind of artificial horizon about the problem, beyond which he dare not tread however great the temptation.

This thesis did not solve GIF, but I believe it provides a new handle on the problem. The remaining failures "have names," and for each of them the framework offers a clear next move: learn or design a better prior, plug it into IKLP, and let inference do the rest. What lies beyond that horizon I do not know, but the Bayesian machinery to get there is now in place. I dearly hope that this work can contribute to a more reliable and holistic use of speech analysis in clinical practice, forensic work, and the scientific study of vocalization as a whole.

Datasets

Dataset 1. EGIFA

EGIFA is the synthetic benchmark introduced by (Chien et al., 2017). It pairs radiated speech waveforms $s(t)$ with matched glottal flow waveforms $u(t)$ generated by a physics-based articulatory synthesizer. In this thesis it is used in two distinct directions. In Chapter 5, the known $u(t)$ is taken seriously as training data and turned into a registered corpus for source-prior calibration. In Chapter 7, the same corpus is turned around into the actual inverse problem: only $s(t)$ is observed, and the task is to recover the registered source signal. EGIFA is therefore not just another dataset chapter. It is the controlled hinge between source learning and source inference in the thesis.

1.1. Corpus

VocalTractLab (Birkholz, 2013) is an open-source articulatory speech synthesizer and the current state of the art in physics-based speech production (Gao et al., 2024). Its glottis model is a self-oscillating, two-mass triangular system coupled to a transmission-line vocal tract; it simulates time-domain finite-difference acoustic wave propagation including nonlinear source-filter interaction. Because $u(t)$ is a direct output of the simulator alongside $s(t)$, every utterance comes with a ground-truth glottal flow.

The EGIFA corpus contains two data sets synthesized with VocalTractLab 2.1.

The vowel subset has a fully crossed factorial structure and is therefore useful for isolating how a method responds to pitch, phonation pressure, and voice quality. The speech subset matters more as a GIF test: it introduces continuous phonatory dynamics and coarticulation, so any prior that works there has to survive outside the tidy factorial setting. Close rounded vowels /o/ and /u/ are empirically the most challenging case for inverse filtering, as noted

Subset	**Files**	**Factors**	**Duration**	**Native f_s**
vowel	750	6 vowels {/i/, /e/, /ɛ/, /ä/, /o/, /u/} × 5 voice qualities × 5 F_0 levels × 5 subglottal pressures	0.6 s each	44.1 kHz
speech	125	5 median F_0 × 5 median pressures × 5 voice qualities	5–10 s each	44.1 kHz

Table 19 | **EGIFA corpus structure.** Voice qualities are pressed, slightly pressed, modal, slightly breathy, and breathy. F_0 levels are 90, 120, 150, 180, and 210 Hz; subglottal pressures are 500, 708, 1000, 1414, and 2000 Pa. The speech subset is derived from a prototype score for the German sentence "Lea und Doreen mögen Bananen" with controlled continuous variation of the glottal parameters. All files are resampled to 20 kHz for processing.

in (Chien et al., 2017), because their narrow first formants bring F_1 close to F_0, making the source-filter boundary hard to locate. Throughout the remainder of the thesis, these two collections are referred to simply as EGIFA/vowel and EGIFA/speech.

1.2. Registration

For source learning, EGIFA is used in the forward direction. The raw glottal-flow recordings are turned into a corpus of registered DGF exemplars

$$\mathcal{D} = \{(\boldsymbol{\tau}_n, \mathbf{u}'_n)\}_{n=1}^{N}, \tag{177}$$

all expressed in shared normalized cycle time τ. LBGID (Algorithm 2) supplies the events, t-PRISM (Algorithm 4) removes unstable stretches under the learned pitch-track prior $\mathcal{P}$, and the retained groups are then warped, smoothed, differentiated, and standardized. This is the source-learning object later used in Chapter 5.

GCI and GOI detection

LBGID (level-based glottal instant detection, Algorithm 2) extracts (GCI, GOI) pairs directly from the continuous $u(t)$ waveform. It works by identifying low-flow regions via a relative amplitude threshold and then selecting, within each such closed phase, the index pair (i, j) that maximizes $E(i, j) = -u'(i)\, u'(j)$ ((242)), i.e., the pair of maximally opposed changes flanking the closure.

EGIFA provides no ground-truth GCI or GOI annotations. We therefore use LBGID rather than the GCI estimation bundled with EGIFA's own evaluation scripts. The EGIFA `area_gci` path segments cycles from the synthesizer's glottal area signal, extracts GCIs via local minima, and then applies a hard offset of $+13$ samples before feeding the result to a separate GOI detector. This produces clustered triggers with no refractory guard, making the implied cycle lengths noisy. LBGID operates directly on $u(t)$ without any area signal or hard offset, and that cleaner event sequence matters downstream: segmentation, registration, and later source calibration all inherit its errors.

Voiced group segmentation

Raw EGIFA recordings contain transient regions — onset glides, pressure ramps, the beginning and end of each utterance — where the glottal cycle is not stationary and the resulting $u(t)$ waveform is not representative of any particular phonatory mode. Including these transients in the learning corpus would corrupt the prior.

t-PRISM (Algorithm 4) segments the detected GCI sequence into voiced groups by conditioning the learned pitch-track prior $\mathcal{P}$ from Dataset 2 on the implied period sequence $T_{\{0,k\}} = \left(n_{\{\mathrm{GCI},k+1\}} - n_{\{\mathrm{GCI},k\}}\right)/f_s$. Equivalently, it conditions the latent log-period function $g(.) \mid \mathcal{P}$ from (187) on the observed log-periods under (188), then uses the resulting weights $w_k = \mathbb{E}[\lambda_k]$ to reject outliers. Maximal contiguous runs with $w_k > 1$ form the voiced groups used for learning, exactly as in the APLAWD segmentation rule of Dataset 2. This is the first point in the thesis where $\mathcal{P}$ is not merely learned but actually used as an operating object. The result is a segmentation rule that is explicit and probabilistic.

Time warp

Within each retained voiced group the GCIs are mapped to the integers,

$$\tau(t) = \texttt{interp}\big(t; n_{\mathrm{GCI},0} \mapsto 0, n_{\mathrm{GCI},1} \mapsto 1, n_{\mathrm{GCI},2} \mapsto 2, \ldots\big), \tag{178}$$

so that one unit of τ corresponds to one glottal cycle. For gated kernels the GOIs are additionally mapped to half-integers,

$$n_{\mathrm{GOI},k} \mapsto k + 1/2, \tag{179}$$

fixing the open-closed boundary at $\tau \in \mathbb{Z} + 1/2$ uniformly across the corpus. This is a registration convention, not a claim about the physical open quotient.

Smoothing, differentiation, and normalization

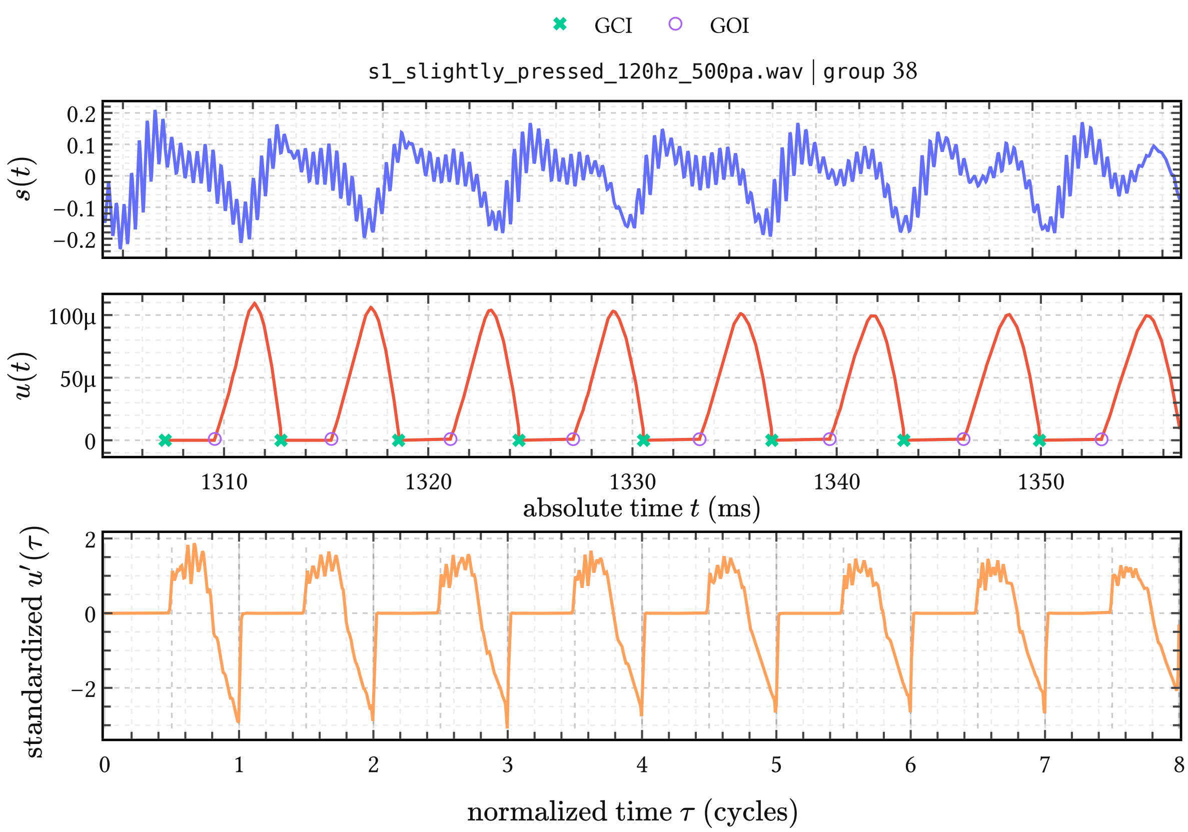


Figure 40 | **Registration of one EGIFA voiced group.** Top: raw speech $s(t)$ in physical time. Middle: raw glottal flow $u(t)$ with LBGID-detected GCI and GOI events. Bottom: the group after differentiation, normalization, and warp to normalized time τ, shown as $u'(\tau)$; GCIs at integers, GOIs at half-integers.

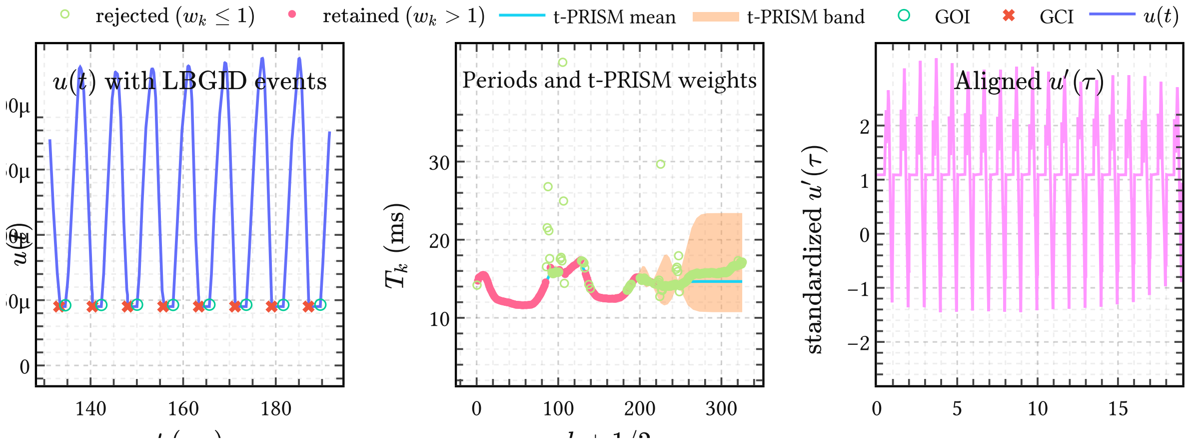


Figure 41 | **The EGIFA registration pipeline on one recording.** (a) Raw glottal flow with LBGID GCI and GOI events. (b) Implied period sequence over the full recording; t-PRISM posterior mean and uncertainty band with retained inliers. (c) One retained voiced group in shared cycle time τ after smoothing, differentiation, GOI-aware registration, and per-group normalization.

Before differentiating, $u(t)$ is smoothed on a physiologically meaningful timescale of 0.1 ms (Appendix D). At 20 kHz this acts mainly as regularization. The derivative $u'(t)$ is then computed by finite differences, registered to τ via (178), and standardized within each group to unit power. The per-group normalization removes inter-utterance amplitude variation that would otherwise dominate the learning signal. What remains is not yet a benchmark score, but the registered source-learning object $\mathcal{D}$ itself.

Implied corpus statistics

The detected GCI and GOI events imply per-cycle pitch and open quotient values:

$$F_{0,k} = f_s / (n_{\mathrm{GCI},k+1} - n_{\mathrm{GCI},k}), \quad \mathrm{OQ}_k = (n_{\mathrm{GOI},k} - n_{\mathrm{GCI},k}) / (n_{\mathrm{GCI},k+1} - n_{\mathrm{GCI},k}) \tag{180}$$

The point of Figure 42 is not descriptive completeness but quality control. The F_0 modes line up with EGIFA's nominal settings and the OQ values remain physiologically plausible, which is enough to justify using the detected events as the backbone of the registration pipeline across the full factorial spread. Chapter Chapter 5 then refines these retained voiced groups further into fixed-width registered frames for calibration.

1.3. The benchmark

The registration pipeline above used EGIFA in the forward direction, starting from known $u(t)$ to build a source-learning corpus. The benchmark flips that logic around. Now only $s(t)$ is observed, and the task is to infer the registered source signal $u'(\tau)$ and compare it against the aligned reference. This is the controlled arena in which BNGIF must face the classical methods on equal terms.

Classic GIF methods

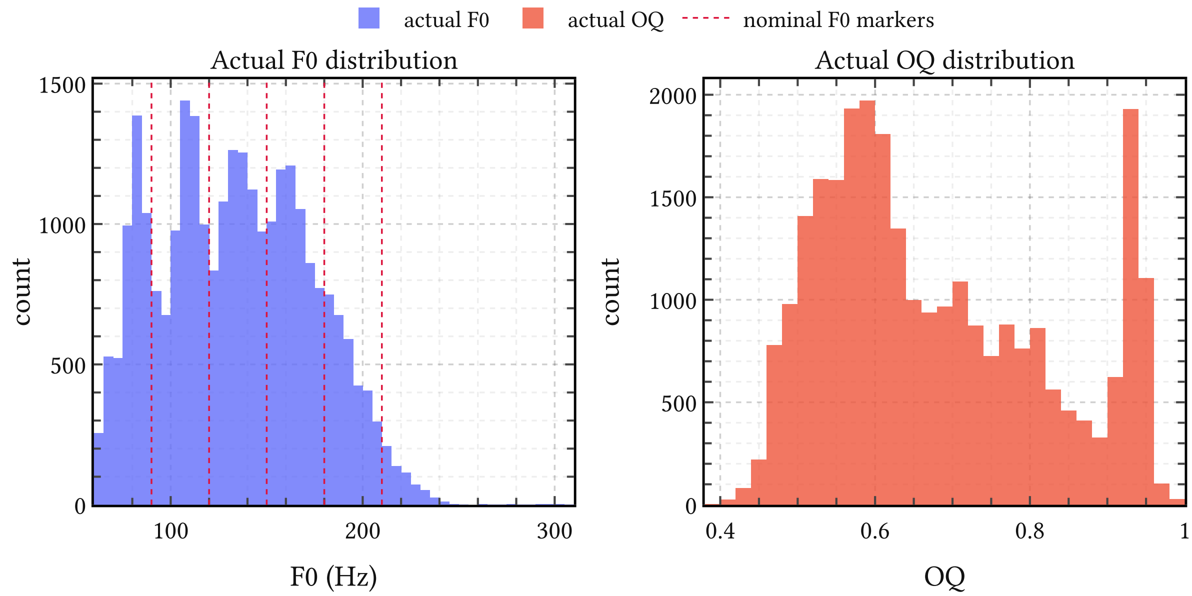


Figure 42 | **Implied F_0 and OQ distributions from LBGID on EGIFA.** Left: frame-level F_0 implied by detected GCIs, with dashed lines at EGIFA's nominal pitch settings. Right: frame-level OQ implied by detected GCI/GOI pairs.

The five methods from (Chien et al., 2017) are the classical baselines used throughout the thesis. All operate frame-by-frame on the speech signal $\mathbf{x} = (x_1, ..., x_M)^\top$ (= discretized $s(t)$, Chapter 2). Four of the five estimate the vocal tract AR coefficients $\mathbf{a}$ by minimizing a weighted LP residual:

$$\hat{\mathbf{a}} = \underset{\mathbf{a}}{\operatorname{argmin}}\, \mathbf{x}^\top \mathbf{\Psi}(\mathbf{a})^\top \mathbf{W} \mathbf{\Psi}(\mathbf{a}) \mathbf{x}, \tag{181}$$

where $\mathbf{\Psi}(\mathbf{a})$ is the AR operator from (7) and $\mathbf{W} = \operatorname{diag}(w_1, ..., w_M) \geq 0$ encodes how much each sample influences the estimate; the inferred DGF is then the LP residual $\varepsilon = \mathbf{\Psi}(\hat{\mathbf{a}})\mathbf{x}$. This makes explicit that most classical methods differ less in overall architecture than in which parts of the waveform they choose to trust. The methods differ only in $\mathbf{W}$:

A practically important detail is that the `wca2` implementation in (Chien et al., 2017) uses the AME weighting scheme introduced for formant estimation of high-pitched vowels (Alku et al., 2013). AME is structurally identical to QCP (quasi-closed phase analysis), which is the current state-of-the-art classical GIF method (Freixes et al., 2023). So throughout this thesis we refer to `wca2` simply as QCP, not as the literal piecewise-linear WLP of (Airaksinen et al., 2014). Likewise, the `wca1` implementation is the smooth Gaussian weighting variant, and we refer to it simply as SLP.

IAIF (Alku, 1992) stands apart in that it requires no GCI input: two LP passes at different orders iteratively separate glottal and tract contributions. CCD (Drugman et al., 2011) stands apart in a different way: it works in the complex cepstral domain, extracting the anticausal

Method	Weight w_m	GCI/GOI required
CPCA	$\mathbb{1}[m \in \text{closed phase}]$ (hard mask)	yes (both)
SLP	$1 - \kappa \sum_l \exp\left(-(m-\gamma_l)^2 / \left(2(\sigma f_s)^2\right)\right)$	yes (GCI)
QCP	AME: quasi-closed phase envelope, piecewise smooth	yes (GCI)
IAIF	$\mathbf{W} = \mathbf{I}$ (two LP passes, iterative)	no
CCD	— (cepstral, not LP-based)	yes (GCI)

Table 20 | **Classic GIF methods in EGIFA, unified under (181).** CPCA uses a hard binary closed-phase mask; SLP uses soft Gaussian attenuation of closed-phase samples; QCP uses the AME (attenuated main excitation) window. In the EGIFA code base these are the methods `cp`, `wca1`, and `wca2`, respectively. IAIF uses uniform weights but applies two LP passes iteratively. CCD is not a weighted LP method: it separates the maximum-phase component of the speech cepstrum and is entirely independent of $\mathbf{W}$.

maximum-phase component of the speech spectrum as the tract estimate and leaving the causal residual as ε.

Evaluation

The original evaluation in (Chien et al., 2017) aligns estimated and reference DGFs per glottal cycle with a fixed 0.65 ms propagation delay and measures the normalized waveform error within each cycle. The arguments of Appendix E show why cycle-level alignment with a fixed delay is inappropriate: different algorithms select different gauge representatives from the source-filter ambiguity, so comparing raw cycle-level samples conflates algorithmic style with algorithmic accuracy.

Throughout this thesis we therefore evaluate all methods — classical and BNGIF — under a single frame-level affine-plus-lag-invariant metric ((345), (347)). This measures shape, not gauge convention, so improvements in the metric transfer directly to downstream tasks. This chapter therefore fixes the comparison contract for the rest of the thesis. Related measures such as NAQ, H1–H2, and HRF remain useful secondary diagnostics in the sense of (Chien et al., 2017), but the primary score throughout this thesis is the gauge-invariant waveform metric because it audits source recovery itself rather than a downstream summary of it.

Finding high quality voiced groups

The practical point of the extraction stage is simple: once EGIFA is used as a source-learning corpus, the retained groups must be homogeneous enough to support later calibration. The combination LBGID + t-PRISM achieves that by putting an explicit pitch-track model on top of the events, rather than relying on ad hoc heuristics alone. Several practical choices make the fit robust across the full corpus:

- **Normalization.** Period sequences are mapped to $(0, 1)$, freezing the GP mean to 0 and kernel variance to 1.

- **Lengthscale initialization.** Starting at ≈ 10 (close to the converged value of ≈ 15) saves iterations without affecting the result.
- **Inducing point placement.** Trajectory lengths are heavy-tailed, so inducing points are initialized from the inverse empirical CDF to concentrate predictive power where data are dense.
- **Degrees of freedom.** ν is set from event statistics rather than optimized (see below); results are insensitive to the exact value as long as $\nu < 10$.

Remaining hyperparameters are optimized with Adam at default settings.

1.3.3.1. Heuristic for setting ν

We set ν from a tail-probability argument. After normalization the GP has unit variance and the learned noise scale is $\sigma \approx 0.15$. Typical spikes have amplitude ≈ 5, giving residuals $r \approx \frac{5}{0.15} \approx 30–35$ noise standard deviations. These should be absorbed by the Student-t tails, not by the GP bending to explain them.

The Student-t tail satisfies $P(|X| > r) \approx C r^{-\nu}$ for large r. Matching this to the empirical spike rate $p_{\text{spike}} \approx 1\%$ gives

$$\nu \approx -\frac{\log(p_{\text{spike}})}{\log(r)}. \tag{182}$$

For $r \approx 35$ and $p_{\text{spike}} \approx 0.01$ this yields $\nu \approx 1–2$, a Cauchy-like regime. Extreme residuals are then downweighted rather than forcing the GP into excessive curvature. The exact value is not sacred; the point is to keep rare event failures in the heavy tails rather than misreading them as genuine pitch curvature.

Dataset 2. APLAWD

APLAWD is a speech corpus with synchronized speech, electroglottograph (EGG), differentiated EGG (DEGG), and hand-corrected GCI reference events (Naylor et al., 2007) (Figure 43). (Serwy, 2017) reports a hand-verified event database over 10,944 waveforms, which is exactly what makes the corpus useful here: APLAWD supplies trustworthy GCI event sequences from which a prior over smooth pitch-period trajectories can be learned.

In this thesis APLAWD is not a glottal-flow learning corpus. It enters only through the geometry of pitch. Pitch (instantaneous fundamental frequency) is a continuous process sampled at discrete glottal closures. Between any two consecutive GCI events the vocal folds complete one cycle, and the elapsed time defines the local period. A sequence of such periods is a variable-length trajectory, naturally modeled as a Gaussian process in cycle time – exactly the GP-on-variable-length-data setup of PRISM on EGIFA (Dataset 1), but for period geometry rather than waveform shape.

We learn a robust pitch-track prior $\mathcal{P}$ from APLAWD via t-PRISM (Algorithm 4), then put it to work in three downstream roles: voiced-group segmentation on EGIFA, time warping for generation in Chapter 6, and analytical prediction of period-to-period jitter with no free parameters.

2.1. From GCI events to log-period tracks

The quantity of interest is not the speech waveform but the inter-GCI geometry. For recording n with GCI sample indices $n_{\text{GCI},n,0}, \ldots, n_{\text{GCI},n,M_n}$ at sampling rate f_s, define the period sequence

$$T_{nm} = 1000 \frac{n_{\text{GCI},n,m+1} - n_{\text{GCI},n,m}}{f_s} \quad (m = 0, \ldots, M_n - 1), \tag{183}$$

measured in milliseconds. We then work with the log-periods

$$\ell_{nm} = \log T_{nm}, \tag{184}$$

which stabilize the scale and turn multiplicative pitch variation into additive variation. These observations live on the cycle-midpoint grid

$$\tau_{nm} = m + \frac{1}{2}, \quad m = 0, \ldots, M_n - 1, \tag{185}$$

so the training data for t-PRISM is the collection of log-period tracks $\{(\boldsymbol{\tau}_n, \ell_n)\}_{n=1}^{N}$ after a global whitening transform. This is the same input format as PRISM's DGF exemplars in Chapter 5 — variable-length sequences, one per recording — but the object being modeled is a one-dimensional pitch trajectory, not a waveform shape.

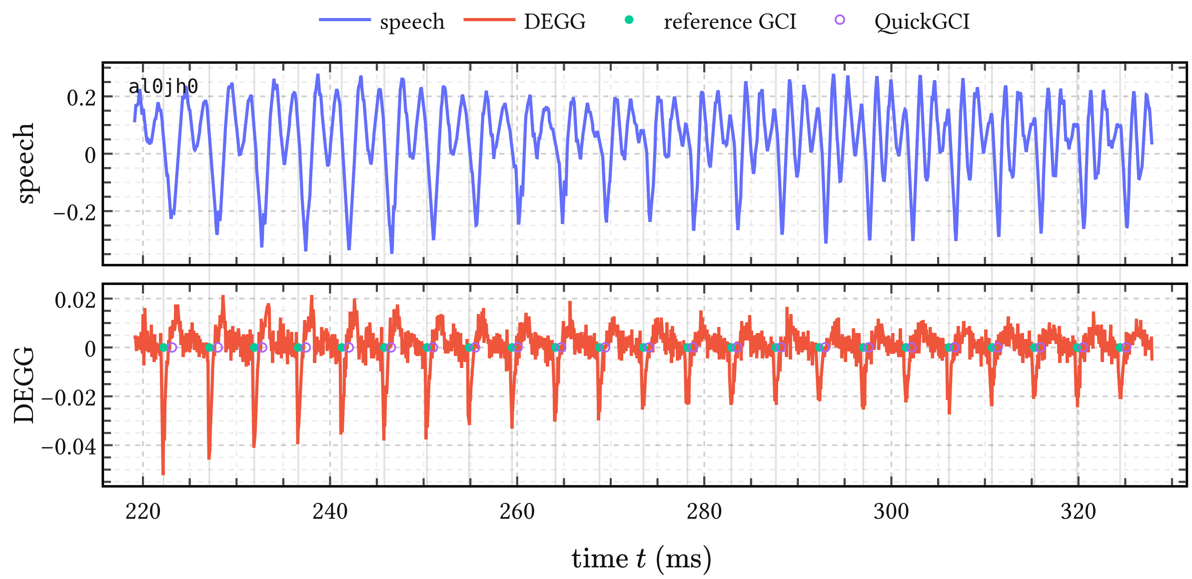


Figure 43 | **One APLAWD excerpt with speech and DEGG.** Top: the acoustic waveform. Bottom: the synchronized DEGG trace with hand-corrected GCI references (green) and QuickGCI estimates (purple).

2.2. Event quality

Because APLAWD contains both speech and hand-corrected GCI references, it provides a natural sanity-check on the geometry we later ask the pitch-track prior to explain. Figure 44 shows two diagnostics. The QuickGCI timing errors against the hand-corrected references are tightly localized, with the mode offset by the expected larynx-to-microphone propagation delay. The local absolute jitter measured over retained voiced groups peaks near the usual clinical pathology threshold, but the bulk of the distribution stays within the physiological range for healthy speech. We return to jitter quantitatively in the last section of this chapter.

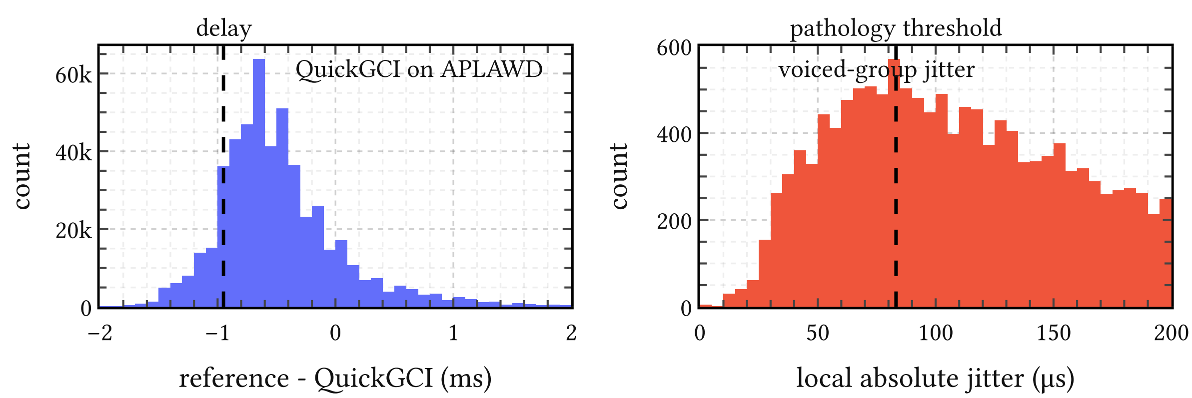


Figure 44 | **Two APLAWD diagnostics.** Left: QuickGCI timing errors against the hand-corrected GCI references; the dashed line marks the expected larynx-to-microphone propagation delay. Right: local absolute jitter measured over retained voiced groups; the dashed line marks the usual clinical pathology threshold.

Not every event needs to be perfect. The thesis needs a corpus where the dominant pitch geometry is trustworthy enough to learn a smooth prior from, and APLAWD meets that standard.

2.3. Experimental design

The 10,944 APLAWD waveforms are preprocessed into variable-length log-period tracks via (183) to (185), globally whitened to zero mean and unit variance, and padded to a common width of 320 samples (the 99th percentile of track length). Padded positions are masked so they do not contribute to the t-PRISM objective. Of these, $N_{\text{train}} = 5000$ tracks are retained for training and $N_{\text{test}} = 1000$ for testing; the average effective track length is roughly 50 cycles.

We train t-PRISM (Algorithm 4) with the Student-t degrees of freedom fixed at $\nu = 1$, giving deliberately heavy tails for gross detector failures and silent gaps. The kernel variance is fixed at 1 (absorbed by the whitening scale), so the free global parameters are the inducing locations $\mathbf{z}$, the kernel lengthscale, and the observation noise scale σ_P. The experimental grid crosses four kernel families — Matérn $\nu \in \{1/2, 3/2, 5/2\}$ and the squared exponential (RBF) — with inducing rank $R \in \{4, 8, 16, 32, 64\}$. Each configuration is run from 16 random initializations. Training uses Adam at learning rate 10^{-2} with batch size 512 complete tracks for 3000 iterations and three inner CAVI steps per minibatch. All runs converge stably in float32 with Cholesky jitter 10^{-4}; no run diverged or required double precision.

A separate experiment with $R = 128$ gave negligible improvement over $R = 64$, consistent with the average effective track length of roughly 50 cycles. Rational-quadratic and learned-ν Matérn kernels (Tronarp et al., 2018) were also tried and did not outperform the RBF. Model selection uses the held-out log likelihood per effective test point, averaged over both test tracks and random initializations.

2.4. Learning the pitch-track prior

The setup mirrors the PRISM calibration on EGIFA in Chapter 5: a variable-length corpus of exemplars sharing a single set of GP hyperparameters, optimized by the collapsed SVI machinery of Algorithm 4. The key difference is the severity of the outliers. A silent region between two detected GCIs defines an absurdly long "period" — not noise at the fringes but a gross violation of the local trend that dominates the residual distribution. This is the precise scenario for which t-PRISM was designed: the Student-t weights absorb such observations instead of forcing the prior to explain them.

Table 21 shows the result. An RBF prior with $R = 64$ inducing points wins decisively and becomes the default pitch-track prior used throughout the thesis. The learned lengthscale is about 15 cycles, the physically interesting quantity: it sets the timescale on which the model expects pitch to vary smoothly. The observation std $\sigma_P = 0.137$ captures residual cycle-to-cycle noise that the smooth GP cannot explain.

We package this winning fit as the learned pitch-track prior object

Kernel	R	Held-out log likelihood	Lengthscale	Obs. std.
`rbf`	64	2.117 ± 0.012	15.25	0.137
`matern:52`	64	2.070 ± 0.009	17.35	0.144
`rbf`	32	2.051 ± 0.011	15.28	0.147
`matern:32`	64	2.029 ± 0.008	18.65	0.148
`matern:52`	32	2.008 ± 0.010	17.23	0.153

Table 21 | **Held-out model selection on APLAWD.** Values are mean test log likelihood per effective test point across random initializations. The winning model is an RBF prior with $R = 64$ inducing points.

$$\mathcal{P} = \left(\mu_P, s_P, k_{\theta_P}, \sigma_P^2, \nu\right), \tag{186}$$

where μ_P and s_P are the global unwhitening location and scale of the log-periods, k_{θ_P} is the learned kernel in cycle time, σ_P^2 is the observation scale, and ν is the fixed Student-t degrees of freedom. This is the APLAWD specialization of the generic t-PRISM model from Algorithm 4.

In unwhitened log-period units, the latent prior under $\mathcal{P}$ is

$$\sim g_n(.) \mid \mathcal{P} \sim \texttt{GaussianProcess}\left(0, k_{\theta_P}(.,.)\right), \quad g_n(\tau) = \mu_P + s_P \sim g_n(\tau), \tag{187}$$

and the observation model is

$$\ell_{nm} \mid g_n, \lambda_{nm}, \mathcal{P} \sim \texttt{Normal}(g_n(\tau_{nm}), \sigma_P^2/\lambda_{nm}), \quad \lambda_{nm} \sim \texttt{Gamma}(\nu/2, \nu/2). \tag{188}$$

So the learned prior over physical periods is simply the image of g under exponentiation,

$$T(\tau) = \exp(g(\tau)). \tag{189}$$

Later chapters reuse exactly this object: EGIFA conditions on $\mathcal{P}$ for voiced-group extraction, while Chapter 6 samples from $\mathcal{P}$ to obtain the time warp.

2.5. Voiced-group segmentation

Once learned, $\mathcal{P}$ can be turned around and used as a robust filter on new GCI-annotated period tracks. The learned pitch model knows what "normal voiced pitch" looks like; anything outside that gets low weight. Given a period sequence $\{T_m\}_{m=0}^{M-1}$ for a single recording, t-PRISM returns local weights

$$w_m = \mathbb{E}[\lambda_m], \tag{190}$$

and we declare period m an inlier when

$$w_m > 1. \tag{191}$$

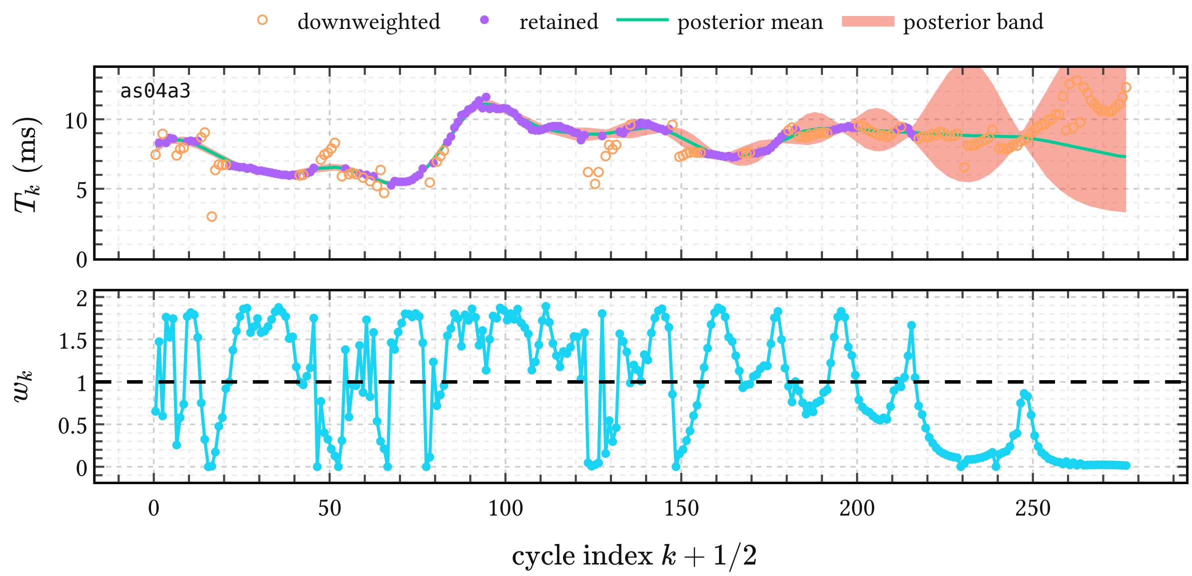


Figure 45 | **Voiced-group selection on one APLAWD pitch track.** Top: the observed inter-GCI periods together with the t-PRISM posterior mean and uncertainty band. Bottom: the inferred weights $w_m = \mathbb{E}[\lambda_m]$. Weight collapses carve the period track into retained voiced runs and rejected gaps.

Contiguous inlier runs are taken as voiced groups. Figure 45 shows the mechanism on one APLAWD recording: weight collapses at silent gaps and detector slips carve the period track into retained voiced runs and rejected regions. This is the segmentation rule reused on EGIFA (Dataset 1).

The same object also serves generation. Registration in Dataset 1 solved the map from physical time to cycle time; generation needs the inverse. A draw $g(.) \mid \mathcal{P}$ yields a smooth period trajectory $T(\tau) = \exp(g(\tau))$ via (189), and integrating gives the monotone warp $t(\tau) = t_0 + \int_0^\tau T(u)\,\mathrm{d}u$. This is the bridge to Section 6.3.1 in Chapter 6, where $\mathcal{P}$ supplies the pitch warp that turns surrogate DGF shapes in cycle time into voiced speech in physical time.

2.6. Predicting jitter

The pitch-track prior $\mathcal{P}$ makes a quantitative prediction about period-to-period jitter with no free parameters beyond those already learned.

Classical local jitter is defined as the expected absolute change in consecutive periods, normalized by the mean period,

$$\text{jitter} := \frac{\langle |T_{k+1} - T_k| \rangle}{\langle T_k \rangle}. \tag{192}$$

In the present framework, jitter is not introduced as an independent noise source. It emerges deterministically from the curvature of the time warp $t(\tau)$, induced by the Gaussian process prior on the log-period $g(\tau) = \log T(\tau)$.

Sampling the process at integer phases $\tau = k$ yields a stationary Gaussian sequence $g_k := g(k)$ with mean μ, marginal variance σ^2, and lag-one correlation

$$\rho(\ell) := \frac{k_\ell(1)}{k_\ell(0)}, \tag{193}$$

which depends only on the kernel family and the lengthscale ℓ measured in cycles.[23]

The observed periods are lognormal, $T_k = \exp(g_k)$, and the pair (T_k, T_{k+1}) is bivariate lognormal. Because g_k and g_{k+1} have equal variance, the sum $S = \frac{g_{k+1}+g_k}{2}$ and difference $D = g_{k+1} - g_k$ are independent Gaussians:

$$S \sim \mathrm{Normal}\left(\mu, \sigma^2 \frac{1+\rho}{2}\right), \quad D \sim \mathrm{Normal}(0, 2\sigma^2(1-\rho)). \tag{194}$$

Using the identity $|\exp(a) - \exp(b)| = 2\exp\left(\frac{a+b}{2}\right) \left|\sinh\left(\frac{a-b}{2}\right)\right|$, the expected absolute period difference factorizes as $\langle |T_{k+1} - T_k| \rangle = 2\langle \exp(S) \rangle \langle |\sinh(D/2)| \rangle$. After normalization by $\langle T_k \rangle = \exp\left(\mu + \frac{\sigma^2}{2}\right)$, all dependence on μ cancels, yielding the exact prediction

$$\langle \text{jitter} \mid \ell \rangle = 4\Phi\left(\sigma\sqrt{\frac{1-\rho(\ell)}{2}}\right) - 2, \tag{195}$$

where Φ denotes the standard normal CDF. For small arguments this reduces to

$$\langle \text{jitter} \mid \ell \rangle \approx \left(2\frac{\sigma}{\sqrt{\pi}}\right)\sqrt{1-\rho(\ell)}, \tag{196}$$

recovering the intuitive result that jitter vanishes as $\rho(\ell) \to 1$, i.e. as the phase warp becomes locally affine.

Note that (195) depends on the marginal log-period variability σ and the kernel-dependent correlation $\rho(\ell)$, but *not* on the observation noise σ_P^2. The predicted jitter is a property of the smooth latent trajectory alone; adding observation noise would increase it further.

Table 22 shows the predicted jitter across kernel families and lengthscales for a representative log-period standard deviation $\sigma = 0.25$, which implies a lognormal pitch distribution with median 150 Hz and 5–95% range of roughly 100–224 Hz — a reasonable spread for adult speech. Normal adult jitter values around 0.5% are only attained for lengthscales of several tens of cycles under smooth kernels; rougher kernels require substantially larger ℓ to suppress cycle-to-cycle variation. In this sense, ℓ admits a direct physiological interpretation as a cycle-to-cycle smoothness parameter governing the temporal regularity of voicing.

[23] Matérn 1/2: $\rho(\ell) = \exp\left(-\frac{1}{\ell}\right)$. Matérn 3/2: $\rho(\ell) = \left(1 + \frac{\sqrt{3}}{\ell}\right)\exp\left(-\frac{\sqrt{3}}{\ell}\right)$. Matérn 5/2: $\rho(\ell) = \left(1 + \frac{\sqrt{5}}{\ell} + \frac{5}{3\ell^2}\right)\exp\left(-\frac{\sqrt{5}}{\ell}\right)$. Squared exponential: $\rho(\ell) = \exp\left(-\frac{1}{2\ell^2}\right)$.

ℓ (cycles)	$\nu = 1/2$	$\nu = 3/2$	$\nu = 5/2$	$\nu = \infty$
5	12.0%	6.2%	5.0%	4.0%
10	8.7%	3.3%	2.6%	2.0%
20	6.2%	1.7%	1.3%	1.0%
40	4.4%	0.85%	0.64%	0.50%
80	3.1%	0.43%	0.32%	0.25%

Table 22 | **Predicted mean period-to-period jitter** as a function of lengthscale ℓ for Matérn kernels with $\nu \in \left\{\frac{1}{2}, \frac{3}{2}, \frac{5}{2}\right\}$ and the squared exponential kernel ($\nu = \infty$). Predictions use a representative $\sigma = 0.25$ (median $F_0 \approx 150$ Hz, 5–95% range $\approx$ 100–224 Hz) and are obtained from (195).

For our learned RBF prior with $\ell = 15.25$ cycles, (195) gives a predicted jitter of approximately 1.3%. This is in the right ballpark: the model captures the order of magnitude of real jitter, which is pleasing for a prediction with no free parameters. At the same time, the value is above the conventional clinical threshold of roughly 1%, suggesting that the prior is somewhat wiggly compared to jitter estimates from controlled sustained-vowel recordings. That discrepancy is expected: conventional jitter norms are measured in steady-state conditions on single speakers, whereas $\mathcal{P}$ was learned across speakers, vowels, and phonatory conditions. A speaker- or condition-specific prior would tighten the lengthscale and bring the prediction down; we leave that investigation to future work.

2.7. Summary

$\mathcal{P}$ is a Gaussian process prior over pitch, learned from APLAWD's hand-corrected GCIs via t-PRISM. It serves three roles in this thesis: segmenting pitch trajectories into voiced groups (Dataset 1), supplying the physical-time warp for source generation (Section 6.3.1 in Chapter 6), and predicting period-to-period jitter from the learned lengthscale alone ((195)). All three fall out of the same object. That is the recurring message of this thesis: Gaussian processes do not just model — they segment, warp, and predict, all from a single learned kernel.

Dataset 3. VTRFormants

VTRFormants (Deng et al., 2006) is the standard public benchmark for quantitative evaluation of formant, or more generally vocal-tract resonance (VTR), tracking. It is a representative TIMIT-derived subset of 538 utterances paired with manually corrected resonance trajectories at 10 ms frame resolution, covering speaker, gender, dialect, and phonetic context. Only the first three frequency tracks are manually corrected, so throughout this thesis VTRFormants means `F1--F3` and nothing beyond that.

Its role in the thesis is complementary to EGIFA. EGIFA asks whether a model can recover glottal source structure when matched source waveforms are available. VTRFormants asks whether the same posterior yields useful tract estimates on real speech. If BNGIF models excitation better, one should expect the inferred vocal tract to improve too.

3.1. The thesis evaluation slice

The raw database is broader than what is needed here. The live benchmark code restricts evaluation to TIMIT vowel events identified from the original phone annotations. This is a narrower slice than the full corpus, but it is the right one for the thesis.

There are three reasons. First, the thesis is about voiced speech, and vowels are the cleanest voiced targets in TIMIT. Second, the VTR tracks are interpolated through silence and other regions where no literal resonance target exists, so those parts should not be trusted for quantitative scoring. Third, vowel events align most cleanly with the voiced-source model used later in BNGIF, avoiding boundary artifacts that would otherwise dominate the comparison. Compared with the broader speech-labeled protocol used by (Mehta et al., 2012), this is a stricter and cleaner evaluation slice.

That stricter slice should still not be overread. (Mehta et al., 2012) note that visible labeling errors remain in the VTR database: the corrected tracks do not always sit on the high-energy regions one would expect from the spectrogram. So in this thesis the VTR benchmark is used mainly for *relative comparison* between algorithms, not as a perfect physical ground truth in the absolute sense.

3.2. Framewise evaluation contract

BNGIF does not operate on the native 10 ms VTR grid. It analyzes overlapping voiced frames in physical time, and each such frame carries whatever corrected VTR points fall inside it. Figure 47 shows the geometry on one example. The selected vowel event lives on the TIMIT waveform in absolute time, while one 32 ms voiced frame inside it is mapped to normalized cycle time τ by its detected GCIs. The same posterior over AR coefficients $\mathbf{a}$ used for GIF then induces posterior formant estimates directly, so no separate formant tracker is bolted on afterward.

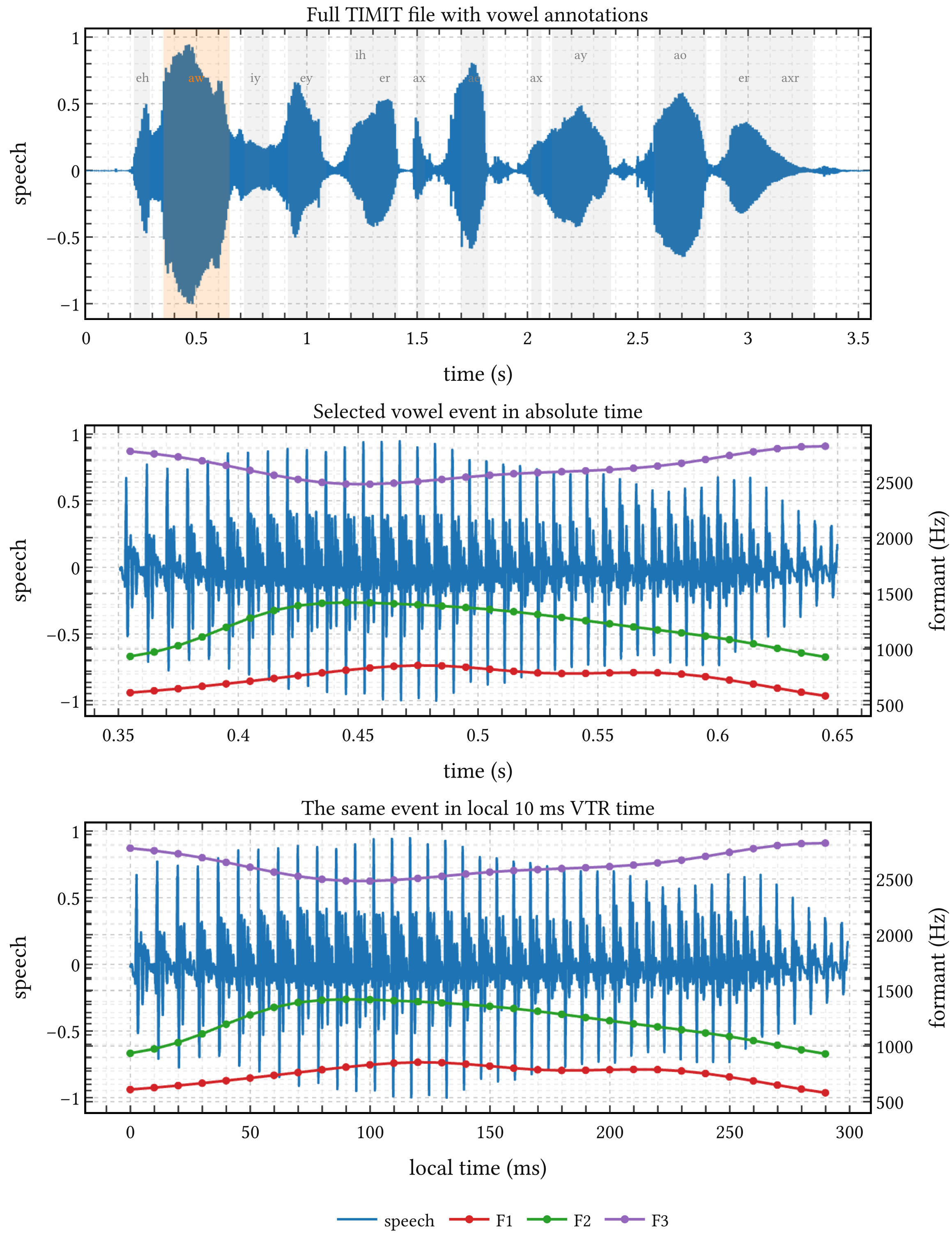


Figure 46 | **One VTRFormants/TIMIT utterance and its corrected vowel-event tracks.** Top: the full TIMIT waveform with vowel events from the original phone annotations. Middle: one selected vowel event in absolute time with the corrected `F1--F3` reference tracks. Bottom: the same event in local time on the native 10 ms VTR grid. The thesis evaluates only these vowel-event portions of the corrected tracks.

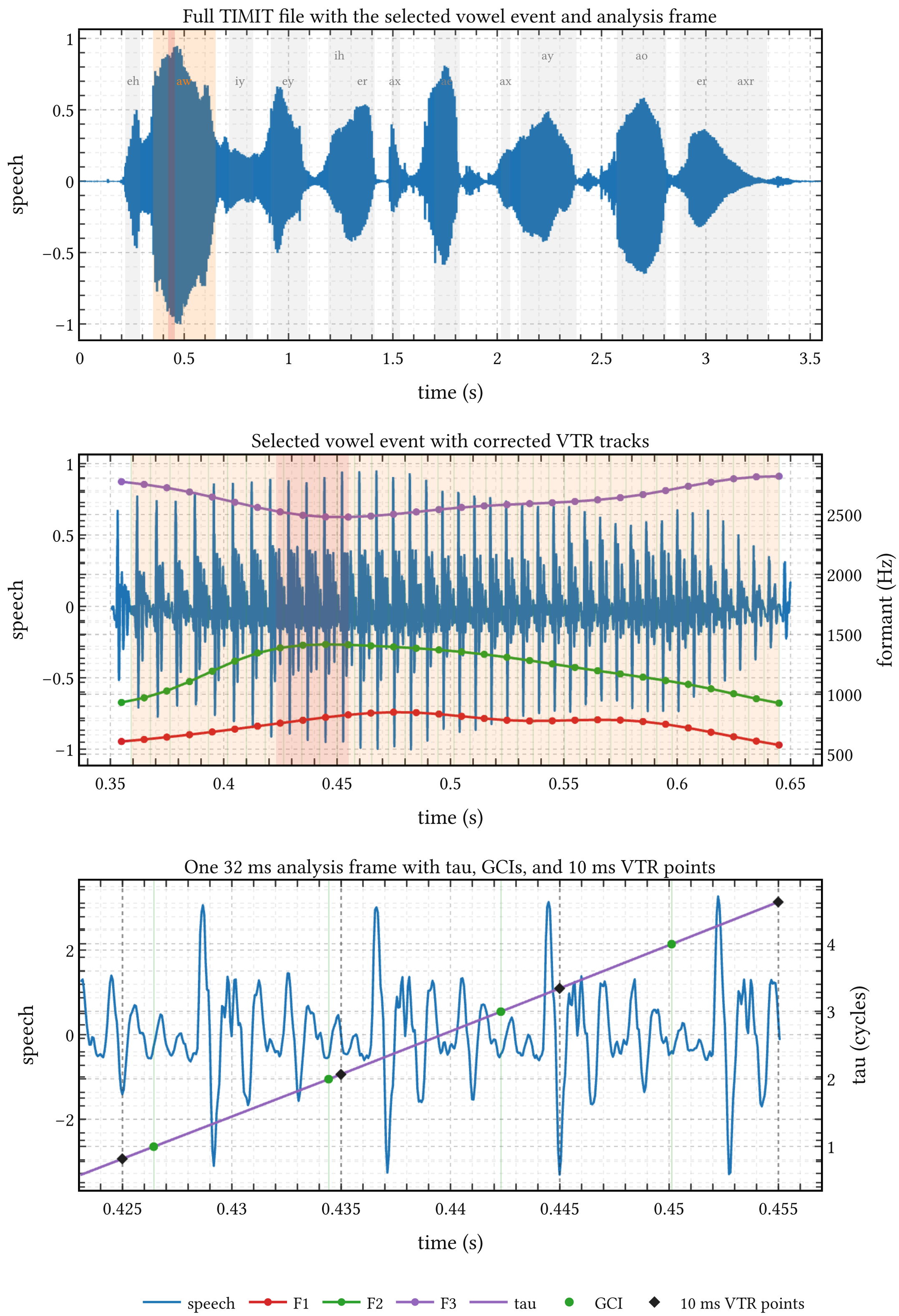


Figure 47 | **How one BNGIF analysis frame sits inside the finer VTR reference grid.** Top: the full TIMIT file with the selected vowel event and one highlighted 32 ms analysis frame. Middle: the vowel event itself with the corrected `F1--F3` tracks. Bottom: the selected analysis frame with its normalized cycle-time coordinate τ, detected GCI events, and the 10

Let t_j denote the native 10 ms VTR reference times, and let frame r produce a posterior formant estimate $\hat{\mathbf{F}}_r$ centered at c_r with effective half-width h_r. The framewise estimates are projected back to the reference grid by the triangular overlap-add rule

$$\hat{\mathbf{F}}(t_j) = \frac{\sum_r w_{jr} \hat{\mathbf{F}}_r}{\sum_r w_{jr}}, \quad w_{jr} = \max\left(1 - |t_j - c_r \frac{|}{h_r}, 0\right). \tag{197}$$

This is exactly the contract reused later in Chapter 7. Formant RMSE is then computed against the corrected `F1--F3` tracks on the retained vowel frames only.

The same setup also explains why BNGIF is not conceptually restricted to perfectly harmonic source segments. Although the learned source bank is calibrated on voiced excitation, the assembled source covariance still contains the broadband term $\nu_e \mathbf{I}$. When harmonic structure weakens, that term can absorb the frame in the same way ordinary LP falls back to broadband excitation. So the framework is not intrinsically incompatible with weakly voiced or nonvoiced material, even though the present benchmark stays deliberately vowel-restricted.

3.3. Spectral tilt and TIMIT

The same TIMIT/VTR pairing also gives a useful side product: an empirical estimate of spectral rolloff on clean vowel material.[24] This is not the main reason VTRFormants appears in the thesis. Its main role is the formant benchmark just defined. But the tilt estimate is still useful because later AR-prior design depends on how much broadband slope should plausibly be attributed to the filter.

What we want is not merely the tilt of voiced speech itself, but an estimate of the tilt contributed by the vocal tract filter. That quantity is not directly observed, and the source-filter split is not unique: depending on the assumed glottal model and smoothing procedure, some broadband slope can be attributed to the source and some to the filter (Fant, 1960). Still, source and radiation physics constrain the plausible range tightly enough to make the residual filter-side term meaningful.

We know that:
1. the glottal flow $u(t)$ itself rolls off by about -12 dB/oct for abrupt closure, and can be steeper, up to about -18 dB/oct, for smoother return phases (Doval et al., 2006);
2. lip radiation contributes about $+6$ dB/oct (Schroeder, 1999);
3. what remains is an effective residual filter tilt, not a directly observed physical primitive, but not entirely arbitrary either;
4. these pieces add up to the overall slope of the observed speech signal.

Estimating the spectral tilt of voiced speech

[24] Here `rolloff`, `spectral tilt`, and `spectral slope` all loosely mean the same thing, although some authors reserve rolloff for asymptotic high-frequency decay and tilt for the overall broadband slope.

The retained vowel events are essentially short quasiperiodic speech fragments. For each such event we estimate the FFT magnitude spectrum in dB and fit the linear trend

$$P(f) = a \log_2 f + b. \tag{198}$$

Then

$$P(2f) - P(f) = a, \tag{199}$$

so the coefficient a itself is the estimated rolloff in dB per octave.

The fit is only performed above about 200 Hz. Below that, the simple additive rolloff picture breaks down because low-frequency source structure dominates. Both Doval et al. (2006) and Fulop & Disner (2011) point to about 200 Hz as a reasonable cutoff for this regime.

Figure 48 shows four representative fits. Averaging over the usable vowel events gives a voiced-speech tilt of about $-8(2)$ dB per octave. Combining this with the source-side heuristics above yields an effective residual filter tilt of about $-2(2)$ dB per octave.

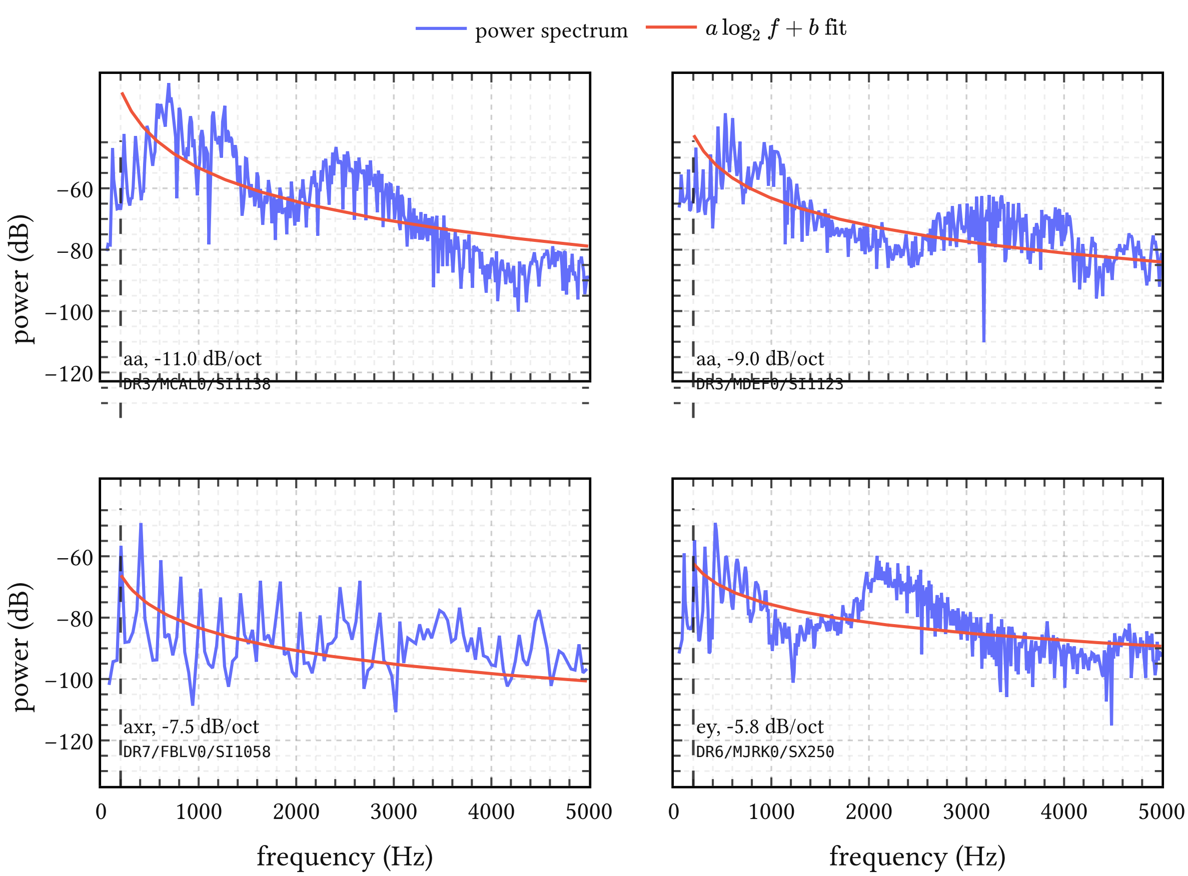


Figure 48 | **Example spectral-tilt fits on retained TIMIT vowel events.** Each panel shows the empirical power spectrum together with the fitted trend $a \log_2 f + b$. The dashed vertical line marks the 200 Hz cutoff above which the fit is performed. This is a supporting estimate derived from the same TIMIT/VTR pairing, not the main benchmark object of the chapter.

That is consistent with mild tilt as predicted by lossy acoustic tubes (de Boer, 2008) and with Chen (2013)'s theoretical estimate of a decay rate near -3 dB/oct for formants. For the thesis, the point is simply that the filter should not be forced into an overly steep low-order shape: if one wants both clear formant peaks and only mild broadband decay, the AR prior needs more flexibility than a rigid three-pole picture.

Algorithms

Algorithm 1. IKLP

Chapter 2 introduces *infinite kernel linear prediction* (IKLP) as the conceptual extension of linear prediction (LP) and multiple kernel linear prediction (MKLP). This chapter gives the technical side deferred there: the variational family, the tractable lower bound, the closed-form CAVI updates, the Woodbury-Cholesky rewrite used in this thesis, and the prior reparameterizations used later. The base model follows Yoshii & Goto (2013) the Woodbury-Cholesky reformulation and generalized AR-prior interface are the thesis-side additions.

1.1. Variational inference

Priors and variational family

To complete the model we place priors on the positive scale parameters and on $\mathbf{a}$. Following Yoshii & Goto (2013), the scales receive gamma priors,

$$\nu_w \sim \texttt{Gamma}(a_w, b_w), \quad \nu_e \sim \texttt{Gamma}(a_e, b_e). \tag{200}$$

For the AR coefficients we generalize the isotropic prior $\mathbf{a} \sim \texttt{Normal}(\mathbf{0}, \lambda\mathbf{I})$ of the original paper to a full multivariate Gaussian,

$$\mathbf{a} \sim \texttt{Normal}(\boldsymbol{\mu}_a, \boldsymbol{\Sigma}_a), \quad \mathbf{Q} := \boldsymbol{\Sigma}_a^{-1}, \tag{201}$$

with precision matrix $\mathbf{Q}$. The isotropic case is $\boldsymbol{\mu}_a = \mathbf{0}$, $\mathbf{Q} = \lambda^{-1}\mathbf{I}$. We need the full form in Chapter 3, where informative priors derived from stability theory and spectral constraints replace the isotropic default.

Exact posterior inference is intractable, so we use a mean-field variational family,

$$q(\boldsymbol{\theta}, \mathbf{a}, \nu_w, \nu_e) = q(\mathbf{a})\, q(\nu_w)\, q(\nu_e) \prod_{i=1}^{I} q(\theta_i), \quad q(\mathbf{a}) = \delta(\mathbf{a} - \mathbf{a}^*), \tag{202}$$

with $\mathbf{a}$ treated as a MAP point estimate. Each remaining factor is updated by coordinate ascent on the ELBO $\mathcal{L}$:

$$\log p(\mathbf{x}) \geq \underbrace{\mathbb{E}[\log p(\mathbf{x} \mid \boldsymbol{\theta}, \mathbf{a}, \nu_w, \nu_e)]}_{\text{expected log-likelihood}} + \sum_z \mathbb{E}[\log p(z)] - \sum_z \mathbb{E}[\log q(z)] =: \mathcal{L}. \tag{203}$$

Compared to MCMC-based GIF approaches — where Auvinen et al. (2014) report runtimes on the order of 2.5 hours of single-CPU time per 25-millisecond analysis frame — variational inference keeps the Bayesian treatment while reducing inference to a sequence of closed-form updates. CAVI is convergence-guaranteed, parallelizable over batches of frames, and compatible with modern autodiff and GPU workflows even when the update steps themselves remain analytic.

Tractable lower bound via matrix inequalities

The expected log-likelihood involves $\mathbb{E}[\log|\mathbf{K}|]$ and $\mathbb{E}[\mathbf{x}^\top \boldsymbol{\Psi}^\top \mathbf{K}^{-1} \boldsymbol{\Psi}\mathbf{x}]$, where $\mathbf{K} = \nu_w \sum_i \theta_i \mathbf{K}_i + \nu_e \mathbf{I}$, and neither expectation is tractable in closed form. Two matrix-variate inequalities provide bounds.

The log-determinant $\log|\mathbf{V}|$ is concave, so a first-order Taylor expansion around any PSD matrix $\boldsymbol{\Omega}$ gives

$$\log|\mathbf{V}| \leq \log|\boldsymbol{\Omega}| + \operatorname{tr}(\boldsymbol{\Omega}^{-1}\mathbf{V}) - M. \tag{204}$$

The quadratic form $\mathbf{z}^\top \mathbf{V}^{-1} \mathbf{z}$ is convex, and an inequality of Sawada et al. (2012) gives

$$\mathbf{z}^\top \left(\sum_i \mathbf{V}_i \right)^{-1} \mathbf{z} \leq \sum_i \mathbf{z}^\top \boldsymbol{\Upsilon}_i^\top \mathbf{V}_i^{-1} \boldsymbol{\Upsilon}_i \mathbf{z}, \tag{205}$$

where $\{\boldsymbol{\Upsilon}_i\}$ are auxiliary matrices summing to the identity.

Applying both inequalities to the expected log-likelihood and optimizing the auxiliary quantities in closed form gives the tractable lower bound $\mathcal{L}' \leq \mathcal{L}$, with optimal values

$$\boldsymbol{\Omega} = \mathbb{E}[\nu_w] \sum_i \mathbb{E}[\theta_i] \mathbf{K}_i + \mathbb{E}[\nu_e] \mathbf{I}, \tag{206}$$

$$\boldsymbol{\Upsilon}_i = \tilde{w}_i \mathbf{K}_i \mathcal{S}^{-1}, \quad \boldsymbol{\Upsilon}_0 = \tilde{\nu}_e \mathcal{S}^{-1}, \tag{207}$$

where the *quadratic operator* $\mathcal{S}$ and harmonic-mean weights are defined by

$$\mathcal{S} = \sum_i \tilde{w}_i \mathbf{K}_i + \tilde{\nu}_e \mathbf{I}, \quad \tilde{\nu}_e = 1/\mathbb{E}[\nu_e^{-1}], \quad \tilde{w}_i = 1/(\mathbb{E}[\nu_w^{-1}]\mathbb{E}[\theta_i^{-1}]). \tag{208}$$

Both $\boldsymbol{\Omega}$ and $\mathcal{S}$ have the form $\nu \mathbf{I} + \sum_i w_i \mathbf{K}_i$; they differ only in which moment or harmonic-mean expectations enter as weights. This shared algebraic structure is what the Woodbury-Cholesky reformulation below exploits.

GIG posteriors and CAVI updates

The sufficient statistics of a gamma prior are $\log z$ and z; the bounds above introduce terms in z and $1/z$. The optimal variational posteriors for θ_i, ν_w, ν_e therefore all take the form of a *generalized inverse Gaussian*,

$$\texttt{GIG}(z \mid \gamma, \rho, \tau) \propto z^{\gamma-1} \exp(-1/2(\rho z + \tau/z)), \quad z > 0, \tag{209}$$

with moments

$$\mathbb{E}[z] = \sqrt{\tau/\rho}\, K_{\gamma+1}(\sqrt{\rho\tau})/K_{\gamma(\sqrt{\rho\tau})}, \quad \mathbb{E}[z^{-1}] = \sqrt{\rho/\tau}\, K_{\gamma-1}(\sqrt{\rho\tau})/K_{\gamma(\sqrt{\rho\tau})}, \tag{210}$$

where K_γ is the modified Bessel function of the second kind.

With the LP residual $\mathbf{r} \coloneqq \boldsymbol{\Psi}(\mathbf{a})\mathbf{x}$, the weighted residual $\mathbf{w} \coloneqq \mathcal{S}^{-1}\mathbf{r}$, and the per-kernel quadratic statistics

$$\mathbf{u}_i := \mathbf{\Phi}_i^\top \mathbf{w}, \quad q_i := \|\mathbf{u}_i\|^2, \quad q_0 := \|\mathbf{w}\|^2, \tag{211}$$

the CAVI parameter updates take the compact form. For the kernel weights:

$$\gamma_{\theta_i} = \alpha / I, \quad \rho_{\theta_i} \leftarrow 2\alpha + \mathbb{E}[\nu_w]\ \mathrm{tr}(\mathbf{\Omega}^{-1}\mathbf{K}_i), \quad \tau_{\theta_i} \leftarrow (1/\mathbb{E}[\nu_w^{-1}])\left(1/\mathbb{E}[\theta_i^{-1}]^2\right) q_i. \tag{212}$$

For the structured scale:

$$\gamma_w = a_w, \quad \rho_w \leftarrow 2b_w + \sum_i \mathbb{E}[\theta_i]\ \mathrm{tr}(\mathbf{\Omega}^{-1}\mathbf{K}_i), \quad \tau_w \leftarrow \sum_i \left(1/\mathbb{E}[\nu_w^{-1}]^2\right)(1/\mathbb{E}[\theta_i^{-1}]) q_i. \tag{213}$$

For the broadband scale:

$$\gamma_e = a_e, \quad \rho_e \leftarrow 2b_e + \mathrm{tr}(\mathbf{\Omega}^{-1}), \quad \tau_e \leftarrow \left(1/\mathbb{E}[\nu_e^{-1}]^2\right) q_0. \tag{214}$$

The AR update is MAP under the generalized prior (201), giving the normal equation

$$(\mathbf{X}^\top \mathcal{S}^{-1}\mathbf{X} + \mathbf{Q})\mathbf{a}^* = \mathbf{X}^\top \mathcal{S}^{-1}\mathbf{x} + \mathbf{Q}\boldsymbol{\mu}_a. \tag{215}$$

All updates are iterated until the ELBO converges or a maximum iteration count is reached.

Posterior decomposition of the excitation After convergence, the posterior does not yield a separate excitation sample directly. We first fix the AR coefficients at the MAP value $\mathbf{a}^*$ and form the observed LP residual

$$\boldsymbol{\varepsilon}^* := \mathbf{\Psi}(\mathbf{a}^*)\mathbf{x}. \tag{216}$$

Writing the excitation as a sum of a structured component and broadband noise,

$$\boldsymbol{\varepsilon} = \mathbf{s} + \mathbf{n}, \tag{217}$$

the converged variational moments define the Gaussian prior

$$\mathbf{s} \sim \texttt{Normal}(\mathbf{0}, \mathbf{\Sigma}_s), \quad \mathbf{n} \sim \texttt{Normal}(\mathbf{0}, \mathbf{\Sigma}_n), \quad \mathbf{\Sigma}_s := \mathbb{E}[\nu_w] \sum_i \mathbb{E}[\theta_i]\mathbf{K}_i, \quad \mathbf{\Sigma}_n := \mathbb{E}[\nu_e]\mathbf{I}, \tag{218}$$

with $\mathbf{\Sigma}_s$ and $\mathbf{\Sigma}_n$ independent and summing to $\mathbf{\Omega}$ of (206). Conditioning this Gaussian sum on the observed residual (216) gives the block posterior

$$\texttt{Normal}\left(\begin{pmatrix}\mathbf{s}\\ \mathbf{n}\end{pmatrix} \mid \boldsymbol{\varepsilon}^*\right) = \texttt{Normal}\left(\begin{pmatrix}\boldsymbol{\mu}_s\\ \boldsymbol{\mu}_n\end{pmatrix}, \begin{pmatrix}\mathbf{\Sigma}_{s|\varepsilon} \mathbf{\Sigma}_{\mathrm{sn}|\varepsilon}\\ \mathbf{\Sigma}_{\mathrm{ns}|\varepsilon} \mathbf{\Sigma}_{n|\varepsilon}\end{pmatrix}\right), \tag{219}$$

with means

$$\boldsymbol{\mu}_s = \mathbf{\Sigma}_s \mathbf{\Omega}^{-1}\boldsymbol{\varepsilon}^*, \quad \boldsymbol{\mu}_n = \mathbf{\Sigma}_n \mathbf{\Omega}^{-1}\boldsymbol{\varepsilon}^*, \tag{220}$$

and covariances

$$\mathbf{\Sigma}_{s|\varepsilon} = \mathbf{\Sigma}_s - \mathbf{\Sigma}_s \mathbf{\Omega}^{-1}\mathbf{\Sigma}_s, \quad \mathbf{\Sigma}_{n|\varepsilon} = \mathbf{\Sigma}_n - \mathbf{\Sigma}_n \mathbf{\Omega}^{-1}\mathbf{\Sigma}_n, \tag{221}$$

$$\boldsymbol{\Sigma}_{\mathrm{sn}|\varepsilon} = -\boldsymbol{\Sigma}_s \boldsymbol{\Omega}^{-1} \boldsymbol{\Sigma}_n, \quad \boldsymbol{\Sigma}_{\mathrm{ns}|\varepsilon} = \boldsymbol{\Sigma}_{\mathrm{sn}|\varepsilon}^{\top}. \tag{222}$$

Since $\boldsymbol{\Sigma}_n = \mathbb{E}[\nu_e]\mathbf{I}$ is isotropic, these simplify to

$$\boldsymbol{\mu}_n = \mathbb{E}[\nu_e]\boldsymbol{\Omega}^{-1}\boldsymbol{\varepsilon}^*, \quad \boldsymbol{\mu}_s = \boldsymbol{\varepsilon}^* - \boldsymbol{\mu}_n. \tag{223}$$

Under the current variational family (202), $q(\mathbf{a})$ is a point mass. Therefore the full excitation posterior is degenerate:

$$q(\boldsymbol{\varepsilon} \mid \mathbf{x}, \xi^*) = \delta(\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^*). \tag{224}$$

In practice we sample only the split into $\mathbf{s}$ and $\mathbf{n}$ under (219); when only the posterior mean is needed, (223) is returned directly.

The dominant costs per iteration are the I trace terms $\operatorname{tr}(\boldsymbol{\Omega}^{-1}\mathbf{K}_i)$ and the operator actions $\mathcal{S}^{-1}\mathbf{r}$ and $\mathcal{S}^{-1}\mathbf{X}$, all requiring solutions to linear systems in matrices of the form $\nu\mathbf{I} + \sum_i w_i \mathbf{K}_i$. The next section reduces all of these to triangular solves on a single small matrix.

1.2. Woodbury-Cholesky reformulation

Both $\boldsymbol{\Omega}$ and $\mathcal{S}$ are instances of the abstract operator

$$\mathbf{S}(\nu, \mathbf{w}) := \nu\mathbf{I}_M + \sum_{i=1}^{I} w_i \mathbf{K}_i \in \mathbb{R}^{M \times M}. \tag{225}$$

The direct approach materializes this $M \times M$ matrix, factors it by Cholesky ($\mathcal{O}(M^3)$), and solves systems against it ($\mathcal{O}(M^2)$ each). For the frame lengths relevant to GIF this is expensive; for large windows it is prohibitive.

Suppose each kernel admits a low-rank factorization $\mathbf{K}_i = \boldsymbol{\Phi}_i \boldsymbol{\Phi}_i^{\top}$ with $\boldsymbol{\Phi}_i \in \mathbb{R}^{M \times R_i}$. In the settings we care about this is exact: the periodic kernels of the original paper are parameterized by Fourier features, and the arc-cosine GP priors of later chapters are defined by their feature matrices $\boldsymbol{\Phi}_i$. Form the weighted stacked factor

$$\tilde{\boldsymbol{\Phi}} := \left[\sqrt{w_1}\boldsymbol{\Phi}_1, ..., \sqrt{w_I}\boldsymbol{\Phi}_I\right] \in \mathbb{R}^{M \times L}, \quad L = \sum_i R_i, \tag{226}$$

so that $\mathbf{S}(\nu, \mathbf{w}) = \nu\mathbf{I}_M + \tilde{\boldsymbol{\Phi}}\tilde{\boldsymbol{\Phi}}^{\top}$. The Woodbury matrix identity gives its inverse as

$$\mathbf{S}^{-1}\mathbf{v} = \nu^{-1}\mathbf{v} - \nu^{-2}\tilde{\boldsymbol{\Phi}}\mathbf{A}^{-1}\tilde{\boldsymbol{\Phi}}^{\top}\mathbf{v}, \quad \mathbf{A} := \mathbf{I}_L + \nu^{-1}\tilde{\boldsymbol{\Phi}}^{\top}\tilde{\boldsymbol{\Phi}} \in \mathbb{R}^{L \times L}. \tag{227}$$

The $L \times L$ *core matrix* $\mathbf{A}$ is the only object that needs to be factored. Its Cholesky decomposition $\mathbf{A} = \mathbf{R}^{\top}\mathbf{R}$ — computed once per operator instance per CAVI iteration — makes the following quantities available through triangular solves alone.

Operator solves To compute $\mathbf{S}^{-1}\mathbf{v}$: form $\mathbf{t} = \tilde{\boldsymbol{\Phi}}^{\top}\mathbf{v} \in \mathbb{R}^L$, solve the triangular systems $\mathbf{R}^{\top}\mathbf{y} = \mathbf{t}$ then $\mathbf{R}\mathbf{u} = \mathbf{y}$ to get $\mathbf{u} = \mathbf{A}^{-1}\mathbf{t}$, then return $\nu^{-1}\mathbf{v} - \nu^{-2}\tilde{\boldsymbol{\Phi}}\mathbf{u}$. No explicit inverse is formed.

Log-determinant By the matrix determinant lemma, $|\mathbf{S}| = \nu^{M}\ |\mathbf{A}|$, so

$$\log|\mathbf{S}| = M \log \nu + 2 \sum_{\ell=1}^{L} \log R_{\ell\ell}. \tag{228}$$

Trace of inverse

$$\operatorname{tr}(\mathbf{S}^{-1}) = (M - L)/\nu + \nu^{-1}\ \operatorname{tr}(\mathbf{A}^{-1}). \tag{229}$$

Per-kernel trace Define $\mathbf{B}_i := \tilde{\mathbf{\Phi}}^{\top} \mathbf{\Phi}_i \in \mathbb{R}^{L \times R_i}$. Then

$$\operatorname{tr}(\mathbf{S}^{-1}\mathbf{K}_i) = \nu^{-1}\ \|\mathbf{\Phi}_i\|_F^2 - \nu^{-2}\ \|\mathbf{R}^{-\top}\mathbf{B}_i\|_F^2, \tag{230}$$

where $\|\mathbf{R}^{-\top}\mathbf{B}_i\|_F^2$ is the squared Frobenius norm of the solution to $\mathbf{R}^{\top}\mathbf{Z} = \mathbf{B}_i$.

AR normal equation With $\mathcal{S}^{-1}$ available as a triangular-solve operator, the $P \times P$ Gram matrix $\mathbf{G} = \mathbf{X}^{\top}\mathcal{S}^{-1}\mathbf{X}$ is assembled column-by-column, and (215) is then a single $P \times P$ Cholesky solve of $\mathbf{G} + \mathbf{Q}$.

Every dense operation in the CAVI loop has been reduced to triangular solves on matrices of size at most $\max(L, P) \ll M$.

Complexity

Let M be the frame length, P the AR order, I the number of kernels, R the rank per kernel, $L = IR$ the total factor dimension, and B the number of frames in a batch.

In the dense baseline, building $\mathbf{S}$ costs $\mathcal{O}(IM^2R)$, factoring it costs $\mathcal{O}(M^3)$, each solve costs $\mathcal{O}(M^2)$, and storing it requires $\mathcal{O}(M^2)$ memory.

In the Woodbury-Cholesky regime, the core matrix $\mathbf{A}$ is assembled in $\mathcal{O}(ML^2)$ and factored in $\mathcal{O}(L^3)$. Each operator solve costs $\mathcal{O}(ML + L^2)$, the log-determinant costs $\mathcal{O}(L)$ after factorization, the per-kernel traces cost $\mathcal{O}(L^2R)$ each, and memory drops to $\mathcal{O}(ML + L^2)$: the $M \times M$ covariance is never stored. The crossover is favorable whenever $L \ll M$, precisely the regime of interest.

For B frames the loop vectorizes linearly in B. Since no dense $M \times M$ matrix is ever formed, the formulation is insensitive to its conditioning. Running in 32-bit floating point gives a substantial throughput gain on accelerators while retaining numerical stability through the Cholesky factorization and a small diagonal jitter on $\mathbf{A}$.

1.3. Prior reparameterizations

The priors (17) and (200) are written in shape-rate form, natural for conjugate analysis but opaque for prior elicitation. We reparameterize both into quantities with direct physical interpretations.

Kernel sparsity: α-calibration

Under (17), define $T := \sum_i \theta_i$ and the normalized weights $p_i = \theta_i / T$. For any finite truncation I, the sum $T \sim \texttt{Gamma}(\alpha, \alpha)$ has $\mathbb{E}[T] = 1$ and $\mathrm{sd}(T) = \alpha^{-\frac{1}{2}}$, while the proportions follow a symmetric Dirichlet:

$$(p_1, ..., p_I) \sim \texttt{Dirichlet}(\alpha/I, ..., \alpha/I). \tag{231}$$

Small α concentrates mass toward the vertices of the probability simplex — one dominant hypothesis — while large α spreads it uniformly.

A natural summary of effective sparsity is the exponentiated Shannon entropy,

$$I_{\text{eff}} := \exp(\mathbb{E}[H(\mathbf{p})]), \quad H(\mathbf{p}) = -\sum_i p_i \log p_i, \tag{232}$$

which counts the number of equally probable hypotheses that would produce the same entropy. Since the Dirichlet expectation of H is known analytically,

$$\mathbb{E}[H(\mathbf{p})] = \psi(\alpha + 1) - \psi(1 + \alpha/I), \tag{233}$$

we obtain

$$I_{\text{eff}(\alpha;I)} = \exp(\psi(\alpha + 1) - \psi(1 + \alpha/I)). \tag{234}$$

As $I \to \infty$ with α fixed, I_{eff} saturates at the constant $\exp(\psi(\alpha + 1) + \gamma_{\text{EM}})$ where γ_{EM} is the Euler-Mascheroni constant. At $\alpha = 1$ and large I this gives $I_{\text{eff}} \to e \approx 2.72$, meaning roughly a handful of simultaneously active hypotheses on average. This matches what Yoshii & Goto (2013) observe empirically at $\alpha = 1$, $I = 400$.

To calibrate α to a desired effective count I^0_{eff}, one solves (233) for α; the solution $\alpha^*(I)$ approaches 1 rapidly for $I^0_{\text{eff}} = e$ as I grows. The local sensitivity of I_{eff} to α near this calibration is described by the log-elasticity

$$b(\alpha; I) := \frac{\partial \log I_{\text{eff}}}{\partial \log \alpha} = \alpha\left[\psi_1(\alpha + 1) - I^{-1}\psi_1(1 + \alpha/I)\right], \tag{235}$$

where ψ_1 is the trigamma function. As $I \to \infty$ this approaches $b \to \psi_1(2) = \pi^2/6 - 1 \approx 0.645$, giving the local approximation

$$I_{\text{eff}(\alpha;I)} \approx e(\alpha/\alpha^*(I))^{0.645}. \tag{236}$$

So α behaves like a concentration parameter whose effect on sparsity is governed by a fixed exponent near the default calibration.

Excitation power decomposition: the (p, s, κ)-reparameterization

When $b_w = b_e =: b$, the total excitation power $\nu_w + \nu_e \sim \texttt{Gamma}(a_w + a_e, b)$ and the structured fraction $\pi := \nu_w/(\nu_w + \nu_e) \sim \texttt{Beta}(a_w, a_e)$ are independent of each other. This motivates the reparameterization

$$a_w = \kappa p, \quad a_e = \kappa(1-p), \quad b_w = b_e = \kappa/s, \tag{237}$$

under which the total power has prior $\nu_w + \nu_e \sim \texttt{Gamma}(\kappa, \kappa/s)$ with

$$\mathbb{E}[\nu_w + \nu_e] = s, \quad \mathrm{Var}(\nu_w + \nu_e) = s^2/\kappa, \tag{238}$$

and the structured fraction has prior $\pi \sim \texttt{Beta}(\kappa p, \kappa(1-p))$ with

$$\mathbb{E}[\pi] = p, \quad \mathrm{Var}(\pi) = p(1-p)/(\kappa+1). \tag{239}$$

The parameter κ controls concentration only — it tightens both distributions simultaneously without moving their means. The parameters (p, s) directly control the prior expected values: s is calibrated to the expected dynamic range of the analysis frame, and p reflects prior beliefs about phonation type. A frame expected to be clearly voiced might receive $p = 0.8$; an unvoiced prior might use $p = 0.1$.

The default of Yoshii & Goto (2013), $(a_w, a_e, b_w, b_e) = (1, 1, 1, 1)$, corresponds to $(p, s, \kappa) = (1/2, 2, 2)$: equal prior probability of structured and broadband excitation, total power expected at 2, with moderate concentration. A unit-power neutral default is $(p, s, \kappa) = (1/2, 1, 1)$.

1.4. Summary

The technical IKLP picture is now complete. The model itself is the sparse kernel-mixture LP likelihood of Chapter 2; the inference layer is a CAVI scheme with closed-form GIG updates; and the Woodbury-Cholesky rewrite removes the cubic bottleneck by reducing every dense step to triangular solves on matrices of size $\max(L, P) \ll M$.

This is why IKLP is useful later in the thesis. The inference engine itself stays fixed while the kernel bank changes. Replacing the periodic kernels of Yoshii and Goto by the arc-cosine GP priors of later chapters requires no change to the CAVI architecture – only the feature matrices $\mathbf{\Phi}_i$ change.

Algorithm 2. LBGID

LBGID stands for *level-based glottal instant detection.*

Glottal closure and opening instants (GCIs and GOIs) define the pseudoperidocity of speech and are important speech features with many practical applications (Drugman et al., 2012). Within the thesis, GCIs and GOIs are used as temporal anchors for the kernels modeling the glottal cycle. Learning these kernels from example $\{u(t)\}$ glottal flow waveforms requires these instants to be known. The task LBGID attempts to solve is to extract consistent (GCI, GOI) pairs from a given $u(t)$, which can then be used downstream for aligning a set of $\{u(t)\}$ waveforms.

In a physics-driven high-quality synthesizer like VocalTractLab (Birkholz, 2013; Krug et al., 2021), glottal closure is not given as a symbolic event and often does not appear as a sharp isolated spike. Instead, it appears as an emergent soft low-flow regime embedded in a continuous waveform whose baseline, amplitude, and degree of breathiness can all vary over time. LBGID extracts the glottal instants based on the following assumption: closure happens where the flow $u(t)$ is below a certain threshold, and the glottal instants are the strongest *opposing changes* in $u(t)$ that flank it.

2.1. Adaptive envelope normalization

A difficulty with glottal airflow signals is that absolute amplitude is rarely interpretable on its own. Slow baseline drift and physical effects like variation in subglottal pressure all alter amplitude to various degrees. To measure effective flow rather than simulator artifacts, LBGID tracks adaptive upper and lower envelopes, denoted by $\mathrm{roof}(t)$ and $\mathrm{floor}(t)$, by detecting peaks and troughs and interpolating between them with a standard peak picking algorithm. These define a smoothly varying local amplitude range,

$$A(t) = \mathrm{roof}(t) - \mathrm{floor}(t) \geq 0, \tag{240}$$

which sets the local scale for the threshold. Note that we assume the polarity of $u(t)$ to be positive; that is, $u(t) \geq 0$ everywhere.

2.2. Level-based closed phase detection

With the local amplitude frame in hand, candidate closed phases are identified via a simple relative threshold,

$$\mathrm{level}(t) = \mathrm{floor}(t) + \alpha(\mathrm{roof}(t) - \mathrm{floor}(t)), \tag{241}$$

for a small constant α, and points satisfying $u(t) \leq \mathrm{level}(t)$ are treated as belonging to a closed phase. This yields contiguous low-flow segments that naturally partition the signal into bounded candidate intervals. In the thesis we use $\alpha = 0.01$ on EGIFA. The three functions in (241) are illustrated in Figure 49 and Figure 50.

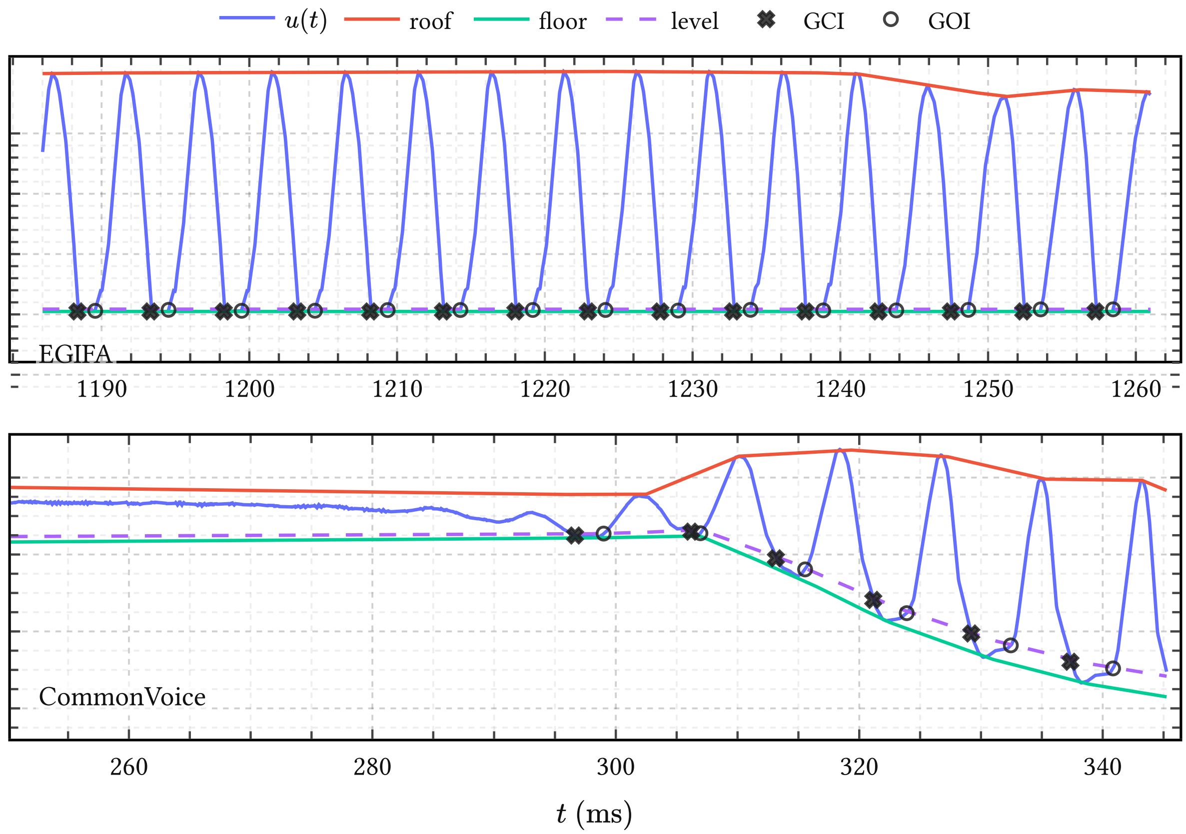


Figure 49 | **LBGID on EGIFA.** Each panel shows $u(t)$ together with roof(t), floor(t), the relative threshold level(t), and the detected (GCI, GOI) pairs.

2.3. Dynamic pairing inside the closed phase

Within each low-flow segment, LBGID searches for a pair of maximally opposed changes in $u(t)$. First, the derivative $u'(t)$ is computed by finite differences. Then algorithm evaluates all pairs of sample indices (i, j), where $i < j$, within the segment by the score

$$E(i, j) = -u'(i)\, u'(j). \tag{242}$$

Among these admissible ordered pairs, the one maximizing $E(i, j)$ is selected. The sample indices of the optimal pair are then output as the glottal closure instant (GCI) and glottal opening instant (GOI), respectively. Equation (242) is thus just a cheap proxy for quantifying the "abruptness" of a candidate glottal instant pair.

Figure 49 shows the result on an excerpt from EGIFA (Dataset 1). EGIFA has well-defined closed phases due to its synthesis quality, and the choice of α is relatively inconsequential.

Figure 50 shows a more detailed view of the algorithm for a given located closed phase. The algorithm first finds the candidate closed phase by amplitude, and then locates the strongest negative-to-positive dynamic pair inside it.

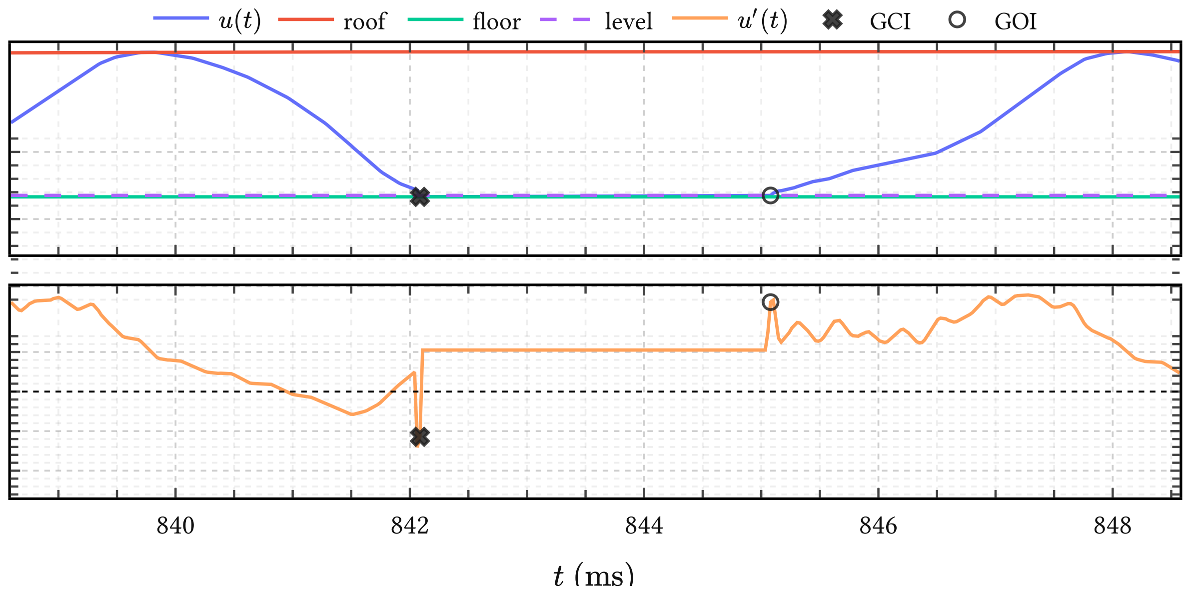


Figure 50 | **Zoom-in of a single closed phase.** Top: $u(t)$ together with roof(t), floor(t) and level(t), with the selected low-flow region shaded. Bottom: $u'(t)$ on the same window, with the chosen GCI and GOI marked.

2.4. Evaluation

There is no direct ground truth for the GCI and GOI estimates themselves, except for manually checking every output (Serwy, 2017). On EGIFA, a proxy for LBGID's performance is simply to look at the implied quantities

$$F_{0,k} = \frac{f_s}{n_{\mathrm{GCI},k+1} - n_{\mathrm{GCI},k}}, \quad \mathrm{OQ}_k = \frac{n_{\mathrm{GOI},k} - n_{\mathrm{GCI},k}}{n_{\mathrm{GCI},k+1} - n_{\mathrm{GCI},k}}, \tag{243}$$

whose distributions over frames in EGIFA are shown in Figure 42. These look reasonable and show a healthy amount of variation.

2.5. Summary

LBGID estimates (GCI, GOI) pairs from given $u(t)$ waveforms via level-based glottal instant detection.

It does this with two heuristics: closed phases correspond to intervals where flow is relatively small, and the GCI and GOI events happen with "maximally opposite abruptness."

These simple rules appear to be sufficient for the registration of the EGIFA dataset.

Algorithm 3. PRISM

PRISM stands for *process-induced surrogate modeling.*

PRISM is an adapter from variable-length datasets to Gaussians in a common amplitude space. The idea is not to fit one GP per waveform and stop there, but to learn a single shared kernel-induced basis and express each dataset in that finite-dimensional basis with calibrated uncertainty. Each dataset is modeled by its own latent function corrupted independently by globally scaled noise.

The name signifies the following. "Process-induced" means that each observation is indexed by an arbitrary input grid and may have arbitrary length, so no ad-hoc resampling or interpolation is required. "Surrogate modeling" refers to the fact that the learned basis defines a cheaper probabilistic surrogate for the original collection of functions, to be optionally refined or compressed by the user. In the language of sparse GPs, PRISM is therefore best viewed as a Nyström-style inducing-feature approximation with analytic Bayesian linear-regression projection (Titsias, 2009; Wilk et al., 2020).

In Chapter 5 this shared-basis view is used for *calibration*: one global kernel is fit to a large registered EGIFA corpus of over 7k waveforms and 6.6M samples. Later chapters use the same representation for downstream surrogate modeling. PRISM is therefore best thought of as the bridge from datasets of functions to tractable Gaussian coordinates in an amplitude space of fixed dimension R.

The mechanism that makes this tractable is an extension of the classic collapsed variational bound proposed by Titsias (2009). For Gaussian likelihoods, the local inducing-variable posteriors can be optimized analytically and substituted back into the objective, so only global quantities need to be trained. Projection to per-dataset Gaussians then becomes a cheap post-training step. This chapter develops that construction explicitly.

3.1. Data and model

We have N independent datasets collected as

$$\mathcal{D} = \{(\mathbf{t}_n, \mathbf{y}_n)\}_{n=1}^{N}. \tag{244}$$

Dataset n consists of an input grid

$$\mathbf{t}_n = \left(t_{n1}, ..., t_{nM_n}\right) \tag{245}$$

and observations

$$\mathbf{y}_n = \left(y_{n1}, ..., y_{nM_n}\right), \tag{246}$$

where M_n may vary freely with n. For simplicity the inputs are written as scalar times, but nothing below depends on that; the same construction works for arbitrary input dimension. Indeed, any model compatible with Titsias (2009) can be handled by PRISM. An important restriction is the assumption of a Gaussian likelihood, which is what makes the local states collapse analytically.

Each dataset is modeled by an independent latent function

$$f_n(.) \sim \texttt{GaussianProcess}(0, k_\theta(.,.)), \tag{247}$$

sharing the same kernel hyperparameters θ across all n but with independent draws of f_n. The kernel k_θ in (247) ties the draws together and acts as a global regularizer in function space.[25] The likelihood is given by

$$\mathbf{y}_n \mid f_n \sim \texttt{Normal}\Big(f_n(\mathbf{t}_n), \sigma^2 \mathbf{I}_{M_n}\Big) \tag{248}$$

which expresses the assumption that observations are conditionally independent given f_n, and imposes additional regularization through the global noise scale σ.

3.2. From kernel to shared basis

Choose R inducing locations

$$\mathbf{z} = (z_1, ..., z_R) \tag{249}$$

in the input domain and treat them as global learnable parameters. More general inducing *features* are also possible (Wilk et al., 2020), but vanilla inducing point evaluations are convenient for more general nonstationary kernels such as g-QPACK (139). For dataset n, define the inducing variables

$$\mathbf{u}_n = f_n(\mathbf{z}) \in \mathbb{R}^R. \tag{250}$$

Since f_n is a GP,

$$\mathbf{u}_n \sim \texttt{Normal}(\mathbf{0}, \mathbf{K}_{zz}), \tag{251}$$

where $\mathbf{K}_{zz} \in \mathbb{R}^{R \times R}$ with $(\mathbf{K}_{zz})_{rj} = k_\theta\big(z_r, z_j\big)$ is the same for all n. The GP conditional on $\mathbf{u}_n$ is

$$f_n(\mathbf{t}_n) \mid \mathbf{u}_n \sim \texttt{Normal}(\mathbf{K}_{nz}\mathbf{K}_{zz}^{-1}\mathbf{u}_n, \mathbf{K}_{nn} - \mathbf{Q}_{nn}), \tag{252}$$

where $\mathbf{K}_{nz} \in \mathbb{R}^{M_n \times R}$ has (m, r) entry $k_\theta(t_{nm}, z_r)$ and

$$\mathbf{Q}_{nn} = \mathbf{K}_{nz}\mathbf{K}_{zz}^{-1}\mathbf{K}_{zn} \tag{253}$$

is the Nyström approximation to $\mathbf{K}_{nn}$.

We emphasize that the column space of $\mathbf{K}_{nz}$ is shared across all datasets because the inducing locations $\mathbf{z}$ are shared. That is the common basis PRISM learns. The local variables $\mathbf{u}_n$ only say how dataset n "sits" in that basis.

[25] Since functions are infinite dimensional, some prior notion of smoothness is typically needed to make the problem well-posed (Sivia & Skilling, 2006).

3.3. Collapsed variational bound

The cost of keeping local parameters A direct sparse variational treatment would introduce one Gaussian variational factor per dataset,

$$q_n(\mathbf{u}_n) = \texttt{Normal}(\boldsymbol{\mu}_n, \boldsymbol{\Sigma}_n), \tag{254}$$

and maximize a separate ELBO for every n. That is workable for a single dataset, but for a large corpus it means carrying a synchronized local Gaussian state for every dataset during training. PRISM avoids this by keeping the local variational problem only long enough to solve it analytically, and then substituting the solution back into the objective.

Collapsed bound Following (Titsias, 2009), the per-dataset variational bound can be written as

$$\begin{aligned}\mathcal{L}_n(q_n) = \mathbb{E}_{q_n}\left[\log \texttt{Normal}\left(\mathbf{y}_n \mid \mathbf{K}_{nz}\mathbf{K}_{zz}^{-1}\mathbf{u}_n, \sigma^2\mathbf{I}_{M_n}\right)\right] \\ -D_{\mathrm{KL}}(q_n(\mathbf{u}_n) \,\|\, p(\mathbf{u}_n)) - \frac{1}{2\sigma^2}\,\mathrm{tr}(\mathbf{K}_{nn} - \mathbf{Q}_{nn}).\end{aligned} \tag{255}$$

For Gaussian likelihoods the optimal factor is available in closed form:

$$q_n(\mathbf{u}_n) = \texttt{Normal}(\boldsymbol{\mu}_n, \boldsymbol{\Sigma}_n), \tag{256}$$

with

$$\mathbf{C}_n = \mathbf{K}_{zz} + \sigma^{-2}\mathbf{K}_{zn}\mathbf{K}_{nz}, \tag{257}$$

$$\boldsymbol{\mu}_n = \sigma^{-2}\mathbf{K}_{zz}\mathbf{C}_n^{-1}\mathbf{K}_{zn}\mathbf{y}_n, \quad \boldsymbol{\Sigma}_n = \mathbf{K}_{zz}\mathbf{C}_n^{-1}\mathbf{K}_{zz}. \tag{258}$$

Substituting this optimum back into (255) gives the collapsed Titsias bound

$$\mathcal{L}_n(\mathbf{z}, \theta, \sigma^2) = \log \texttt{Normal}\left(\mathbf{y}_n \mid \mathbf{0}, \mathbf{Q}_{nn} + \sigma^2\mathbf{I}_{M_n}\right) - \frac{1}{2\sigma^2}\,\mathrm{tr}(\mathbf{K}_{nn} - \mathbf{Q}_{nn}). \tag{259}$$

The first term is the low-rank marginal likelihood and the second penalizes the discarded covariance.[26]

The PRISM ELBO is then just the sum over independent datasets,

$$\mathcal{L}(\mathbf{z}, \theta, \sigma^2) = \sum_{n=1}^{N} \mathcal{L}_n(\mathbf{z}, \theta, \sigma^2), \tag{260}$$

which is seen to depend only on the global parameters $(\mathbf{z}, \theta, \sigma^2)$.

Evaluating the bound In practice we whiten the basis so the local prior becomes isotropic. Let $\mathbf{K}_{zz} = \mathbf{L}\mathbf{L}^\top$ and define the amplitude coordinates

[26]The newer bounds of (Titsias, 2025) can also be used here. For BNGIF they did not yield better models in practice and were less efficient to batch, so we stayed with the original bound of (Titsias, 2009).

$$\mathbf{a}_n = \mathbf{L}^{-1}\mathbf{u}_n, \tag{261}$$

so that $\mathbf{a}_n \sim \texttt{Normal}(\mathbf{0}, \mathbf{I}_R)$. The corresponding feature map is

$$\varphi(t) = \mathbf{L}^{-1}\mathbf{k}_z(t), \quad \mathbf{k}_z(t) = (k_\theta(z_1, t), ..., k_\theta(z_R, t))^\top, \tag{262}$$

and the design matrix for dataset n is

$$\mathbf{\Phi}_n = \begin{pmatrix} \varphi(t_{n1})^\top \\ \vdots \\ \varphi\left(t_{nM_n}\right)^\top \end{pmatrix} \in \mathbb{R}^{M_n \times R}. \tag{263}$$

Then the sparse GP mean becomes a BLR mean,

$$\mathbf{y}_n = \mathbf{\Phi}_n \mathbf{a}_n + \varepsilon_n, \quad \varepsilon_n \sim \texttt{Normal}\left(\mathbf{0}, \sigma^2 \mathbf{I}_{M_n}\right). \tag{264}$$

Define

$$\mathbf{A}_n = \frac{\mathbf{\Phi}_n^\top}{\sigma} \in \mathbb{R}^{R \times M_n}, \quad \mathbf{B}_n = \mathbf{I}_R + \mathbf{A}_n \mathbf{A}_n^\top, \quad \mathbf{v}_n = \mathbf{A}_n \mathbf{y}_n \in \mathbb{R}^R. \tag{265}$$

Both terms in (259) are then computed efficiently from $\mathbf{A}_n$, $\mathbf{B}_n$, and $\mathbf{v}_n$, so the cost per dataset scales as $\mathcal{O}(R^2 M_n)$ and the $M_n \times M_n$ matrix $\mathbf{Q}_{nn} + \sigma^2 \mathbf{I}$ is never formed.

For the log marginal term, write $\mathbf{Q}_{nn} + \sigma^2 \mathbf{I} = \sigma^2 \left(\mathbf{I}_{M_n} + \mathbf{A}_n^\top \mathbf{A}_n\right)$. By the matrix determinant lemma,

$$\log\det(\mathbf{Q}_{nn} + \sigma^2 \mathbf{I}) = M_n \log \sigma^2 + \log\det(\mathbf{B}_n). \tag{266}$$

The Woodbury identity gives

$$(\mathbf{Q}_{nn} + \sigma^2 \mathbf{I})^{-1} = \sigma^{-2}\left(\mathbf{I}_{M_n} - \mathbf{A}_n^\top \mathbf{B}_n^{-1} \mathbf{A}_n\right), \tag{267}$$

so the quadratic form is

$$\mathbf{y}_n^\top (\mathbf{Q}_{nn} + \sigma^2 \mathbf{I})^{-1} \mathbf{y}_n = \sigma^{-2}(\|\mathbf{y}_n\|^2 - \mathbf{v}_n^\top \mathbf{B}_n^{-1} \mathbf{v}_n). \tag{268}$$

For the trace correction, $\operatorname{tr}(\mathbf{Q}_{nn}) = \|\mathbf{\Phi}_n\|_F^2$, so

$$-\frac{1}{2\sigma^2} \operatorname{tr}(\mathbf{K}_{nn} - \mathbf{Q}_{nn}) = -\frac{1}{2\sigma^2}(\operatorname{tr}(\mathbf{K}_{nn}) - \|\mathbf{\Phi}_n\|_F^2). \tag{269}$$

Every quantity in (259) is thus computable from the global parameters and the current dataset alone.

3.4. Stochastic optimization

Because the datasets are independent, an unbiased estimate of (260) is available from any minibatch $\mathcal{B} \subset \{1, ..., N\}$,

$$\hat{\mathcal{L}} = \frac{N}{|\mathcal{B}|} \sum_{n \in \mathcal{B}} \mathcal{L}_n(\mathbf{z}, \theta, \sigma^2). \tag{270}$$

The minibatch unit is a *complete dataset*, not an individual observation. This is what keeps irregular grids and unequal lengths harmless: each dataset contributes one collapsed term, regardless of its internal geometry. In practice we optimize (270) with Adam following (Orvieto & Gower, 2025) at a learning rate of 10^{-3}. We found this stable and amenable to float32 optimization, which gives a substantial speed gain over float64.

The contrast with the ELBO of stochastic variational inference (Hensman et al., 2013, Eq. 3) is interesting here. There, stochastic optimization over very large datasets is obtained by introducing and maintaining an explicit variational state of size $\mathcal{O}(R^2)$ in order to have the objective factorize over datapoints (and allow more general likelihoods beyond (248)). We avoid that move here because the factorization is already over complete datasets. The local inducing posteriors are collapsed analytically and only the global parameters $(\mathbf{z}, \theta, \sigma^2)$ are updated.

3.5. Projection to amplitude space

Once $(\mathbf{z}, \theta, \sigma^2)$ are trained, PRISM maps any dataset of arbitrary length M_n to a Gaussian in the fixed amplitude space $\mathbb{R}^R$. At the fitted global parameters, the local inducing posterior is simply the optimal factor (258) evaluated for the new dataset:

$$\mathbf{u}_n \mid \mathbf{y}_n \sim \texttt{Normal}(\boldsymbol{\mu}_n, \boldsymbol{\Sigma}_n). \tag{271}$$

Passing to amplitude coordinates via $\mathbf{a}_n = \mathbf{L}^{-1}\mathbf{u}_n$ gives

$$\mathbf{a}_n \mid \mathbf{y}_n \sim \texttt{Normal}(\mathbf{m}_n, \mathbf{S}_n), \tag{272}$$

$$\mathbf{S}_n = \mathbf{B}_n^{-1}, \quad \mathbf{m}_n = \mathbf{B}_n^{-1}\mathbf{v}_n. \tag{273}$$

This reuses $\mathbf{B}_n$ and $\mathbf{v}_n$ from the collapsed bound, so projection is essentially for free.

PRISM therefore learns the shared basis (262) and returns the projected Gaussian coordinates

$$\mathcal{D}_R = \{(\mathbf{m}_n, \mathbf{S}_n)\}_{n=1}^{N}. \tag{274}$$

This is a collection of Gaussian blobs in a common $\mathbb{R}^R$. Together with the learned feature map $\boldsymbol{\varphi}(t)$, it defines the empirical amplitude-space density

$$p(\mathbf{a} \mid \mathcal{D}_R) = \frac{1}{N} \sum_{n=1}^{N} \texttt{Normal}(\mathbf{a} \mid \mathbf{m}_n, \mathbf{S}_n), \tag{275}$$

which is the object handed to downstream methods.

3.6. Surrogate modeling

Importantly, PRISM alone does not yet produce a good *global* surrogate model of the corpus. What it learns is the regularized basis (262) together with the Gaussian collection (274) of the observed data in that basis. This is already enough to define a set of *local* Gaussian surrogate posteriors. For dataset n, one can sample amplitudes

$$\mathbf{a}_n^* \sim \texttt{Normal}(\mathbf{m}_n, \mathbf{S}_n) \tag{276}$$

cheaply, push them through the synthesis map $\mathbf{\Phi}_n \mathbf{a}_n^*$, and thereby resample from the local surrogate posterior induced by the shared basis. The posterior covariances are what make this cheap resampling possible.

To obtain a better corpus-level surrogate model, extra structure must be learned on top of the empirical amplitude density $p(\mathbf{a} \mid \mathcal{D}_R)$ of (275). One may ignore the covariances and apply ordinary tools such as PCA, clustering, or Gaussian mixtures to the posterior means. Or one may keep the covariances and use errors-in-variables (uncertain-input) variants of those same ideas. For very large collections one generally also wants to quantize or compress the number of Gaussian blobs, for example by hierarchical clustering or by the latent-variable model of the next chapter. In this thesis we take the latter route and hand the amplitude-space object $\mathcal{D}_R$ off to q-GPLVM (Algorithm 5), which learns a more compact generative model in the same shared amplitude space.

3.7. Summary

PRISM learns a shared, global kernel-induced basis from a possibly large collection of independent variable-length time series without storing or updating per-dataset variational parameters. In terms of GP approximation, PRISM is a reduced-rank projection of the GP prior k_θ to a rank-R BLR. The infinite-dimensional function space is projected onto the span of the learned basis, and the remaining degrees of freedom are absorbed by the observation noise σ^2.

After training, each dataset is projected to a Gaussian amplitude posterior (272) in a common, R-dimensional amplitude space. The returned collection $\mathcal{D}_R$ of (274), together with the shared basis (262), can then be used as-is for cheap surrogate sampling or passed to more structured downstream models.

Algorithm 4. t-PRISM

The Gaussian likelihood in PRISM (Algorithm 3) is analytically convenient, but inappropriate when data contain outliers. For learning a pitch-track model from APLAWD (Dataset 2), for example, the observations are inter-GCI period measurements, where a single detector failure, silent gap, or irregular closure can create an isolated gross outlier. More generally, any functional series may contain observations that should be downweighted rather than fitted exactly.

To address this, we introduce t-PRISM here as the "robust" version of PRISM. It replaces the Gaussian likelihood by a Student-t while retaining the same collapsed basis-learning structure. Tractability is maintained by representing the Student-t as a Gaussian scale mixture with one local precision variable per observation. Conditional on those precisions the likelihood is Gaussian again, so the collapsed bound from Algorithm 3 applies with the single modification that the noise covariance becomes diagonally weighted. The result is again a minibatch algorithm over complete datasets, and like PRISM it outputs Gaussian amplitude posteriors in a shared space.

Figure 51 illustrates the principle. Under a Gaussian likelihood, the posterior mean is pulled toward the locally irregular periods because the model must explain every residual at full strength. t-PRISM instead preserves the smooth trend and incorporates outliers by reducing their corresponding local weights $w_m \ll 1$.

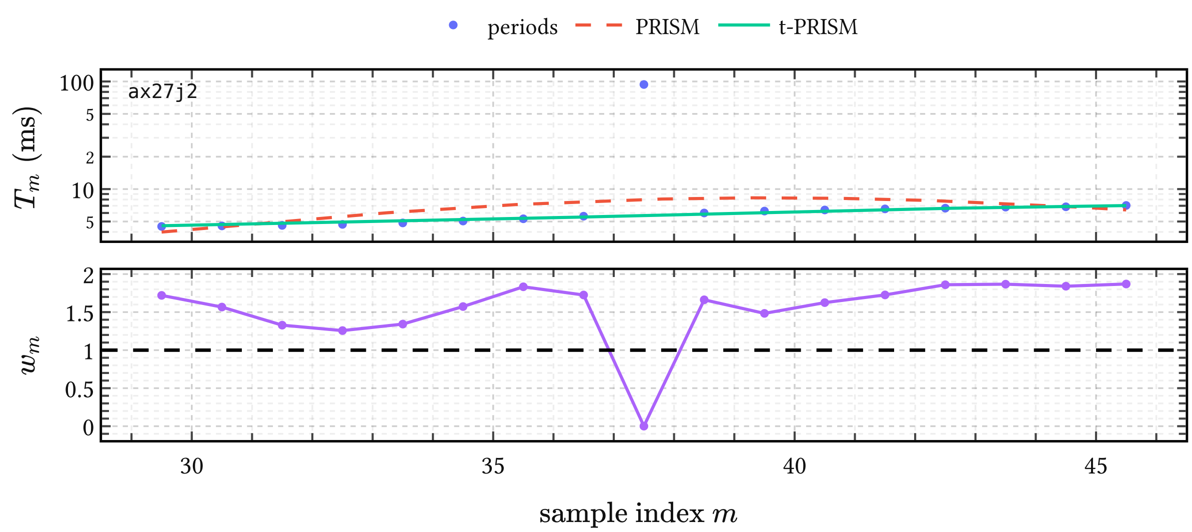


Figure 51 | **t-PRISM on a local APLAWD period track.** The top panel compares PRISM and t-PRISM on one excerpt of inter-GCI periods from the APLAWD pitch-track corpus, using the learned pitch track model of Dataset 2. The bottom panel shows the inferred precision weights $w_m = \mathbb{E}[\lambda_m]$. Weights $w_m \geq 1$ mark observations consistent with the local trend; smaller weights $w_m < 1$ signal outliers. Note that t-PRISM ignores the outlier at $m = 37.5$.

4.1. Student-t as a scale mixture

Recall the PRISM setup from Algorithm 3. Given our kernel of choice k_θ, we are presented with N independent datasets $\{(\mathbf{t}_n, \mathbf{y}_n)\}_{n=1}^N$, where dataset n has length M_n, latent function

$$f_n(.) \sim \texttt{GaussianProcess}(0, k_\theta(.,.)), \tag{277}$$

and inducing variables $\mathbf{u}_n = f_n(\mathbf{z})$ at shared inducing locations $\mathbf{z}$. In PRISM this prior is paired with the Gaussian observation model

$$\mathbf{y}_n \mid f_n \sim \texttt{Normal}\Big(f_n(\mathbf{t}_n), \sigma^2 \mathbf{I}_{M_n}\Big), \tag{278}$$

as in (248). t-PRISM leaves this prior side untouched. The only change is that the homoscedastic Gaussian likelihood is replaced by a locally weighted one: we introduce one latent precision variable per observation and integrate it out to obtain a Student-t likelihood.

The Student-t likelihood with degrees of freedom $\nu > 0$ and scale σ^2 is recovered from the standard Normal-Gamma augmentation

$$\lambda_{nm} \sim \texttt{Gamma}(\nu/2, \nu/2), \quad y_{nm} \mid f_{nm}, \lambda_{nm} \sim \texttt{Normal}(f_{nm}, \sigma^2/\lambda_{nm}). \tag{279}$$

Marginalizing over λ_{nm} recovers $\texttt{StudentT}(y_{nm} \mid f_{nm}, \nu, \sigma^2)$ exactly. For dataset n, stack the per-sample precisions into

$$\mathbf{\Lambda}_n = \operatorname{diag}\Big(\lambda_{n1}, ..., \lambda_{nM_n}\Big). \tag{280}$$

The augmented likelihood is then Gaussian with diagonal but heteroscedastic noise covariance:

$$\mathbf{y}_n \mid f_n, \mathbf{\Lambda}_n \sim \texttt{Normal}(f_n(\mathbf{t}_n), \sigma^2 \mathbf{\Lambda}_n^{-1}). \tag{281}$$

Large λ_{nm} means that sample (n, m) is trusted; small λ_{nm} means that its residual is downweighted. The prior mean of every λ_{nm} is one, so if the data contain no clear outliers the posterior expected precisions remain close to one. Then $\mathbf{W}_n \approx \mathbf{I}$ below, and the weighted formulas reduce continuously back to the PRISM expressions.

Figure 52 shows the intended fallback behavior. When there is no clear evidence for contamination, robustness costs essentially nothing: t-PRISM simply behaves like PRISM.

4.2. Variational family and objective

We introduce a mean-field variational family (Blei et al., 2017) over the latent precisions,

$$q_n(\mathbf{\Lambda}_n) = \prod_{m=1}^{M_n} q_{nm}(\lambda_{nm}), \quad q_{nm}(\lambda_{nm}) = \texttt{Gamma}(\alpha_{nm}, \beta_{nm}), \tag{282}$$

while the GP part is kept exactly as in the PRISM derivation. That is, for fixed precisions we still introduce a Gaussian factor $q_n(\mathbf{u}_n)$ over the inducing variables, use the exact GP

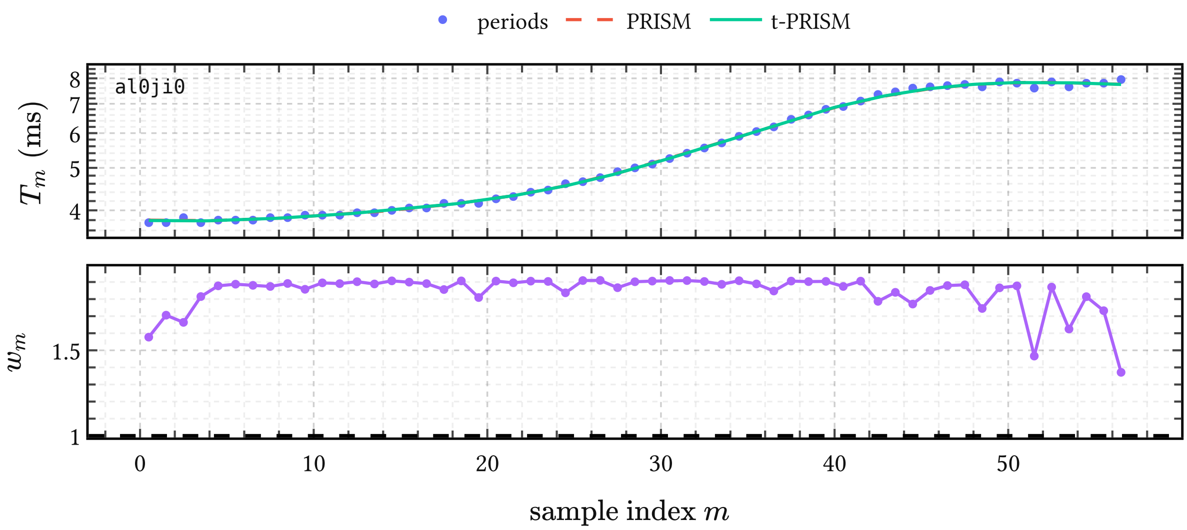


Figure 52 | **t-PRISM reducing to PRISM in the absence of outliers.** On this local APLAWD period track the observations are already consistent with a smooth trend. The PRISM and t-PRISM fits are nearly indistinguishable, and all inferred weights $w_m \geq 1$ and $O(1)$.

conditional $p(f_n \mid \mathbf{u}_n)$, and collapse the local Gaussian state analytically as in (255) and (259). The per-dataset ELBO before collapse is

$$\begin{aligned}\mathcal{L}_n^t = \mathbb{E}_{q_n}[\log p(\mathbf{y}_n \mid f_n, \mathbf{\Lambda}_n)] - D_{\mathrm{KL}}(q_n(\mathbf{u}_n) \parallel p(\mathbf{u}_n)) \\ - \sum_{m=1}^{M_n} D_{\mathrm{KL}}(q_{nm}(\lambda_{nm}) \parallel p(\lambda_{nm})).\end{aligned} \tag{283}$$

Define the expected precision and expected log-precision under q_{nm}:

$$\begin{aligned}w_{nm} &= \mathbb{E}[\lambda_{nm}] = \alpha_{nm}/\beta_{nm} \\ \ell_{nm} &= \mathbb{E}[\log \lambda_{nm}] = \psi(\alpha_{nm}) - \log \beta_{nm},\end{aligned} \tag{284}$$

where ψ is the digamma function (MacKay & Bauman Peto, 1994).

4.3. Weighted collapsed variational bound

Given fixed expected precisions, write

$$\mathbf{W}_n = \mathrm{diag}\big(w_{n1}, ..., w_{nM_n}\big). \tag{285}$$

Conditional on $\mathbf{W}_n$, the likelihood is Gaussian with noise covariance $\sigma^2 \mathbf{W}_n^{-1}$. This is exactly the PRISM setting with weighted observation noise, so the collapsed bound applies again.

Define the weighted matrices

$$
\begin{aligned}
\mathbf{A}_n^W &= \frac{1}{\sigma}\boldsymbol{\Phi}_n^\top \mathbf{W}_n^{1/2} \in \mathbb{R}^{R\times M_n} \\
\mathbf{B}_n^W &= \mathbf{I}_R + \mathbf{A}_n^W (\mathbf{A}_n^W)^\top \in \mathbb{R}^{R\times R} \\
\mathbf{v}_n^W &= \mathbf{A}_n^W \mathbf{W}_n^{1/2}\mathbf{y}_n \in \mathbb{R}^R.
\end{aligned}
\tag{286}
$$

In the unweighted case $\mathbf{W}_n = \mathbf{I}$ these reduce exactly to $\mathbf{A}_n$, $\mathbf{B}_n$, and $\mathbf{v}_n$ from (265).

The collapsed Gaussian contribution becomes

$$
\mathcal{L}_n^{\text{coll}}(\mathbf{W}_n) = \log \mathtt{Normal}(\mathbf{y}_n \mid \mathbf{0}, \mathbf{Q}_{nn} + \sigma^2 \mathbf{W}_n^{-1}) - \frac{1}{2\sigma^2}\operatorname{Tr}(\mathbf{W}_n(\mathbf{K}_{nn} - \mathbf{Q}_{nn})). \tag{287}
$$

The weighted predictive moments needed below are

$$
\begin{aligned}
m_{nm} &= \boldsymbol{\varphi}(t_{nm})^\top (\mathbf{B}_n^W)^{-1}\mathbf{v}_n^W, \\
v_{nm} &= k(t_{nm}, t_{nm}) - \boldsymbol{\varphi}(t_{nm})^\top \Big(\mathbf{I} - (\mathbf{B}_n^W)^{-1}\Big)\boldsymbol{\varphi}(t_{nm}).
\end{aligned}
\tag{288}
$$

The only new part is the expected log-likelihood under the random precisions. At sample (n, m), conditioning on λ_{nm} gives a Gaussian variance σ^2/λ_{nm}. Its normalization therefore contributes

$$
-\frac{1}{2}\log\left(\frac{\sigma^2}{\lambda_{nm}}\right) = -\frac{1}{2}\log\sigma^2 + \frac{1}{2}\log\lambda_{nm}, \tag{289}
$$

so after expectation the log-precision enters through $\ell_{nm} = \mathbb{E}[\log\lambda_{nm}]$. The quadratic penalty is weighted by $w_{nm} = \mathbb{E}[\lambda_{nm}]$. Substituting the moments from (288) gives, up to constants,

$$
\mathbb{E}[\log p(\mathbf{y}_n \mid f_n, \boldsymbol{\Lambda}_n)] = -\frac{1}{2}\sum_{m=1}^{M_n}\left(\log\sigma^2 - \ell_{nm} + \frac{w_{nm}}{\sigma^2}\Big((y_{nm} - m_{nm})^2 + v_{nm}\Big)\right), \tag{290}
$$

and the full per-dataset t-PRISM ELBO is

$$
\begin{aligned}
\mathcal{L}_n^t(\mathbf{W}_n, \ell_n) = \mathcal{L}_n^{\text{coll}}(\mathbf{W}_n) &+ \frac{1}{2}\sum_{m=1}^{M_n}\ell_{nm} \\
&- \sum_{m=1}^{M_n} D_{\text{KL}}(q_{nm}(\lambda_{nm}) \,\|\, \mathtt{Gamma}(\nu/2, \nu/2)).
\end{aligned}
\tag{291}
$$

Local CAVI updates for the precision variables Differentiating (291) gives the optimal values for tightening the ELBO:

$$
\alpha_{nm}^* = (\nu+1)/2, \quad \beta_{nm}^* = \frac{1}{2}\Big(\nu + \Big((y_{nm} - m_{nm})^2 + v_{nm}\Big)/\sigma^2\Big). \tag{292}
$$

The shape α is fixed once ν is chosen and does not depend on the data. In the deployed APLAWD model of this thesis we use the conservative choice $\nu = 1$, giving deliberately heavy tails for occasional gross detector failures or silent gaps. As noted in Dataset 2, a learned-ν extension was also tried, but did not improve held-out score over this simpler fixed choice. Only the rate β changes, increasing when the current residual $(y_{nm} - m_{nm})^2 + v_{nm}$ is large relative to σ^2. Large residuals therefore drive w_{nm} down in the next weighted collapse. That is the mechanism driving "robustness."

4.4. Training

The full t-PRISM objective is

$$\mathcal{L}^t = \sum_{n=1}^{N} \mathcal{L}_n^t. \tag{293}$$

For a minibatch $\mathcal{B}$ of complete datasets, the stochastic estimate is

$$\widehat{\mathcal{L}}^t = \frac{N}{|\mathcal{B}|} \sum_{n \in \mathcal{B}} \mathcal{L}_n^t. \tag{294}$$

For each dataset n in the minibatch:

1. initialize $w_{nm} = 1$ for all m,
2. repeat for a small fixed number of steps:
   - run the weighted collapsed GP computation with current $\mathbf{W}_n$, obtaining (m_{nm}, v_{nm}) via (288),
   - update $(\alpha_{nm}, \beta_{nm})$ via (292) and refresh (w_{nm}, ℓ_{nm}) via (284),
3. evaluate $\mathcal{L}_n^t$ via (291).

Then take a gradient step in the global parameters $(\mathbf{z}, \theta, \sigma^2, \nu)$. The Gamma parameters are local to the current minibatch pass and discarded afterwards. As in PRISM, no large per-dataset state persists across iterations. In the experiments of this thesis we use a fixed three inner CAVI steps, do not differentiate through those local updates, and optimize the global parameters with the same Adam setup and float32 arithmetic as in PRISM.

4.5. Projection to amplitude space

After training, each dataset is projected to a Gaussian amplitude posterior exactly as in Algorithm 3.5., now using the final converged weights from the inner loop. The posterior formulas are weighted versions of (272) and (273):

$$\begin{aligned} \mathbf{a}_n \mid \mathbf{y}_n &\sim \texttt{Normal}(\mathbf{m}_n, \mathbf{S}_n) \\ \mathbf{S}_n &= \left(\mathbf{B}_n^W\right)^{-1} \\ \mathbf{m}_n &= \left(\mathbf{B}_n^W\right)^{-1} \mathbf{v}_n^W. \end{aligned} \tag{295}$$

This again reuses quantities already computed during the inner loop. So t-PRISM returns the same interface object as PRISM,

$$\mathcal{D}_R = \{(\mathbf{m}_n, \mathbf{S}_n)\}_{n=1}^{N}, \tag{296}$$

but now obtained from a robust fit. The weight vectors $\{\mathbf{w}_n\}_{n=1}^{N}$ can be retained separately as auxiliary diagnostics when one wants local inlier scores in the original observation domain.

4.6. Summary

t-PRISM relaxes the Gaussian noise assumption of PRISM, which allows it to "robustly" learn a shared basis for a set of exemplars contaminated with outliers. This is achieved by replacing the Gaussian likelihood with a Student-t likelihood written as a Gaussian scale mixture.

Conditional on the local precision variables, the model is Gaussian again, so the same collapsed bound can be reused with weighted noise covariance $\sigma^2 \mathbf{W}_n^{-1}$. The local CAVI updates (292) then shrink the weights of observations whose residuals are too large for the current smooth fit. Training therefore remains a minibatch procedure over complete datasets, and the final output remains the same Gaussian collection $\mathcal{D}_R$ as in PRISM.

In this thesis the fitted weight vectors $\mathbf{w}_n$ are used both indirectly, through the robust amplitude posteriors, and directly, as local inlier scores on period trajectories. In Dataset 2 the generic fit is specialized to log-period tracks and the learned global parameter tuple is packaged as the pitch-track prior object $\mathcal{P}$. The motivation for developing t-PRISM came from needing to solve the following problems:

- learning a pitch-track model from the pitch-trajectory exemplars of APLAWD in Dataset 2;
- and segmenting pitch trajectories into voiced groups for the EGIFA dataset (Dataset 1).

Algorithm 5. q-GPLVM

As we established in Algorithm 3, PRISM projects a collection of variable-length waveforms into Gaussians in a shared amplitude space $\mathbb{R}^R$. More precisely, PRISM learns the shared feature map $\boldsymbol{\varphi}(t)$ and returns the Gaussian collection

$$\mathcal{D}_R = \{(\mathbf{m}_n, \mathbf{S}_n)\}_{n=1}^N \tag{297}$$

where waveform n is projected from data space to amplitude space through the posterior

$$\mathbf{a}_n \mid \mathbf{y}_n \sim \texttt{Normal}(\mathbf{m}_n, \mathbf{S}_n). \tag{298}$$

This induces the empirical amplitude-space density

$$p(\mathbf{a} \mid \mathcal{D}_R) = \frac{1}{N} \sum_{n=1}^N \texttt{Normal}(\mathbf{a} \mid \mathbf{m}_n, \mathbf{S}_n), \tag{299}$$

which is not yet a compact global generative model. The density (275) remains tied to the training set, and scoring a new waveform against it still requires keeping all N components around.

q-GPLVM compresses (275) into a compact generative prior. It stands for *quantized Gaussian process latent variable model.* It first learns a Bayesian GPLVM in a lower-dimensional manifold space, then quantizes the aggregated manifold posterior into a small Gaussian mixture, and finally pushes that mixture forward through the learned GP map back to amplitude space. The result is a mixture of K Gaussians in $\mathbb{R}^R$ that summarizes the training distribution without reference to individual training waveforms. After composition with the PRISM synthesis map, this becomes a compact mixture of BLR models in data space. The key idea is that a nonlinear manifold model can be converted into a mixture of linear Gaussian models via moment matching, and that this mixture is a cheap surrogate model for waveforms learned by PRISM.

5.1. Prior art

This idea was first introduced by Nickisch & Rasmussen (2010), who showed that fitting a GMM in the latent space of a GPLVM and pushing the mixture through the GP map via the moment-matching formulas of Girard et al. (2002) yields a tractable mixture density in observation space. They also showed that this improved generalization capability compared to a GMM in the raw high-dimensional input space in the context of density scoring. The main difference here is that for q-GPLVM we “upgrade” their usage of vanilla MAP GPLVM and GMM to Bayesian GPLVM and XDGMM, respectively (Bovy et al., 2011; Titsias & Lawrence, 2010). This is done in the interest of retaining enough variability in the learned components and contributes to downstream uncertainty calibration.

5.2. Data, amplitude, and manifold space

q-GPLVM is an input-output pipeline with three levels.

Level 0: data space This is the space of observed waveforms. The input to PRISM is the collection

$$\mathcal{D} = \{(\mathbf{t}_n, \mathbf{y}_n)\}_{n=1}^{N}, \tag{300}$$

with waveform n sampled at its own grid $\mathbf{t}_n$ and values $\mathbf{y}_n$. After all compression steps, q-GPLVM returns a prior that again lives in this same variably-dimensioned space.

Level 1: amplitude space PRISM maps each dataset $(\mathbf{t}_n, \mathbf{y}_n) \in \mathcal{D}$ to a Gaussian in amplitude space $\mathbb{R}^R$ and collects those posteriors as $\mathcal{D}_R$ in (297). Its shared feature map $\boldsymbol{\varphi}(t)$ (262) implies the linear synthesis model

$$y(t) \approx \boldsymbol{\varphi}(t)^\top \mathbf{a}, \quad \mathbf{a} \sim \texttt{Normal}(\mathbf{0}, \mathbf{I}), \tag{301}$$

so Level 1 is summarized exactly by the density $p(\mathbf{a} \mid \mathcal{D}_R)$ of (275). This is a *linear* dimensionality reduction: variable-length waveforms become Gaussian amplitude coordinates in fixed dimension R.

Level 2: manifold space The Bayesian GPLVM introduces manifold coordinates $\mathbf{x}_n \in \mathbb{R}^Q$ with $Q \ll R$ and learns the posterior

$$\mathcal{D}_Q = \{(\boldsymbol{\mu}_n^x, \mathbf{S}_n^x)\}_{n=1}^{N}, \tag{302}$$

equivalently

$$q(\mathbf{X} \mid \mathcal{D}_Q) = \prod_{n=1}^{N} \texttt{Normal}(\mathbf{x}_n \mid \boldsymbol{\mu}_n^x, \mathbf{S}_n^x). \tag{303}$$

Here the covariance $\mathbf{S}_n^x$ is diagonal, as in the mean-field variational family of (Titsias & Lawrence, 2010). This is a *nonlinear* dimensionality reduction from amplitude space to manifold space. Quantization then replaces the aggregated posterior by a small mixture

$$q(\mathbf{x} \mid \mathcal{M}_Q) = \sum_{k=1}^{K} \pi_k \, \texttt{Normal}(\mathbf{x} \mid \mathbf{c}_k, \mathbf{C}_k), \qquad \sum_{k=1}^{K} \pi_k = 1, \tag{304}$$

which is later pushed back to Level 1 as the amplitude-space density $p(\mathbf{a} \mid \mathcal{M}_R)$ and then composed with PRISM via (333) to return to Level 0.

5.3. The Bayesian GPLVM

The theoretical foundation is the Bayesian GP latent variable model of Titsias & Lawrence (2010), which we now review. We refer to (Titsias & Lawrence, 2010) for the full derivations and details; here we record only the pieces needed later for quantization and push-forward.

Observations and notation PRISM returns the Gaussian amplitude collection $\mathcal{D}_R$. The Bayesian GPLVM, however, is formulated only for point observations in amplitude space, subject to homoscedastic observation noise. We therefore apply one global affine whitening transform

$$\tilde{\mathbf{a}} = \mathbf{W}(\mathbf{a} - \mathbf{b}), \tag{305}$$

with empirical center

$$\mathbf{b} = \frac{1}{N} \sum_{n=1}^{N} \mathbf{m}_n, \tag{306}$$

and a fixed global whitening matrix $\mathbf{W}$ chosen once from the PRISM amplitude cloud. To avoid a copious amount of tildes, we continue to write $\mathbf{a}_n$ for the transformed posterior mean and fit the standard BGPLVM to these point estimates.

Collect them into the amplitude matrix

$$\mathbf{A} = \begin{pmatrix} \mathbf{a}_1^\top \\ \vdots \\ \mathbf{a}_N^\top \end{pmatrix} \in \mathbb{R}^{N \times R}, \tag{307}$$

so that a_{nr} denotes the r-th amplitude coordinate of waveform n. This matrix plays the role of the observed data matrix in standard GPLVM usage.

Latent variable model For each amplitude coordinate $r = 1, \ldots, R$, introduce an independent latent function $g_r(\cdot)$ with shared GP prior

$$g_r(\cdot) \sim \texttt{GaussianProcess}(0, k(\cdot, \cdot)), \tag{308}$$

and an observation model

$$a_{nr} \mid g_r, \mathbf{x}_n \sim \texttt{Normal}(g_r(\mathbf{x}_n), \beta^{-1}), \tag{309}$$

where $\mathbf{x}_n \in \mathbb{R}^Q$ is the manifold coordinate of waveform n and β^{-1} is the learned isotropic GPLVM noise precision. The latent prior is isotropic:

$$p(\mathbf{x}_n) = \texttt{Normal}\big(\mathbf{0}, \mathbf{I}_Q\big). \tag{310}$$

The kernel is ARD squared exponential (MacKay, 1998),

$$k(\mathbf{x}, \mathbf{x}') = \sigma_g^2 \exp\left(-\frac{1}{2} \sum_{q=1}^{Q} \alpha_q \big(x_q - x_q'\big)^2\right), \tag{311}$$

with per-dimension inverse lengthscales α_q. When the model drives $\alpha_q \to 0$ during training, dimension q is effectively switched off, providing automatic selection of the intrinsic latent

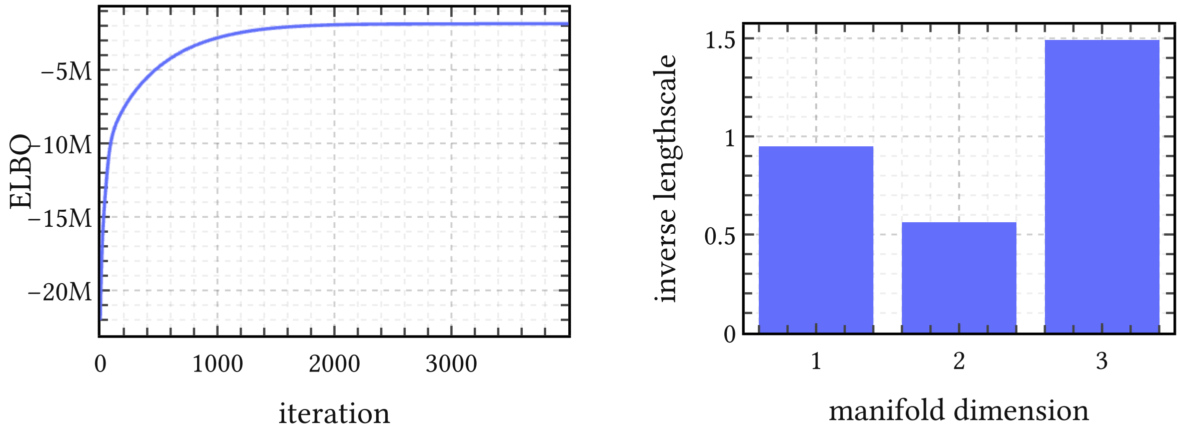


Figure 53 | **Representative objective and ARD on EGIFA.** Left: ELBO trace for one typical q-GPLVM run. Right: the learned inverse lengthscales in manifold space. The optimization stabilizes cleanly, and the ARD kernel keeps all three manifold dimensions active, though not equally: the third carries the sharpest variation and the second is the smoothest.

dimensionality. In the q-GPLVM fits to DGF waveforms we set $Q = 3$ and all three manifold dimensions remain active, though to different degrees as seen in Figure 53.

Variational inference Integrating out the manifold coordinates from the marginal likelihood is intractable because they enter the kernel matrix nonlinearly. Following (Titsias & Lawrence, 2010), introduce P inducing inputs $\mathbf{Z} \in \mathbb{R}^{P \times Q}$ in manifold space and optimize the factorized Gaussian posterior (302), with diagonal latent covariance

$$\mathbf{S}_n^x = \operatorname{diag}\left(S_{n1}^x, ..., S_{nQ}^x\right). \tag{312}$$

The integer P is the number of inducing points; importantly later it also bounds the rank of the structured covariance terms produced by push-forward. For these point observations, the collapsed variational bound has the standard form

$$\mathcal{L}^q = \tilde{\mathcal{L}}^q - D_{\mathrm{KL}}(q(\mathbf{X}) \parallel p(\mathbf{X})), \tag{313}$$

where $\tilde{\mathcal{L}}^q$ is the sparse-GP expected log-likelihood and the KL term is analytic. The inducing variables are again collapsed analytically, exactly as in Algorithm 3. The variational parameters $\{\boldsymbol{\mu}_n^x, \mathbf{S}_n^x\}_{n=1}^N$, the inducing inputs $\mathbf{Z}$, and the kernel and noise parameters $\left\{\sigma_g^2, \alpha_1, ..., \alpha_Q, \beta\right\}$ are all optimized jointly by gradient ascent on (313).

The Ψ statistics All tractable computation in the Bayesian GPLVM reduces to these expectations of kernel quantities under $q(\mathbf{x}_n)$:

$$
\begin{aligned}
\psi_0 &= \sum_{n=1}^{N} \mathbb{E}_{q(\mathbf{x}_n)}[k(\mathbf{x}_n, \mathbf{x}_n)], \\
[\mathbf{\Psi}_1]_{np} &= \mathbb{E}_{q(\mathbf{x}_n)}\left[k\left(\mathbf{x}_n, \mathbf{z}_p\right)\right], \\
[\mathbf{\Psi}_2]_{pp'} &= \sum_{n=1}^{N} \mathbb{E}_{q(\mathbf{x}_n)}\left[k\left(\mathbf{x}_n, \mathbf{z}_p\right)k\left(\mathbf{x}_n, \mathbf{z}_{p'}\right)\right],
\end{aligned}
\tag{314}
$$

with $\mathbf{\Psi}_1 \in \mathbb{R}^{N \times P}$ and $\mathbf{\Psi}_2 \in \mathbb{R}^{P \times P}$. Note that $\mathbf{\Psi}_2$ is *not* $\mathbf{\Psi}_1^\top \mathbf{\Psi}_1$: the integration couples the two kernel evaluations through the shared uncertain input $\mathbf{x}_n$.

For the ARD RBF kernel (311) with $q(\mathbf{x}_n) = \texttt{Normal}(\boldsymbol{\mu}_n^x, \mathbf{S}_n^x)$, (Titsias & Lawrence, 2010) show that all three statistics admit closed forms. The first statistic is trivial: $\psi_0 = N\sigma_g^2$ since $k(\mathbf{x}, \mathbf{x}) = \sigma_g^2$ for any $\mathbf{x}$. For $\mathbf{\Psi}_1$,

$$
[\mathbf{\Psi}_1]_{np} = \sigma_g^2 \prod_{q=1}^{Q} \left(\alpha_q S_{nq}^x + 1\right)^{-1/2} \exp\left(-\frac{\alpha_q\left(\mu_{nq}^x - z_{pq}\right)^2}{2\left(\alpha_q S_{nq}^x + 1\right)}\right),
\tag{315}
$$

which is a kernel evaluation at $\boldsymbol{\mu}_n^x$ with inflated lengthscales $\alpha_q^{-1} + S_{nq}^x$. When $\mathbf{S}_n^x \to \mathbf{0}$ (no input uncertainty), this recovers the standard kernel evaluation $k\left(\boldsymbol{\mu}_n^x, \mathbf{z}_p\right)$. $\mathbf{\Psi}_2$ is likewise explicit:

$$
[\mathbf{\Psi}_2]_{pp'} = \sigma_g^4 \sum_{n=1}^{N} \prod_{q=1}^{Q} \left(2\alpha_q S_{nq}^x + 1\right)^{-1/2} \exp\left(-\alpha_q \frac{\left(z_{pq} - z_{p'q}\right)^2}{4} - \alpha_q \frac{\left(\mu_{nq}^x - \frac{z_{pq} + z_{p'q}}{2}\right)^2}{2\alpha_q S_{nq}^x + 1}\right)
\tag{316}
$$

Later, in the push-forward step, the quantized manifold components carry full covariances $\mathbf{C}_k$ rather than diagonal $\mathbf{S}_n^x$. The same Gaussian-integral mechanism extends to any Gaussian input covariance, so no new derivations are needed (Damianou et al., 2014).

5.4. Quantization

After training, the Bayesian GPLVM provides a variational posterior (302) for each training waveform. Define the empirical manifold density induced by $\mathcal{D}_Q$ as

$$
q\left(\mathbf{x} \mid \mathcal{D}_Q\right) = \frac{1}{N} \sum_{n=1}^{N} \texttt{Normal}(\mathbf{x} \mid \boldsymbol{\mu}_n^x, \mathbf{S}_n^x).
\tag{317}
$$

This is already a generative model in manifold space, but with N components it still encodes the entire training set, and is bound to generalize poorly. Quantization replaces it by the compact K-component approximation ($K \ll N$)

$$
q\left(\mathbf{x} \mid \mathcal{M}_Q\right) = \sum_{k=1}^{K} \pi_k \, \texttt{Normal}(\mathbf{x} \mid \mathbf{c}_k, \mathbf{C}_k), \qquad \sum_{k=1}^{K} \pi_k = 1,
\tag{318}
$$

where $\mathbf{c}_k \in \mathbb{R}^Q$ and $\mathbf{C}_k \in \mathbb{R}^{Q \times Q}$ are the mean and full covariance of component k. Full covariances are retained because local curvature and anisotropy in manifold space are precisely the structure we wish to preserve for downstream uncertainty calibration.

GMM with uncertain inputs A standard GMM fit to the posterior means $\{\boldsymbol{\mu}_n^x\}$ alone would ignore the latent uncertainty and overfit to noise in the mean estimates. The standard algorithm to take into account the posterior uncertainty in $\{\mathbf{S}_n^x\}$ is XDGMM, or extreme deconvolution Gaussian mixture models (Bovy et al., 2011). XDGMM is a variant of EM for Gaussian mixtures that accounts for known, heterogeneous Gaussian uncertainty on the inputs, and is easily fit to (317) by treating each posterior mean $\boldsymbol{\mu}_n^x$ as an uncertain observation with known covariance $\mathbf{S}_n^x$.

The output of the quantization step is the compact object

$$\mathcal{M}_Q = \{(\pi_k, \mathbf{c}_k, \mathbf{C}_k)\}_{k=1}^K, \tag{319}$$

which parameterizes the mixture $q(\mathbf{x} \mid \mathcal{M}_Q)$ in (318) and stands in for the entire training collection. On EGIFA, the learned manifold for $Q = 3$ is already visibly organized by open quotient, and the fitted ellipsoids with $K = 5$ then provide a compact cover of that structure, as shown in Figure 54.

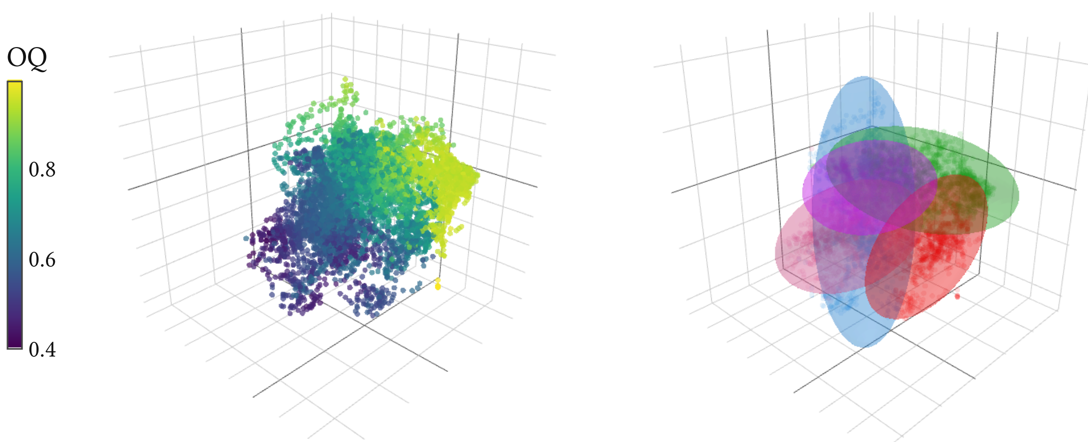


Figure 54 | **Manifold structure and quantization on EGIFA.** Left: the three-dimensional manifold space learned by q-GPLVM, with each waveform colored by open quotient. Right: the $K = 5$ fitted Gaussian ellipsoids used for quantization in that same manifold space (319). The left panel shows that open quotient varies smoothly over the manifold rather than appearing as disconnected clusters, while the right panel shows how XDGMM compresses the empirical mixture $q(\mathbf{x} \mid \mathcal{D}_Q)$ of (317) into the $K = 5$ approximation $q(\mathbf{x} \mid \mathcal{M}_Q)$ of (318).

5.5. Push-forward

We now have the quantized manifold mixture $q(\mathbf{x} \mid \mathcal{M}_Q)$ from (318) and a trained GP map from manifold space $\mathbb{R}^Q$ to amplitude space $\mathbb{R}^R$. Each component of $\mathcal{M}_Q$ must be pushed through that map. One can either sample from the manifold mixture and propagate those samples through the nonlinear GP directly (Figure 55), or construct the moment-matched Gaussian approximation in amplitude space first and then synthesize waveforms from that. The latter method allows deriving surrogate priors directly in data space.

Recovered inducing parameters During training, the inducing variables were eliminated analytically from the bound. For the push-forward we need them back. Since the GPLVM uses R independent GPs that share the same kernel but have independent function draws, the collapsed sparse GP posterior of (Titsias, 2009) gives a shared inducing covariance

$$\hat{\mathbf{S}}_u = \mathbf{K}_{ZZ}(\mathbf{K}_{ZZ} + \beta\mathbf{\Psi}_2^{\mathrm{tr}})^{-1}\mathbf{K}_{ZZ} \in \mathbb{R}^{P\times P}, \tag{320}$$

which depends only on the kernel and the Ψ statistics, and is therefore the same for all output dimensions. The inducing means, however, depend on the observed amplitudes for each dimension r separately. Collecting them into the matrix $\widehat{\mathbf{M}} \in \mathbb{R}^{P\times R}$ gives

$$\widehat{\mathbf{M}} = \beta\,\mathbf{K}_{ZZ}(\mathbf{K}_{ZZ} + \beta\mathbf{\Psi}_2^{\mathrm{tr}})^{-1}[\mathbf{\Psi}_1^{\mathrm{tr}}]^{\top}\mathbf{A}, \tag{321}$$

where $\mathbf{\Psi}_1^{\mathrm{tr}}$ and $\mathbf{\Psi}_2^{\mathrm{tr}}$ denote the trained versions of the statistics defined in (314) and $\mathbf{A} \in \mathbb{R}^{N\times R}$ is the amplitude matrix. Thus $\widehat{\mathbf{M}}$ encodes how the training amplitudes project onto the inducing basis, one column per output dimension.

Moment matching For one manifold component $\mathbf{x} \sim \mathtt{Normal}(\mathbf{c}, \mathbf{C})$, the induced distribution of the amplitude vector

$$\mathbf{a} = (g_1(\mathbf{x}), ..., g_R(\mathbf{x}))^{\top} \tag{322}$$

under the GP is non-Gaussian due to the nonlinearity of the map. The moment-matched Gaussian approximation $\mathtt{Normal}(\boldsymbol{\mu}_a, \mathbf{S}_a)$ whose first two moments equal those of the true push-forward was derived by Girard et al. (2002) for GP prediction at uncertain inputs, applied to GPLVM density modeling by Nickisch & Rasmussen (2010), and is standard in GP uncertainty propagation (Deisenroth & Rasmussen, 2011). We use the sparse-GP variant of those formulas, where the Ψ statistics of Damianou et al. (2014) replace the full-GP kernel expectations.

The required expectations are the same Ψ statistics of (314), now evaluated at the single GMM component $(\mathbf{c}, \mathbf{C})$ rather than summed over the training posteriors. Since the ARD RBF formula (315) remains valid for any Gaussian input, including one with full covariance $\mathbf{C}$, no new derivations are needed. We write $\boldsymbol{\psi}_1(\mathbf{c}, \mathbf{C}) \in \mathbb{R}^P$ and $\mathbf{\Psi}_2(\mathbf{c}, \mathbf{C}) \in \mathbb{R}^{P\times P}$ for the per-component versions of these statistics.

Mean The sparse GP predictive mean at a fixed manifold point $\mathbf{x}$ is $\widehat{\mathbf{M}}^\top \mathbf{K}_{ZZ}^{-1} \mathbf{k}_Z(\mathbf{x})$, where $\mathbf{k}_Z(\mathbf{x}) = (k(\mathbf{x}, \mathbf{z}_1), ..., k(\mathbf{x}, \mathbf{z}_P))^\top$. Taking the expectation over $\mathbf{x} \sim \texttt{Normal}(\mathbf{c}, \mathbf{C})$ replaces $\mathbf{k}_Z(\mathbf{x})$ by $\boldsymbol{\psi}_1(\mathbf{c}, \mathbf{C})$:

$$\boldsymbol{\mu}_a = \widehat{\mathbf{M}}^\top \mathbf{K}_{ZZ}^{-1} \boldsymbol{\psi}_1(\mathbf{c}, \mathbf{C}) \in \mathbb{R}^R. \tag{323}$$

Covariance The law of total variance gives

$$\mathbf{S}_a = \underbrace{\mathbb{E}_{\mathbf{x}}[\texttt{Cov}(\mathbf{a} \mid \mathbf{x})]}_{\text{averaged predictive variance}} + \underbrace{\texttt{Cov}_{\mathbf{x}}(\mathbb{E}[\mathbf{a} \mid \mathbf{x}])}_{\text{input uncertainty}}. \tag{324}$$

The first term averages the GP predictive variance (which is the same for every amplitude coordinate r since all outputs share the kernel) under the input distribution. Given the latent function values at a fixed $\mathbf{x}$, the output coordinates are conditionally independent, so this term is isotropic:

$$\mathbb{E}_{\mathbf{x}}[\texttt{Cov}(\mathbf{a} \mid \mathbf{x})] = \gamma \, \mathbf{I}_R, \tag{325}$$

where

$$\gamma = \psi_0 + \beta^{-1} - \operatorname{Tr}(\mathbf{K}_{ZZ}^{-1} \boldsymbol{\Psi}_2(\mathbf{c}, \mathbf{C})) + \operatorname{Tr}\left(\mathbf{K}_{ZZ}^{-1} \hat{\mathbf{S}}_u \mathbf{K}_{ZZ}^{-1} \boldsymbol{\Psi}_2(\mathbf{c}, \mathbf{C})\right). \tag{326}$$

The four contributions are respectively: the prior variance at $\mathbf{x}$, the observation noise, minus the variance explained by the inducing inputs, plus the posterior uncertainty on those inducing inputs. Thus the isotropic term $\gamma \, \mathbf{I}_R$ collects averaged predictive uncertainty and observation noise. Cross-output correlations reappear only after integrating over inducing-variable uncertainty and latent uncertainty, which is exactly what the second term below captures.

The second term in (324) captures how the predictive mean varies as $\mathbf{x}$ moves over the component. It introduces cross-output correlations via

$$\texttt{Cov}_{\mathbf{x}}(\mathbb{E}[\mathbf{a} \mid \mathbf{x}]) = \widehat{\mathbf{M}}^\top \mathbf{K}_{ZZ}^{-1} (\boldsymbol{\Psi}_2(\mathbf{c}, \mathbf{C}) - \boldsymbol{\psi}_1(\mathbf{c}, \mathbf{C}) \boldsymbol{\psi}_1(\mathbf{c}, \mathbf{C})^\top) \mathbf{K}_{ZZ}^{-1} \widehat{\mathbf{M}}, \tag{327}$$

which has rank at most P.

Combining, the moment-matched covariance is

$$\mathbf{S}_a = \gamma \, \mathbf{I}_R + \widehat{\mathbf{M}}^\top \mathbf{K}_{ZZ}^{-1} (\boldsymbol{\Psi}_2(\mathbf{c}, \mathbf{C}) - \boldsymbol{\psi}_1(\mathbf{c}, \mathbf{C}) \boldsymbol{\psi}_1(\mathbf{c}, \mathbf{C})^\top) \mathbf{K}_{ZZ}^{-1} \widehat{\mathbf{M}}. \tag{328}$$

The push-forward covariance thus decomposes into an isotropic term, which reflects averaged GP uncertainty and observation noise, and a structured term, which carries the geometry induced by the manifold component.

The mixture weights The weights π_k of the manifold-space mixture $q(\mathbf{x} \mid \mathcal{M}_Q)$ are a prior on manifold space. Under the push-forward, a simple approximation is to retain those same

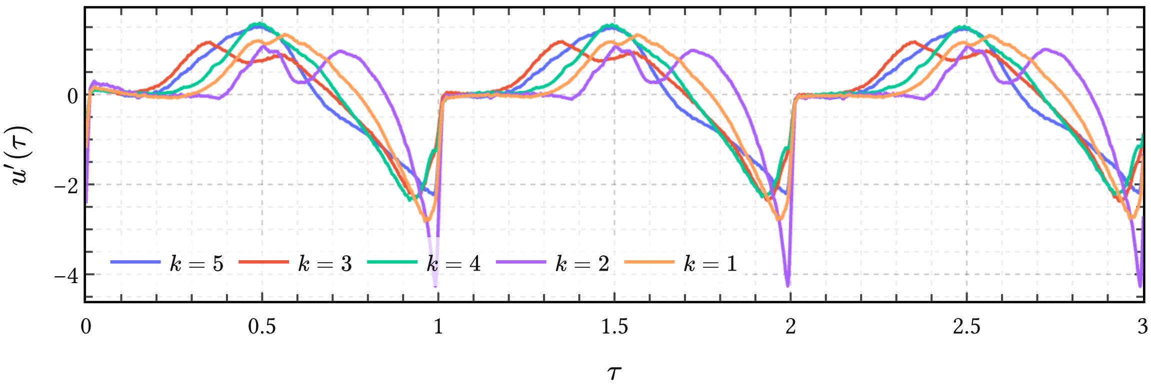


Figure 55 | **Direct samples from the quantized manifold mixture in data space** for the data in Figure 54. Here one first samples a component and manifold coordinate from $q(\mathbf{x} \mid \mathcal{M}_Q)$ in (318), then pushes that sample through the GPLVM map described in Algorithm 5.5.2. and finally through the PRISM synthesis map (301) to Level 0. This shows the nonlinear generative mechanism itself, before moment matching replaces it by a Gaussian approximation in amplitude space.

weights in amplitude space:[27] Define the corresponding compact amplitude-space object as

$$\mathcal{M}_R = \{(\pi_k, \boldsymbol{\mu}_k^a, \mathbf{S}_k^a)\}_{k=1}^K. \tag{329}$$

The induced amplitude-space density is then

$$p(\mathbf{a} \mid \mathcal{M}_R) \approx \sum_{k=1}^{K} \pi_k \, \texttt{Normal}(\mathbf{a} \mid \boldsymbol{\mu}_k^a, \mathbf{S}_k^a), \tag{330}$$

where $(\boldsymbol{\mu}_k^a, \mathbf{S}_k^a)$ is the moment-matched push-forward of the k-th component via (323) and (328).

5.6. Composing with PRISM

The density $p(\mathbf{a} \mid \mathcal{M}_R)$ of (330) lives in the PRISM amplitude space $\mathbb{R}^R$. To obtain a prior over observations on a grid $\mathbf{t}$, let $\boldsymbol{\Phi}(\mathbf{t}) \in \mathbb{R}^{M \times R}$ denote the design matrix obtained by evaluating the PRISM feature map (262) on that grid. Then PRISM synthesizes via the linear map

$$\mathbf{y} = \boldsymbol{\Phi}(\mathbf{t})\mathbf{a} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \texttt{Normal}(\mathbf{0}, \sigma^2 \mathbf{I}_M), \tag{331}$$

Because this step is linear, pushing a Gaussian in amplitude space through it preserves Gaussianity:

$$\mathbf{y} \mid \mathbf{t}, k, \mathcal{M}_R \sim \texttt{Normal}(\boldsymbol{\Phi}(\mathbf{t})\boldsymbol{\mu}_k^a, \quad \boldsymbol{\Phi}(\mathbf{t})\mathbf{S}_k^a\boldsymbol{\Phi}(\mathbf{t})^\top + \sigma^2 \mathbf{I}_M), \tag{332}$$

and the full prior over data space is

[27] More specialized analyses of pushing Gaussians through nonlinear maps, including higher-order corrections beyond moment matching, are given by (Zhang & Shin, 2021) and (Zhang & Shin, 2022).

$$p(\mathbf{y} \mid \mathbf{t}, \mathcal{M}_R) \approx \sum_{k=1}^{K} \pi_k \, \texttt{Normal}(\mathbf{y} \mid \mathbf{\Phi}(\mathbf{t})\boldsymbol{\mu}_k^a, \mathbf{\Phi}(\mathbf{t})\mathbf{S}_k^a\mathbf{\Phi}(\mathbf{t})^\top + \sigma^2\mathbf{I}_M). \tag{333}$$

This is a finite mixture of BLR models in data space, all sharing the same synthesis map $\boldsymbol{\varphi}(t)$ inherited from PRISM.

Samples from (333) are shown in Figure 56 for the same data from Figure 54 where $Q = 3$ and $K = 5$. It is seen that each component k effectively learns a DGF waveform type with variations in open quotient (OQ) and GCI spike intensity. Compared with Figure 55, these draws are visibly rougher because the nonlinear push-forward has been replaced by a Gaussian approximation in amplitude space, whose covariance includes the extra isotropic uncertainty term of (326) to compensate for the approximation.

Low-rank structure In amplitude space, (328) is the sum of a full-rank isotropic term $\gamma\,\mathbf{I}_R$ and a structured term of rank at most P. The latent dimension Q controls the geometry of that structured term through the kernel and the posterior over manifold positions, but it does not itself give a literal rank bound after passage through the inducing-feature map.

After composition with PRISM on a fixed grid $\mathbf{t}$, the data-space covariance becomes

$$\mathbf{\Phi}(\mathbf{t})\mathbf{S}_k^a\mathbf{\Phi}(\mathbf{t})^\top + \sigma^2\mathbf{I}_M = \sigma^2\mathbf{I}_M + \gamma\,\mathbf{\Phi}(\mathbf{t})\mathbf{\Phi}(\mathbf{t})^\top + \mathbf{\Phi}(\mathbf{t})\mathbf{S}_{\text{str}}\mathbf{\Phi}(\mathbf{t})^\top, \tag{334}$$

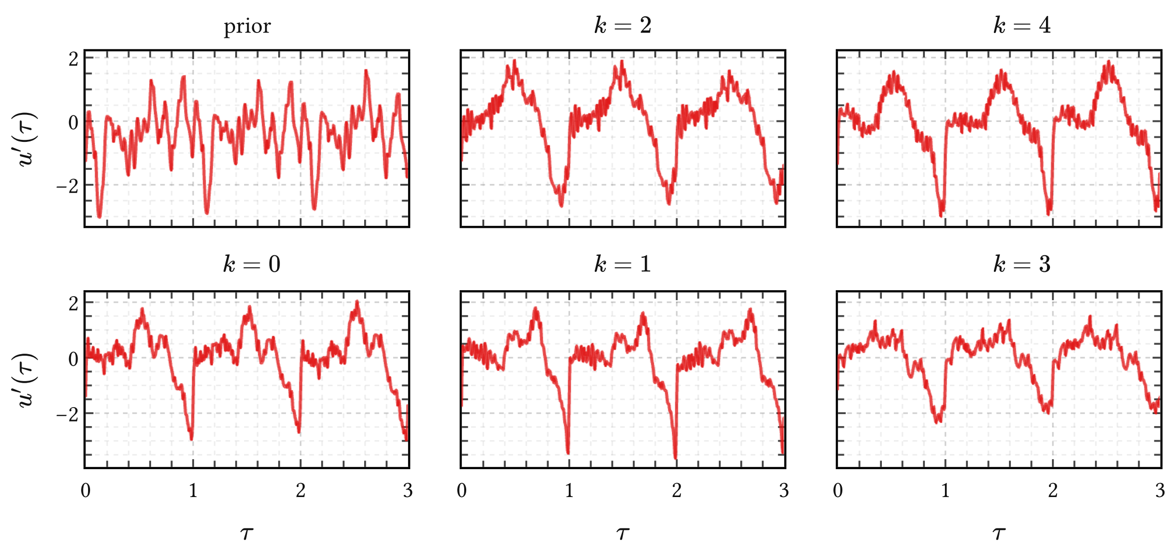


Figure 56 | **Samples from the moment-matched prior in data space** for the data in Figure 54. The “prior” panel at the top left draws from the PRISM prior (331) on the displayed grid, which are just generic combinations of the learned PRISM basis. The other panels condition on individual mixture components in (333) and have learned to model the density in amplitude space, each mixture component k modeling different types of DGF waveforms. Note how much more structured and DGF-like these are in comparison to the “prior” panel.

where $\mathbf{S}_{\text{str}}$ denotes the structured part of (328). Hence the only full-rank term in data space is the nugget $\sigma^2 \mathbf{I}_M$, while the nontrivial covariance structure remains low rank, with rank bounded by $\min(R, P)$.

So the dimensionality reduction story is this: PRISM reduces the original function space to amplitude space of dimension R; the GPLVM then organizes those amplitudes through a manifold of dimension Q; and the final waveform prior keeps only a low-rank structured covariance on top of the nugget. The entire computation required to produce that structure is amortized at training time.

5.7. Summary

q-GPLVM is a three-stage compression pipeline that converts the PRISM object $\mathcal{D}_R$ into a compact prior over waveforms.

1. Bayesian GPLVM discovers the low-dimensional manifold structure of the PRISM amplitude means, producing the posterior collection $\mathcal{D}_Q$ of (302) in a Q-dimensional manifold space.
2. XDGMM quantizes the aggregated posterior into the compact parameter set $\mathcal{M}_Q$ of (319), using the latent-space uncertainties $\mathbf{S}_n^x$ in an uncertain-inputs formulation.
3. Moment matching pushes each component of $\mathcal{M}_Q$ through the trained GP map, giving the compact amplitude-space object $\mathcal{M}_R$ of (329) and the density $p(\mathbf{a} \mid \mathcal{M}_R)$ of (330). Because PRISM synthesis is linear, this extends immediately to the analytically tractable data-space prior $p(\mathbf{y} \mid \mathbf{t}, \mathcal{M}_R)$ in (333).

The entire computation is amortized: once trained, only the small parameter sets $\mathcal{M}_Q$ and $\mathcal{M}_R$ must be retained. In this thesis PRISM followed by q-GPLVM is used to learn priors for DGF waveforms directly from many synthetic examples. These priors are mixtures of cheap BLR surrogates which are used as generative models in their own right or as kernel mixture priors in IKLP for GIF.

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Appendices

Appendix A. Gaussian processes as Bayesian linear models

Gaussian processes (GPs) appear throughout this thesis, but mostly in one operative form: as Gaussian priors on the coefficients of a finite feature expansion. This is the viewpoint that matters for the main text. Every GP that enters the thesis is either already written in sparse finite-rank form or immediately approximated that way, so inference reduces to Bayesian linear algebra (Murphy, 2022; Rasmussen & Williams, 2006).

The basic object is a Bayesian linear model,

$$f_R(t) = \boldsymbol{\varphi}(t)^\top \mathbf{a}, \quad \mathbf{a} \sim \texttt{Normal}(\mathbf{0}, \mathbf{I}_R), \tag{335}$$

where $\boldsymbol{\varphi}(t) \in \mathbb{R}^R$ is a feature map and R is the rank. For any grid $\mathbf{t} = (t_1, \cdots, t_M)$ this gives

$$\mathbf{f}_R = f_R(\mathbf{t}) = \mathbf{\Phi}(\mathbf{t})\mathbf{a}, \tag{336}$$

with design matrix $\mathbf{\Phi}(\mathbf{t})$ having rows $\boldsymbol{\varphi}(t_m)^\top$. It follows immediately that

$$\mathbf{f}_R \sim \texttt{Normal}\left(\mathbf{0}, \mathbf{\Phi}(\mathbf{t})\mathbf{\Phi}(\mathbf{t})^\top, \right), \tag{337}$$

so the induced covariance is

$$k_R(t, t') = \boldsymbol{\varphi}(t)^\top \boldsymbol{\varphi}(t'). \tag{338}$$

This is the kernel view: the same finite-rank Bayesian linear model, expressed through the covariance it induces on function values. An especially useful special case is the Karhunen–Loève form

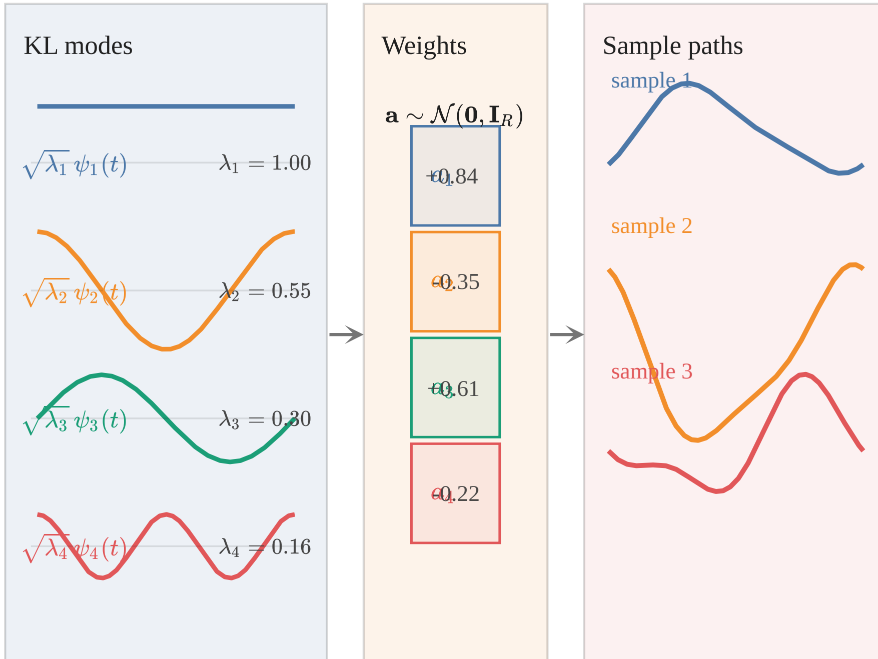
$$f_R(t) = \sum_{r=1}^{R} a_r \sqrt{\lambda_r} \psi_r(t), \quad a_r \sim \texttt{Normal}(0, 1) \quad (r = 1, ..., R), \tag{339}$$

where the functions ψ_r are orthonormal and the coefficients $\lambda_r \geq 0$ set the variance carried by each mode. Then

$$k_R(t, t') = \sum_{r=1}^{R} \lambda_r \psi_r(t) \psi_r(t'). \tag{340}$$

For finite R this expansion is a degenerate or sparse GP. If (340) has a well-defined limit when $R \to \infty$ it is called a nondegenerate GP.

The conceptual direction in this thesis is mostly from left to right in Figure 57: start with a finite-rank Gaussian model in weight space, then read off the corresponding kernel. The pitch-track prior (Dataset 2), the source priors (Algorithm 3, Algorithm 4), the inducing machinery (Algorithm 5), and the low-rank kernel bank (Algorithm 1) all follow this pattern.



$$f_R(t) = \sum_{r=1}^{R} a_r \sqrt{\lambda_r}\, \psi_r(t)$$

finite R: sparse / degenerate GP; if the KL expansion converges as $R \to \infty$, one obtains an ordinary GP

Figure 57 | **Finite-rank Gaussian priors on functions as Bayesian linear models.** Left: the Karhunen-Loève modes $\sqrt{\lambda_r}\psi_r(t)$. Middle: Gaussian coefficients $\mathbf{a} \sim \texttt{Normal}(\mathbf{0}, \mathbf{I}_R)$. Right: sample paths obtained by the linear combination $f_{R(t)} = \sum_r a_r \sqrt{\lambda_r}\psi_{r(t)}$. The induced kernel is $k_R(t, t') = \sum_r \lambda_r \psi_r(t)\psi_r(t')$. In this thesis that finite-rank object is the operative form of a GP.

The reader can therefore safely think of every GP in this thesis as a Bayesian linear model in disguise.

Appendix B. Regularization and priors

As its name suggests, GIF belongs to the class of *inverse problems* (Le Besnerais & Giovannelli, 2010). Inverse problems confront us when we have observations of some effects [noisy oservations of $s(t)$ in our case] of which we want to infer the causes $u(t)$ and $h(t)$ within the context of some model $s(t) = u(t) * h(t) * r(t)$.

In the applied sciences it is very often the case for inverse problems that the observed data are quite consistent with a broad and sometimes colorful class of possible solutions rather than a unique and logically deductible solution. From the viewpoint of pure mathematics therefore such inverse problems are often deemed 'ill-posed', but perhaps a more productive viewpoint[28] is that such problems are simply underdetermined.

Underdetermined problems are ubiquitous in a wide range of applications, from medical diagnosis (what could be causing the patient's persistent cough) to astronomy (what is perturbing the orbit of Uranus). Bayesian theory by itself does not require any particular structure on the "feasible set" of plausible solutions and as such underdetermined problems have no special status within it, nor pose any special challenge to it (Jaynes, 2003). In practice however, we would like to have a more well determined solution to underdetermined problems as our computing budgets (and patience) are limited. For example, it would most certainly help if we can already discard those solutions that lack to some degree a set of required properties we established beforehand. This is where *regularization* and *priors* come into play.

In general, *regularization* is the process of constraining the solution space of some problem to obtain a more stable and perhaps unique solution; in other words, to convert an underdetermined problem into something more 'well-determined' and, hopefully, more amenable to computational optimization. For example, ridge regression is a widely used regularization technique to numerically stabilize linear regression in high dimensions.[29]

In the Bayesian framework regularization is imposed simply through careful assignment of the probability distributions describing a specific problem – of which the bulk of the effort is traditionally concentrated on the prior probabilities, or *priors* (Sivia & Skilling, 2006). Assuming our prior information about the problem at hand is correct and the assignment of the priors accurately reflect that information (and we got away with the approximations made in our inference methods) we will have automatically implemented just the right degree of regularization for our problem; indeed, specially catered to the one data set we happened to observe, because we express everything in terms of posterior probabilities that are conditioned on the observed data.

[28] And closer to the historical origin of the epithet "mal posée" (Jaynes, 1984).

[29] Ridge regression (also known as ℓ^2 regularization) can be derived from Bayesian theory as standard linear regression with Gaussian priors on the amplitudes (Murphy, 2022, Sec. 11.3). This principled view allows us to go much further than numerical stabilization. We will show in the next chapters that ridge regression can (approximately) express intricate pieces of prior information such as definite signal polarity, the differentiability class to which a function belongs, and power constraints on waveforms.

For simple inverse problems where intuition can see the answer clearly, Bayesian regularization leads to answers that comply with common sense (MacKay, 2005). This is simply because Bayesian probability theory is a formalized theory of plausible reasoning itself (Skilling, 2005), which humans undertake every day. But regularization in the Bayesian framework can go beyond our intuition: it will of course continue to work for complex problems that require considerable analysis, and this consistency makes it a valuable tool for attacking inverse problems (Calvetti & Somersalo, 2018).

Appendix C. More comments on the nonparametric glottal flow model

Some aspects of the nonparametric GFM derived in Chapter 4 are discussed in more detail here.

Infinite precision? In the GP view, increasing H increases the *Monte Carlo resolution* with which the finite neural network model samples the arc cosine kernel — the deviation from the limiting kernel vanishes as $\mathcal{O}\left(1/\sqrt{H}\right)$ by (116). The nonparametric limit replaces explicit random changepoints with their continuous Gaussian measure, and the prior over $u'(t)$ becomes a GP whose covariance is the arc cosine kernel restricted to the open phase.

There is a caveat though. We gained full rank — the prior now has support over an infinite-dimensional function space — but we lost the ability to represent true discontinuities. Arc cosine GP sample paths are continuous; they cannot jump.[30] This is the analogue of the amplitude prior constraining the ellipsoid in the finite-H case (footnote 17 of Chapter 4): there we traded a large discrete family of possible changepoint configurations for a continuous Gaussian prior that concentrates mass inside a smooth ellipsoid; here we trade a distribution over Poisson-style jump processes for a Gaussian prior that concentrates mass on smooth function classes. The best the arc cosine GP can do is approximate a sharp GCI with a steep ramp.

Whether this is a problem in practice is an empirical question. The closed-phase GCI is physiologically sharp but not infinitely so, and the DGF data from VocalTractLab are sampled at finite rate, so the distinction between "true jump" and "ramp steep enough that it spans one sample" may not be detectable.

Why this matters for the choice of modeled signal The continuity constraint has a direct implication for which signal we should model with the arc cosine GP. We model $u'(t)$ — the DGF — because the GCI appears as a sharp negative peak there, which is easier to detect and align than the integral $u(t)$. But if the arc cosine GP is continuous, then $u'(t)$ is smooth, which means $u(t)$ is differentiable — and real glottal flow does not have a differentiable closure.

One way around this is to model $u(t)$ directly with the arc cosine GP instead of $u'(t)$. Then the GP sample paths are continuous (which $u(t)$ should be) and discontinuities can be pushed into the *derivative* implicitly via the radiation factor: the combined vocal tract and radiation impulse response is modeled as the AR filter, and the combined source signal becomes $u(t) * r(t)$ rather than $u'(t)$ alone. This is, at bottom, what classical LPC does — it posits $s(t) = e(t) * h(t)$ and allows $h(t)$ to absorb the radiation effect $r(t)$, since differentiation corresponds to a zero in the transfer function, not a pole, so $h(t)$ can remain all-pole. From this viewpoint the present approach generalizes LPC in exactly that way.

[30] Specifically, sample paths of a GP with a kernel of degree d are d-times continuously differentiable with probability one (Rasmussen & Williams, 2006). For $d = 0$ this gives continuous but nowhere-differentiable paths; for $d \geq 1$, paths are smoother still. In either case there are no jumps.

The alternative of obtaining $u'(t)$ implicitly — from the spectral domain or from the AR pole structure — is also possible and may allow the sharp-closure structure to be recovered without committing the GP to model it directly.

Why not go infinite depth? Increasing depth has a different character from increasing width. For a deep GP with Gaussian kernels at each layer, letting the depth diverge drives the process toward one that is independent of its inputs (Diaconis & Freedman, 1999) — a known pathology in practice, where deep GPs tend to "forget" their inputs and produce nearly constant sample paths. Finite but large depth mitigates this, but one might note that infinite width carries a vaguely analogous cost: in the GP limit, the basis functions are frozen at their prior expectation and no longer adapt to the data as drawn features. What is learned is entirely encoded in the kernel hyperparameters. This makes kernel learning — and in particular the multiple-kernel structure of IKLP — the mechanism through which the model does acquire data-driven spectral features, rather than learning them through random feature adaptation.

Why not spline models? Piecewise spline models are well-studied and effective in low-dimensional nonparametric regression. The issue here is resolution dependence. As GP priors, spline kernels produce posterior means that are splines with knots fixed at the observed input locations (MacKay, 1998, p. 6): add more data, add more knots. The arc cosine kernel, by contrast, learns the effective number of hingepoints from the data through its hyperparameters, and this number can remain $\mathcal{O}(1)$ even as the number of observations grows. For glottal flow this matters: a DGF waveform sampled at 16 kHz and one sampled at 44 kHz should both require $\mathcal{O}(1)$ effective changepoints to describe the open phase, not a number proportional to the sampling rate.

Why not Lévy processes? These encode Poisson-style jumps, $\mathcal{O}(1)$ in number, which is exactly the right prior for a signal with a small number of hard discontinuities. The problem is inference: the posterior over a Lévy process is indexed by the jump locations, and marginalizing those out requires MCMC or particle methods that scale with the number of jumps. The GP approach achieves fast closed-form posterior inference precisely because everything — changepoints, amplitudes, timing — has been absorbed into the kernel. Trading exact jumps for steep smooth ramps is the price of that tractability.

Appendix D. On derivative time scales

The EGIFA dataset presents us with ground-truth $u(t)$ but we need $u'(t)$ for learning. Naive finite difference of $u(t)$,

$$u'_\Delta(t) = \frac{u(t+\Delta) - u(t)}{\Delta}, \qquad \Delta = \frac{1}{f_s}, \tag{341}$$

often yields spikes in $u'(t)$ which effectively defeat kernel modeling because the increased observation noise will prevent learning fine structure. Luckily, we are saved by the physiology of the speech production process.

Glottal closure is not an instantaneous discontinuity but a short transition extending over a finite time interval. High-speed imaging and synchronized EGG measurements show that opening and closing events occupy $O(0.1)$ ms durations and may exhibit small temporal offsets depending on how closure propagates along the folds (e.g. zipper-like anterior-posterior contact), as described by Herbst et al. (2014) and Orlikoff et al. (2012). Consequently, a derivative operator that reacts to arbitrarily small sample-scale fluctuations does not reflect the underlying physiology; it instead amplifies numerical artefacts or simulator-specific microstructure. A meaningful derivative therefore requires an explicit temporal scale reflecting the duration over which closure dynamics remain physically interpretable.

Since we already model $u(t)$ by Gaussian processes,[31] we are implicitly asserting that first and second moments alone carry most of the important information. Therefore, we propose to use a *Gaussian-smoothed derivative* to measure the slope of a *locally averaged signal*

$$u'_\sigma(t) := \partial_t (G_\sigma * u)(t), \qquad G_\sigma(t) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{t^2}{\sigma^2}\right), \tag{342}$$

rather than differentiating raw samples directly.[32] This type of differentiation results in another GP, since differentiation and convolution (smearing) are closed operators in GP space.

Selecting σ such that the effective smoothing width corresponds to approximately 0.1 ms (about four to five samples at $f_s = 44.1$ kHz) aligns the operator with the expected duration of glottal closure transitions. At smaller scales the derivative is dominated by high-frequency noise and discretization effects; at larger scales the transition itself becomes blurred. The resulting derivative can therefore be interpreted as a slope measured at the physically relevant temporal resolution rather than as a raw numerical difference, and this removes the occurence of large spikes in $u'(t)$ almost completely; an example is shown in Figure 58.

[31]More precisely, we model its derivative $u'(t)$ as a GP, but the integration in (6) preserves GP-ness.

[32]Gaussian smoothing also turns out to be the unique linear kernel that preserves causality in scale-space, meaning that increasing scale removes fine structure without introducing spurious extrema (Lindeberg, 2013).

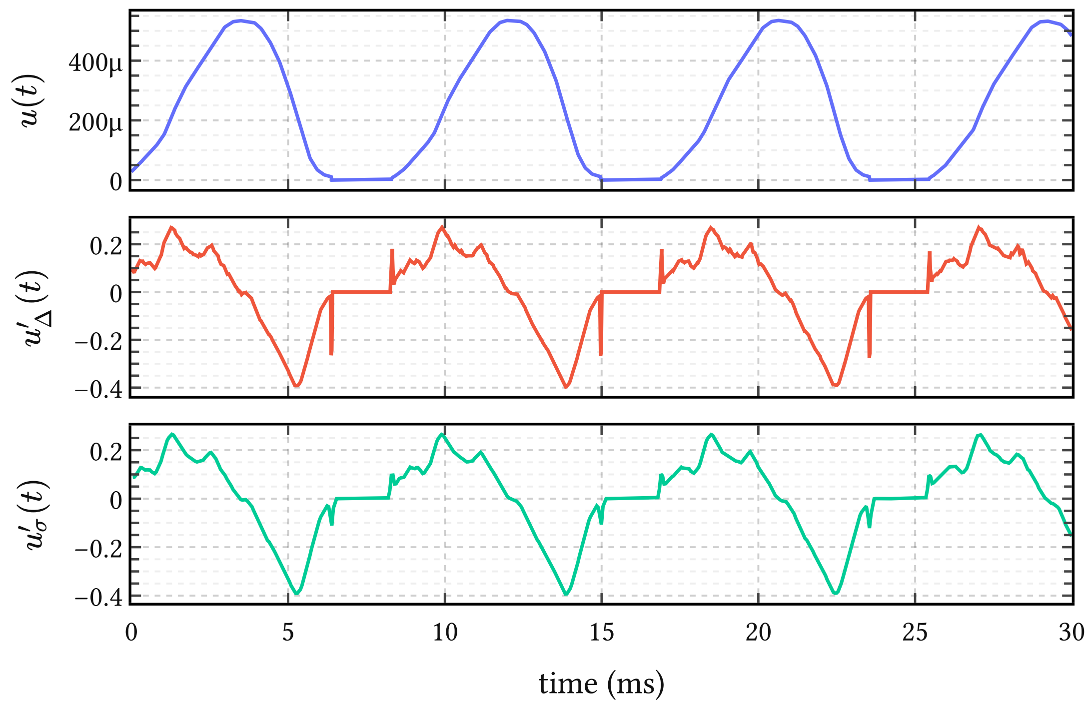


Figure 58 | **Smooth differentiation dampens spikes in $u'(t)$.** The top panel shows the ground truth glottal flow $u(t)$ extracted from a random EGIFA file. The middle panel contains its derivative $u'_\Delta(t)$ obtained by finite differences (341), and the bottom panel the Gaussian-smoothed derivative $u'_\sigma(t)$ proposed in (342). It can be seen that raw differentiation amplifies sample-scale fluctuations into large spikes, whereas the smoothed derivative retains the closure structure at the intended temporal resolution.

Appendix E. Gauge invariance in glottal inverse filtering evaluation

Comparing two DGF waveforms sample by sample is less straightforward than it sounds. The source-filter convolution $s(t) = (u'(t) * h(t)) + \varepsilon(t)$ does not uniquely determine the pair (u', h): for any nonzero α and any delay τ, the pair $(\alpha\, u'(t - \tau),;\, h(t + \tau)/\alpha)$ produces the same observed signal. Scale and time-translation are therefore *gauge freedoms* of the model. Different inverse filtering algorithms resolve these freedoms differently — LP-based methods anchor on closed-phase energy, cepstral methods anchor on spectral phase — so any evaluation that compares raw waveform representatives rather than correcting for this ambiguity will conflate algorithmic style with algorithmic error.

This appendix develops the evaluation metric used throughout the experimental chapters. The idea is simple: align each estimated frame to the ground truth by fitting an affine map and a lag jointly, then measure shape agreement by cosine similarity. The motivation, the formula, and its relationship to the EGIFA benchmark of Chien et al. (2017) are given below.

E.1. Source-filter ambiguity

The speech production model is convolutional:

$$s(t) = (u'(t) * h(t)) + \varepsilon(t), \tag{343}$$

where $u'(t)$ is the glottal flow derivative, $h(t)$ is the vocal tract impulse response including radiation, and $\varepsilon(t)$ is noise. The inverse filtering problem asks: given $s(t)$, recover $u'(t)$ and $h(t)$.

Two transformations leave $s(t)$ invariant. First, *scale*: for any nonzero α, the pair $\left(\alpha u'(t), \frac{h(t)}{\alpha}\right)$ produces exactly the same signal. Absolute source amplitude is not recoverable from acoustics alone. Second, *time-translation*: for any delay τ, the pair $(u'(t - \tau), h(t + \tau))$ again convolves to the same $s(t)$. Together these form the equivalence class

$$[(u', h)] = \{(\alpha\, u'(t - \tau),;\, h(t + \tau)/\alpha) \mid \alpha \in \mathbb{R} \setminus \{0\}, \tau \in \mathbb{R}\}. \tag{344}$$

Inverse filtering selects one representative from this class, and different algorithms make different selections.

A separate physical effect compounds the ambiguity. The vocal tract is a resonant system whose group delay $\tau_g(\omega) = -\left(\frac{\mathrm{d}}{\mathrm{d}\omega}\right) \arg H\left(e^{j\omega}\right)$ varies with frequency: narrow formants store and release energy over extended time windows, smearing the temporal relationship between excitation and observed pressure. This smearing varies across vowels — close rounded vowels with narrow F_1 bandwidths are the most challenging (Chien et al., 2017) — and no single "correct" delay compensates it uniformly.

E.2. Frame-level affine alignment

The EGIFA benchmark (Chien et al., 2017) accounts for propagation delay with a fixed 0.65 ms shift, motivated by an assumed 22 cm radiation distance, and evaluates in a per-cycle fashion using GCI-delimited segments. The fixed shift is a reasonable first-order correction, but it applies one number to a phenomenon that varies with vowel, pitch, and voice quality. Per-cycle alignment introduces a reference frame that the generative model does not privilege: the filter $h(t)$ is estimated from a windowed segment spanning many cycles, so the effective gauge it selects is a frame-level property, not a cycle-level one.

We therefore align at frame level. Each estimated frame is aligned to the corresponding ground-truth frame by jointly fitting an affine map and a nonnegative integer lag:

$$(a^*, b^*, \tau^*) = \underset{a,b,\tau}{\operatorname{argmin}} \| \mathbf{x}_{\text{est}[\tau:]} \cdot a + b - \mathbf{x}_{\text{true}[:n-\tau]} \|^2, \tag{345}$$

where $\tau \in \{0, 1, ..., \tau_{\text{max}}\}$ is a nonnegative integer lag measured in samples.[33] The scalar a absorbs scale and polarity; the scalar b absorbs any mean offset from pre-emphasis or measurement convention; the lag τ handles the frame-level delay that varies across algorithms and physical conditions.

The aligned estimate is

$$\tilde{\mathbf{x}}_{\text{est}} = a^* \mathbf{x}_{\text{est}[\tau^*:]} + b^*, \tag{346}$$

defined on the overlap $[: n - \tau^*]$. Performance is then measured by centered cosine similarity:

$$\text{score} = \frac{\langle \tilde{\mathbf{x}}_{\text{est}} - \overline{\tilde{\mathbf{x}}_{\text{est}}}, \mathbf{x}_{\text{true}} - \overline{\mathbf{x}}_{\text{true}} \rangle}{\| \tilde{\mathbf{x}}_{\text{est}} - \overline{\tilde{\mathbf{x}}_{\text{est}}} \| \| \mathbf{x}_{\text{true}} - \overline{\mathbf{x}}_{\text{true}}} \|). \tag{347}$$

Centering before taking the inner product removes the b degree of freedom, making the score invariant to the bias correction. After optimal affine alignment the centered cosine similarity equals $\sqrt{1 - \text{NRMSE}^2}$, since $R^2 = 1 - \text{NRMSE}^2$ and the affine slope shares the sign of the correlation. The score is bounded in $[0, 1]$ and monotonically decreasing in NRMSE, which makes it convenient for averaging across frames of different lengths.

Table 23 summarizes the gauge sources and their treatment.

E.3. Summary

The source-filter convolution has a gauge group — scale and time-translation — that makes absolute amplitude and absolute timing unobservable from acoustics alone. Different algorithms resolve this ambiguity differently, so comparing raw waveforms conflates gauge choice with signal quality. Frame-level affine alignment with a lag search ((345)), followed by centered cosine similarity ((347)), corrects for all gauge sources uniformly and is the evaluation procedure used throughout this thesis.

[33]The lag is constrained to be nonnegative, reflecting the physical direction of propagation delay: the acoustic observation always lags the source. Allowing both directions would conflate delay correction with arbitrary time-warping.

Gauge source	Origin	EGIFA treatment	This work
Scale / polarity	Source-filter non-identifiability	Orthogonal projection per cycle	Affine fit per frame (345)
Absolute delay	Source-filter non-identifiability	Fixed 0.65 ms shift	Lag search per frame (345)
Dispersive group delay	Vocal tract resonances	Fixed 0.65 ms shift (partial)	Lag search per frame
Algorithmic gauge	LP vs. cepstral estimation	Not corrected	Corrected uniformly
Mean offset	Pre-emphasis / measurement	Not corrected	Affine fit (*b* term)

Table 23 | **Gauge sources in GIF evaluation** and their treatment in the EGIFA benchmark of Chien et al. (2017) versus this work. The EGIFA fixed delay addresses one physical effect; affine alignment at frame level handles all sources uniformly.

Appendix F. Related contributions

Van Soom & de Boer (2019) Measuring formant frequency $\mathbf{F}$ and bandwidth $\mathbf{B}$ from steady-state vowel waveforms, taking into account the low-frequency effects of the DGF $u'(t)$ during the open phase of the pitch period. Inference is implemented using a maximum-a-posteriori (MAP) approximation. In BNGIF these low-frequency effects are taken into account automatically.

Van Soom & de Boer (2020) Same as the above, but now using robust nested sampling inference instead of a MAP approximation. A new heuristic derivation of the model function from source-filter theory is added. In BNGIF variational inference is the preferred inference method.

Van Soom & De Boer (2022) Theoretical derivation of a new weakly informative prior for resonance frequencies. Shown to have superior performance in predicting steady-state vowel waveforms compared to the usual priors in inferring the pole frequency $\mathbf{x}$ and bandwidth $\mathbf{y}$, while being roughly equally (un)informative.